Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-14 23:59:01
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 3
rollout_epoch_start.................... 2
rollout_epoch_end...................... 40
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_64x128/best_model.pt
input_data_path........................ data/processed_64x128/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_64x128/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

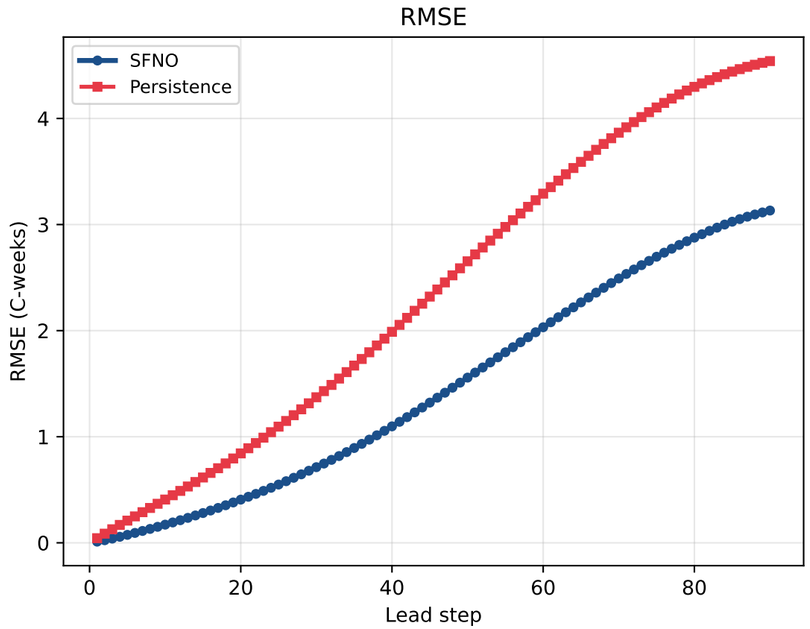
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

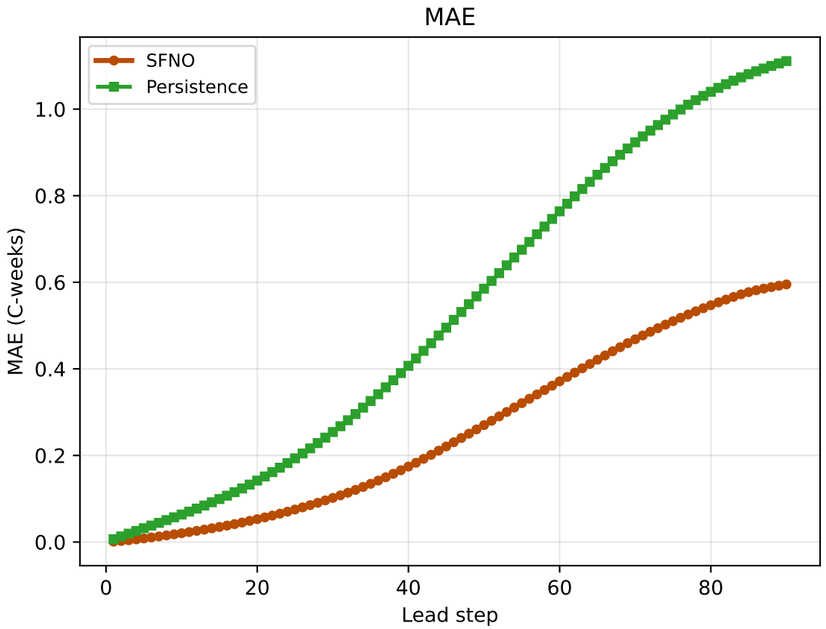
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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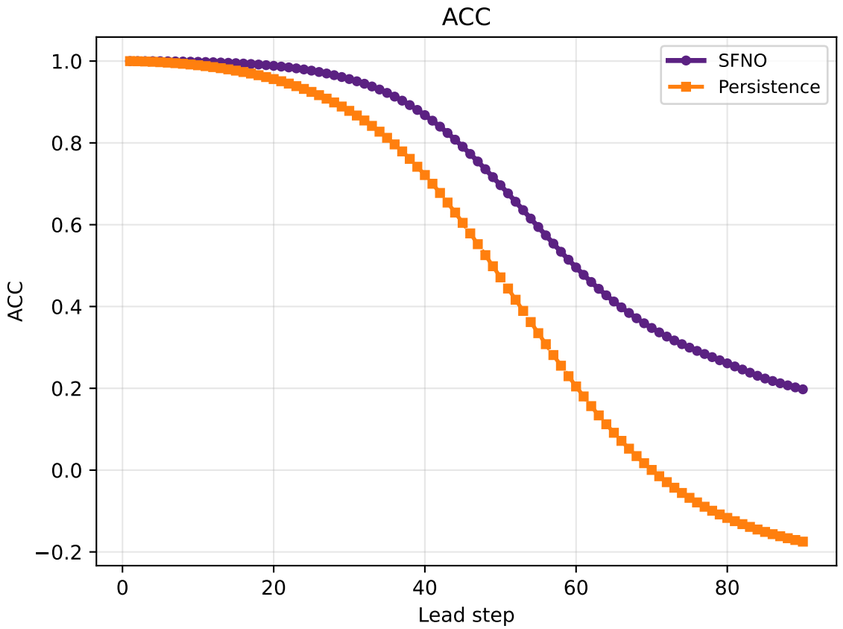
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

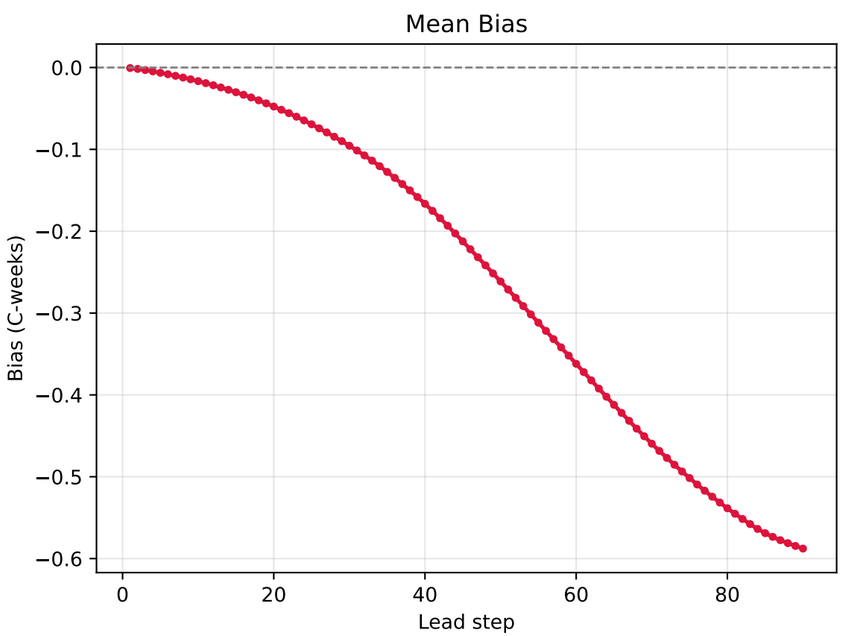
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0112	0.0437	74.3%	0.0009	0.0065	-0.0006	1.0000	0.9999	1.0000
2	0.0244	0.0862	71.7%	0.0024	0.0130	-0.0017	1.0000	0.9996	0.9999
3	0.0399	0.1278	68.8%	0.0042	0.0193	-0.0031	0.9999	0.9990	0.9998
4	0.0569	0.1687	66.3%	0.0062	0.0257	-0.0047	0.9998	0.9983	0.9997
5	0.0749	0.2091	64.2%	0.0083	0.0319	-0.0064	0.9997	0.9973	0.9994
6	0.0932	0.2491	62.6%	0.0106	0.0381	-0.0082	0.9995	0.9962	0.9991
7	0.1119	0.2888	61.3%	0.0129	0.0443	-0.0101	0.9992	0.9949	0.9987
8	0.1311	0.3285	60.1%	0.0153	0.0507	-0.0122	0.9990	0.9934	0.9982
9	0.1509	0.3680	59.0%	0.0178	0.0571	-0.0144	0.9986	0.9917	0.9976
10	0.1712	0.4079	58.0%	0.0204	0.0635	-0.0166	0.9982	0.9898	0.9969
11	0.1919	0.4484	57.2%	0.0230	0.0702	-0.0191	0.9977	0.9877	0.9961
12	0.2131	0.4893	56.4%	0.0258	0.0772	-0.0216	0.9972	0.9853	0.9952
13	0.2352	0.5310	55.7%	0.0287	0.0844	-0.0244	0.9965	0.9827	0.9941
14	0.2579	0.5733	55.0%	0.0317	0.0918	-0.0272	0.9958	0.9798	0.9928
15	0.2808	0.6158	54.4%	0.0348	0.0993	-0.0302	0.9949	0.9767	0.9914
16	0.3044	0.6590	53.8%	0.0380	0.1071	-0.0333	0.9939	0.9733	0.9898
17	0.3288	0.7031	53.2%	0.0414	0.1152	-0.0366	0.9928	0.9695	0.9879
18	0.3541	0.7484	52.7%	0.0450	0.1237	-0.0401	0.9915	0.9654	0.9858
19	0.3804	0.7951	52.2%	0.0489	0.1326	-0.0438	0.9900	0.9609	0.9834
20	0.4071	0.8429	51.7%	0.0529	0.1419	-0.0476	0.9884	0.9560	0.9807
21	0.4340	0.8913	51.3%	0.0569	0.1515	-0.0516	0.9866	0.9507	0.9778
22	0.4612	0.9406	51.0%	0.0611	0.1614	-0.0557	0.9845	0.9450	0.9746
23	0.4895	0.9912	50.6%	0.0655	0.1717	-0.0601	0.9822	0.9387	0.9709
24	0.5188	1.0427	50.2%	0.0702	0.1824	-0.0646	0.9796	0.9319	0.9668
25	0.5491	1.0951	49.9%	0.0750	0.1933	-0.0694	0.9766	0.9246	0.9622
26	0.5808	1.1486	49.4%	0.0801	0.2046	-0.0743	0.9733	0.9167	0.9571
27	0.6130	1.2029	49.0%	0.0852	0.2163	-0.0793	0.9696	0.9081	0.9514
28	0.6457	1.2585	48.7%	0.0906	0.2285	-0.0846	0.9656	0.8989	0.9452
29	0.6792	1.3152	48.4%	0.0962	0.2411	-0.0900	0.9610	0.8889	0.9383
30	0.7130	1.3721	48.0%	0.1019	0.2541	-0.0956	0.9560	0.8783	0.9309
31	0.7471	1.4298	47.7%	0.1078	0.2675	-0.1014	0.9506	0.8669	0.9229
32	0.7819	1.4883	47.5%	0.1140	0.2813	-0.1074	0.9446	0.8547	0.9141
33	0.8177	1.5482	47.2%	0.1204	0.2956	-0.1138	0.9379	0.8416	0.9045
34	0.8551	1.6090	46.9%	0.1273	0.3104	-0.1205	0.9305	0.8274	0.8939
35	0.8937	1.6707	46.5%	0.1345	0.3257	-0.1276	0.9222	0.8123	0.8823
36	0.9328	1.7328	46.2%	0.1419	0.3414	-0.1348	0.9131	0.7963	0.8697
37	0.9726	1.7960	45.8%	0.1496	0.3573	-0.1423	0.9032	0.7792	0.8562
38	1.0135	1.8599	45.5%	0.1576	0.3737	-0.1501	0.8924	0.7609	0.8416
39	1.0554	1.9244	45.2%	0.1658	0.3902	-0.1582	0.8806	0.7417	0.8260
40	1.0978	1.9890	44.8%	0.1743	0.4070	-0.1665	0.8679	0.7214	0.8094
41	1.1407	2.0544	44.5%	0.1831	0.4241	-0.1751	0.8543	0.7001	0.7920
42	1.1846	2.1205	44.1%	0.1921	0.4416	-0.1841	0.8398	0.6776	0.7737
43	1.2296	2.1870	43.8%	0.2014	0.4592	-0.1933	0.8242	0.6541	0.7545
44	1.2755	2.2540	43.4%	0.2110	0.4771	-0.2027	0.8078	0.6297	0.7346
45	1.3216	2.3209	43.1%	0.2207	0.4951	-0.2124	0.7907	0.6045	0.7142
46	1.3680	2.3878	42.7%	0.2305	0.5132	-0.2221	0.7729	0.5787	0.6935
47	1.4149	2.4546	42.4%	0.2403	0.5313	-0.2318	0.7546	0.5524	0.6724
48	1.4623	2.5213	42.0%	0.2502	0.5494	-0.2417	0.7356	0.5255	0.6509
49	1.5097	2.5876	41.7%	0.2601	0.5673	-0.2515	0.7163	0.4985	0.6294
50	1.5573	2.6535	41.3%	0.2700	0.5852	-0.2613	0.6965	0.4712	0.6078
51	1.6048	2.7188	41.0%	0.2801	0.6031	-0.2713	0.6765	0.4439	0.5863
52	1.6527	2.7840	40.6%	0.2902	0.6211	-0.2814	0.6561	0.4166	0.5649
53	1.7007	2.8487	40.3%	0.3004	0.6391	-0.2915	0.6356	0.3892	0.5436
54	1.7485	2.9131	40.0%	0.3106	0.6572	-0.3016	0.6150	0.3620	0.5226
55	1.7963	2.9772	39.7%	0.3208	0.6752	-0.3117	0.5943	0.3348	0.5020
56	1.8436	3.0409	39.4%	0.3309	0.6931	-0.3218	0.5739	0.3079	0.4819
57	1.8908	3.1042	39.1%	0.3409	0.7110	-0.3318	0.5537	0.2813	0.4622
58	1.9378	3.1668	38.8%	0.3510	0.7287	-0.3419	0.5338	0.2552	0.4432
59	1.9849	3.2292	38.5%	0.3611	0.7463	-0.3519	0.5143	0.2296	0.4247
60	2.0319	3.2912	38.3%	0.3712	0.7638	-0.3620	0.4955	0.2045	0.4070
61	2.0790	3.3528	38.0%	0.3813	0.7813	-0.3721	0.4774	0.1801	0.3901
62	2.1262	3.4139	37.7%	0.3913	0.7985	-0.3822	0.4599	0.1565	0.3739
63	2.1731	3.4741	37.4%	0.4013	0.8155	-0.3923	0.4433	0.1338	0.3585
64	2.2196	3.5329	37.2%	0.4112	0.8320	-0.4022	0.4274	0.1121	0.3440
65	2.2662	3.5910	36.9%	0.4211	0.8483	-0.4121	0.4122	0.0913	0.3302
66	2.3126	3.6481	36.6%	0.4309	0.8641	-0.4219	0.3978	0.0715	0.3171
67	2.3583	3.7045	36.3%	0.4405	0.8795	-0.4316	0.3842	0.0524	0.3047
68	2.4037	3.7596	36.1%	0.4500	0.8945	-0.4412	0.3713	0.0343	0.2930
69	2.4482	3.8136	35.8%	0.4593	0.9091	-0.4505	0.3590	0.0169	0.2819
70	2.4916	3.8659	35.5%	0.4684	0.9233	-0.4596	0.3475	0.0005	0.2715
71	2.5343	3.9166	35.3%	0.4772	0.9370	-0.4684	0.3367	-0.0149	0.2618
72	2.5761	3.9656	35.0%	0.4857	0.9503	-0.4770	0.3265	-0.0294	0.2526
73	2.6172	4.0132	34.8%	0.4940	0.9631	-0.4854	0.3170	-0.0431	0.2440
74	2.6572	4.0591	34.5%	0.5022	0.9755	-0.4936	0.3079	-0.0558	0.2358
75	2.6966	4.1036	34.3%	0.5102	0.9875	-0.5016	0.2995	-0.0678	0.2281
76	2.7350	4.1466	34.0%	0.5180	0.9991	-0.5094	0.2915	-0.0790	0.2208
77	2.7720	4.1876	33.8%	0.5255	1.0101	-0.5170	0.2839	-0.0895	0.2139
78	2.8079	4.2264	33.6%	0.5329	1.0206	-0.5244	0.2762	-0.0993	0.2070
79	2.8427	4.2630	33.3%	0.5400	1.0305	-0.5315	0.2685	-0.1084	0.2001
80	2.8765	4.2978	33.1%	0.5469	1.0400	-0.5385	0.2610	-0.1168	0.1934
81	2.9090	4.3305	32.8%	0.5536	1.0491	-0.5452	0.2535	-0.1248	0.1867
82	2.9404	4.3612	32.6%	0.5600	1.0576	-0.5516	0.2459	-0.1321	0.1800
83	2.9708	4.3898	32.3%	0.5662	1.0657	-0.5578	0.2382	-0.1388	0.1732
84	3.0000	4.4167	32.1%	0.5722	1.0733	-0.5638	0.2306	-0.1452	0.1665
85	3.0269	4.4418	31.9%	0.5772	1.0805	-0.5689	0.2239	-0.1511	0.1606
86	3.0515	4.4649	31.7%	0.5815	1.0873	-0.5734	0.2179	-0.1565	0.1553
87	3.0741	4.4865	31.5%	0.5854	1.0937	-0.5775	0.2123	-0.1616	0.1505
88	3.0947	4.5062	31.3%	0.5888	1.0998	-0.5811	0.2072	-0.1664	0.1461
89	3.1142	4.5244	31.2%	0.5920	1.1055	-0.5845	0.2023	-0.1708	0.1419
90	3.1320	4.5408	31.0%	0.5950	1.1109	-0.5878	0.1975	-0.1748	0.1379

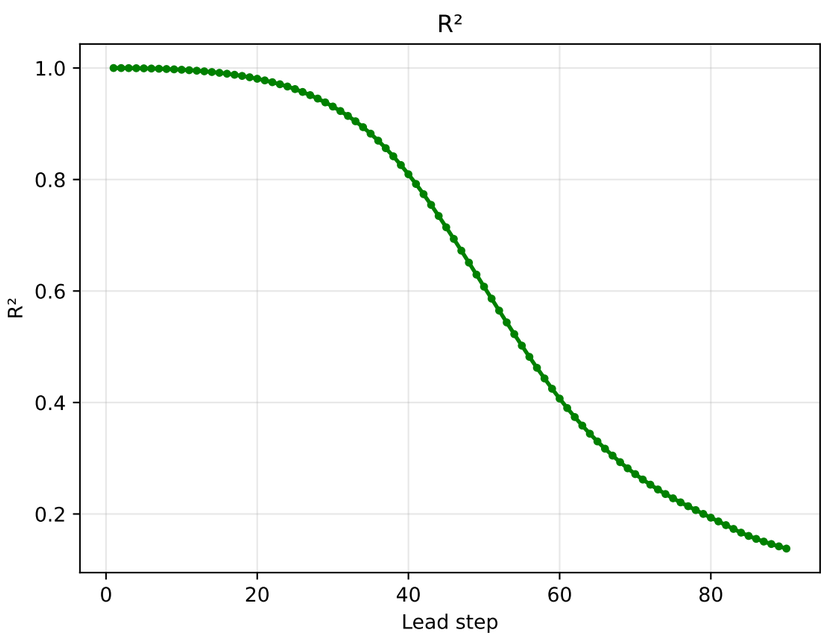
DHW Forecast Skill vs Lead Step - Test

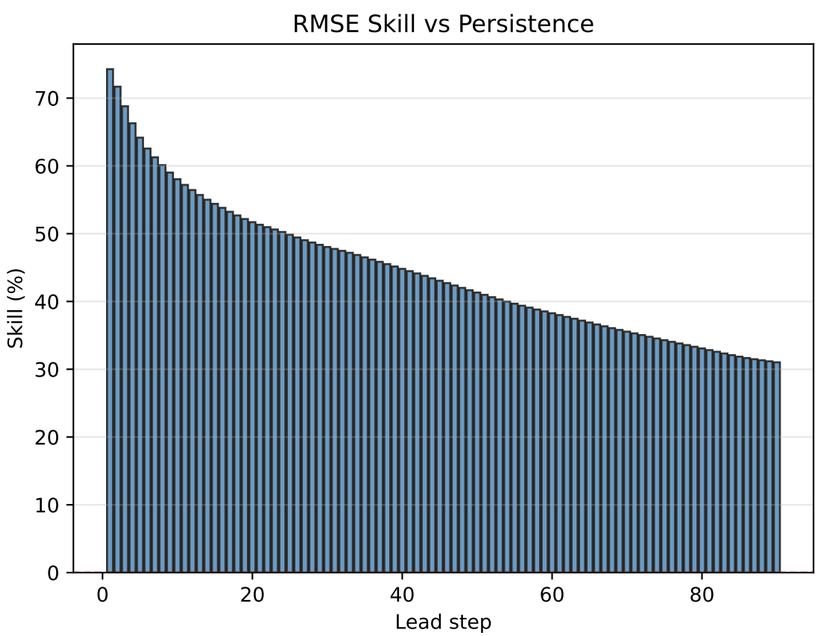


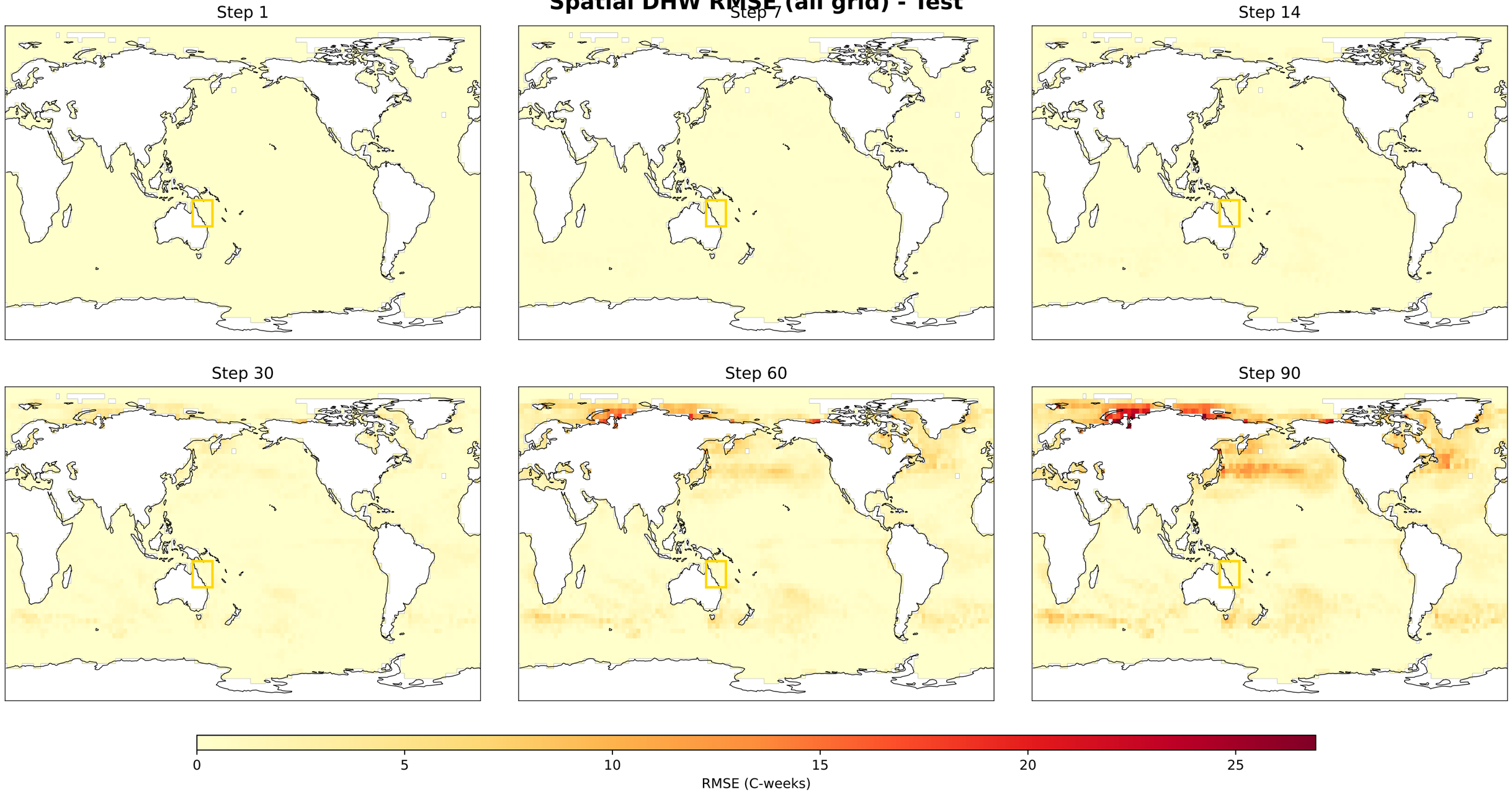


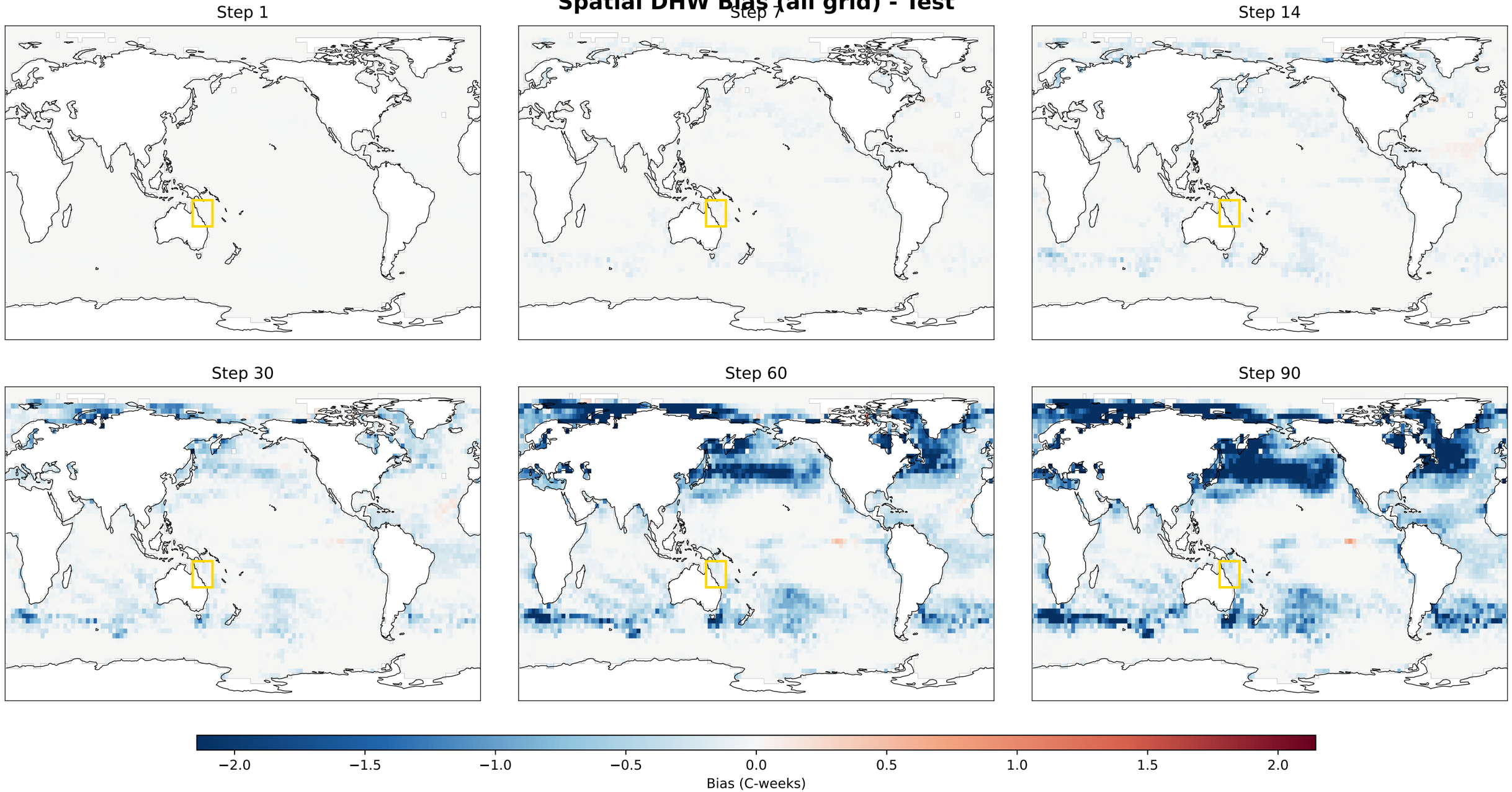


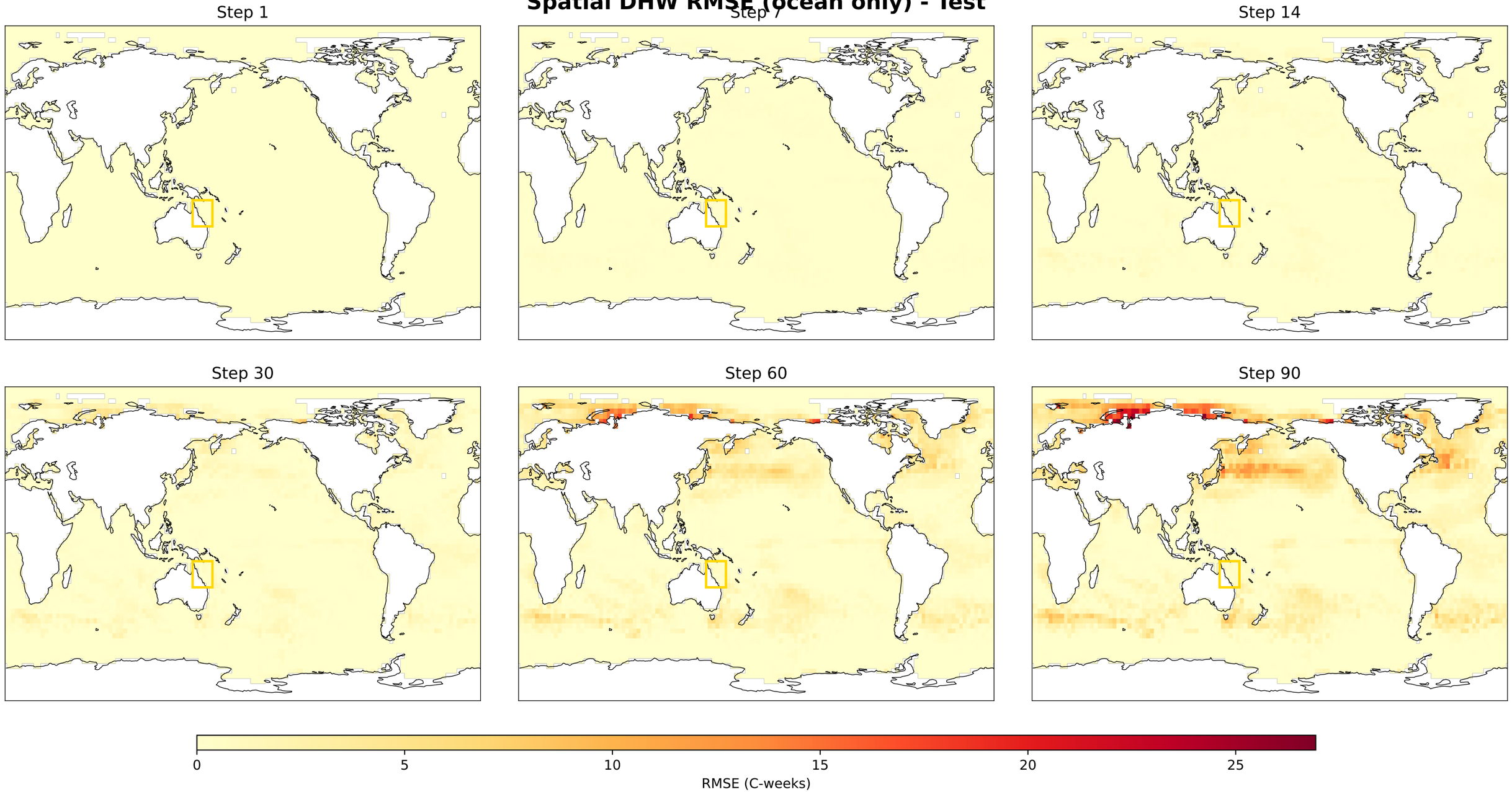


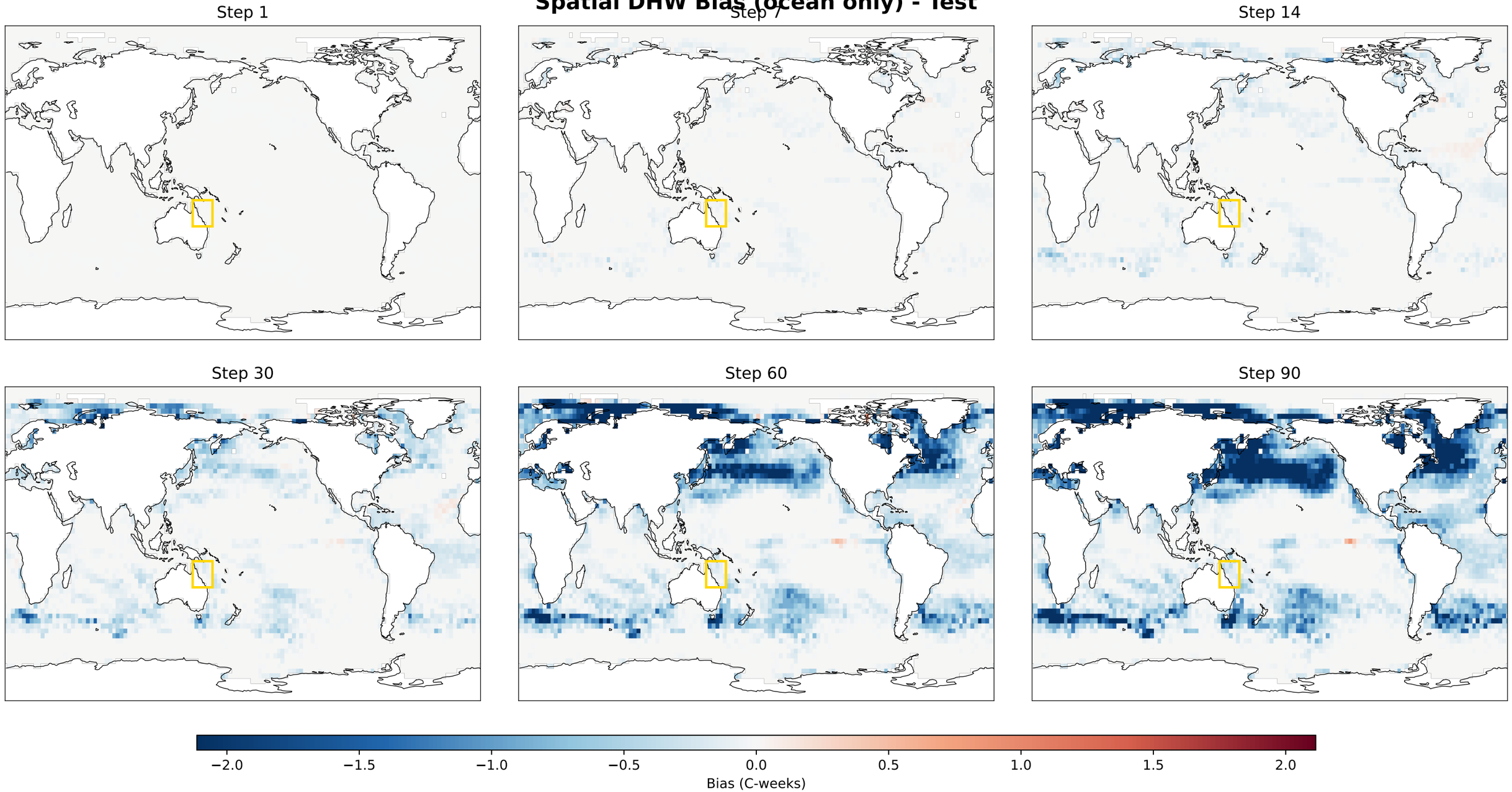





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

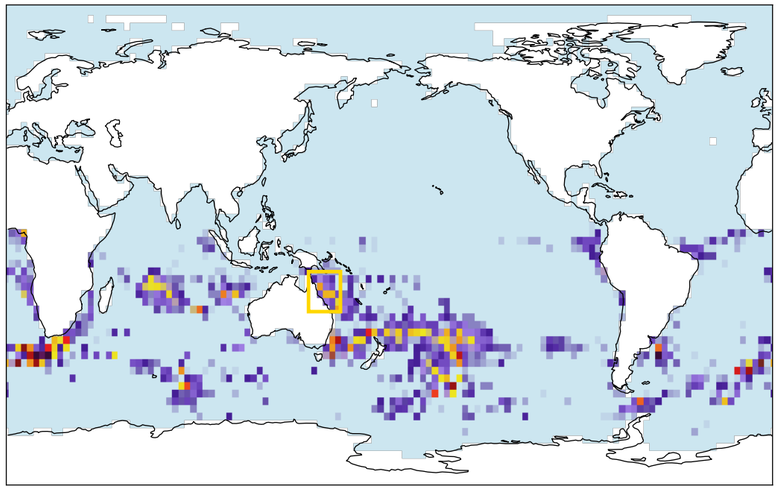

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

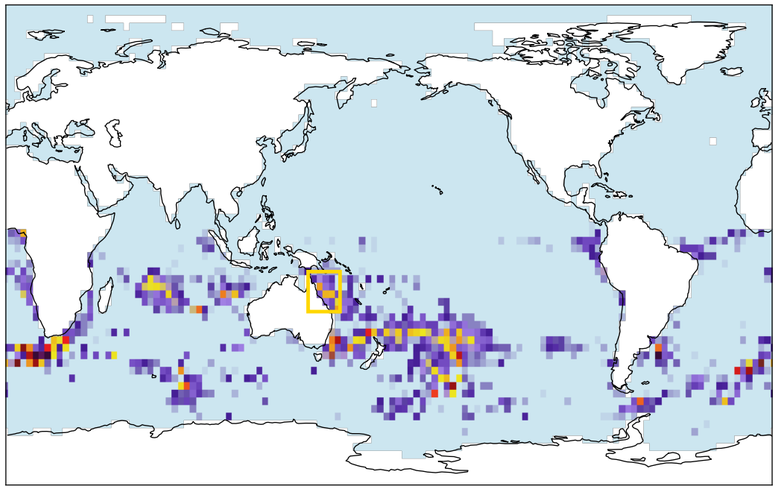

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

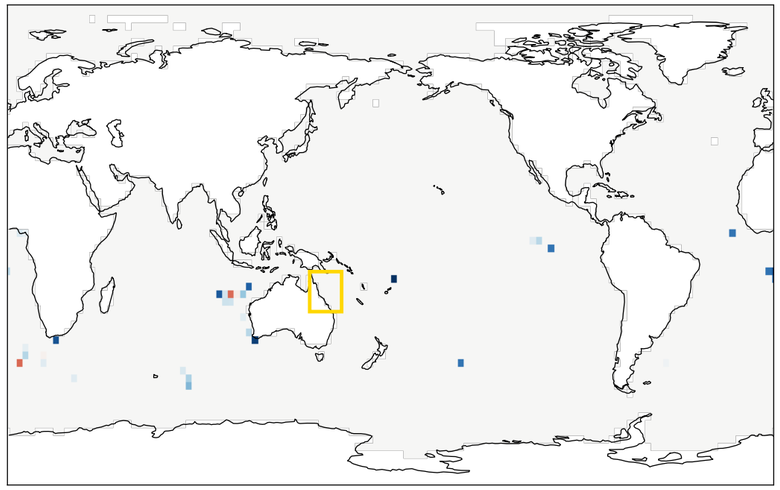
Actual DHW - Step 1



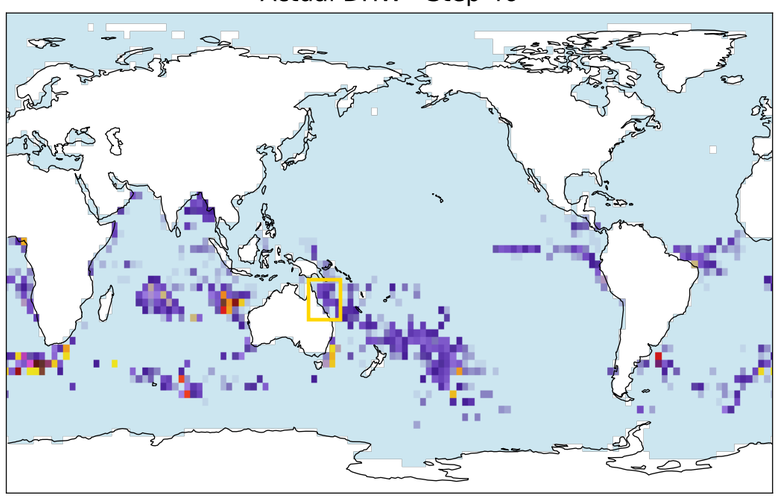
Predicted DHW - Step 1



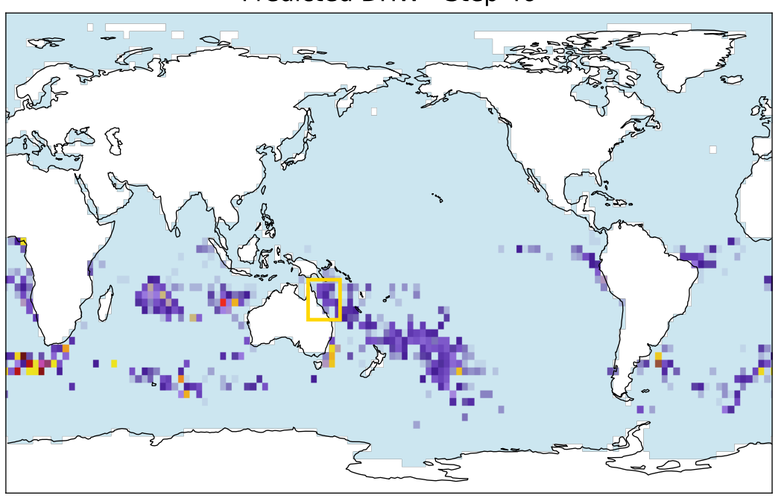
Error (Pred - Actual) - Step 1 [scale ±0.24]



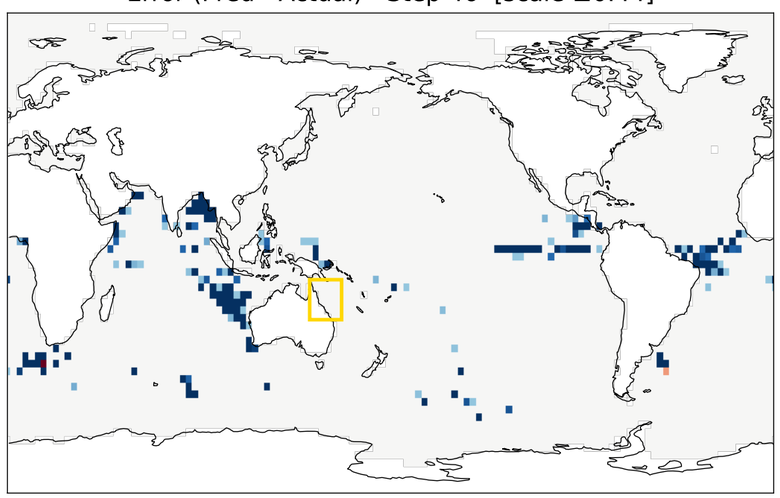
Actual DHW - Step 46



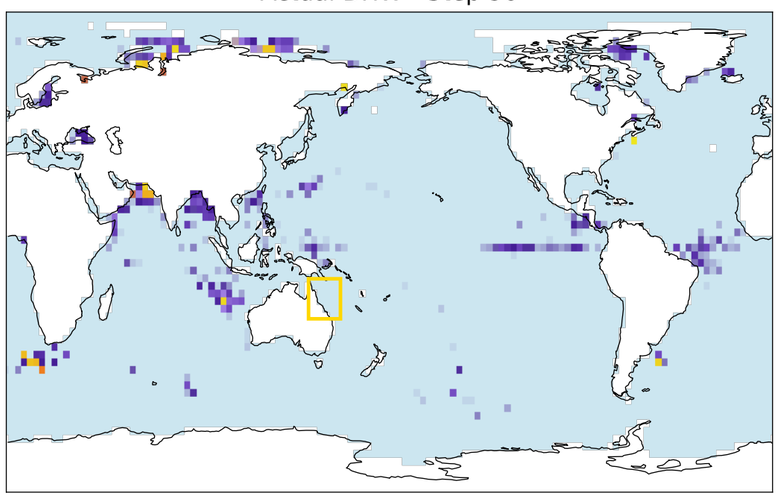
Predicted DHW - Step 46



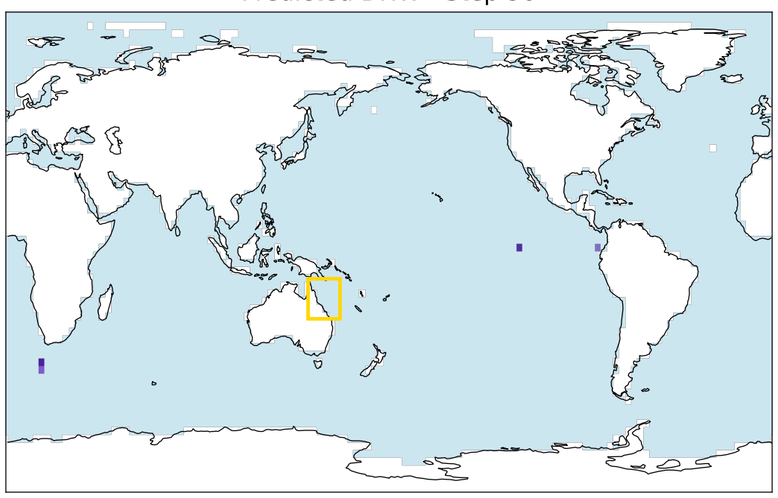
Error (Pred - Actual) - Step 46 [scale ±0.44]



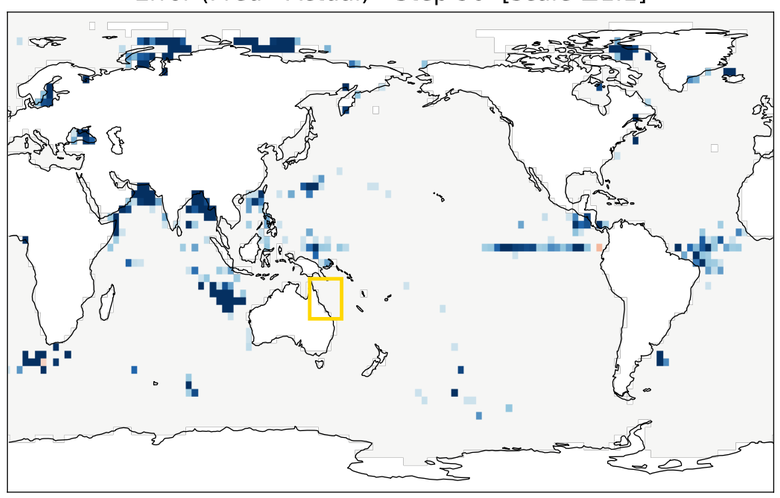
Actual DHW - Step 90

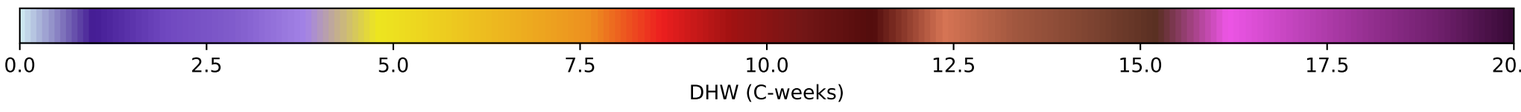


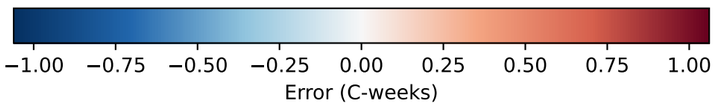
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.1]

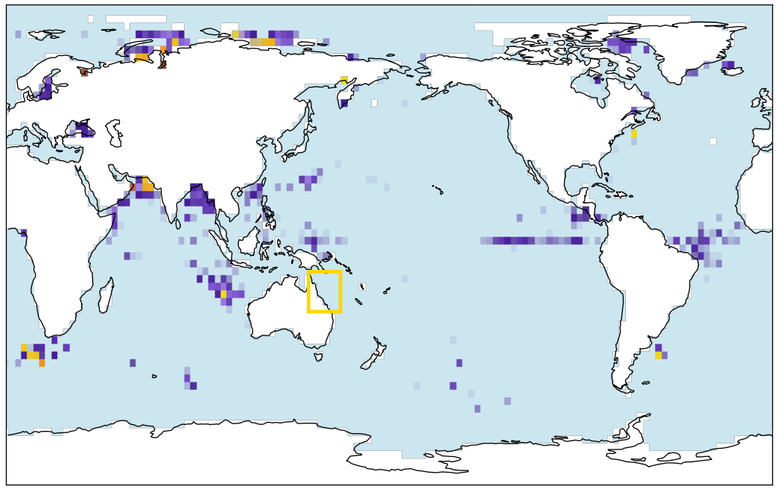




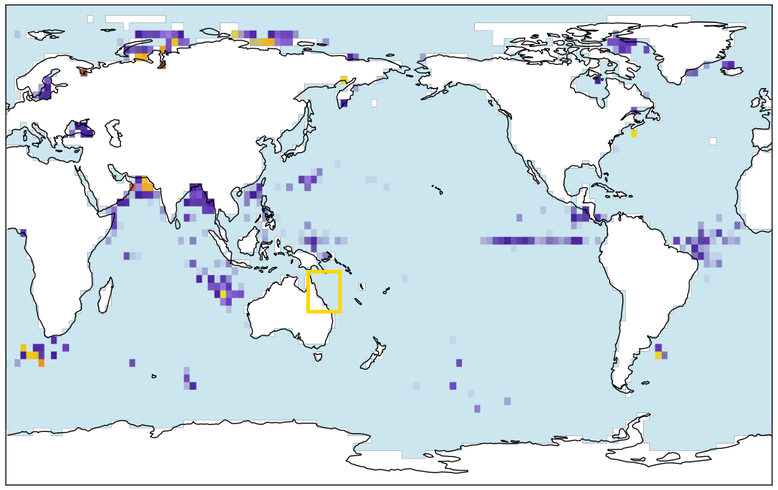


DHW Chain init: 2020-07-09 - Test (C-weeks)

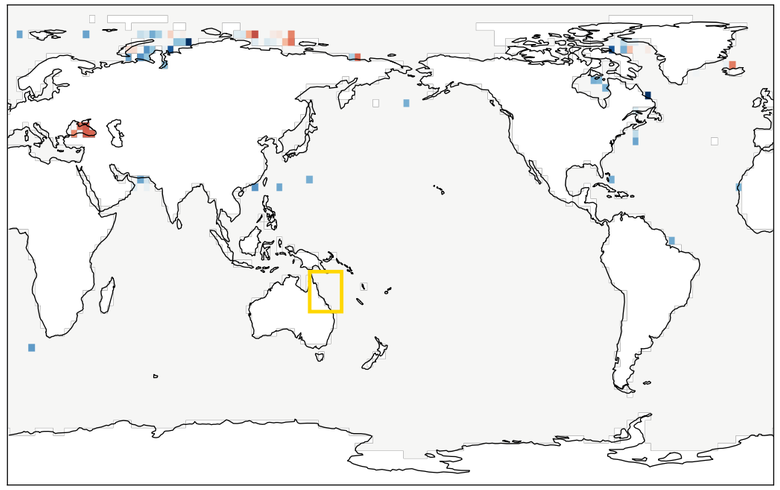
Actual DHW - Step 1



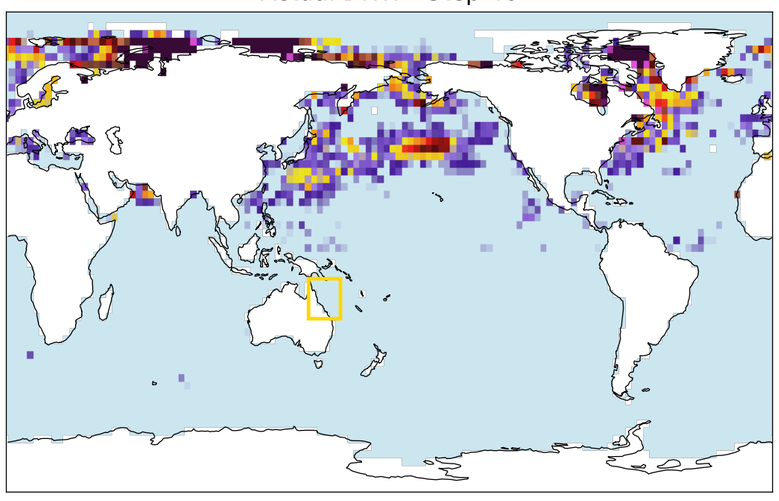
Predicted DHW - Step 1



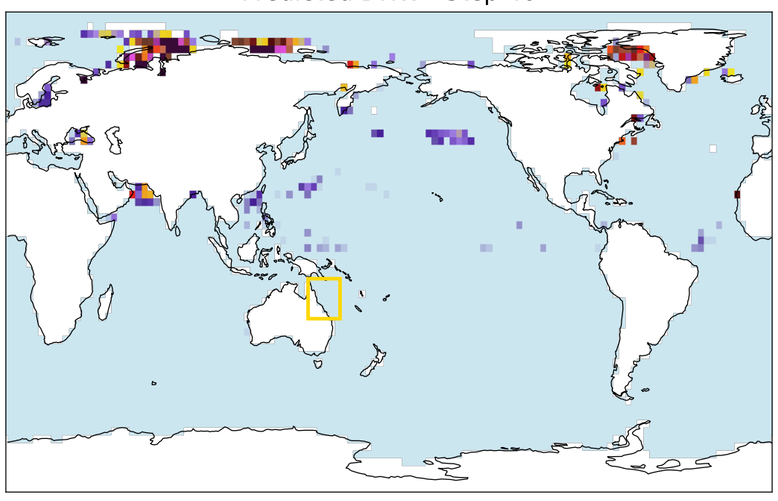
Error (Pred - Actual) - Step 1 [scale ±0.37]



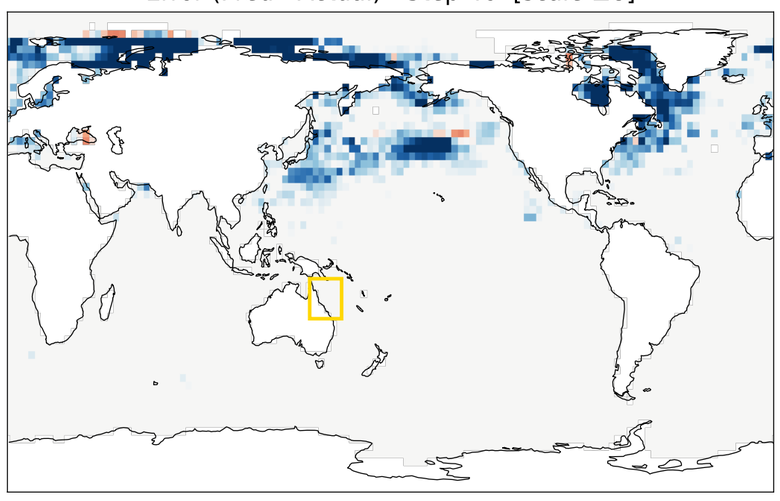
Actual DHW - Step 46



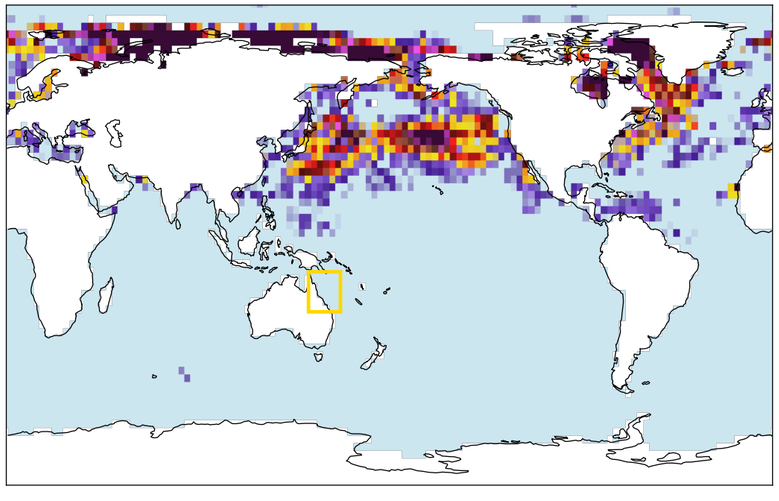
Predicted DHW - Step 46



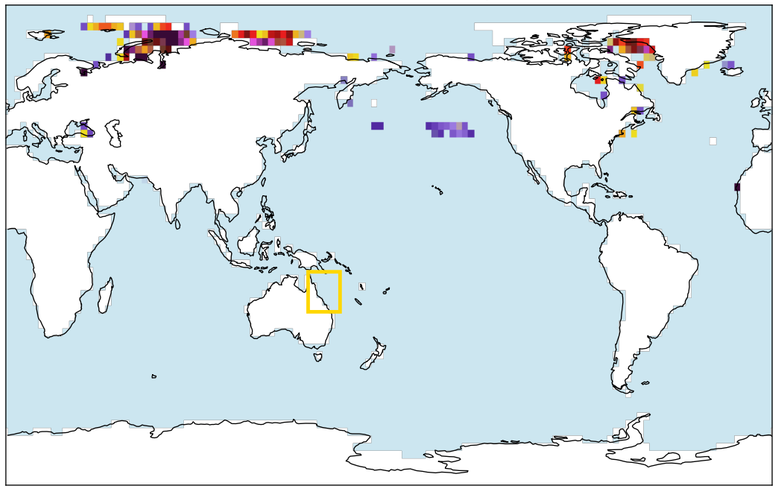
Error (Pred - Actual) - Step 46 [scale ±8]



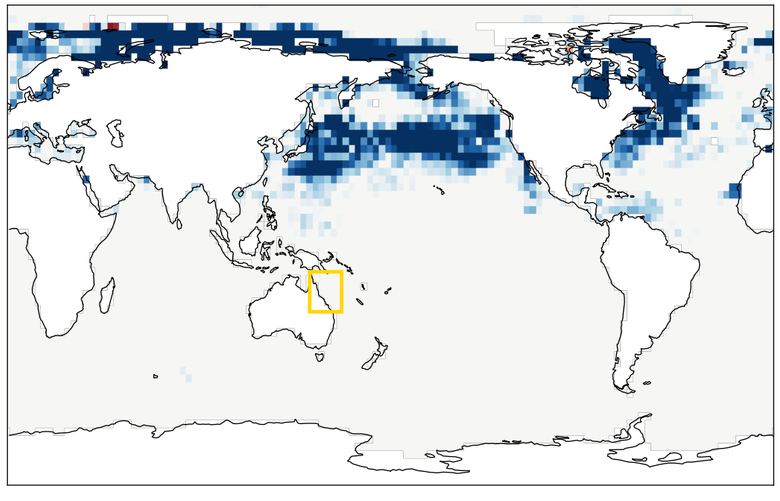
Actual DHW - Step 90

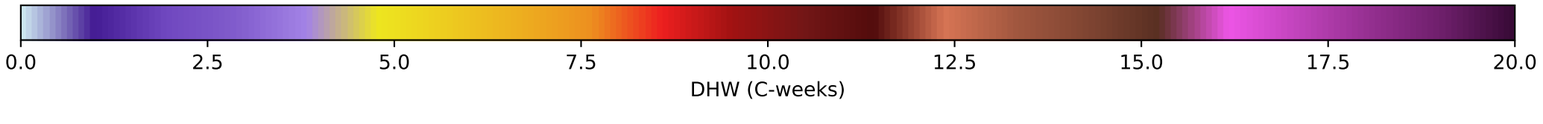


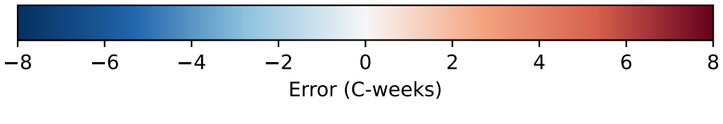
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

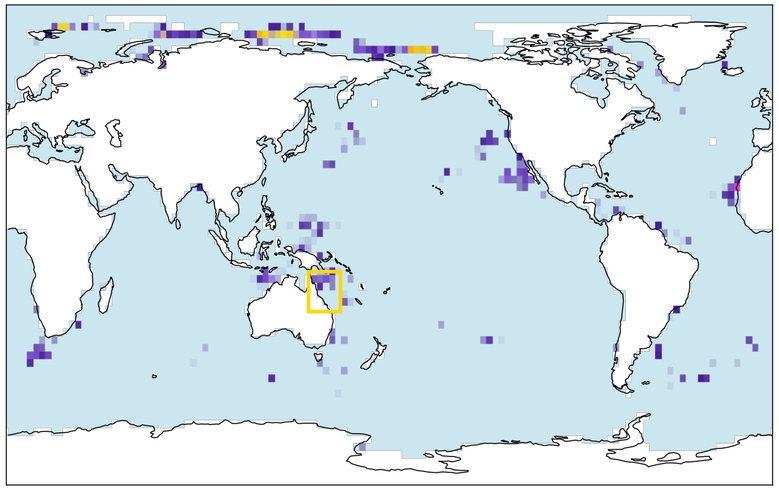




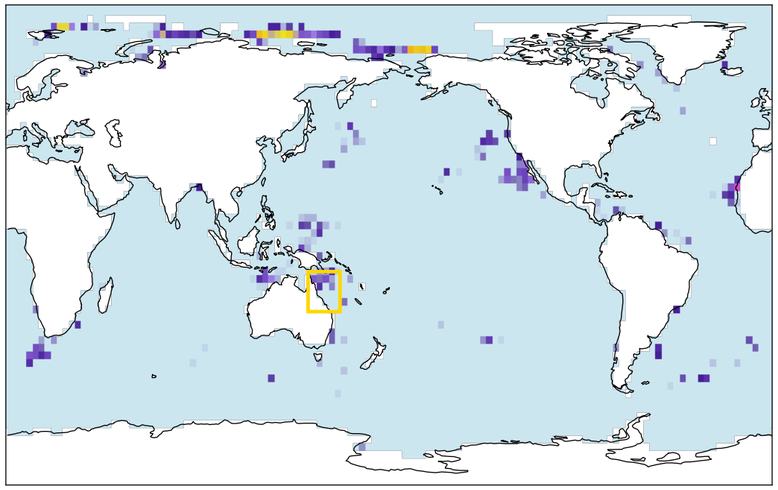


DHW Chain init: 2021-01-06 - Test (C-weeks)

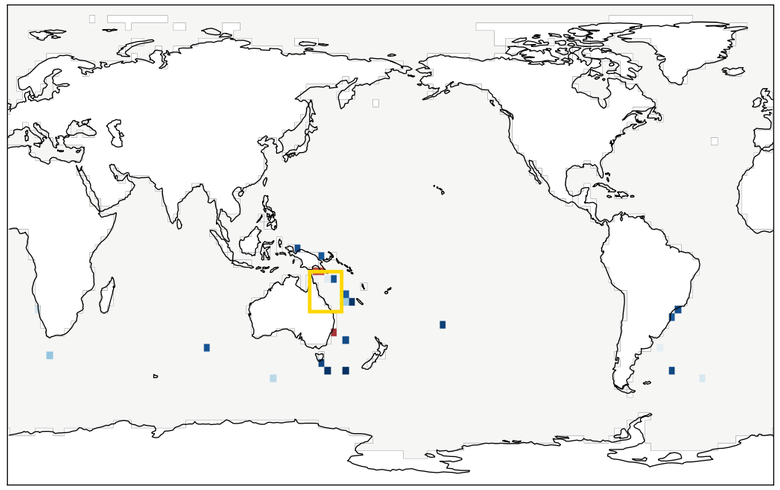
Actual DHW - Step 1



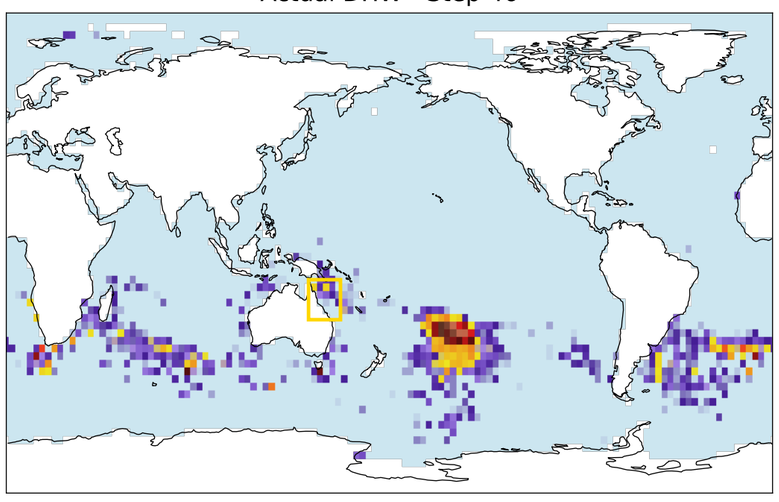
Predicted DHW - Step 1



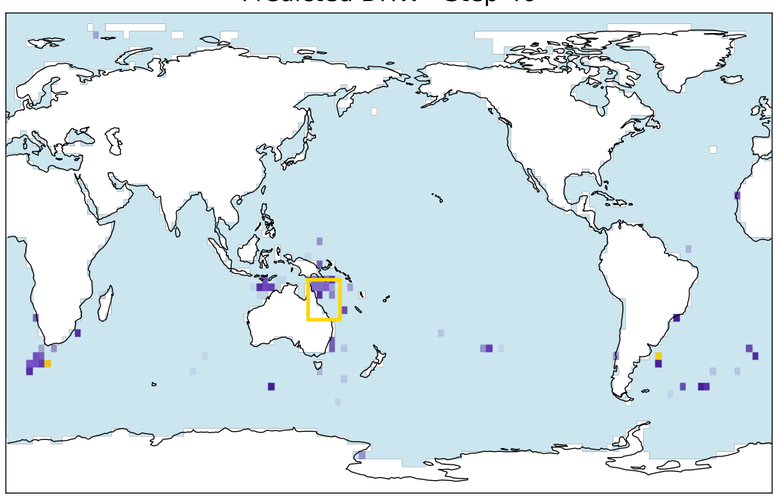
Error (Pred - Actual) - Step 1 [scale ±0.2]



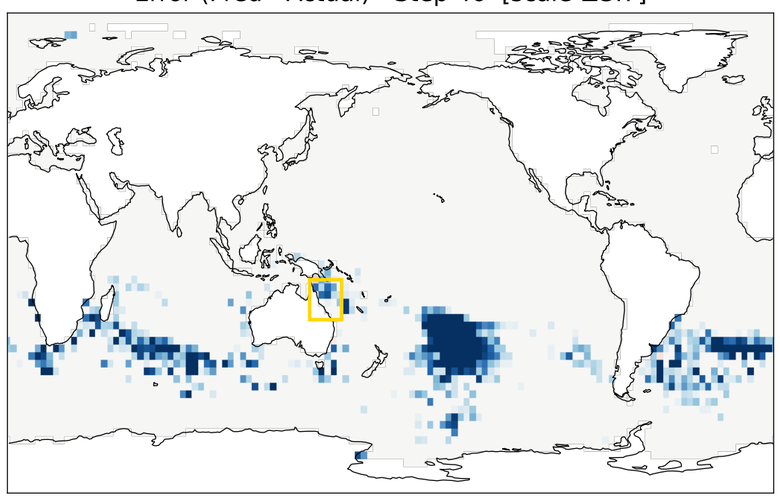
Actual DHW - Step 46



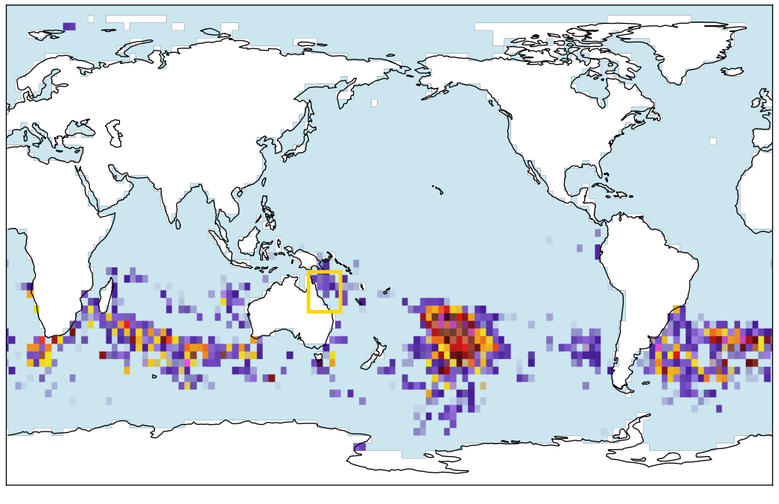
Predicted DHW - Step 46



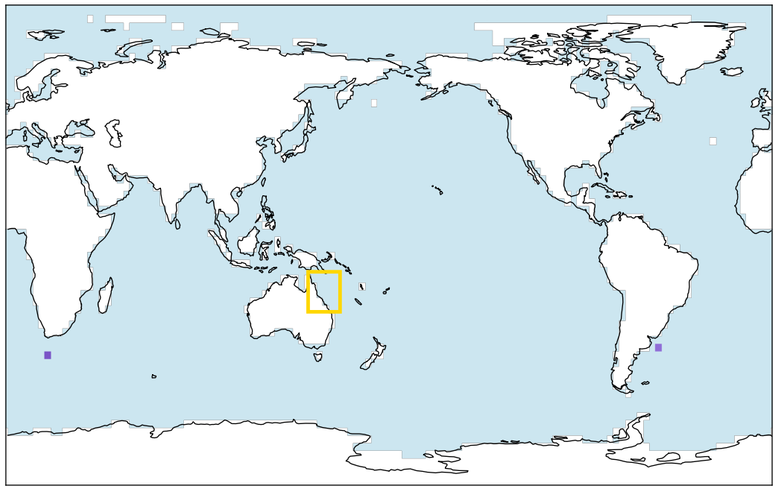
Error (Pred - Actual) - Step 46 [scale ±3.7]



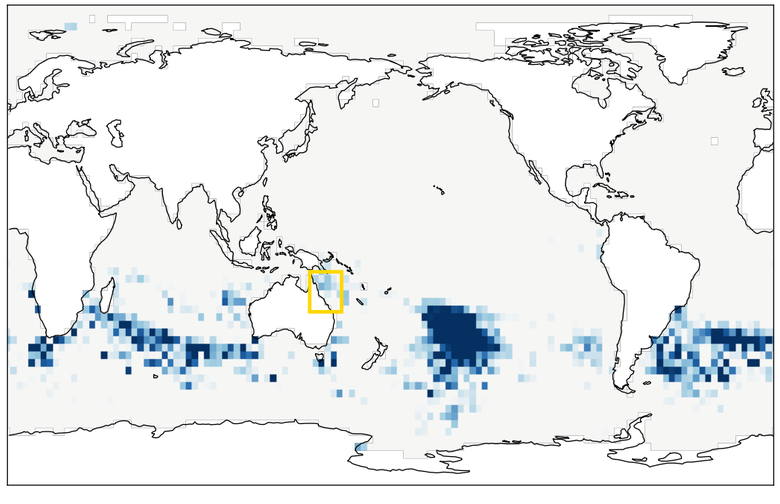
Actual DHW - Step 90

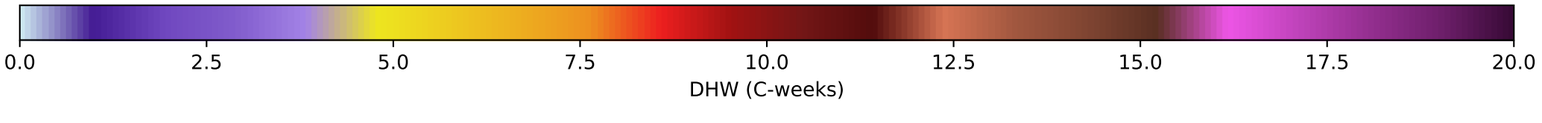


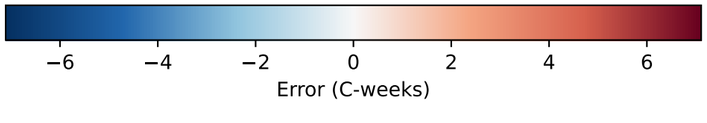
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

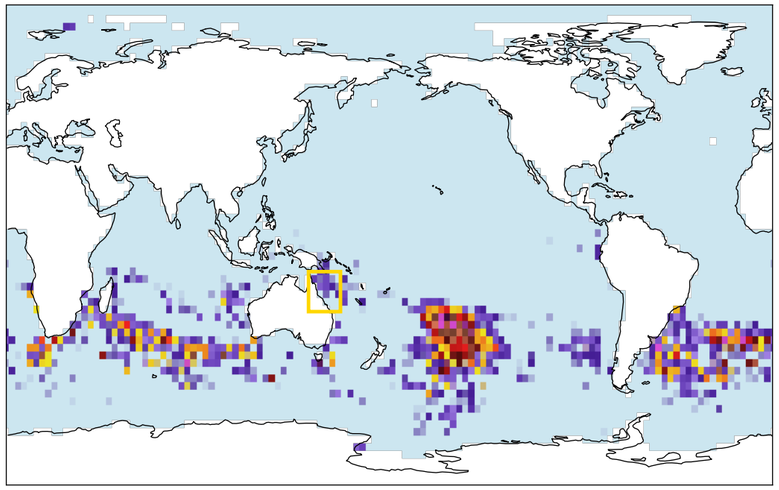




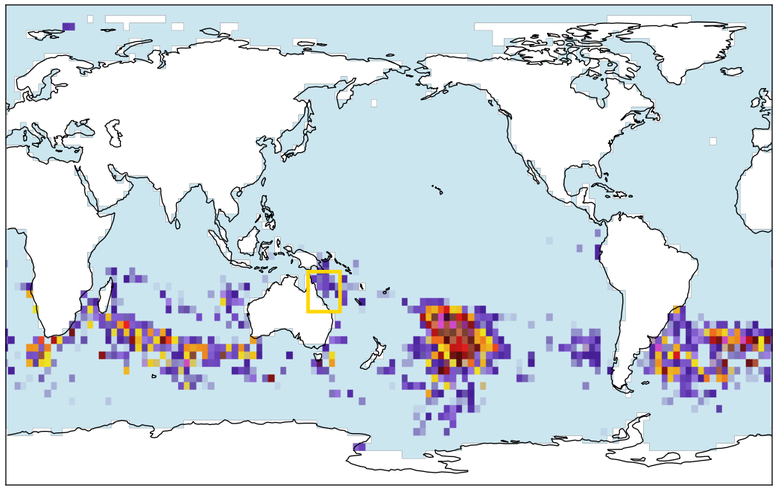


DHW Chain init: 2021-04-06 - Test (C-weeks)

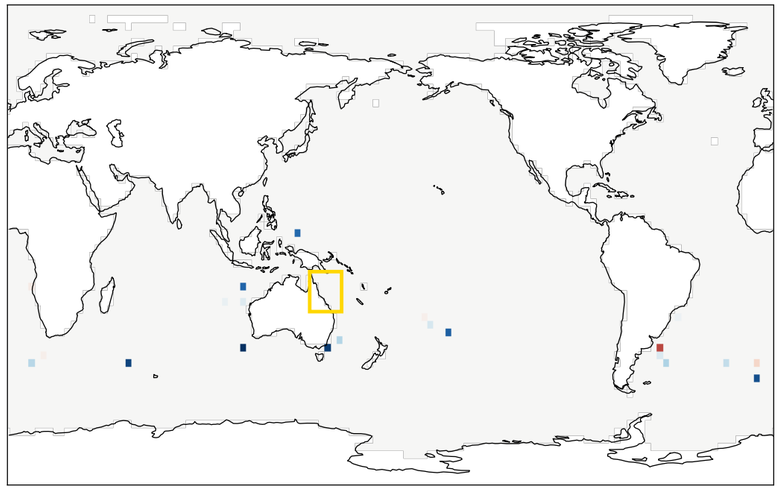
Actual DHW - Step 1



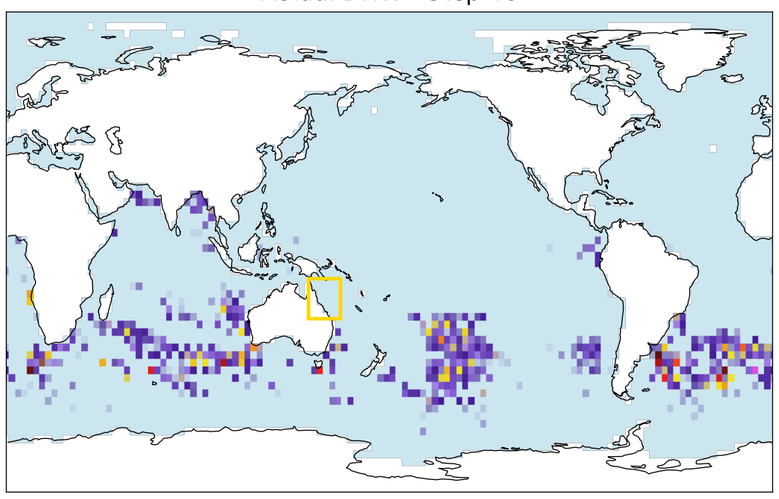
Predicted DHW - Step 1



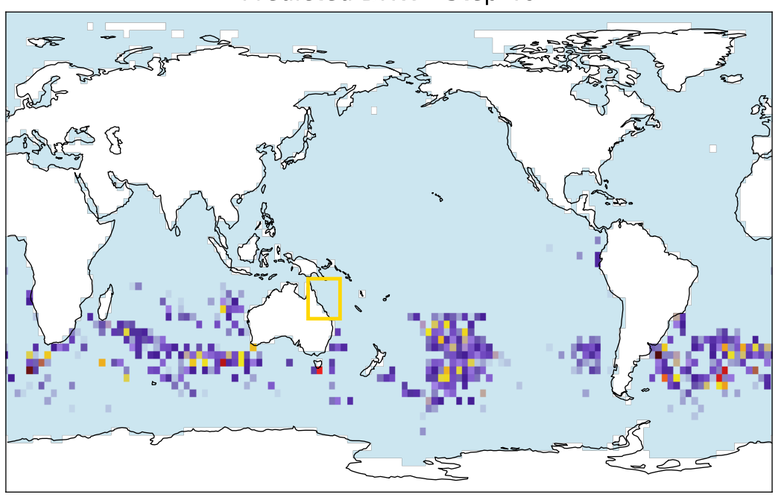
Error (Pred - Actual) - Step 1 [scale ±0.23]



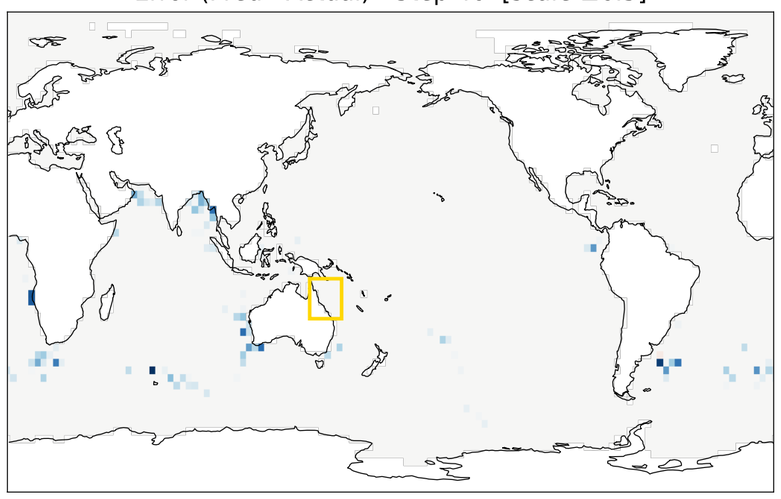
Actual DHW - Step 46



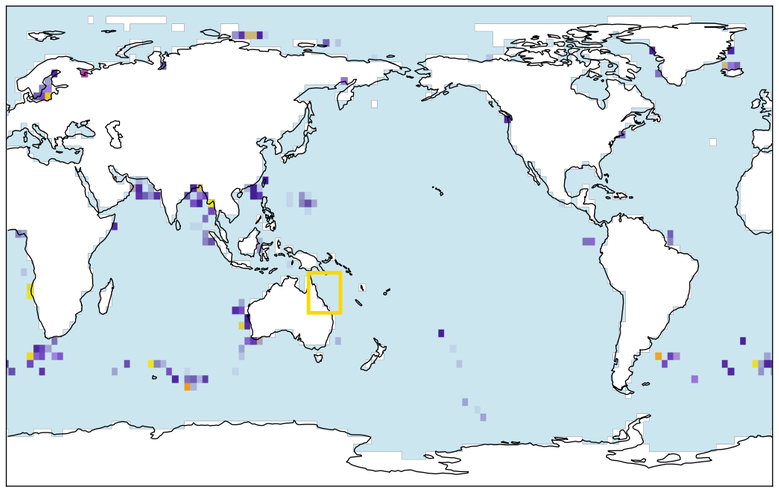
Predicted DHW - Step 46



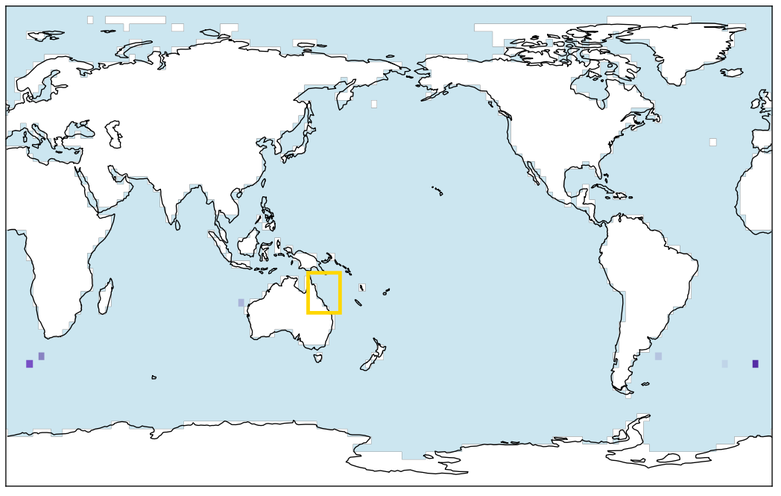
Error (Pred - Actual) - Step 46 [scale ±6.5]



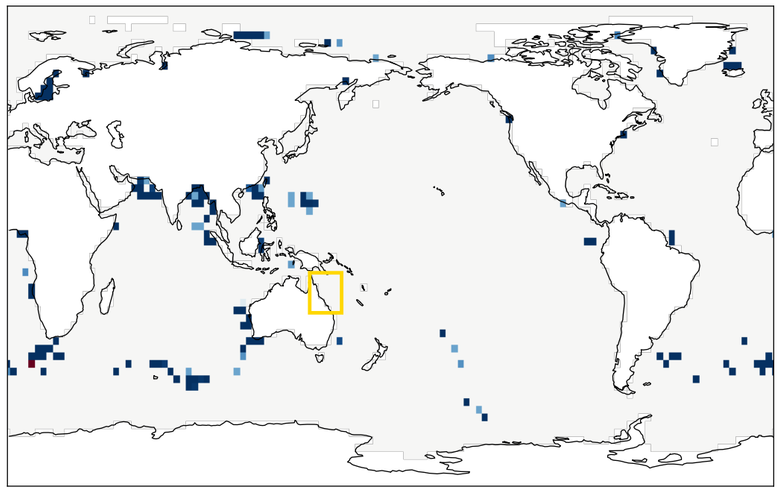
Actual DHW - Step 90

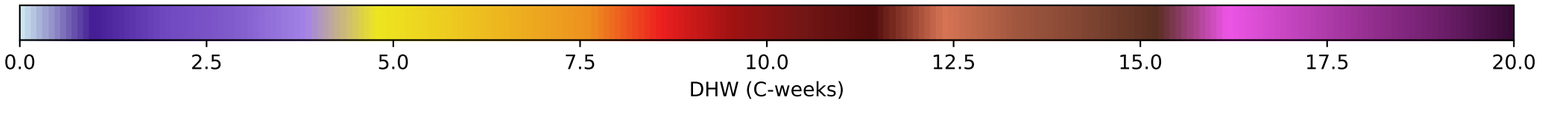


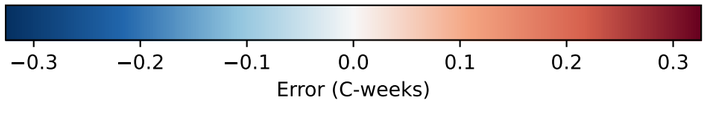
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

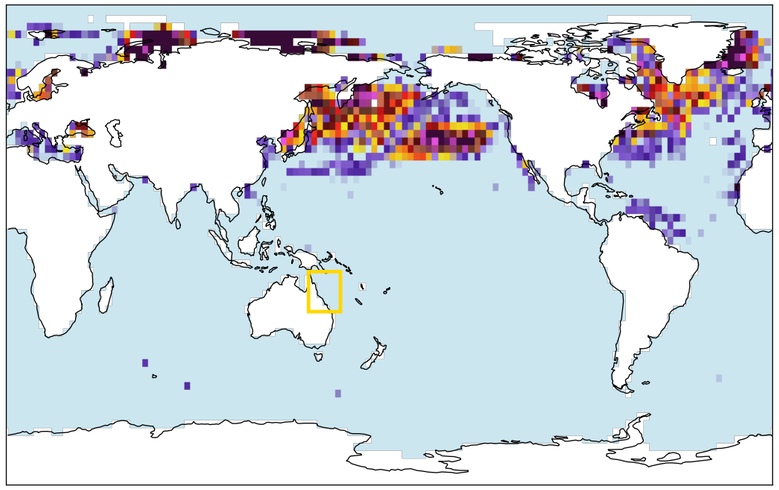




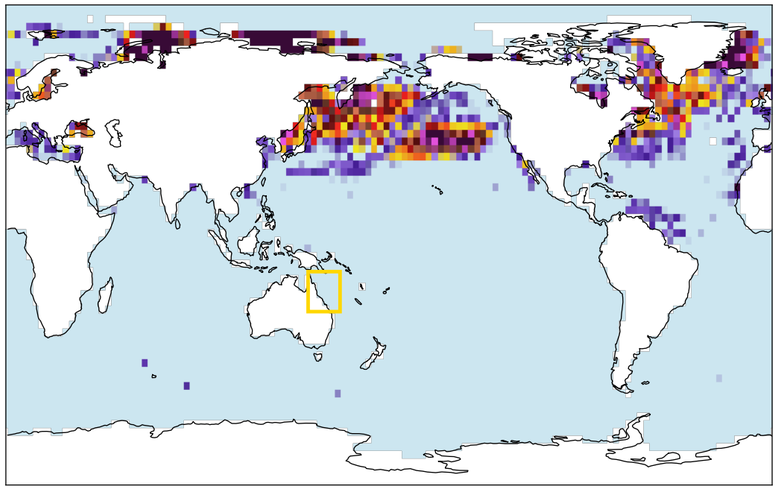


DHW Chain init: 2021-10-03 - Test (C-weeks)

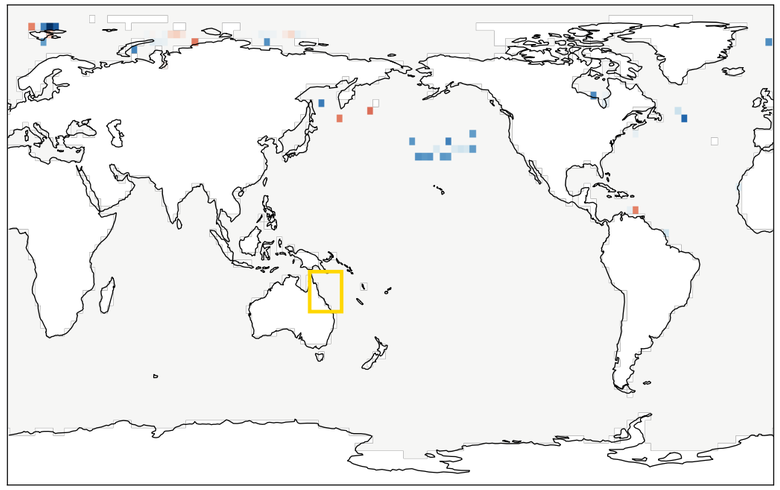
Actual DHW - Step 1



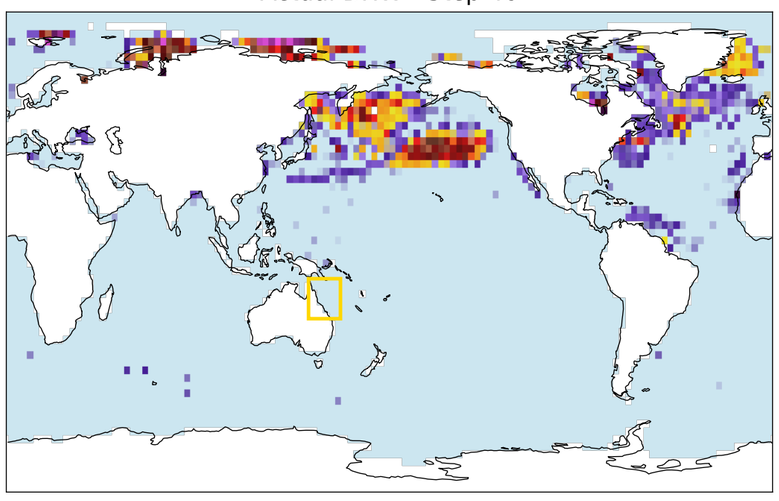
Predicted DHW - Step 1



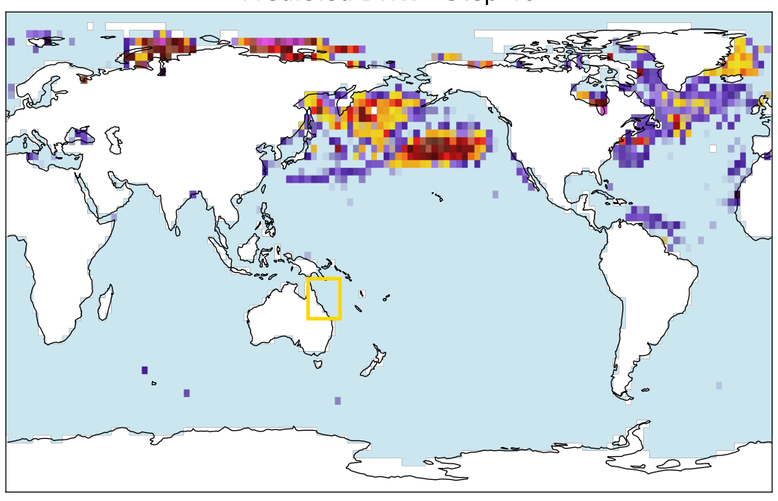
Error (Pred - Actual) - Step 1 [scale ±0.31]



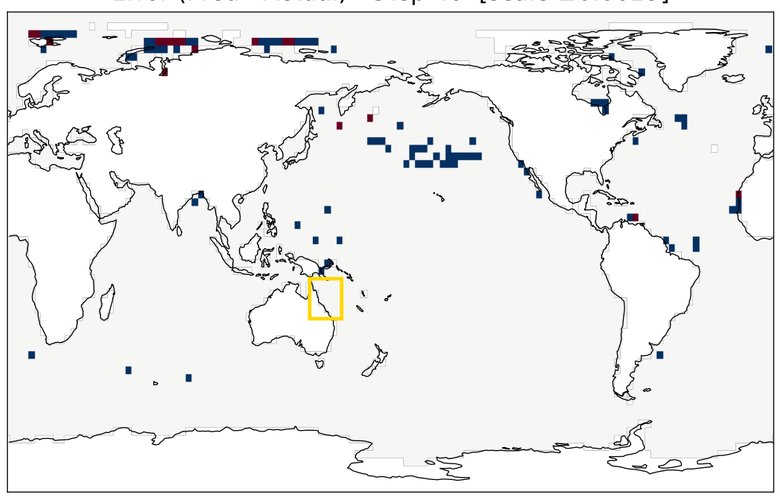
Actual DHW - Step 46



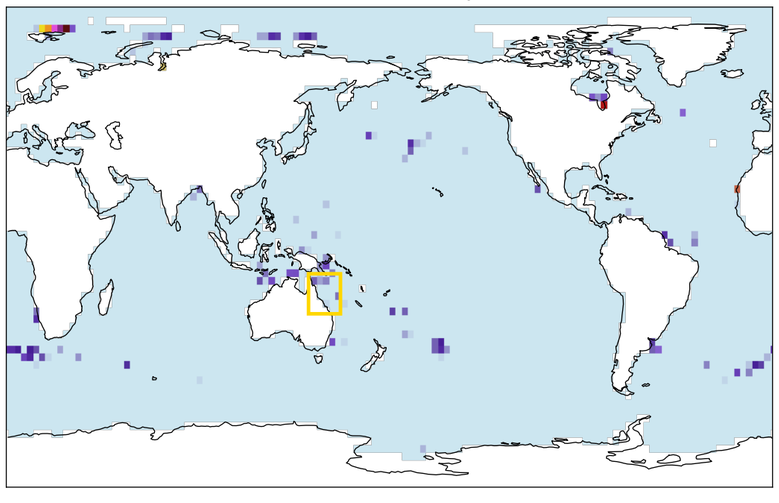
Predicted DHW - Step 46



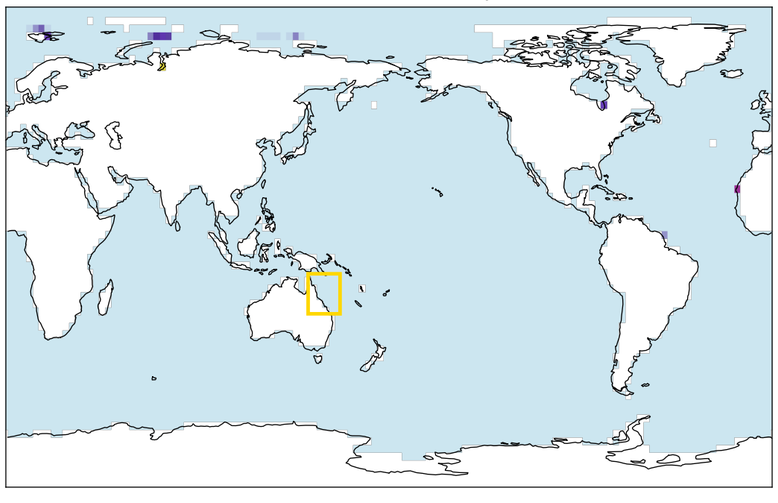
Error (Pred - Actual) - Step 46 [scale ±0.0029]



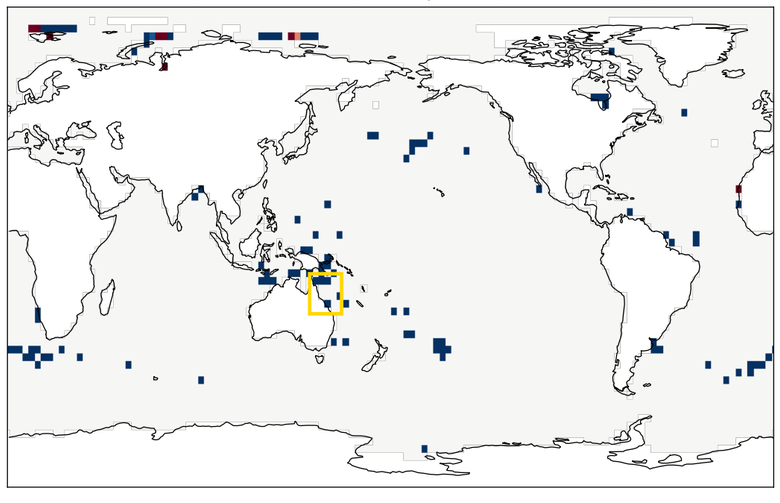
Actual DHW - Step 90

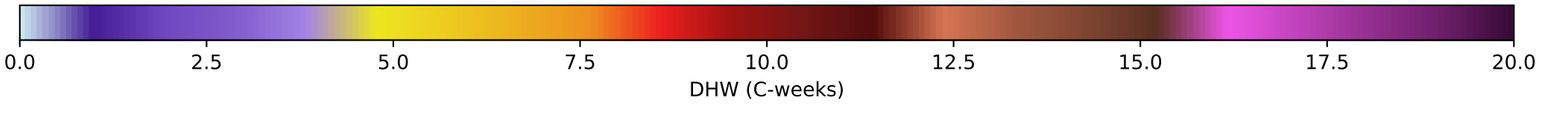


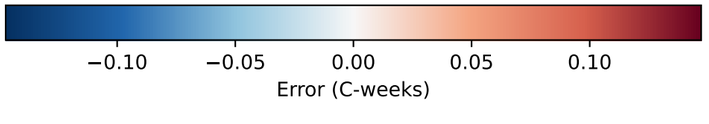
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

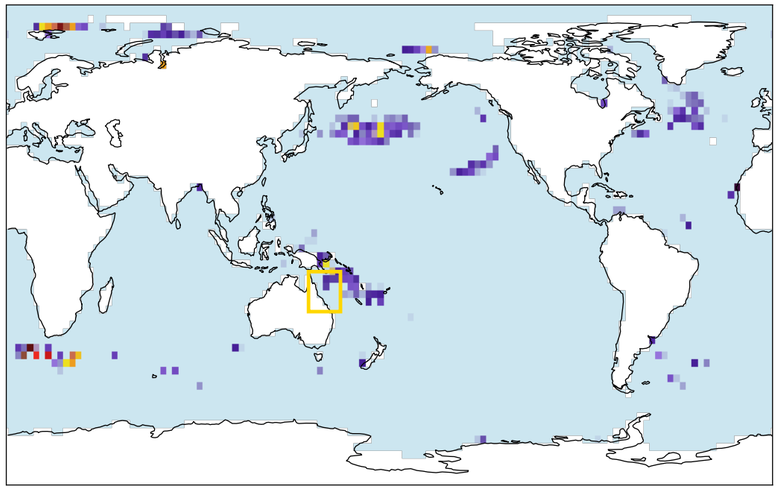




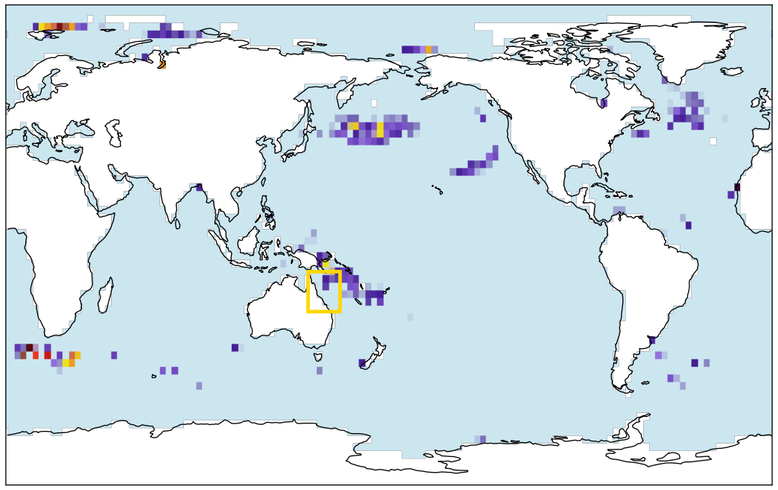


DHW Chain init: 2022-12-27 - Test (C-weeks)

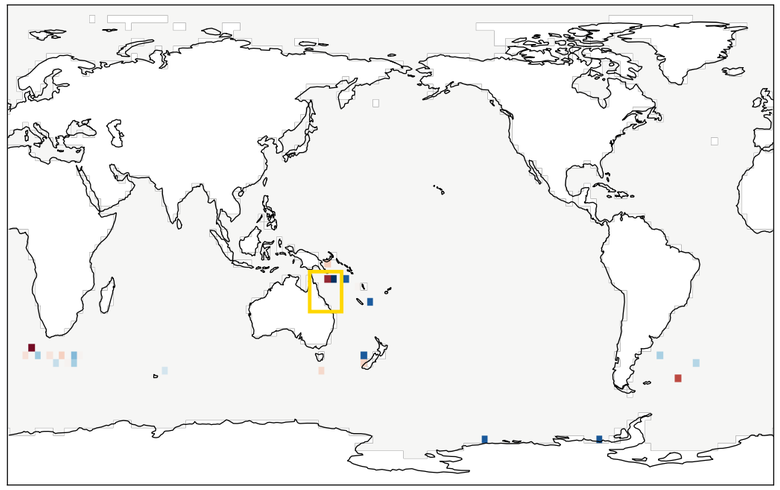
Actual DHW - Step 1



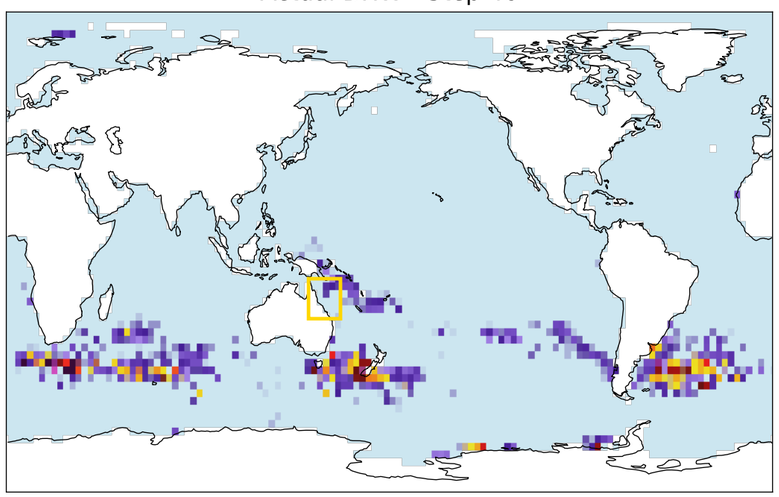
Predicted DHW - Step 1



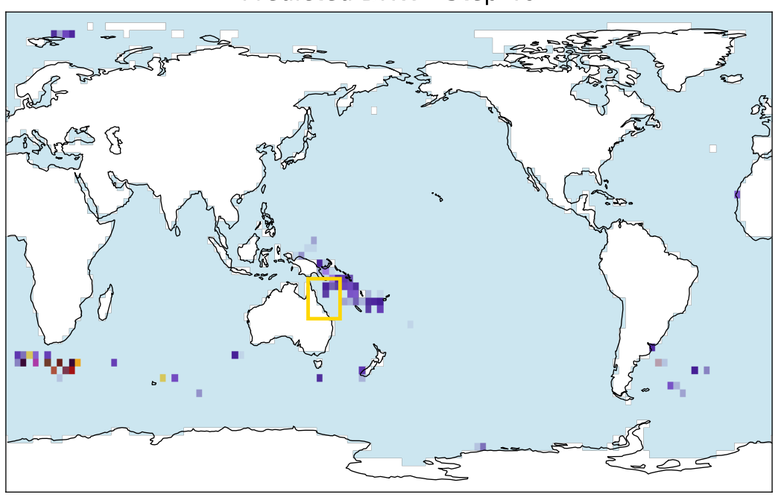
Error (Pred - Actual) - Step 1 [scale ±0.2]



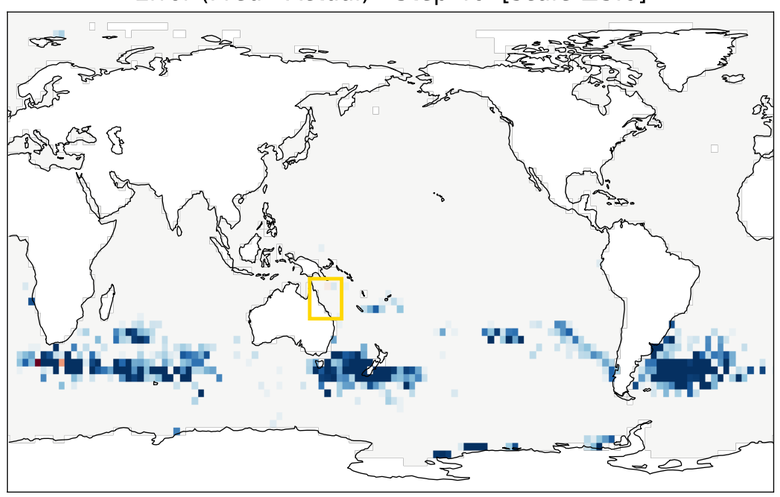
Actual DHW - Step 46



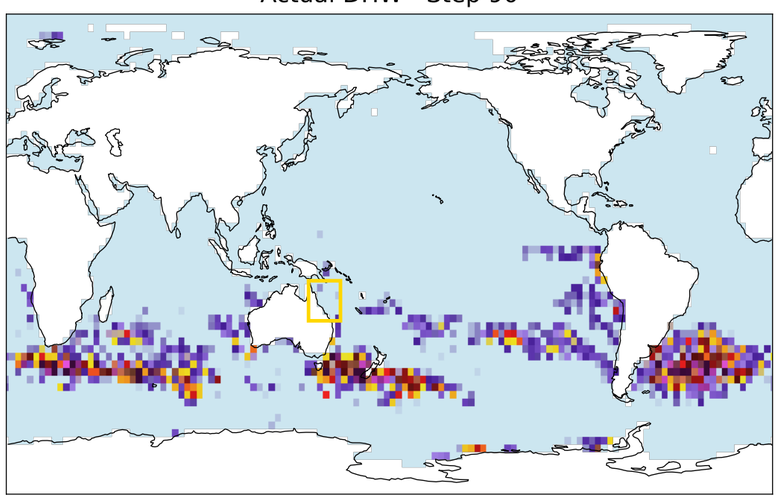
Predicted DHW - Step 46



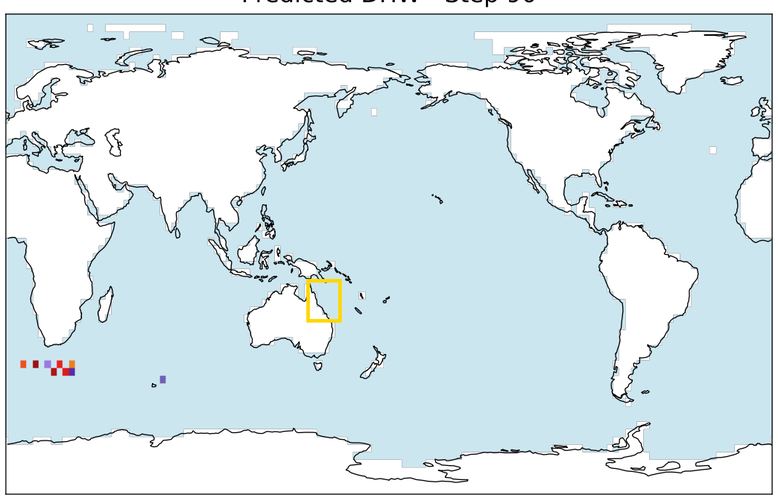
Error (Pred - Actual) - Step 46 [scale ±3.6]



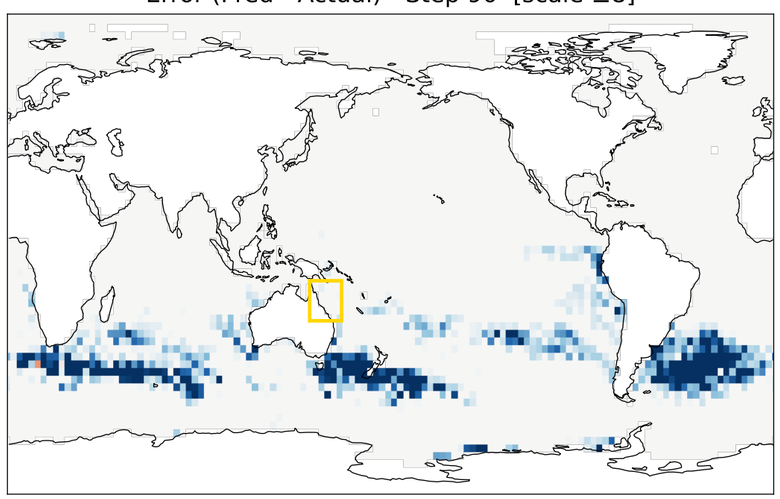
Actual DHW - Step 90

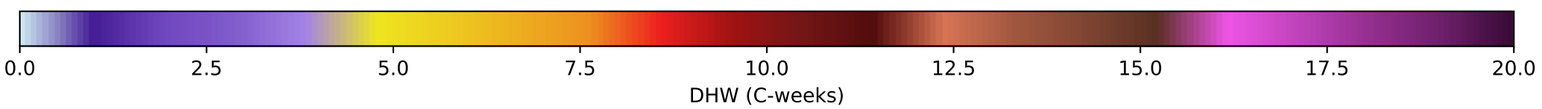


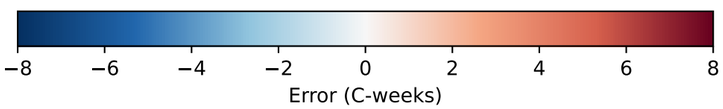
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

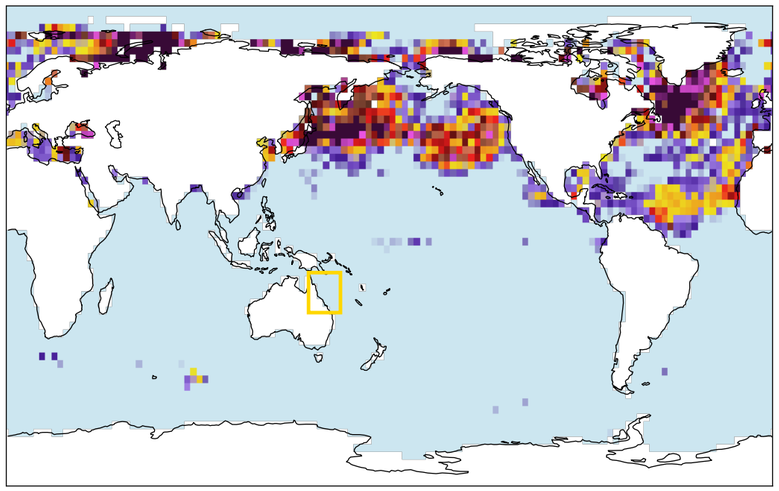




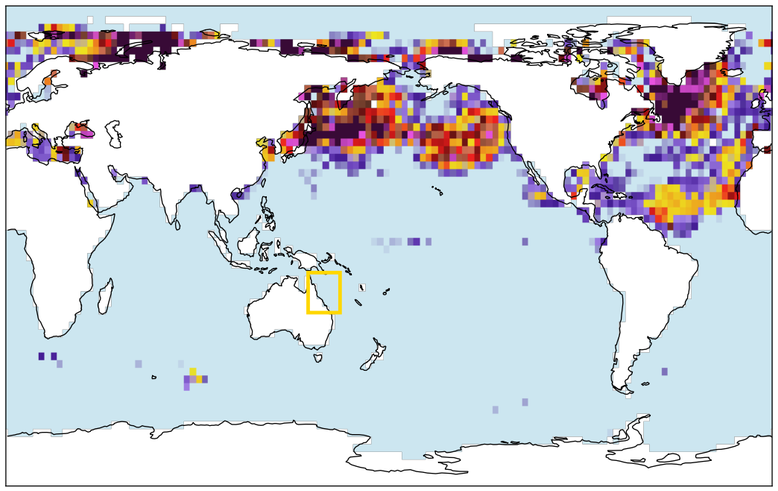


DHW Chain init: 2023-09-23 - Test (C-weeks)

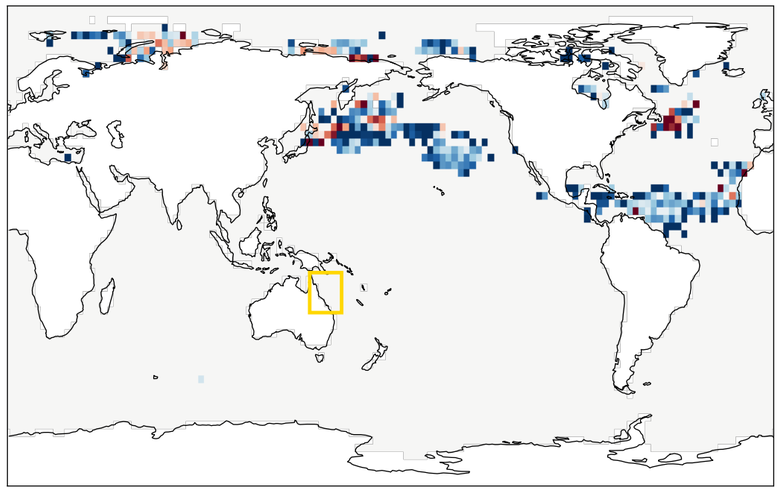
Actual DHW - Step 1



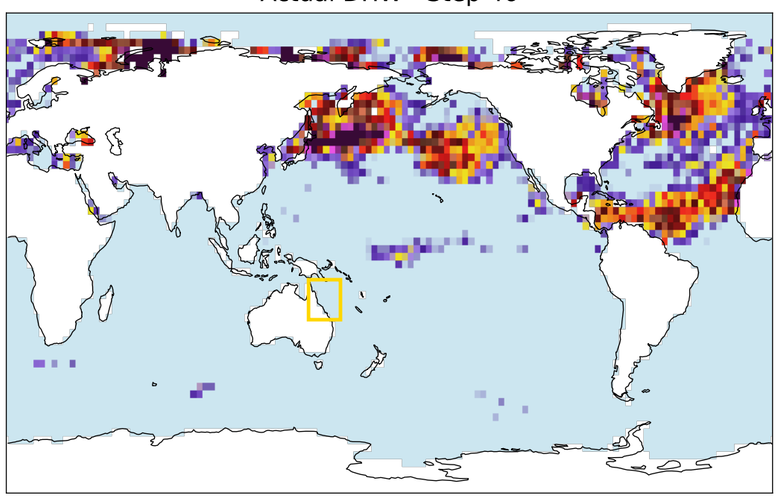
Predicted DHW - Step 1



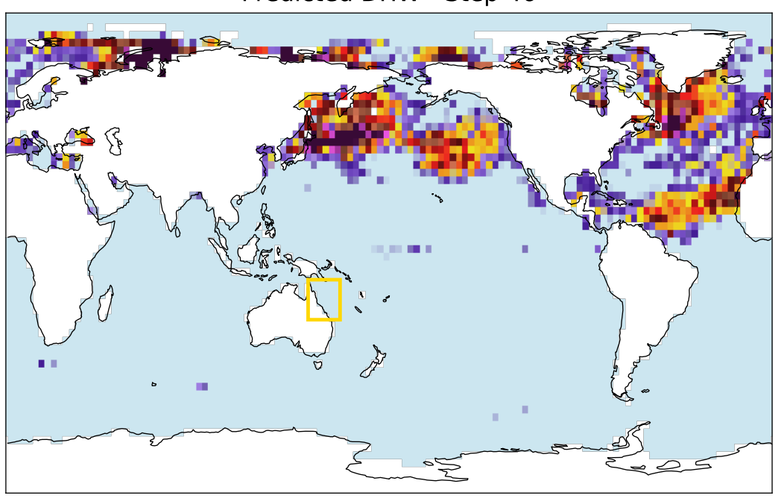
Error (Pred - Actual) - Step 1 [scale ±0.038]



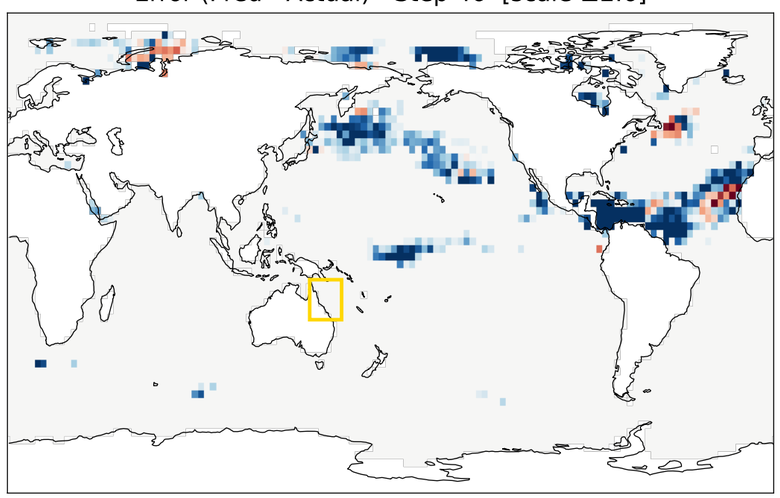
Actual DHW - Step 46



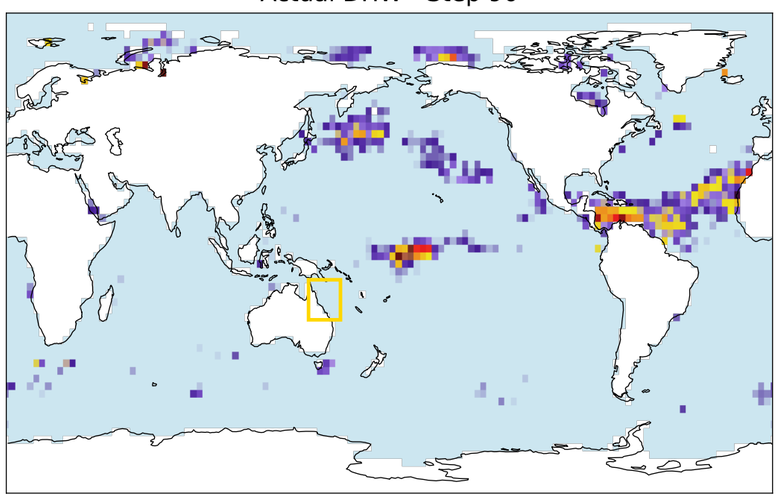
Predicted DHW - Step 46



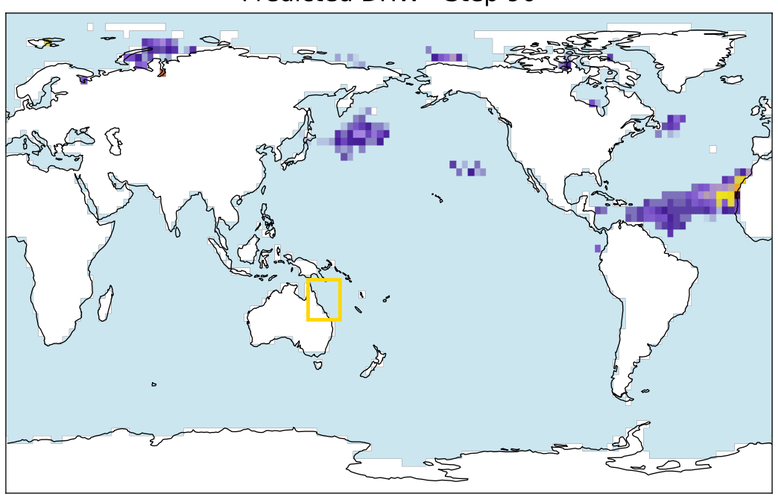
Error (Pred - Actual) - Step 46 [scale ±2.6]



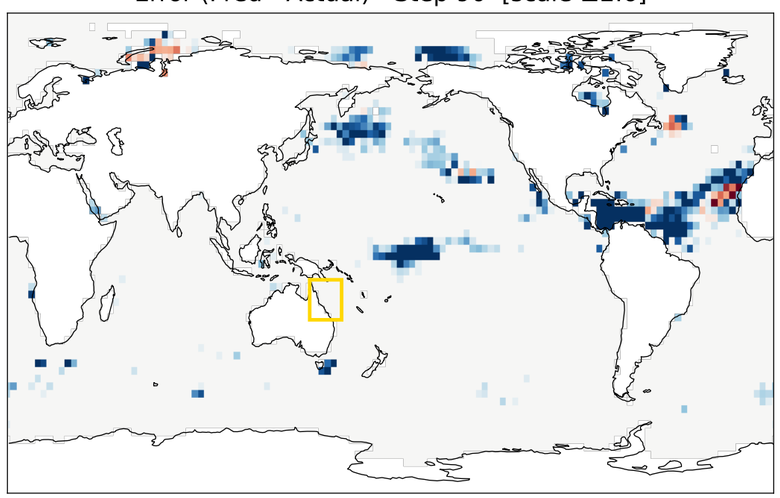
Actual DHW - Step 90

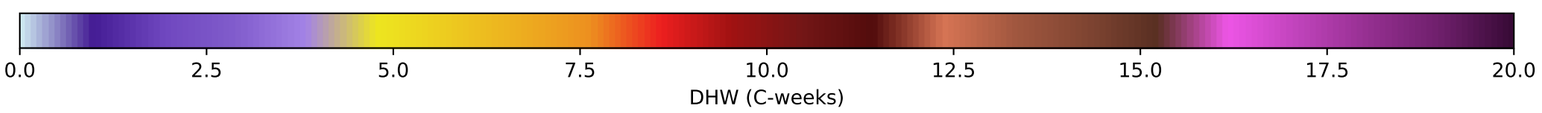


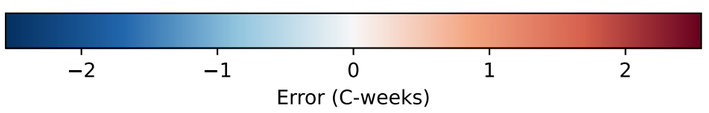
Predicted DHW - Step 90



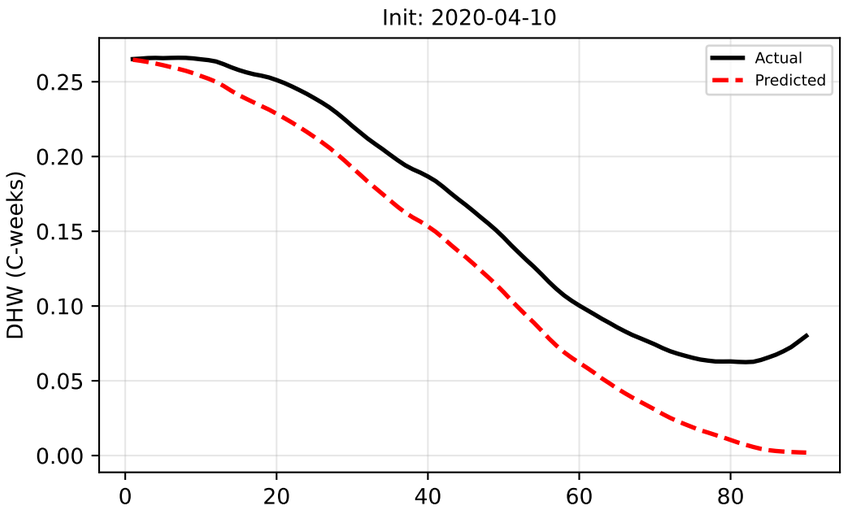
Error (Pred - Actual) - Step 90 [scale ±2.6]

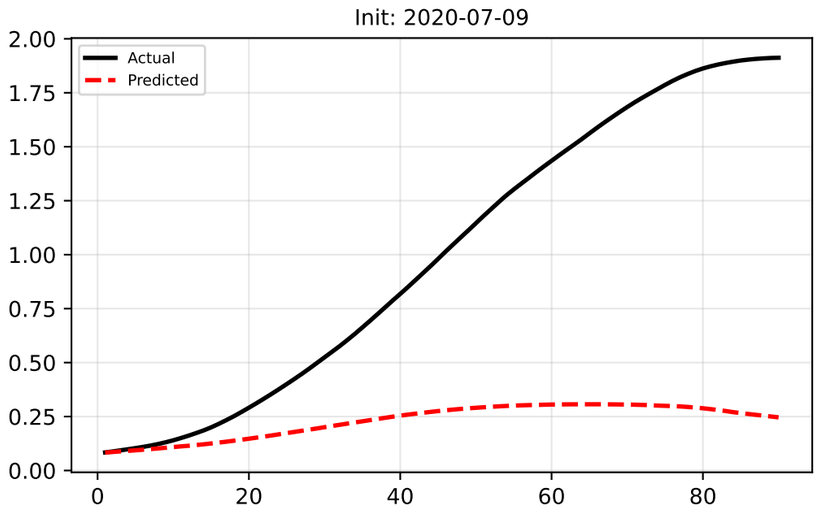


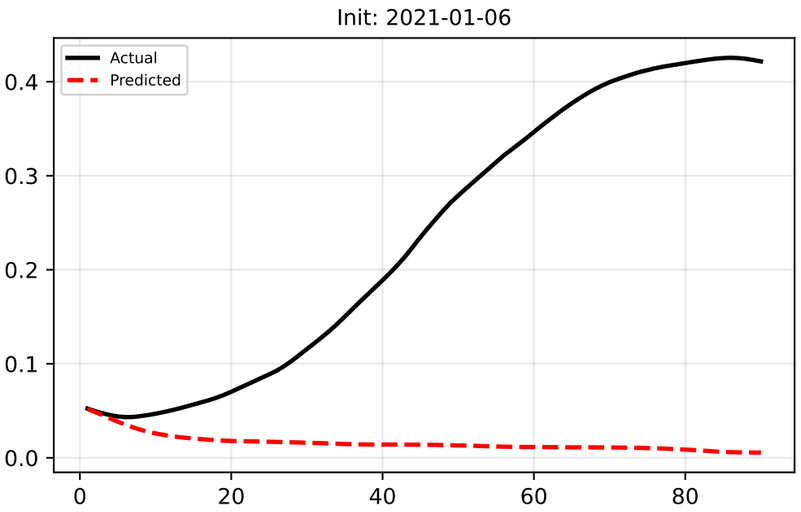


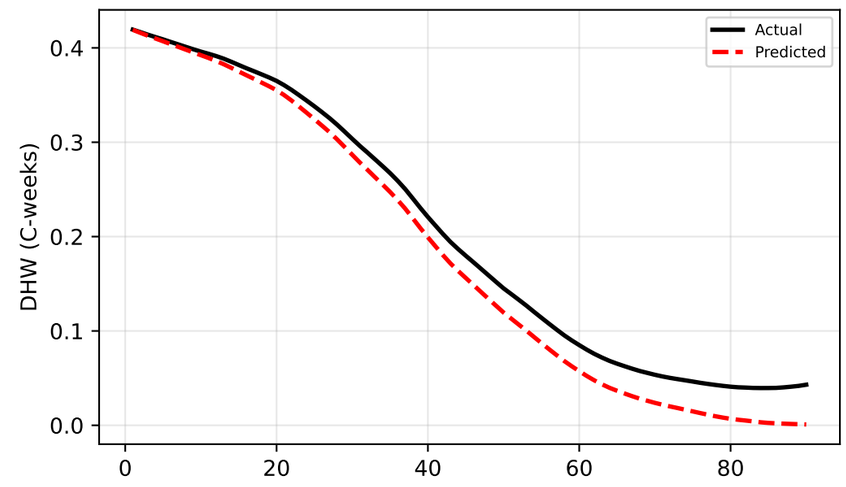


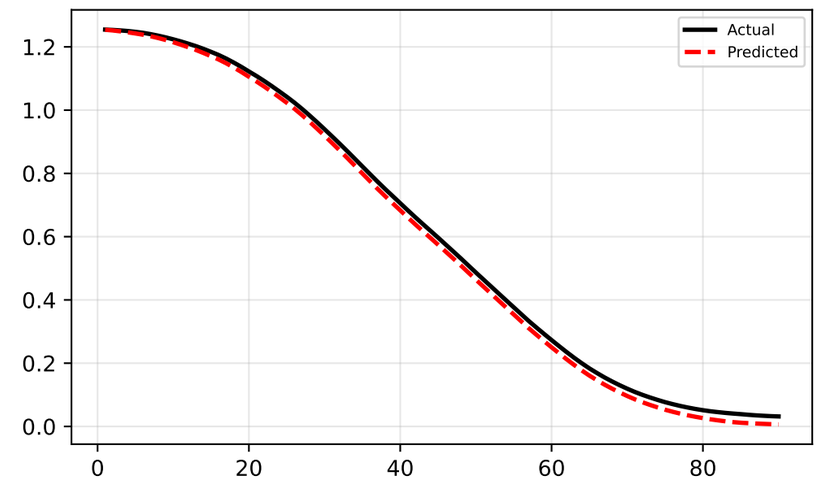
Global-Mean DHW - Test (C-weeks)

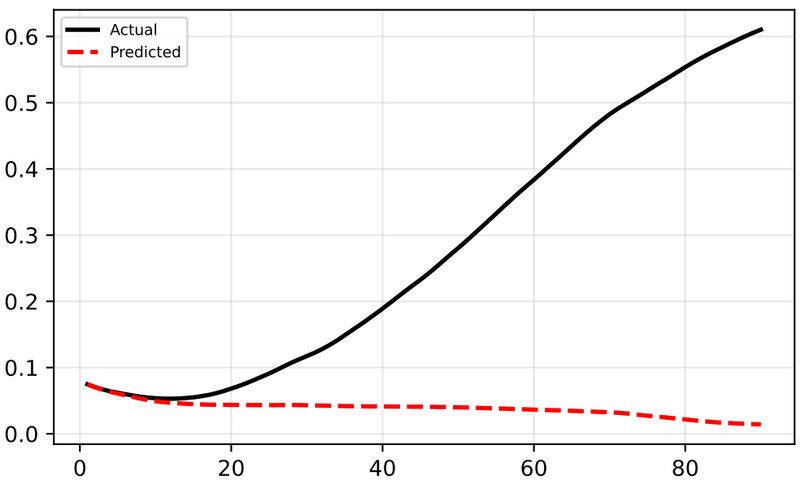


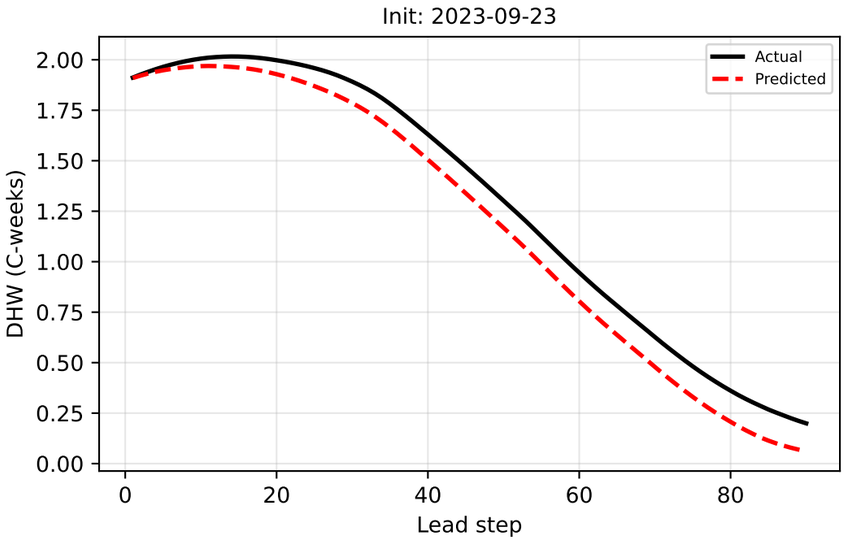




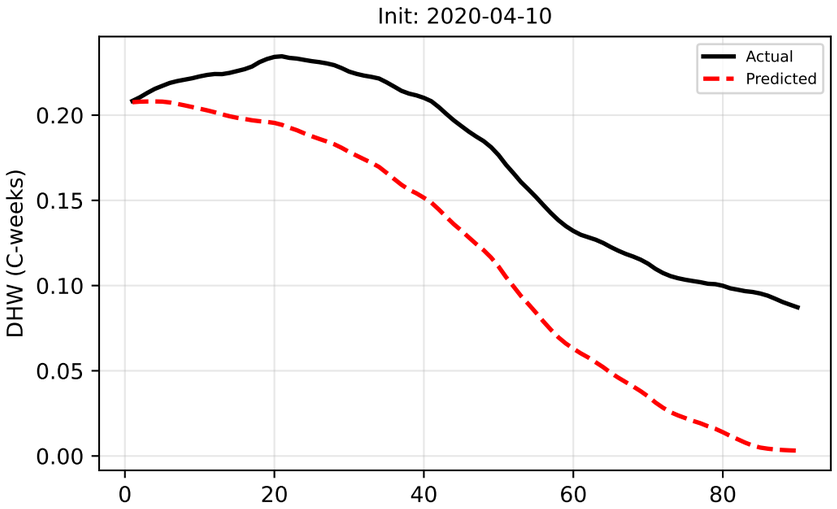


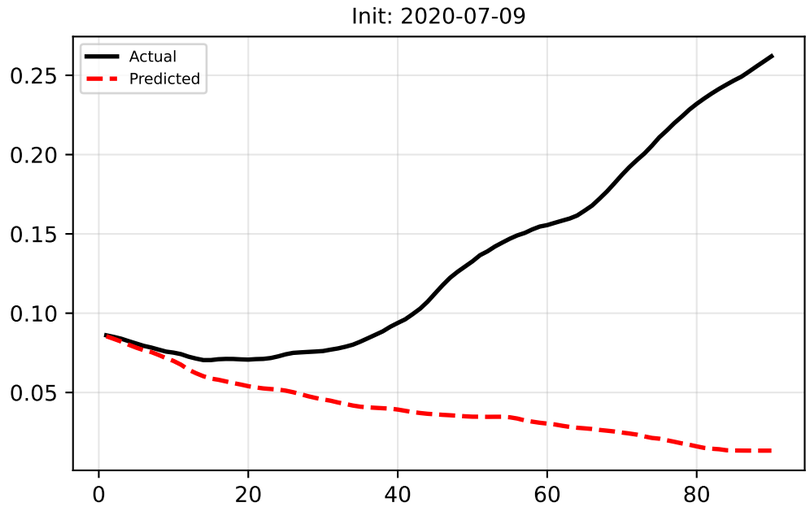


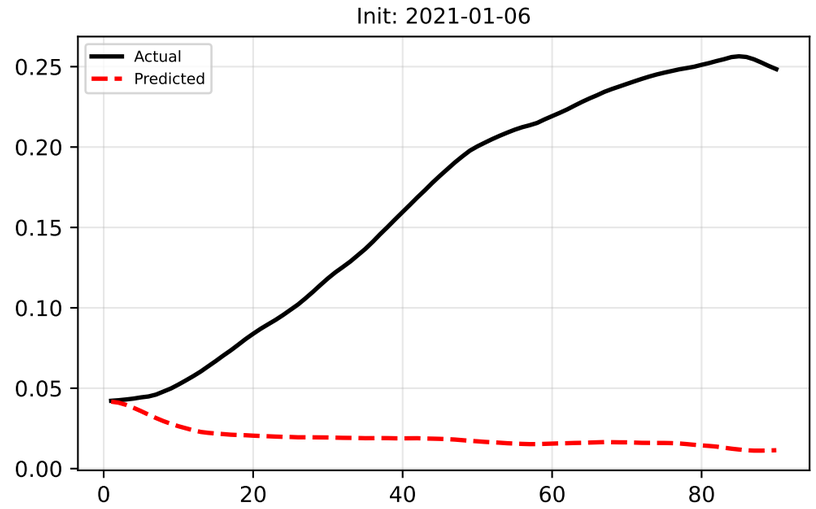


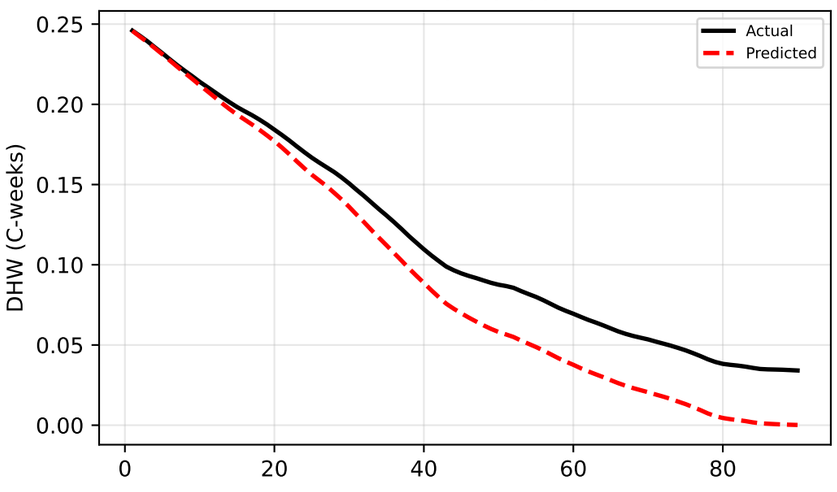


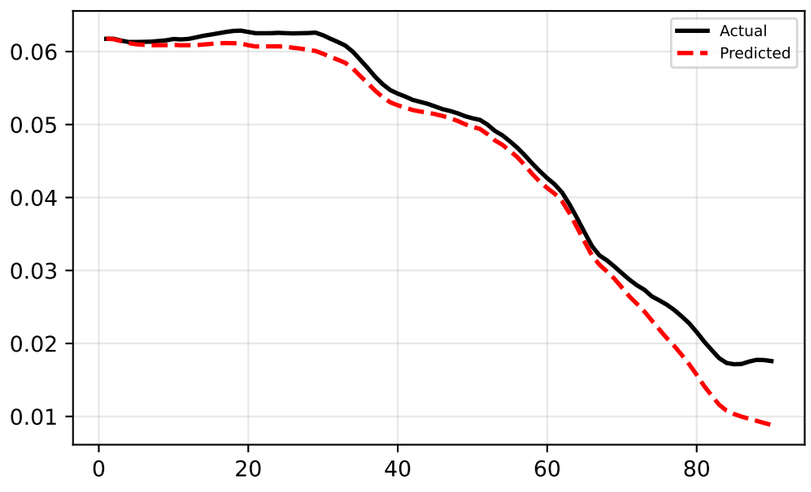
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

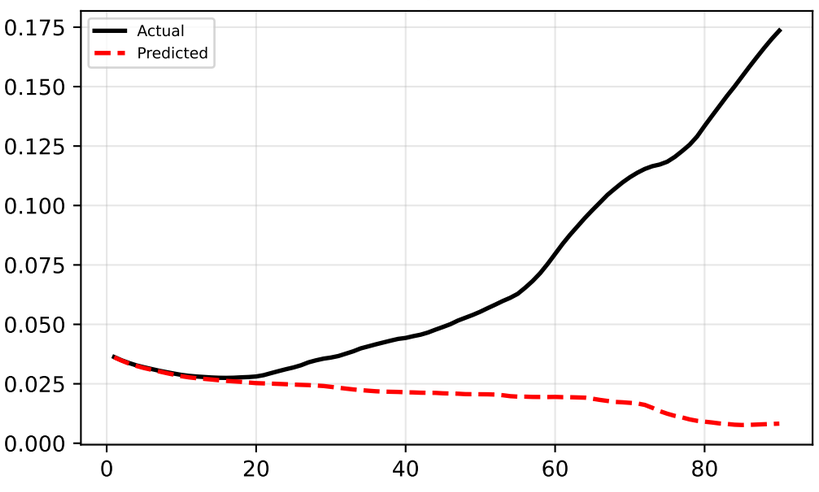


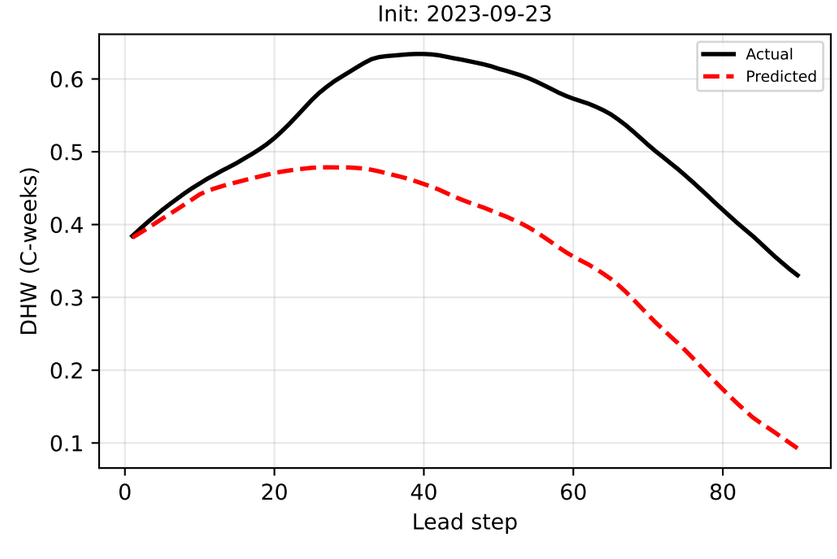


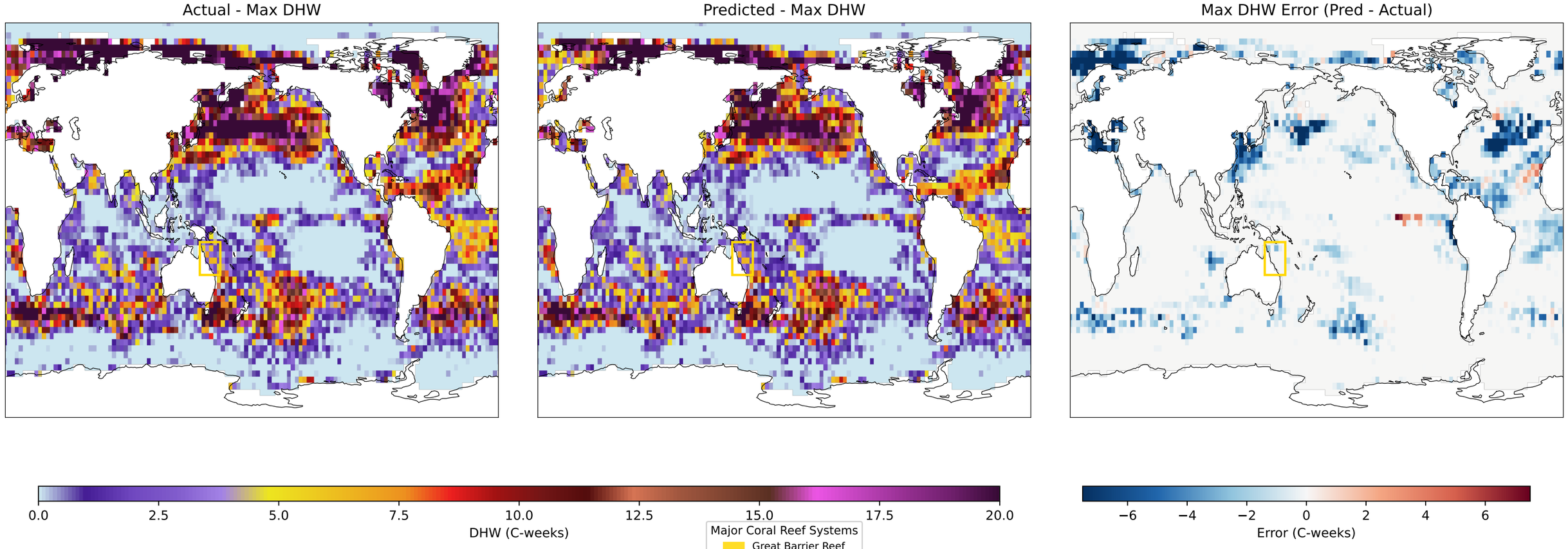


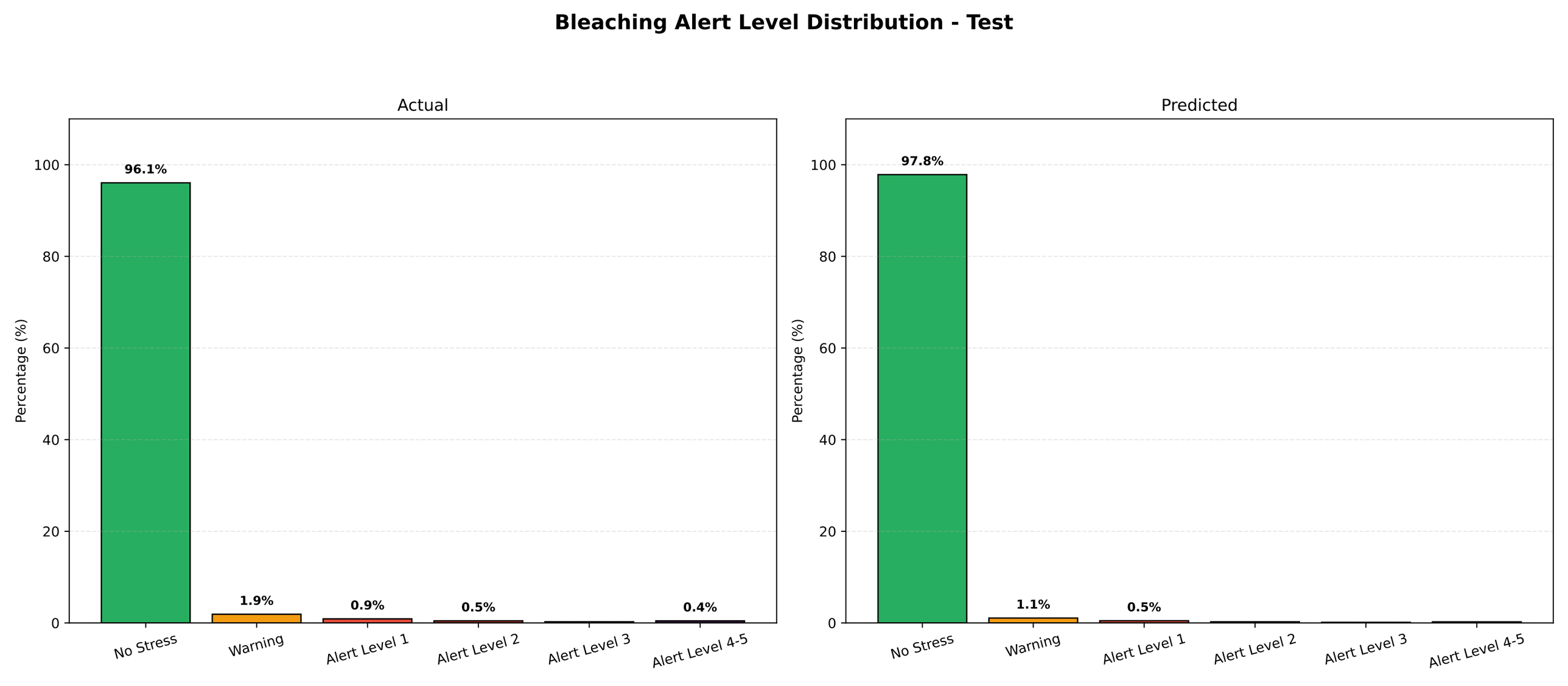






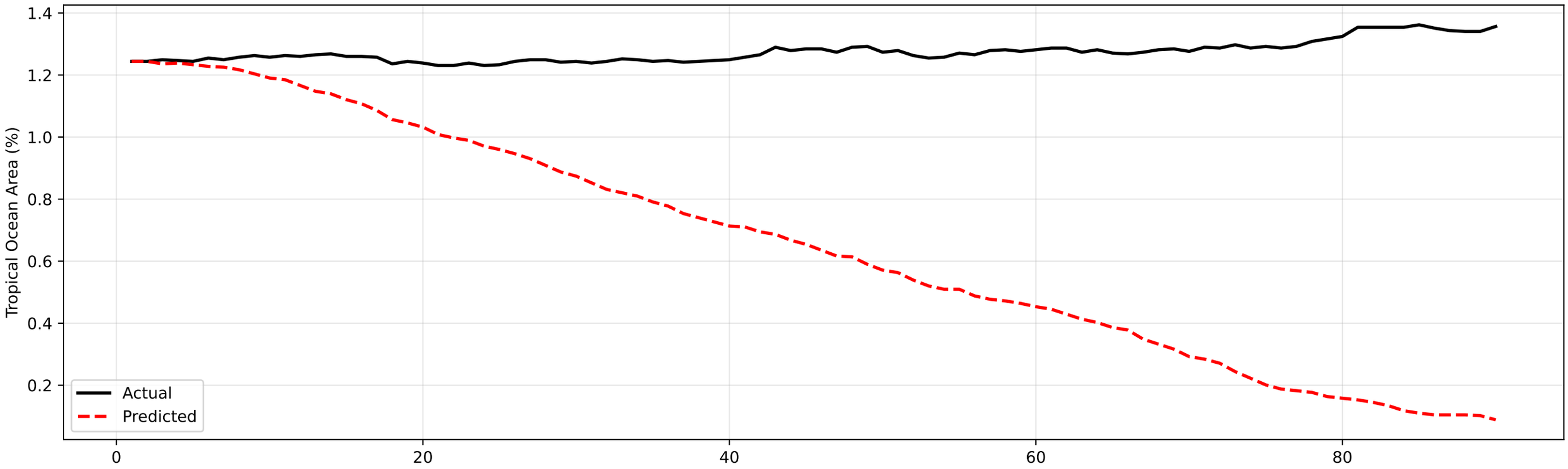



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−6
−4
−2
0
2
4
6
Error (C-weeks)

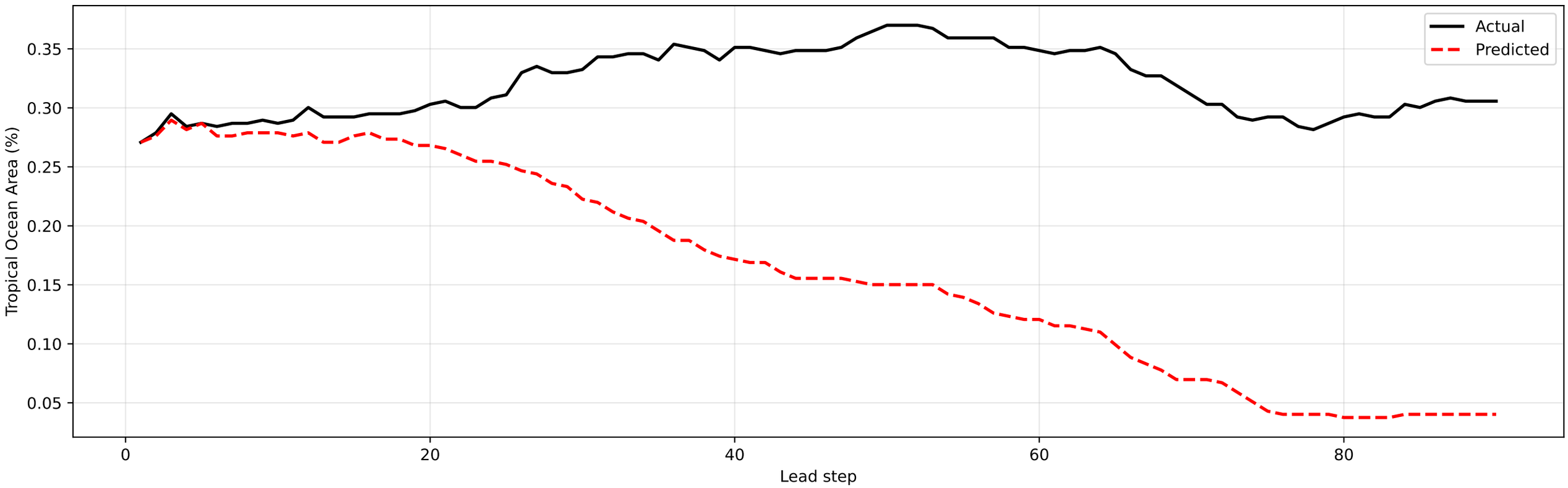

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.8%
1.1%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



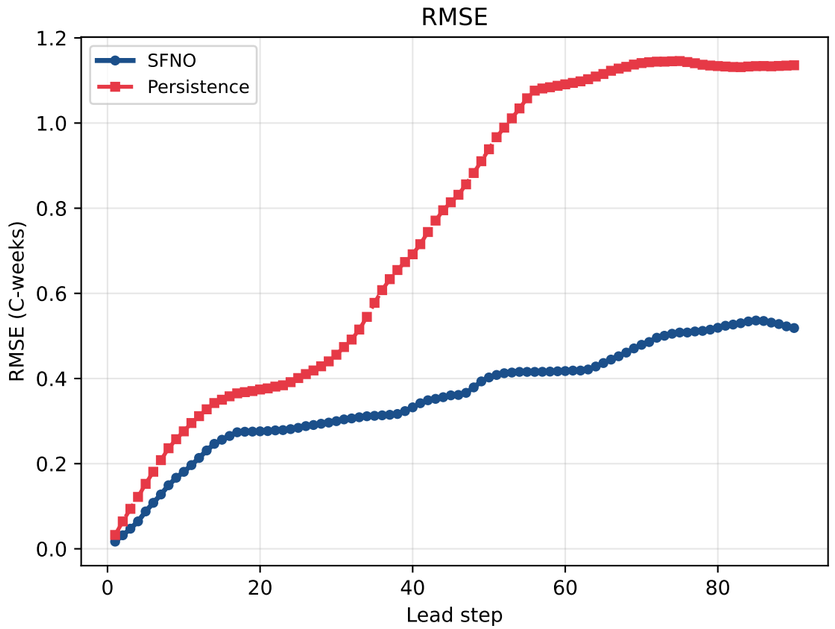
Great Barrier Reef Analysis - Test

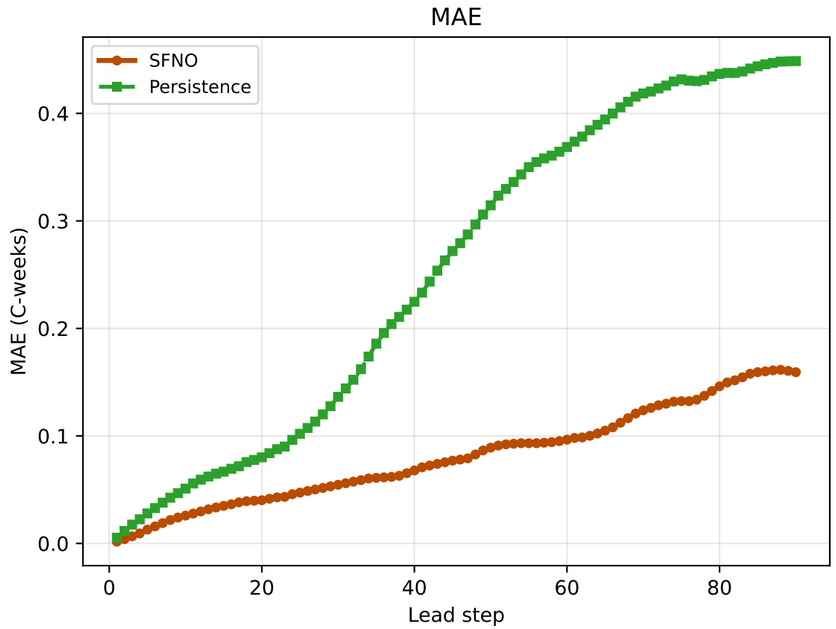
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

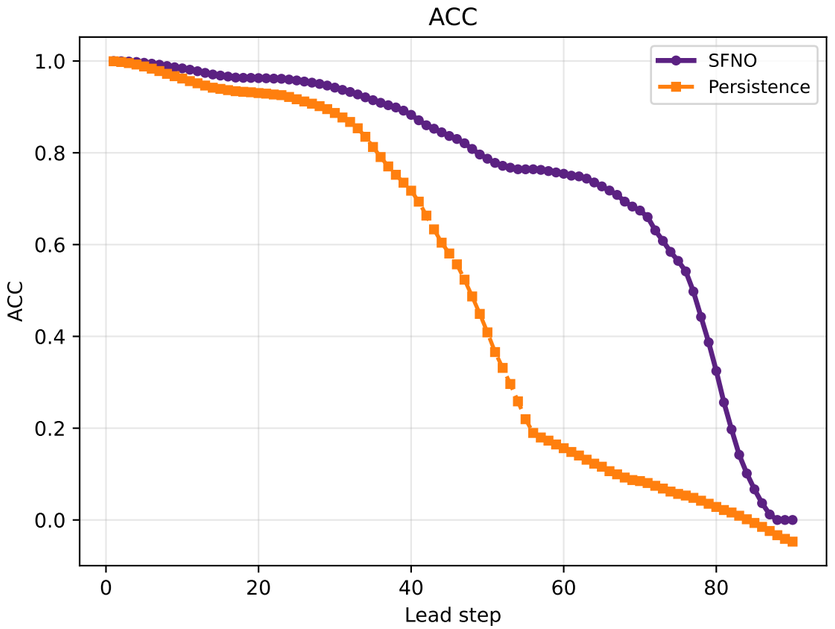
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

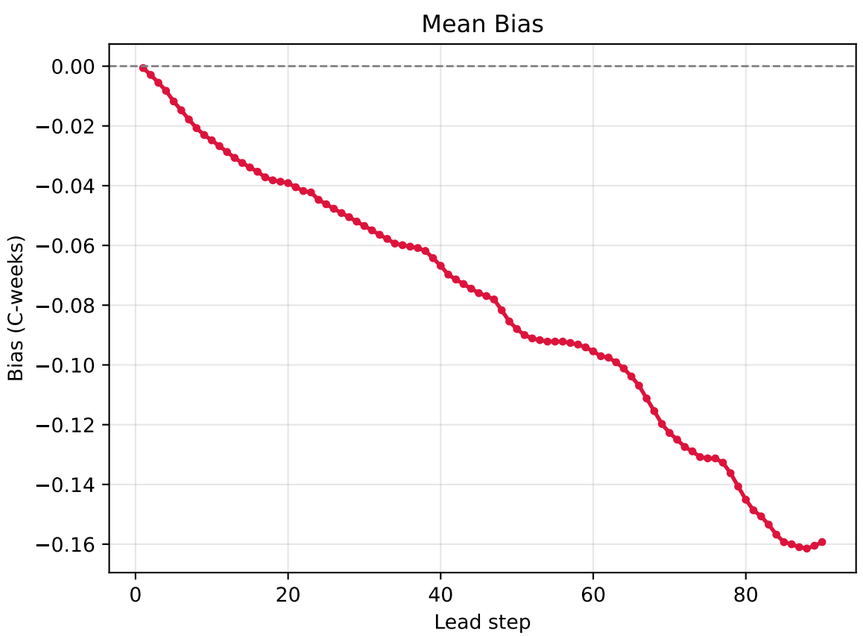
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0169	0.0325	48.0%	0.0017	0.0054	-0.0006	0.9999	0.9995	0.9997
2	0.0318	0.0643	50.6%	0.0040	0.0116	-0.0029	0.9995	0.9979	0.9990
3	0.0473	0.0939	49.7%	0.0066	0.0175	-0.0055	0.9989	0.9955	0.9978
4	0.0644	0.1219	47.2%	0.0093	0.0224	-0.0082	0.9979	0.9924	0.9959
5	0.0878	0.1525	42.4%	0.0128	0.0279	-0.0118	0.9961	0.9881	0.9925
6	0.1080	0.1810	40.3%	0.0158	0.0329	-0.0148	0.9942	0.9834	0.9887
7	0.1277	0.2083	38.7%	0.0189	0.0379	-0.0179	0.9919	0.9781	0.9843
8	0.1493	0.2362	36.8%	0.0218	0.0425	-0.0207	0.9890	0.9721	0.9787
9	0.1669	0.2574	35.2%	0.0241	0.0468	-0.0230	0.9864	0.9669	0.9736
10	0.1808	0.2759	34.5%	0.0259	0.0510	-0.0248	0.9840	0.9619	0.9689
11	0.1966	0.2955	33.5%	0.0278	0.0558	-0.0267	0.9810	0.9562	0.9632
12	0.2134	0.3116	31.5%	0.0298	0.0593	-0.0287	0.9778	0.9515	0.9569
13	0.2310	0.3277	29.5%	0.0317	0.0622	-0.0307	0.9741	0.9466	0.9498
14	0.2465	0.3424	28.0%	0.0335	0.0649	-0.0324	0.9706	0.9419	0.9432
15	0.2562	0.3502	26.9%	0.0350	0.0668	-0.0339	0.9683	0.9393	0.9387
16	0.2650	0.3582	26.0%	0.0364	0.0694	-0.0353	0.9661	0.9366	0.9346
17	0.2737	0.3652	25.0%	0.0382	0.0718	-0.0372	0.9639	0.9343	0.9305
18	0.2750	0.3680	25.3%	0.0393	0.0757	-0.0382	0.9634	0.9329	0.9294
19	0.2753	0.3705	25.7%	0.0397	0.0777	-0.0387	0.9632	0.9319	0.9291
20	0.2760	0.3742	26.3%	0.0402	0.0801	-0.0391	0.9628	0.9302	0.9282
21	0.2766	0.3768	26.6%	0.0416	0.0840	-0.0405	0.9624	0.9289	0.9275
22	0.2782	0.3810	27.0%	0.0428	0.0876	-0.0418	0.9617	0.9271	0.9263
23	0.2788	0.3841	27.4%	0.0433	0.0901	-0.0423	0.9613	0.9256	0.9256
24	0.2812	0.3914	28.2%	0.0458	0.0963	-0.0447	0.9596	0.9217	0.9225
25	0.2841	0.4011	29.2%	0.0473	0.1020	-0.0462	0.9577	0.9167	0.9189
26	0.2882	0.4104	29.8%	0.0488	0.1074	-0.0477	0.9553	0.9118	0.9145
27	0.2908	0.4189	30.6%	0.0502	0.1133	-0.0492	0.9529	0.9070	0.9101
28	0.2937	0.4287	31.5%	0.0516	0.1201	-0.0505	0.9501	0.9017	0.9050
29	0.2964	0.4402	32.7%	0.0531	0.1277	-0.0520	0.9465	0.8952	0.8987
30	0.2999	0.4558	34.2%	0.0545	0.1363	-0.0535	0.9420	0.8868	0.8906
31	0.3040	0.4739	35.9%	0.0560	0.1442	-0.0550	0.9371	0.8770	0.8818
32	0.3062	0.4915	37.7%	0.0575	0.1522	-0.0564	0.9325	0.8672	0.8737
33	0.3090	0.5150	40.0%	0.0589	0.1621	-0.0578	0.9272	0.8535	0.8641
34	0.3114	0.5448	42.8%	0.0604	0.1738	-0.0594	0.9209	0.8349	0.8531
35	0.3124	0.5777	45.9%	0.0610	0.1858	-0.0599	0.9146	0.8127	0.8420
36	0.3135	0.6076	48.4%	0.0615	0.1958	-0.0604	0.9089	0.7908	0.8320
37	0.3148	0.6333	50.3%	0.0619	0.2040	-0.0609	0.9037	0.7703	0.8231
38	0.3167	0.6548	51.6%	0.0629	0.2107	-0.0619	0.8988	0.7521	0.8147
39	0.3235	0.6735	52.0%	0.0653	0.2175	-0.0643	0.8919	0.7349	0.8031
40	0.3322	0.6918	52.0%	0.0679	0.2248	-0.0668	0.8827	0.7175	0.7876
41	0.3419	0.7155	52.2%	0.0708	0.2332	-0.0698	0.8709	0.6935	0.7678
42	0.3489	0.7441	53.1%	0.0724	0.2436	-0.0714	0.8600	0.6632	0.7491
43	0.3522	0.7709	54.3%	0.0740	0.2537	-0.0729	0.8525	0.6331	0.7357
44	0.3561	0.7952	55.2%	0.0755	0.2632	-0.0745	0.8446	0.6042	0.7215
45	0.3604	0.8139	55.7%	0.0770	0.2720	-0.0760	0.8367	0.5806	0.7066
46	0.3614	0.8315	56.5%	0.0780	0.2793	-0.0769	0.8301	0.5570	0.6943
47	0.3663	0.8560	57.2%	0.0791	0.2874	-0.0781	0.8208	0.5236	0.6768
48	0.3791	0.8827	57.1%	0.0827	0.2967	-0.0817	0.8084	0.4870	0.6540
49	0.3932	0.9103	56.8%	0.0865	0.3059	-0.0854	0.7962	0.4490	0.6317
50	0.4025	0.9384	57.1%	0.0890	0.3145	-0.0880	0.7871	0.4087	0.6141
51	0.4083	0.9668	57.8%	0.0911	0.3234	-0.0900	0.7781	0.3658	0.5983
52	0.4124	0.9892	58.3%	0.0922	0.3297	-0.0911	0.7716	0.3313	0.5875
53	0.4140	1.0111	59.1%	0.0927	0.3362	-0.0917	0.7676	0.2963	0.5795
54	0.4155	1.0344	59.8%	0.0932	0.3432	-0.0922	0.7641	0.2584	0.5720
55	0.4155	1.0582	60.7%	0.0932	0.3501	-0.0922	0.7641	0.2195	0.5695
56	0.4155	1.0761	61.4%	0.0932	0.3548	-0.0922	0.7641	0.1896	0.5677
57	0.4158	1.0812	61.5%	0.0937	0.3581	-0.0926	0.7628	0.1797	0.5632
58	0.4163	1.0841	61.6%	0.0942	0.3607	-0.0932	0.7602	0.1728	0.5574
59	0.4170	1.0875	61.7%	0.0952	0.3643	-0.0941	0.7572	0.1645	0.5505
60	0.4175	1.0909	61.7%	0.0965	0.3688	-0.0955	0.7543	0.1564	0.5439
61	0.4186	1.0943	61.7%	0.0981	0.3738	-0.0971	0.7502	0.1481	0.5343
62	0.4187	1.0980	61.9%	0.0986	0.3785	-0.0975	0.7486	0.1404	0.5281
63	0.4212	1.1030	61.8%	0.1002	0.3843	-0.0991	0.7438	0.1313	0.5202
64	0.4282	1.1092	61.4%	0.1022	0.3894	-0.1012	0.7353	0.1227	0.5092
65	0.4363	1.1152	60.9%	0.1049	0.3943	-0.1039	0.7268	0.1160	0.4993
66	0.4444	1.1228	60.4%	0.1080	0.3999	-0.1069	0.7178	0.1063	0.4889
67	0.4522	1.1280	59.9%	0.1123	0.4056	-0.1112	0.7083	0.0994	0.4784
68	0.4609	1.1321	59.3%	0.1165	0.4108	-0.1155	0.6936	0.0924	0.4614
69	0.4707	1.1374	58.6%	0.1208	0.4156	-0.1198	0.6829	0.0870	0.4504
70	0.4791	1.1409	58.0%	0.1238	0.4186	-0.1228	0.6742	0.0847	0.4414
71	0.4856	1.1428	57.5%	0.1261	0.4203	-0.1250	0.6600	0.0805	0.4256
72	0.4958	1.1442	56.7%	0.1285	0.4232	-0.1275	0.6310	0.0744	0.3941
73	0.5004	1.1443	56.3%	0.1300	0.4259	-0.1289	0.6081	0.0684	0.3677
74	0.5051	1.1451	55.9%	0.1319	0.4297	-0.1308	0.5842	0.0619	0.3430
75	0.5080	1.1456	55.7%	0.1323	0.4316	-0.1313	0.5645	0.0567	0.3229
76	0.5080	1.1433	55.6%	0.1323	0.4305	-0.1313	0.5416	0.0532	0.3022
77	0.5105	1.1405	55.2%	0.1338	0.4300	-0.1327	0.4977	0.0482	0.2662
78	0.5118	1.1370	55.0%	0.1373	0.4312	-0.1362	0.4424	0.0419	0.2244
79	0.5148	1.1351	54.7%	0.1417	0.4344	-0.1407	0.3870	0.0353	0.1855
80	0.5192	1.1339	54.2%	0.1462	0.4368	-0.1451	0.3246	0.0284	0.1421
81	0.5239	1.1328	53.7%	0.1497	0.4377	-0.1486	0.2561	0.0216	0.0949
82	0.5267	1.1317	53.5%	0.1517	0.4376	-0.1507	0.1973	0.0162	0.0537
83	0.5299	1.1313	53.2%	0.1545	0.4390	-0.1535	0.1420	0.0092	0.0141
84	0.5340	1.1328	52.9%	0.1578	0.4417	-0.1568	0.1013	0.0015	-0.0158
85	0.5362	1.1337	52.7%	0.1593	0.4438	-0.1593	0.0668	-0.0065	-0.0428
86	0.5351	1.1340	52.8%	0.1600	0.4457	-0.1600	0.0366	-0.0150	-0.0671
87	0.5313	1.1333	53.1%	0.1610	0.4469	-0.1610	0.0120	-0.0240	-0.0904
88	0.5279	1.1341	53.5%	0.1615	0.4482	-0.1615	0.0000	-0.0332	-0.1032
89	0.5225	1.1349	54.0%	0.1605	0.4484	-0.1605	-0.0000	-0.0412	-0.1042
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

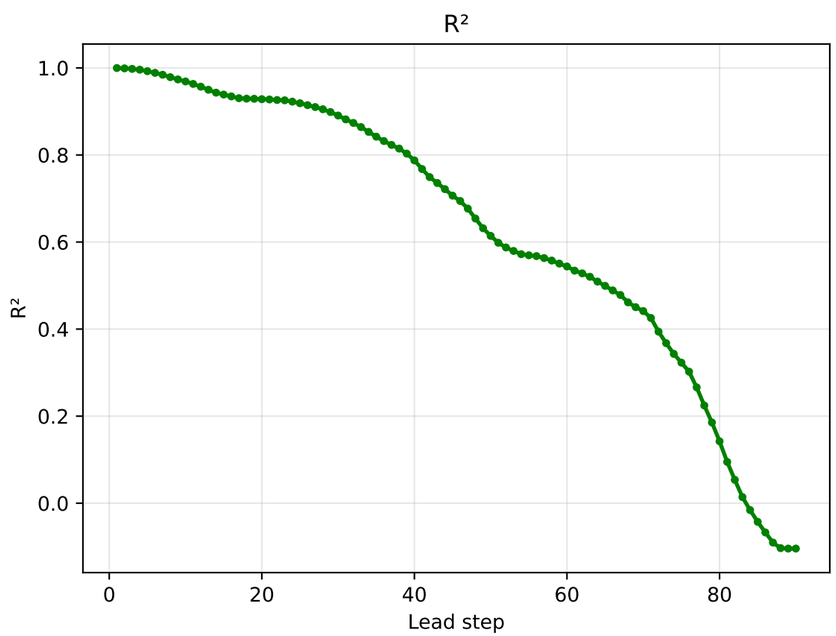
DHW Forecast Skill vs Lead Step - Test - GBR

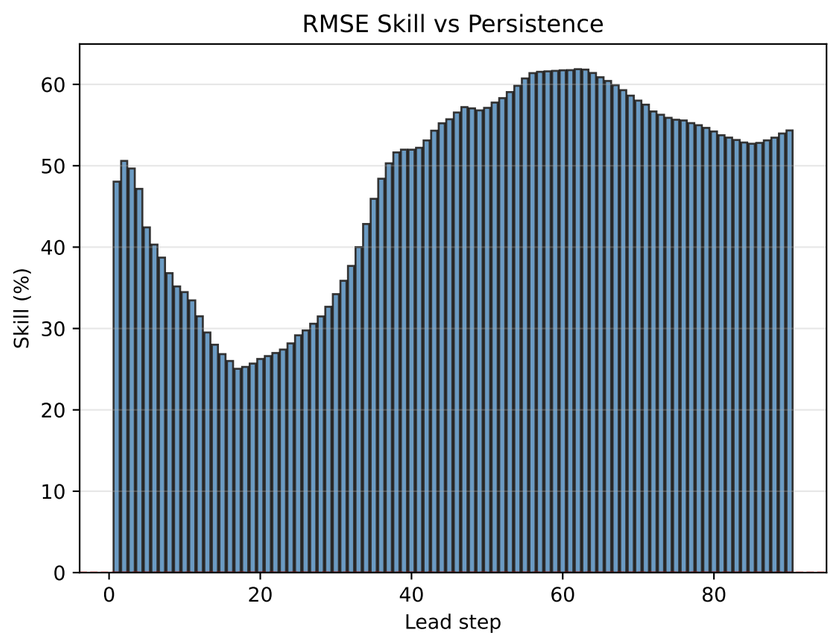


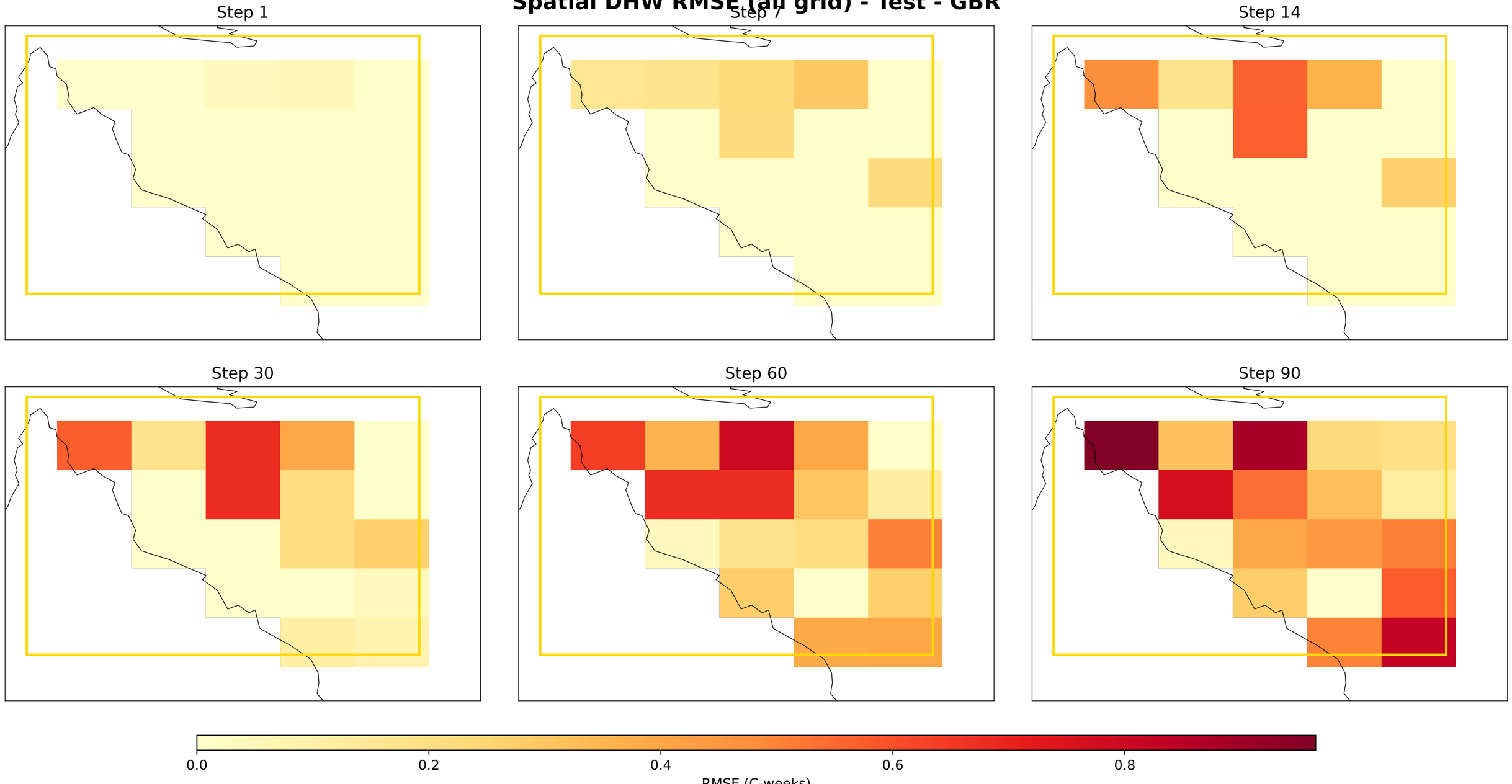


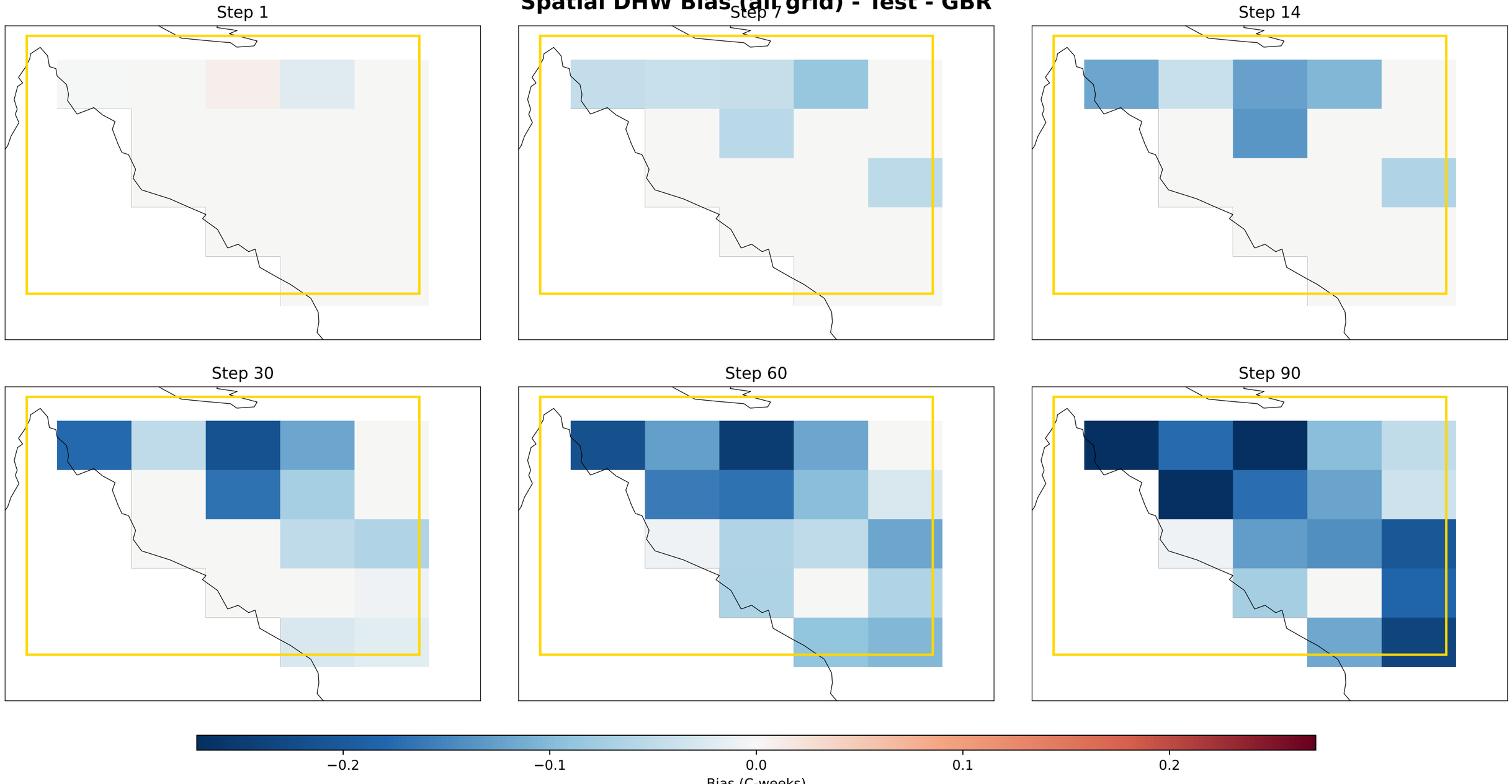


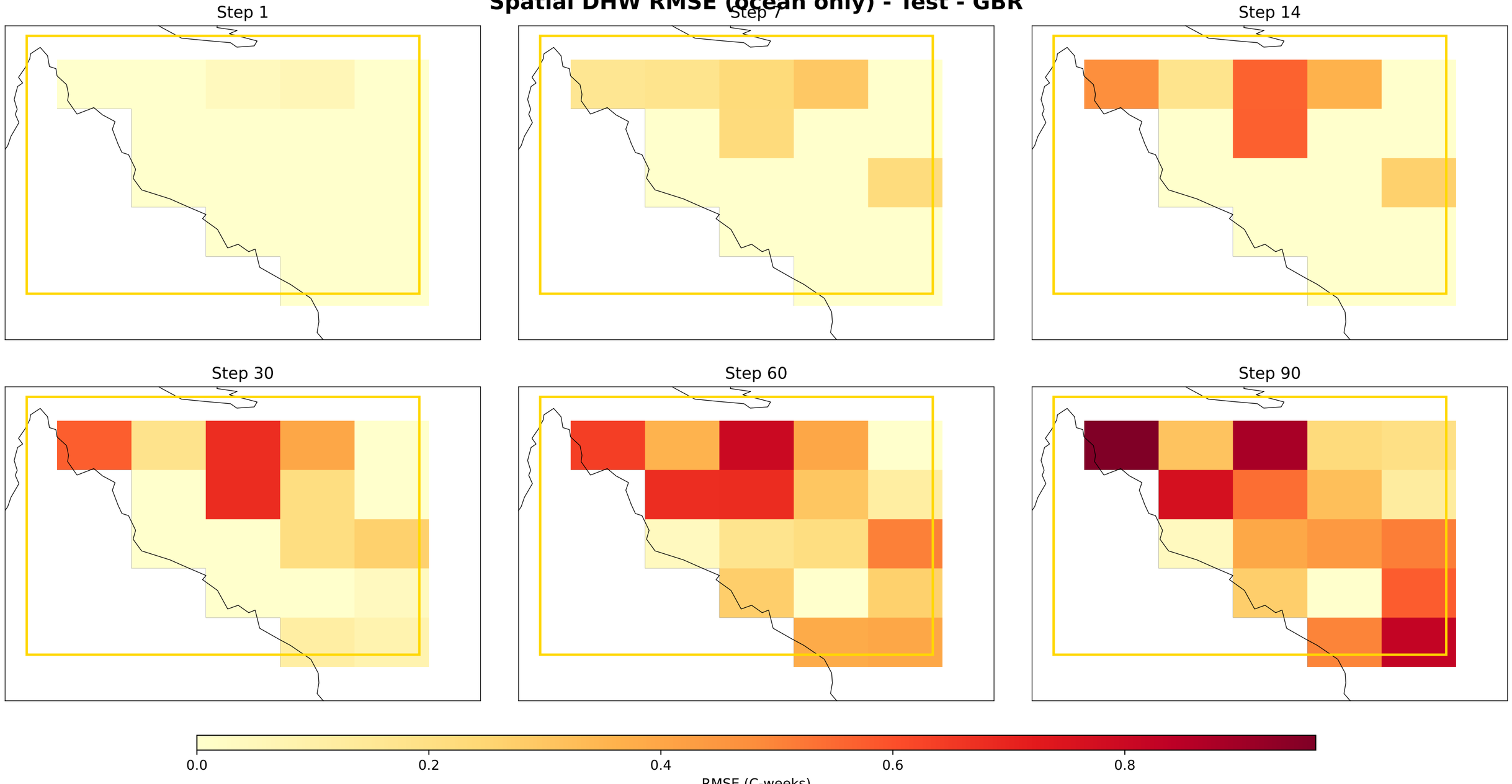


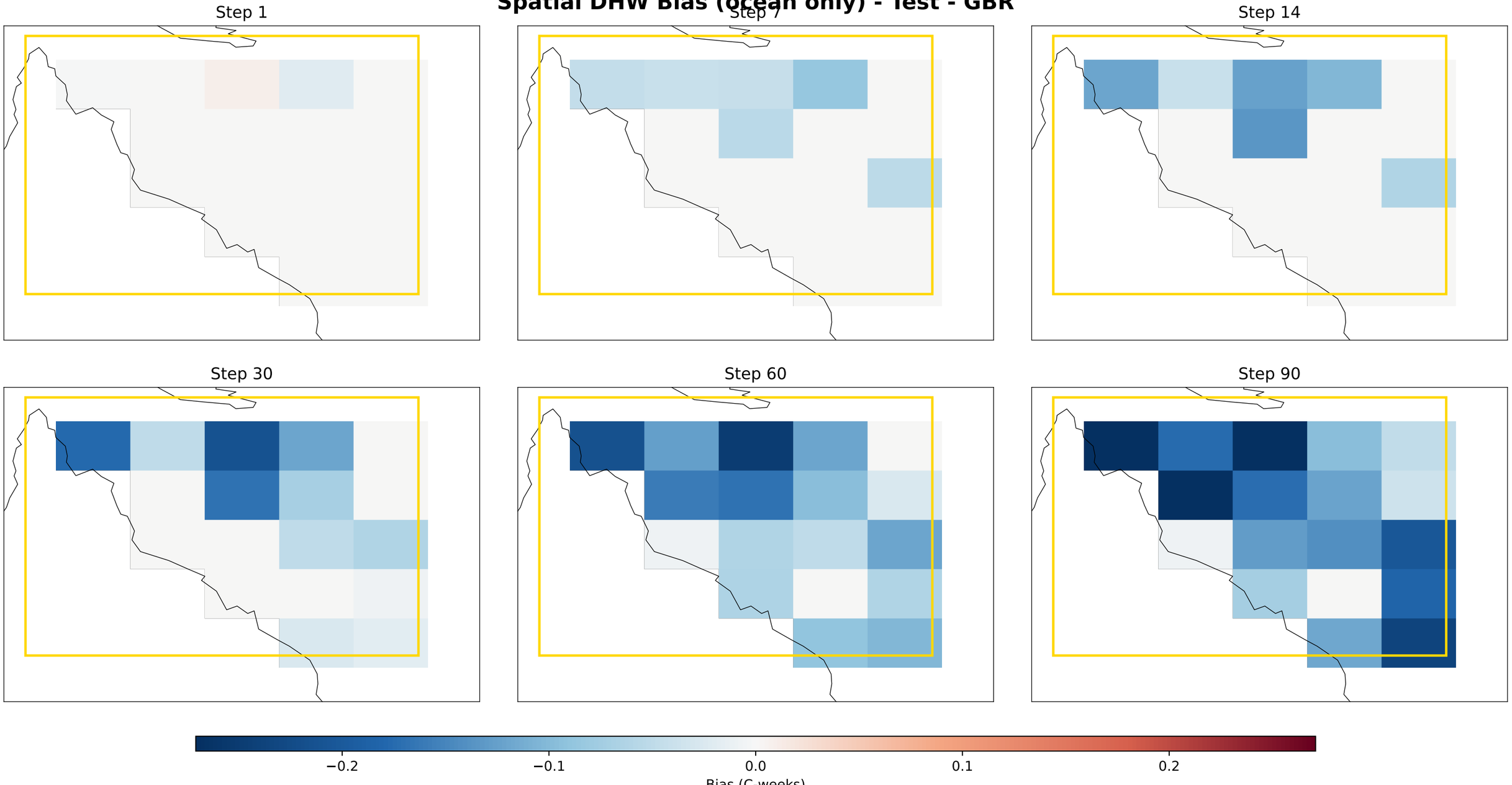





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

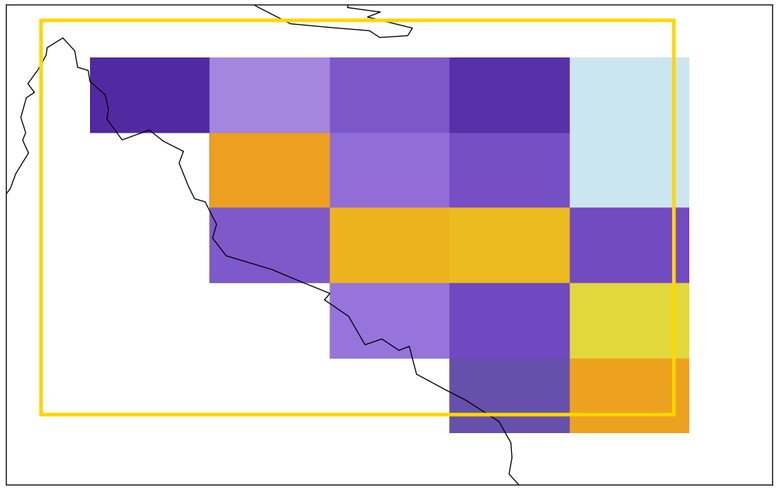

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

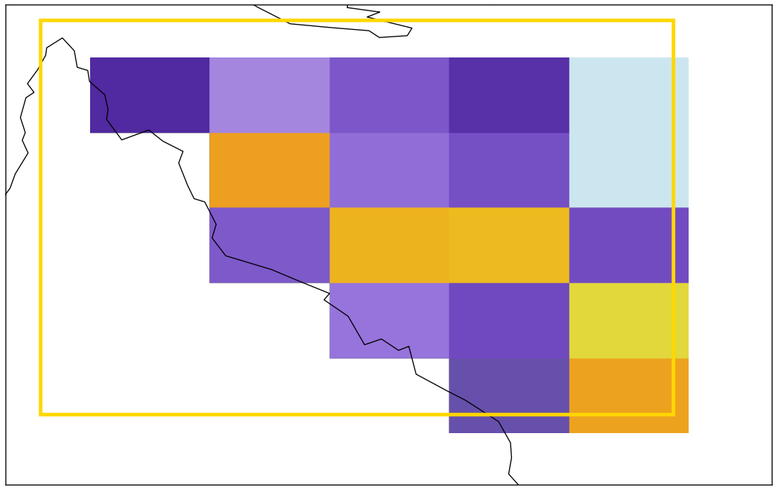

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

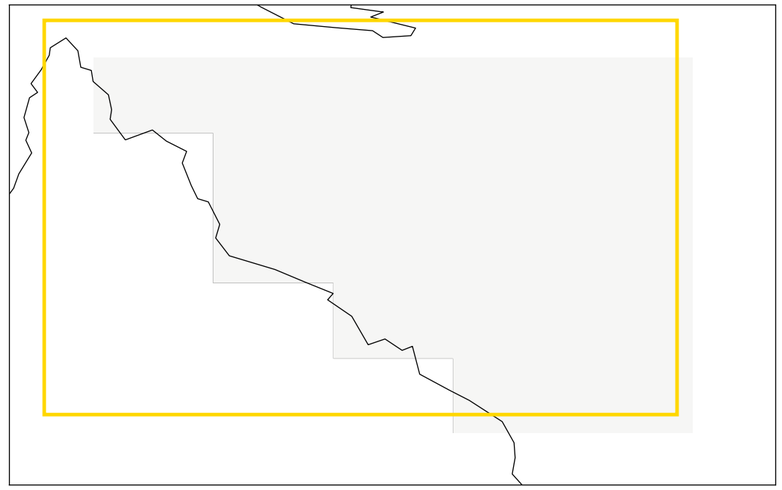
Actual DHW - Step 1



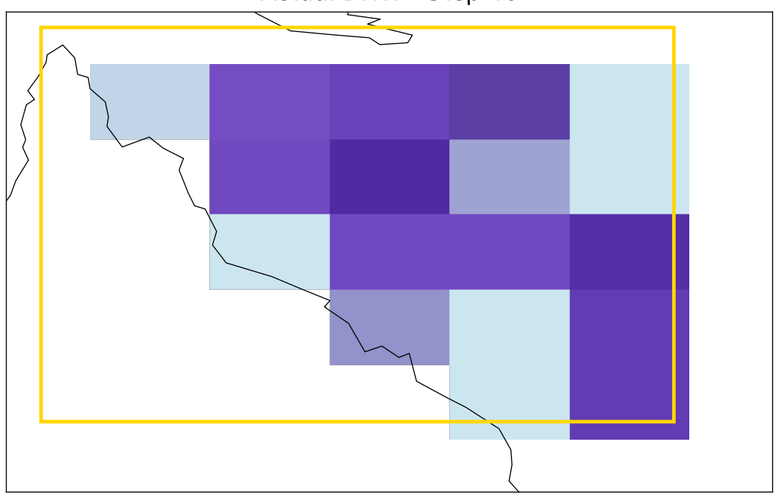
Predicted DHW - Step 1



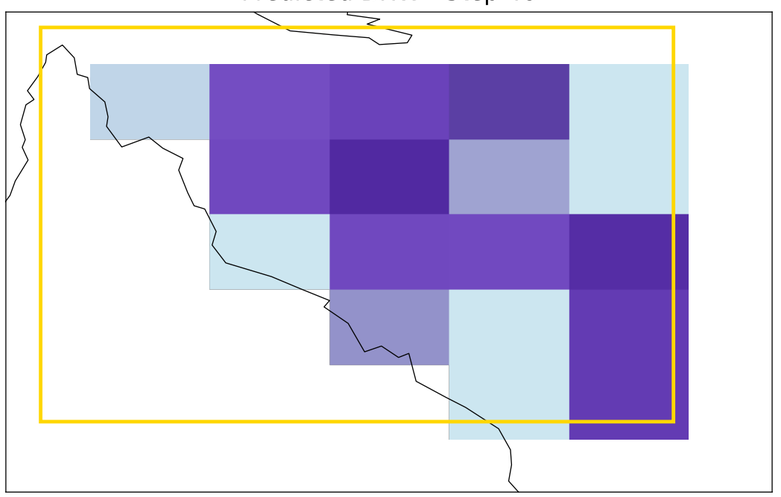
Error (Pred - Actual) - Step 1 [scale ±8]



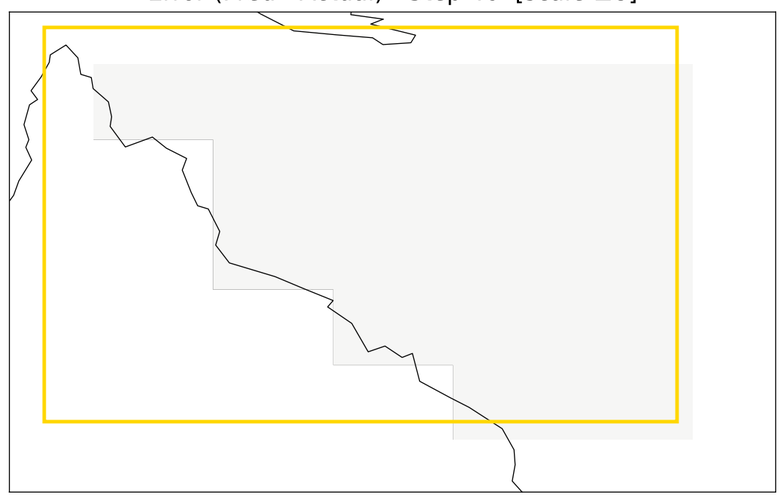
Actual DHW - Step 46



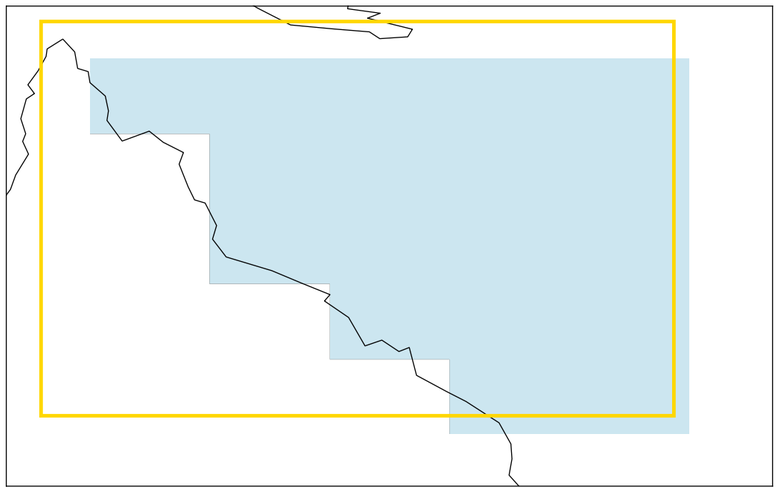
Predicted DHW - Step 46



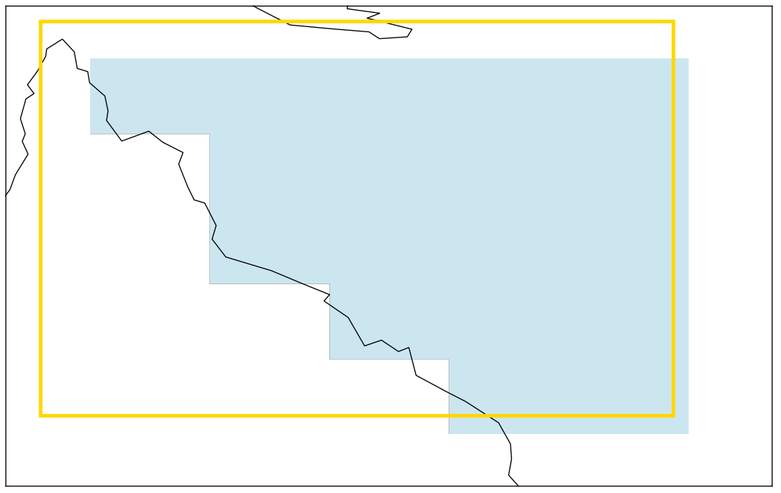
Error (Pred - Actual) - Step 46 [scale ±8]



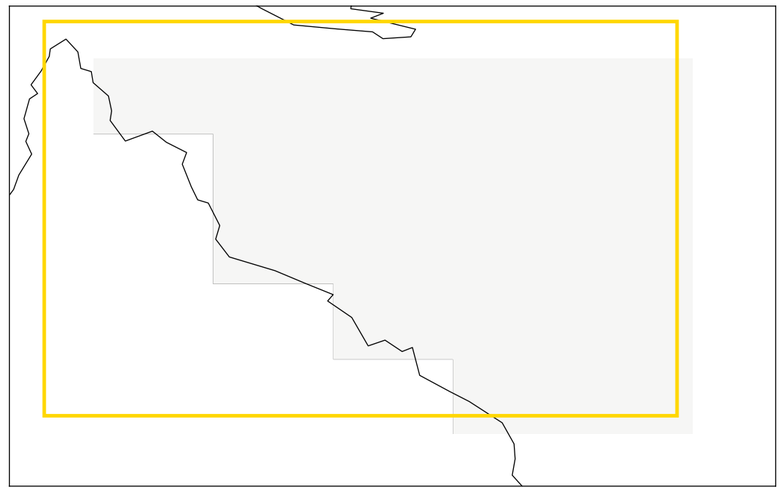
Actual DHW - Step 90

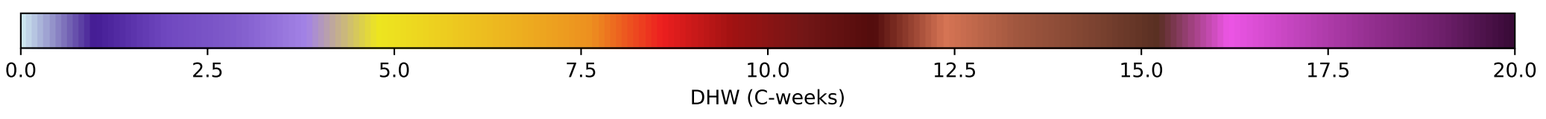


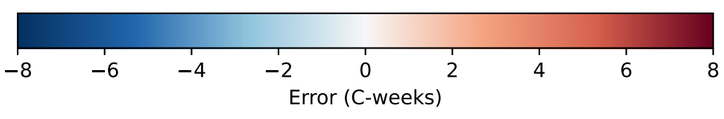
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

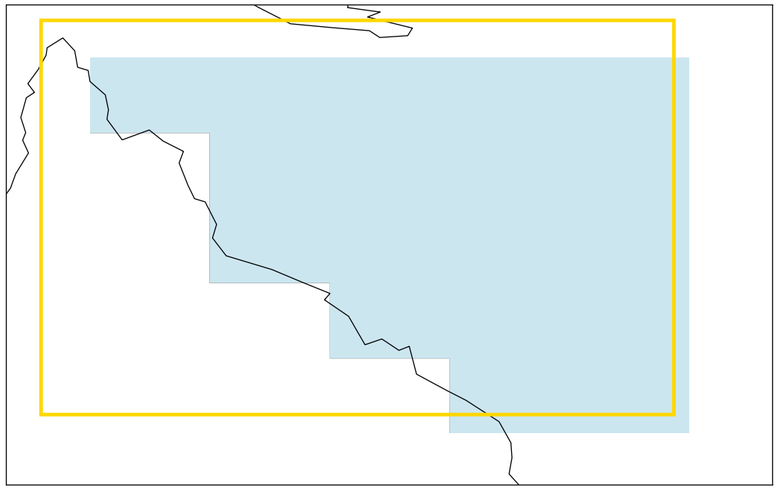




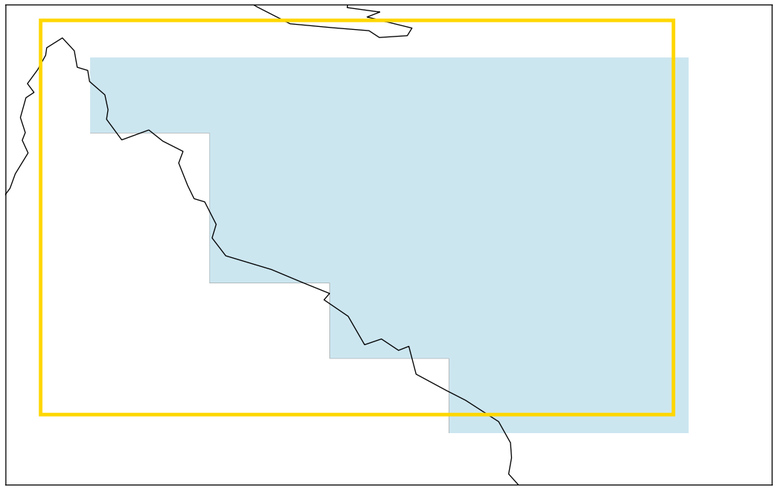


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

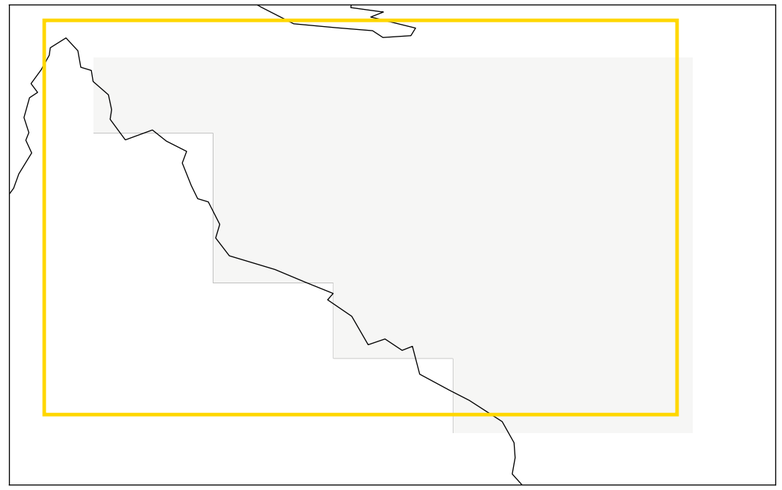
Actual DHW - Step 1



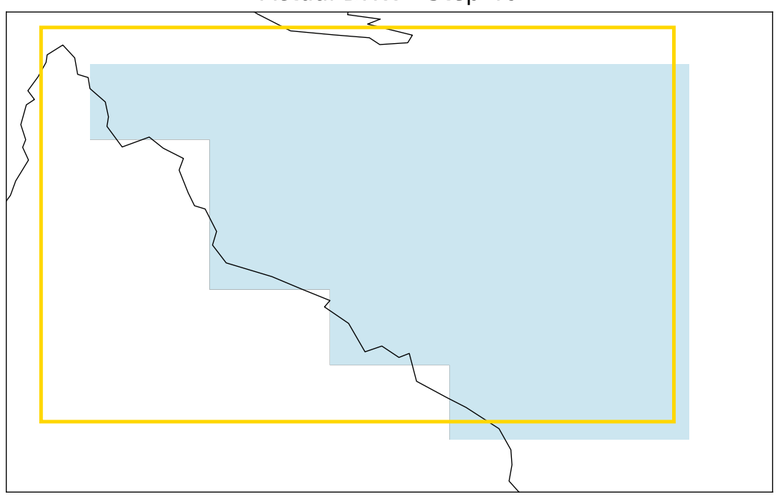
Predicted DHW - Step 1



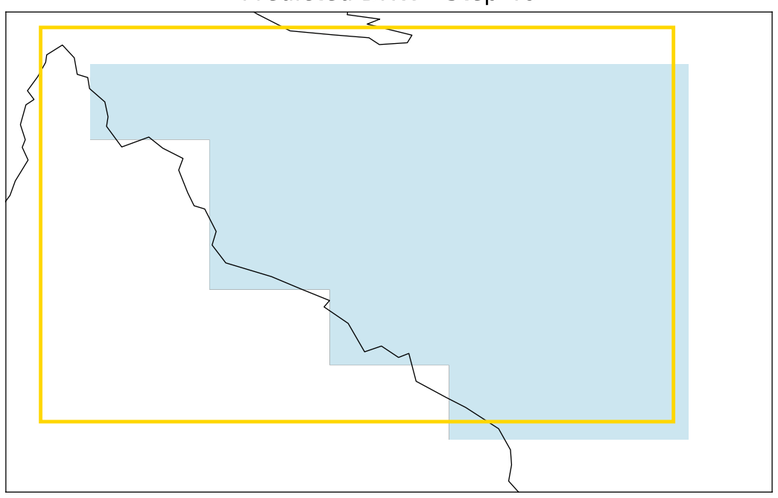
Error (Pred - Actual) - Step 1 [scale ±8]



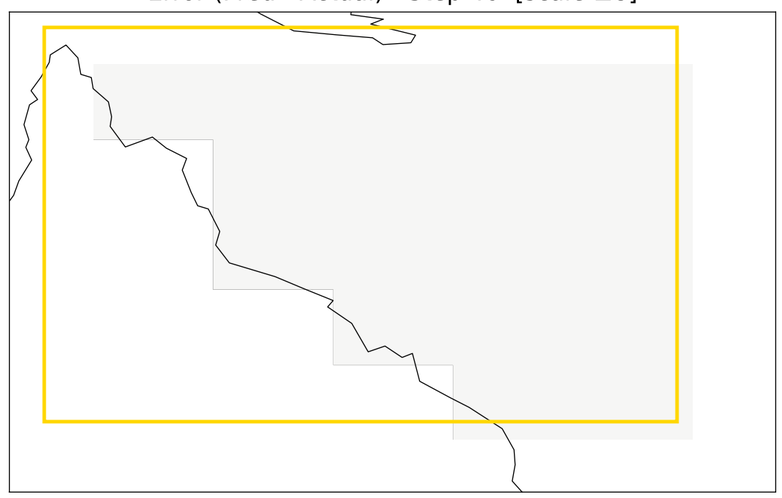
Actual DHW - Step 46



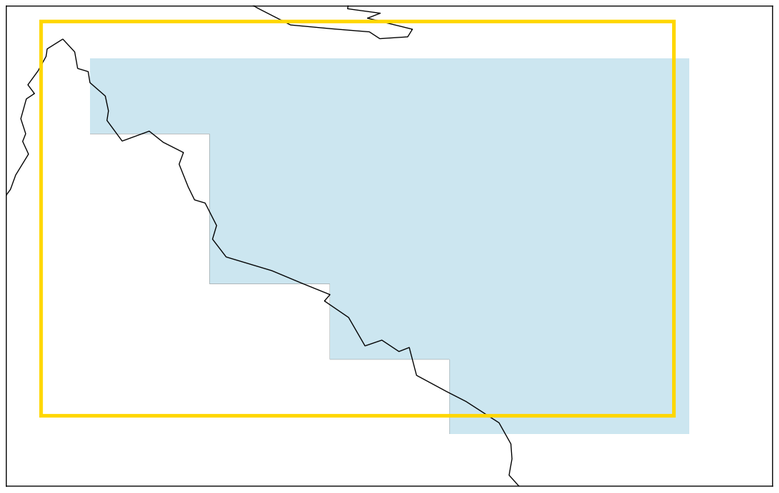
Predicted DHW - Step 46



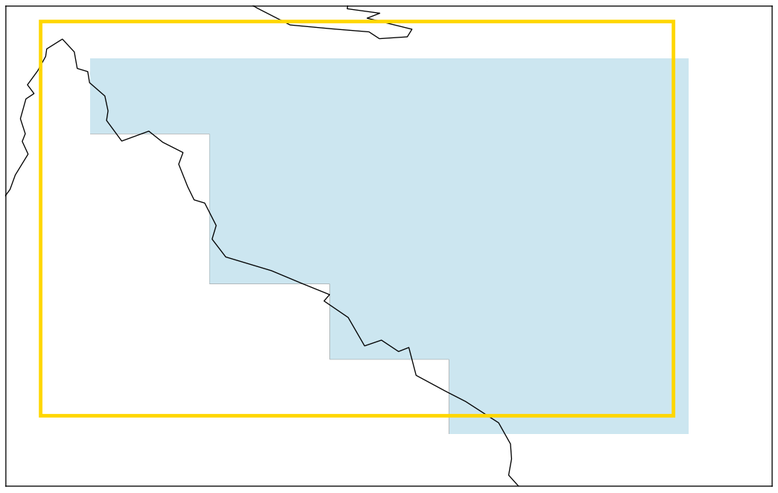
Error (Pred - Actual) - Step 46 [scale ±8]



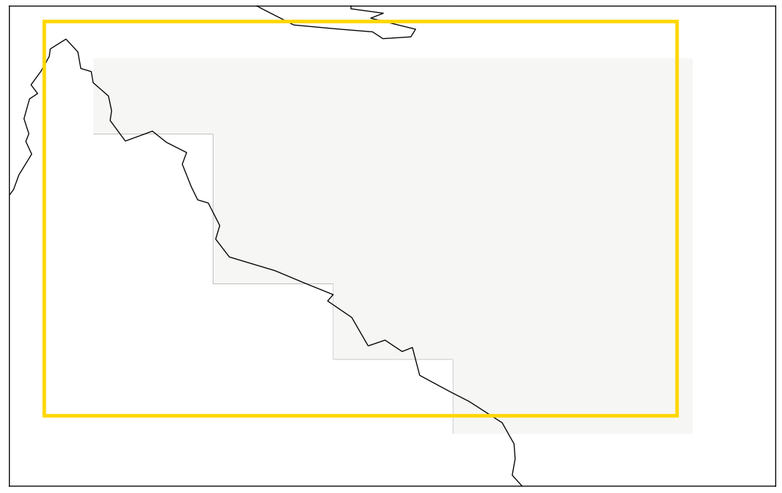
Actual DHW - Step 90

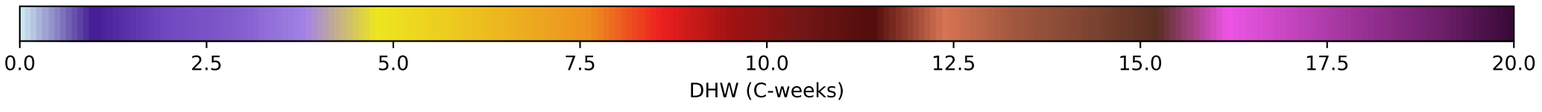


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

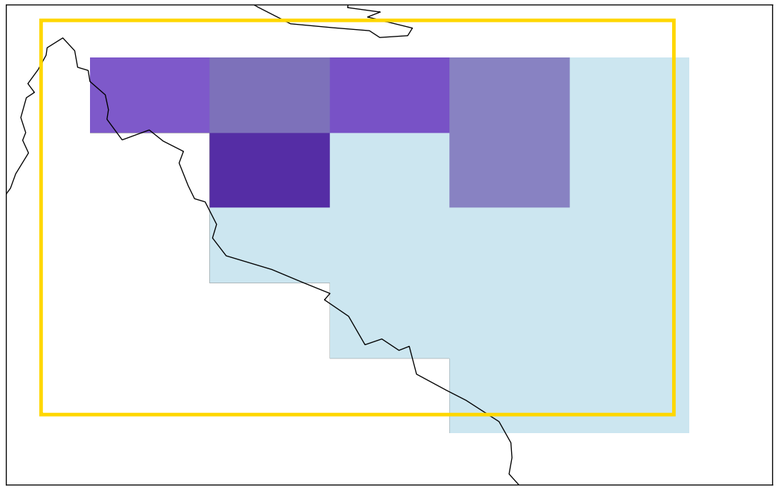




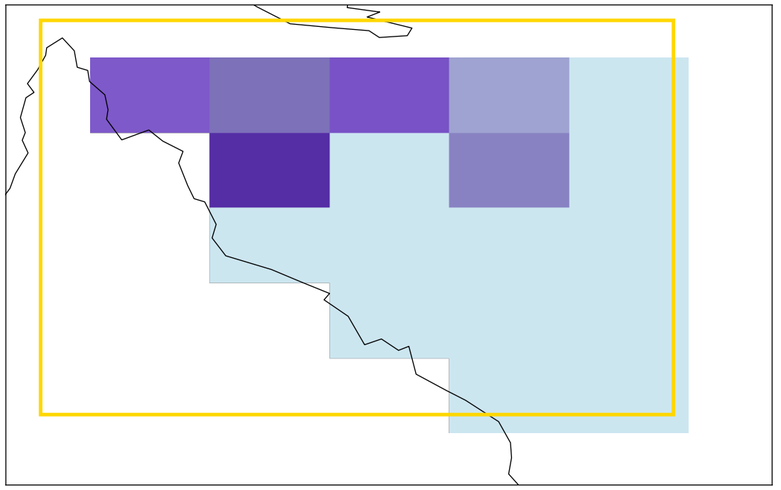


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

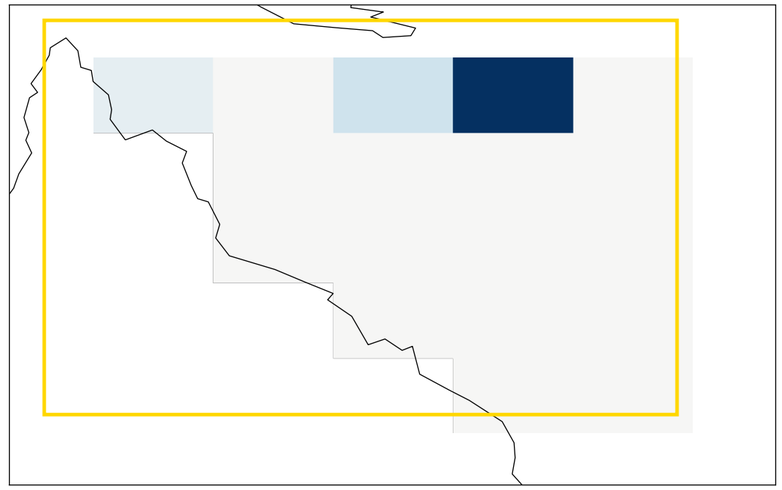
Actual DHW - Step 1



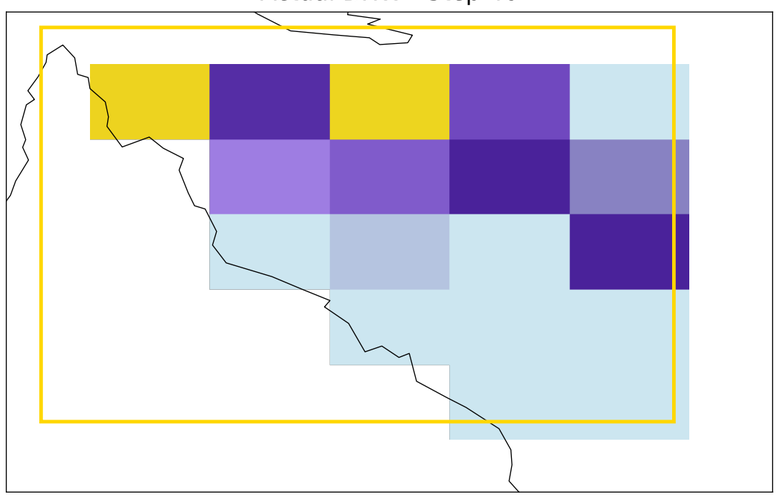
Predicted DHW - Step 1



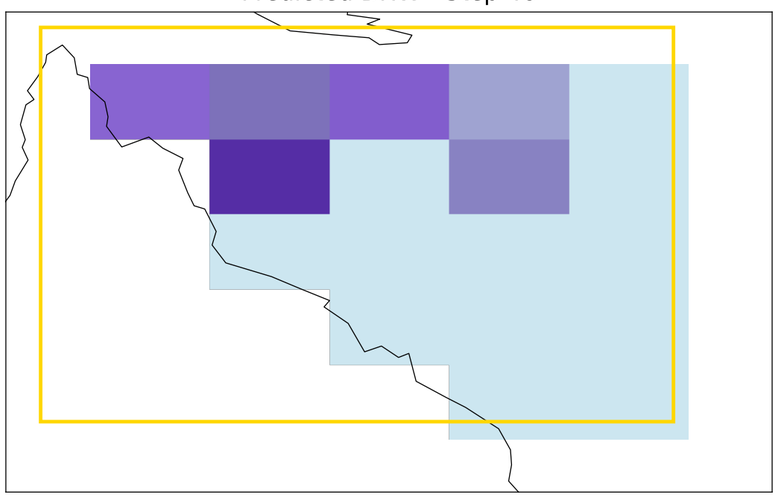
Error (Pred - Actual) - Step 1 [scale ±0.11]



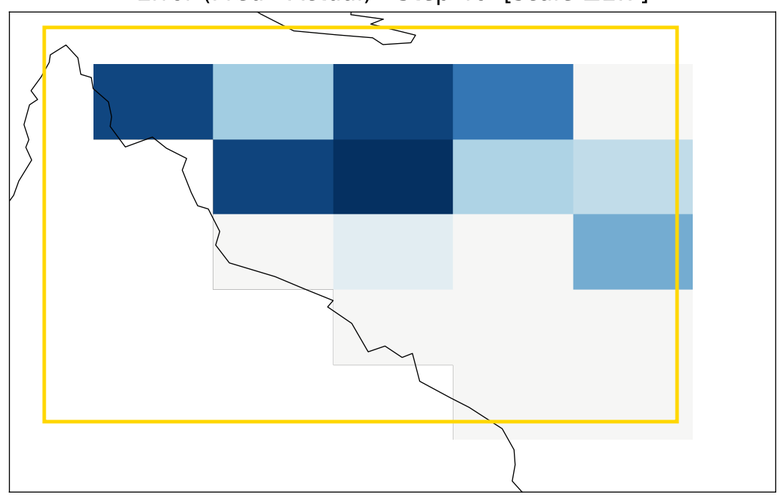
Actual DHW - Step 46



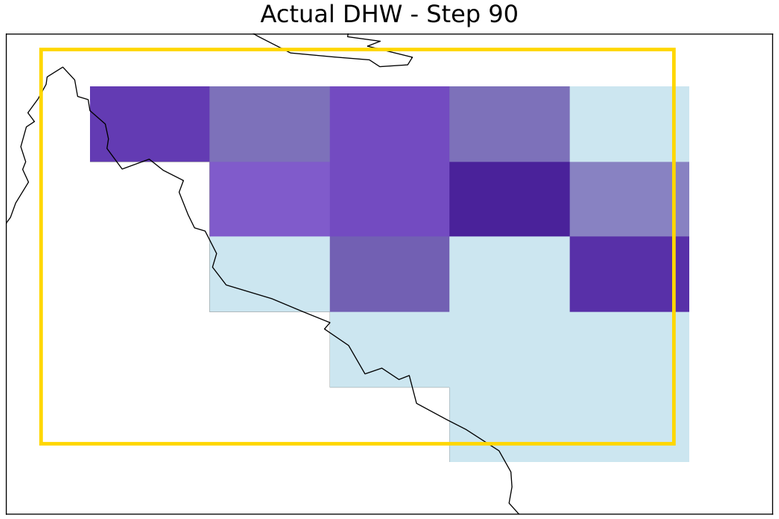
Predicted DHW - Step 46



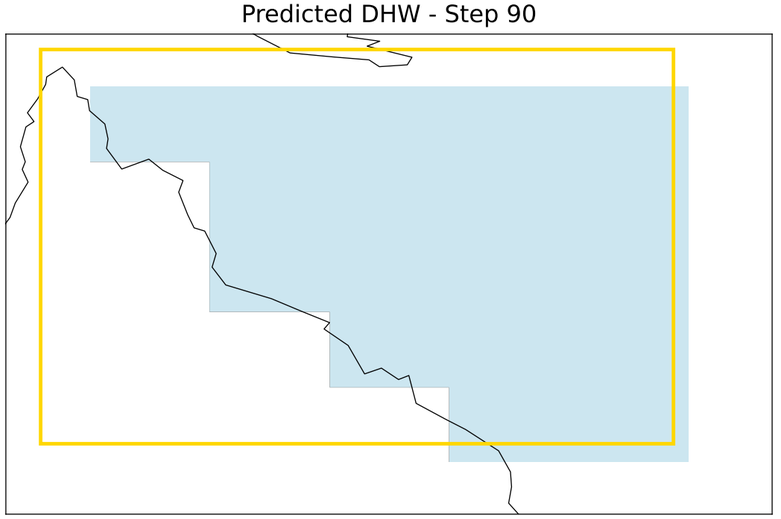
Error (Pred - Actual) - Step 46 [scale ±2.7]



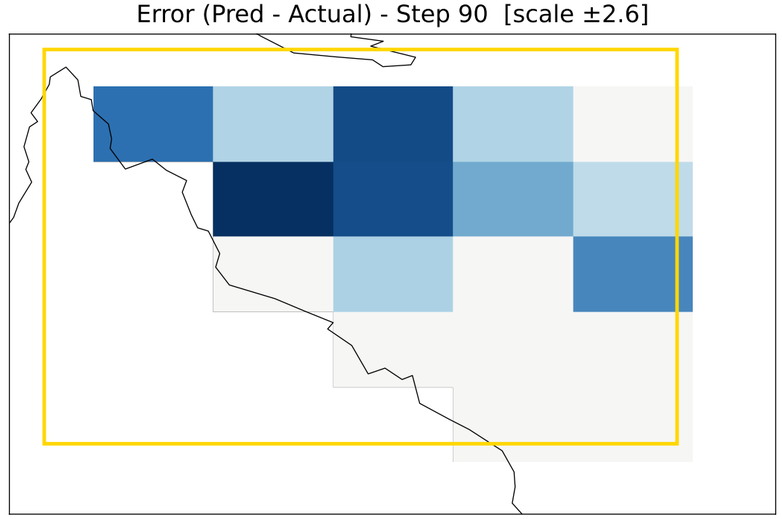
Actual DHW - Step 90

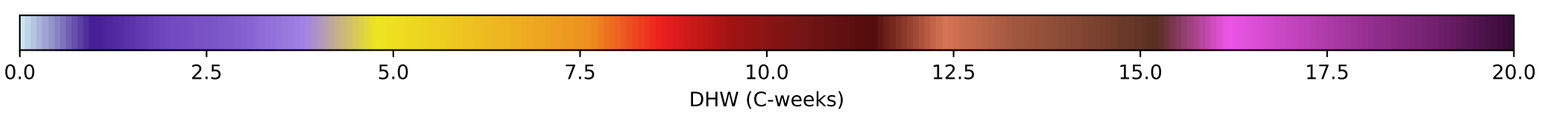


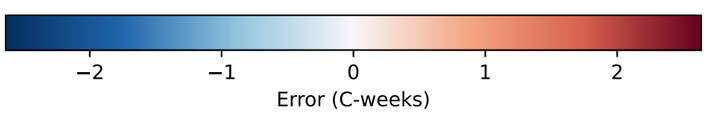
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

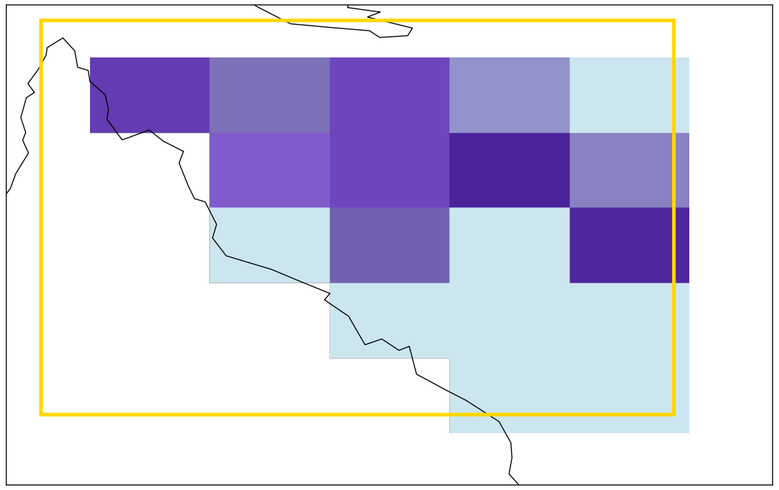




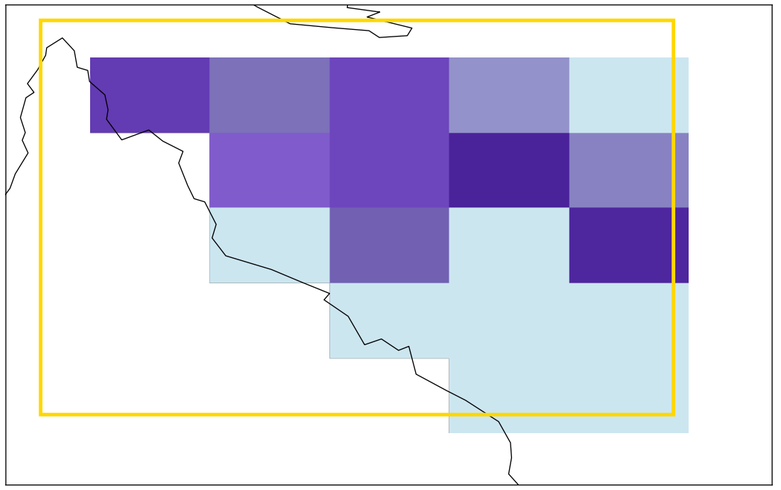


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

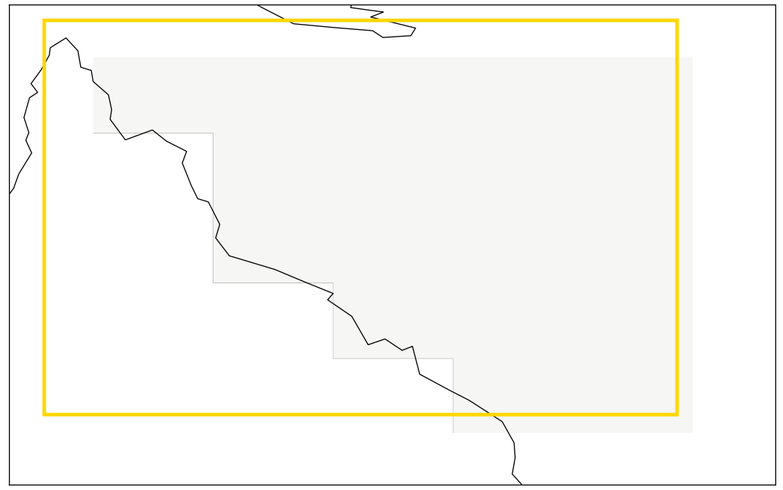
Actual DHW - Step 1



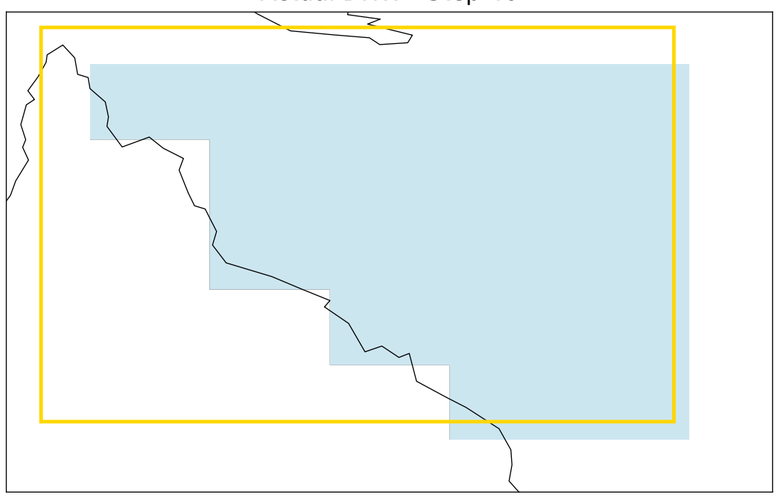
Predicted DHW - Step 1



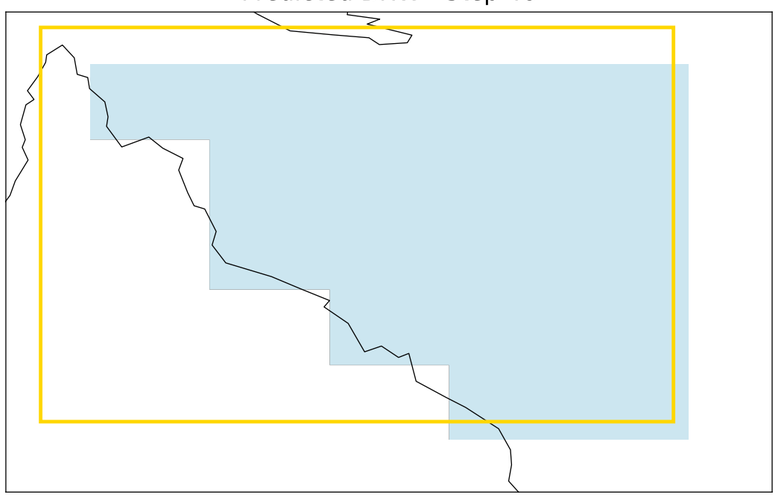
Error (Pred - Actual) - Step 1 [scale ±8]



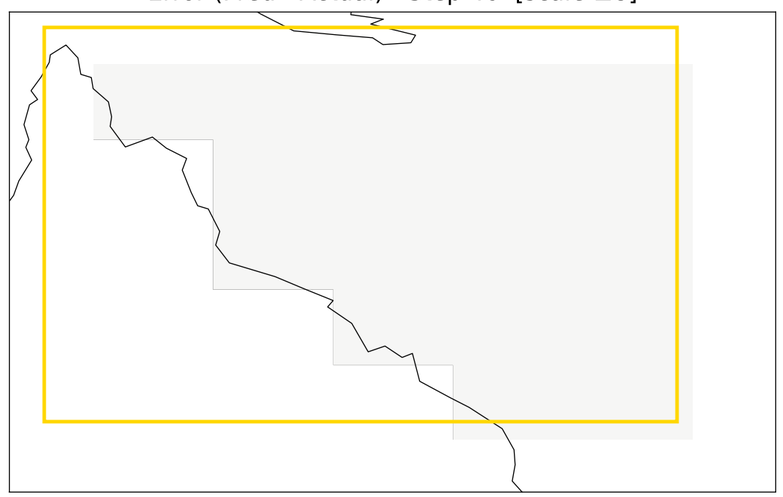
Actual DHW - Step 46



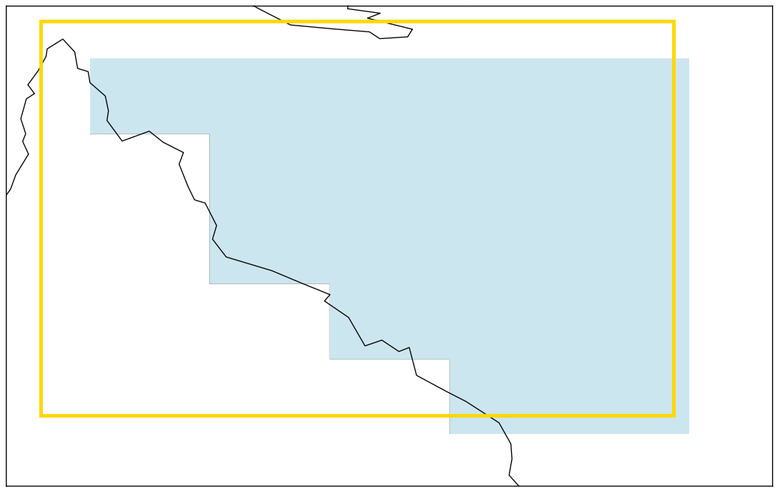
Predicted DHW - Step 46



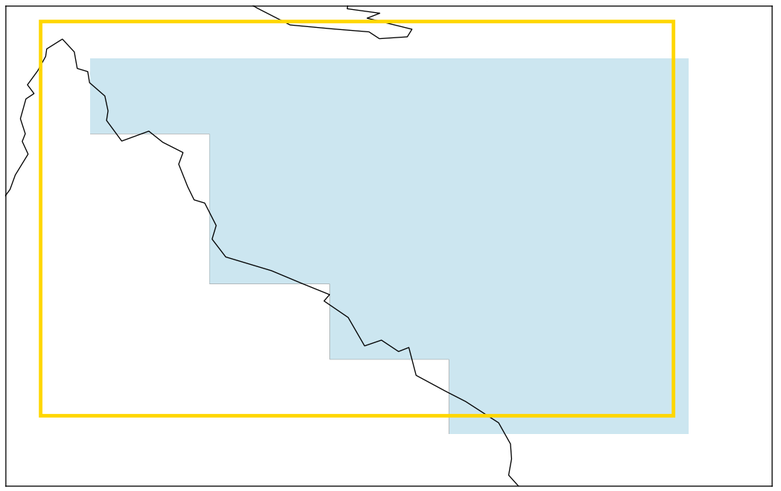
Error (Pred - Actual) - Step 46 [scale ±8]



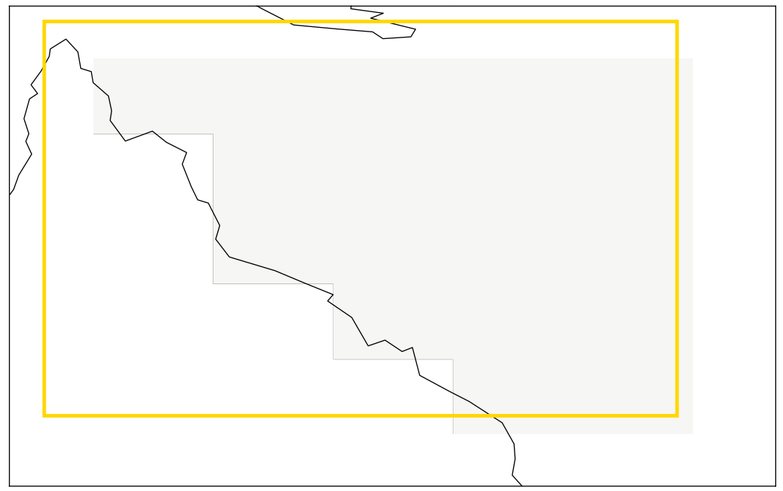
Actual DHW - Step 90

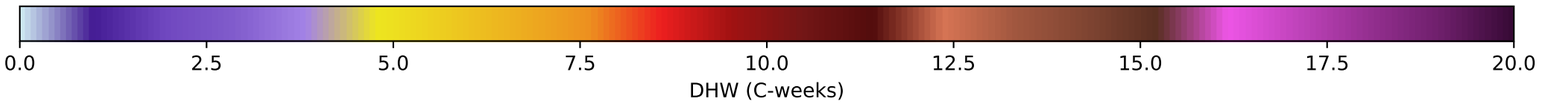


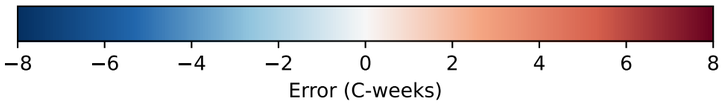
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

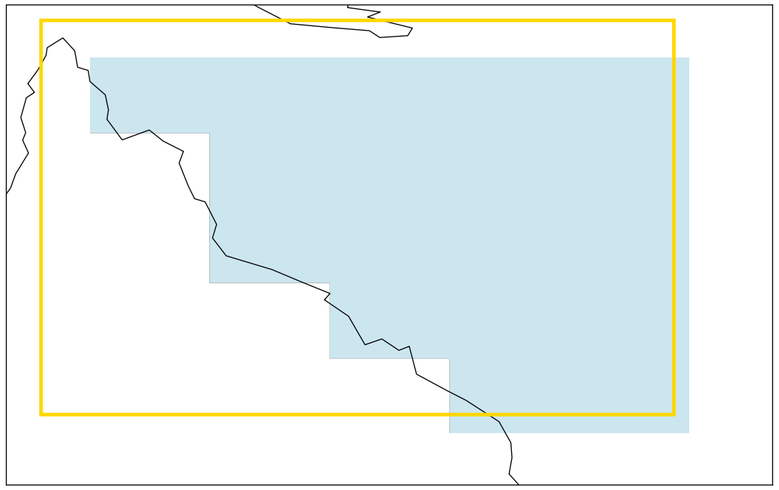




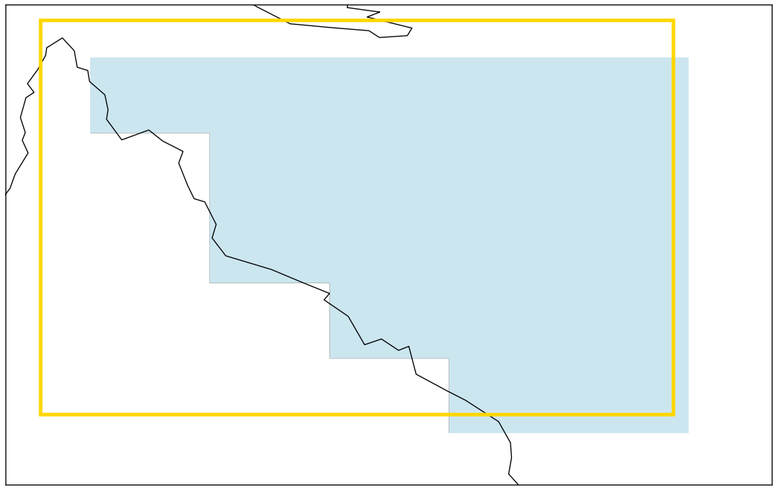


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

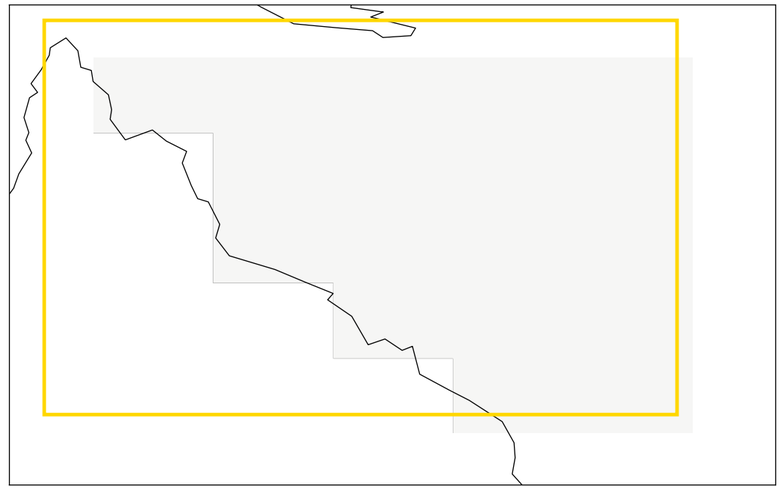
Actual DHW - Step 1



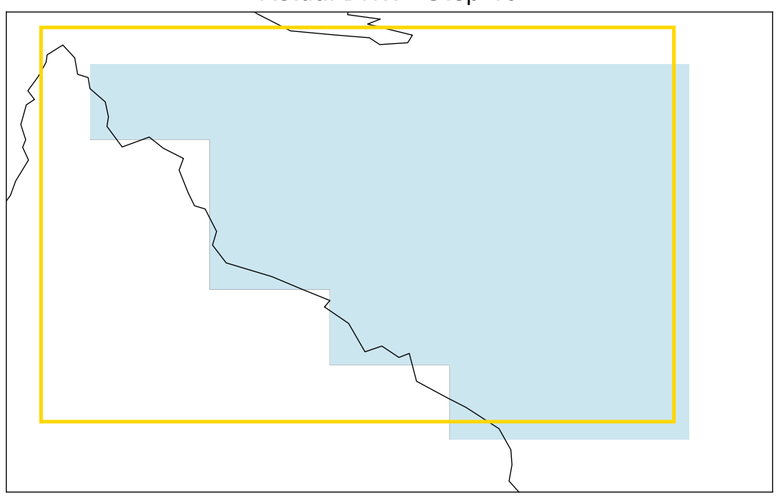
Predicted DHW - Step 1



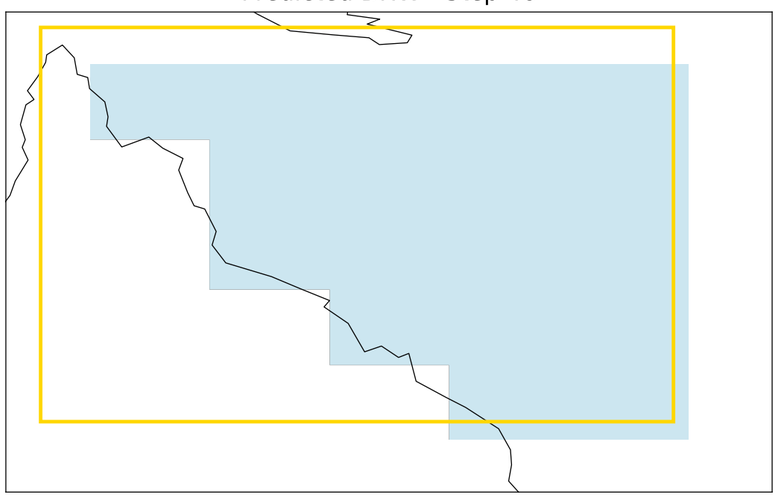
Error (Pred - Actual) - Step 1 [scale ±8]



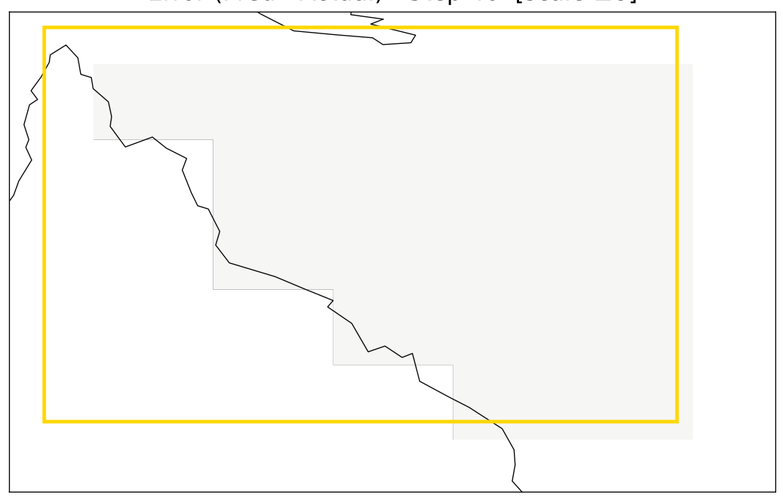
Actual DHW - Step 46



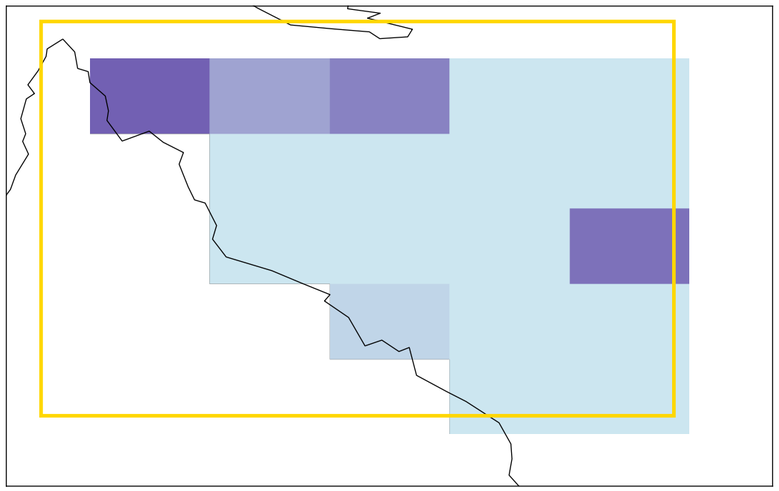
Predicted DHW - Step 46



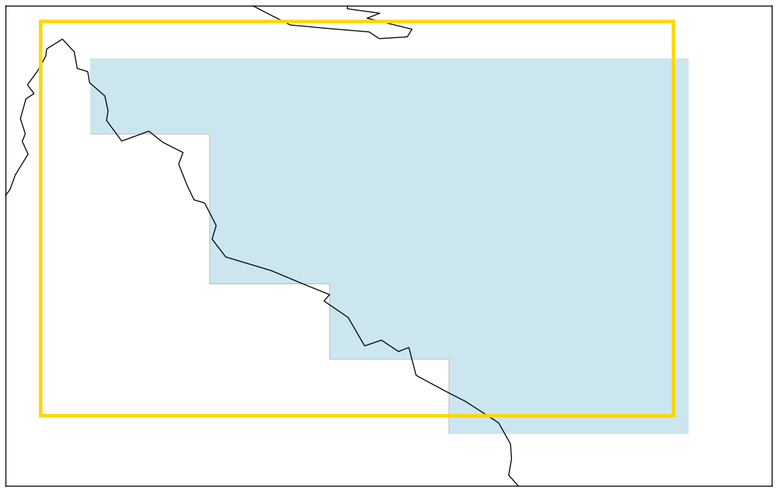
Error (Pred - Actual) - Step 46 [scale ±8]



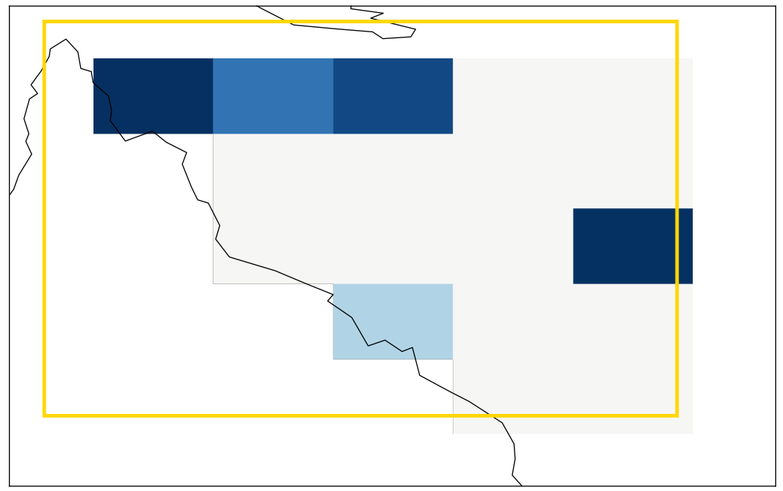
Actual DHW - Step 90

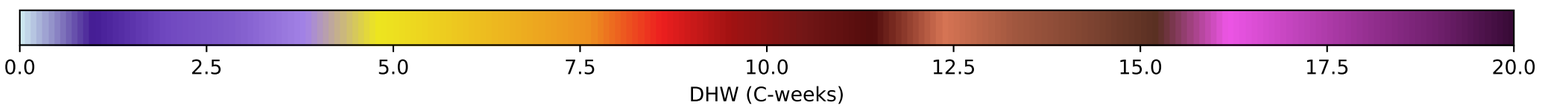


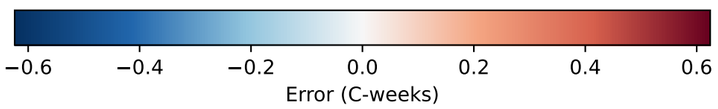
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

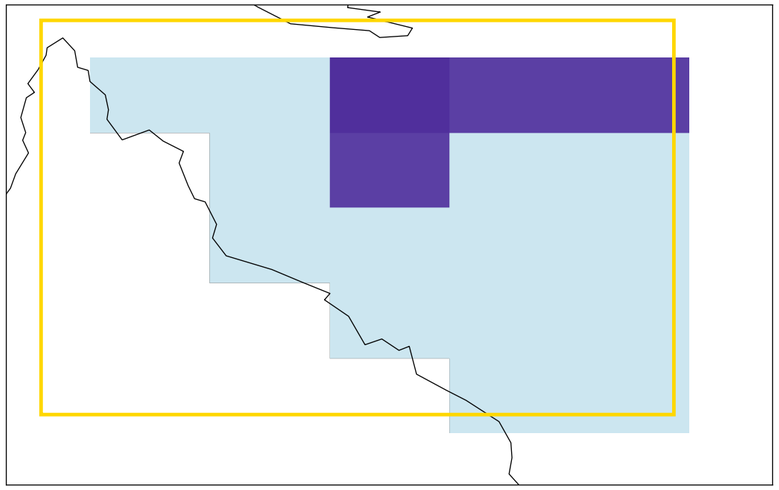




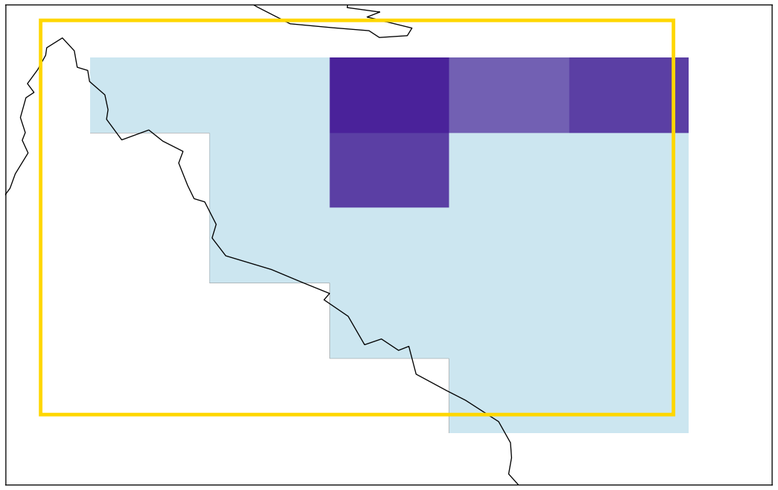


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

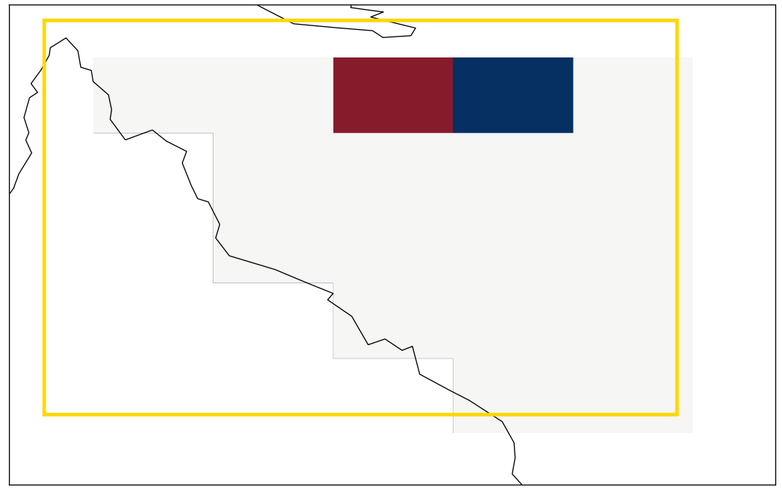
Actual DHW - Step 1



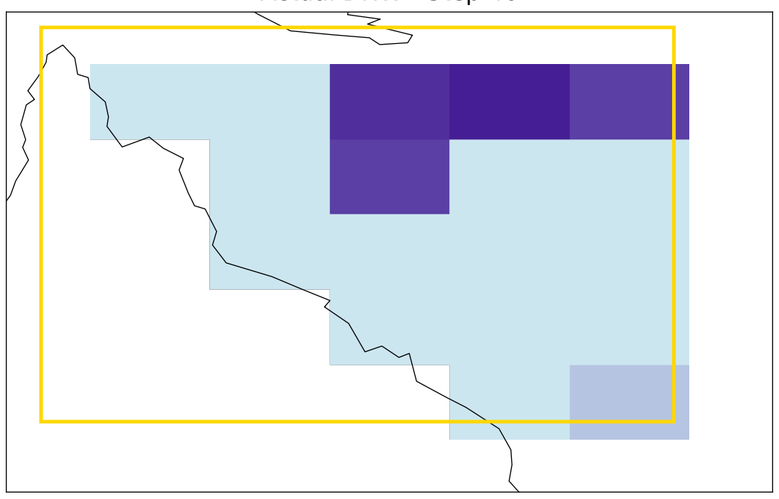
Predicted DHW - Step 1



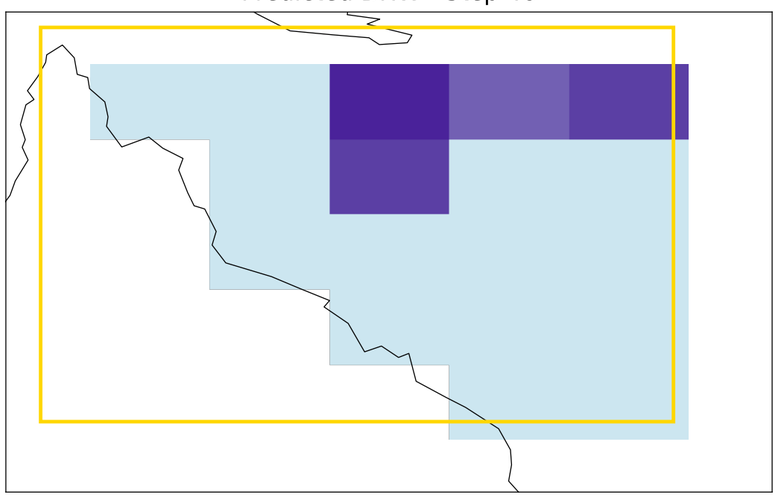
Error (Pred - Actual) - Step 1 [scale ±0.19]



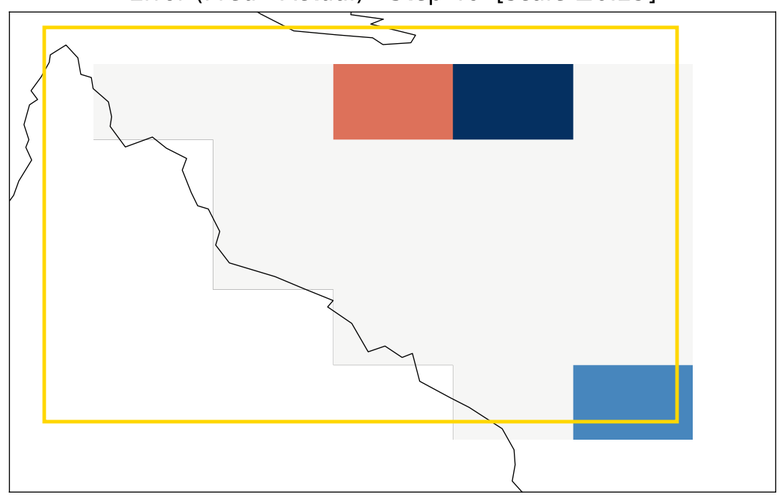
Actual DHW - Step 46



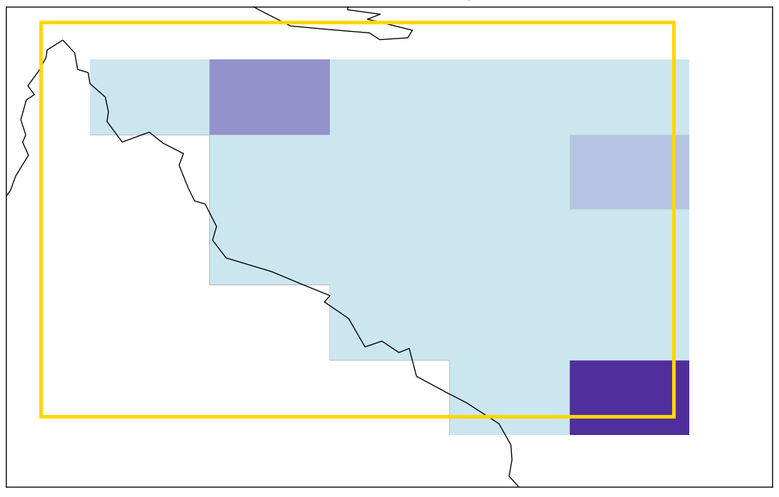
Predicted DHW - Step 46



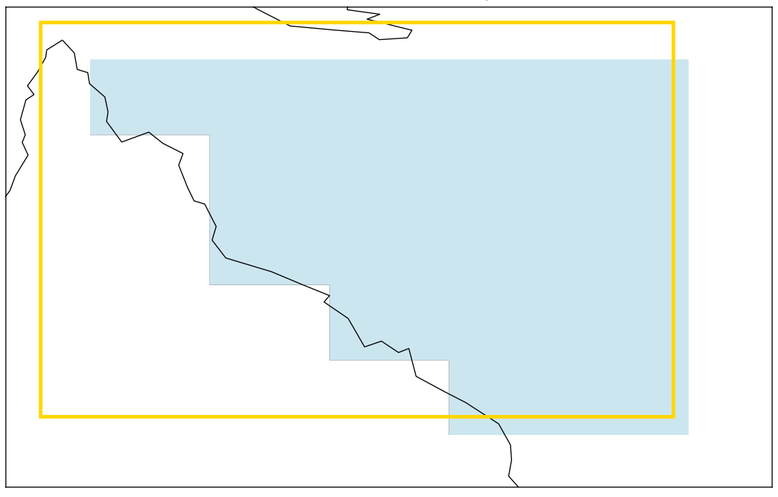
Error (Pred - Actual) - Step 46 [scale ±0.29]



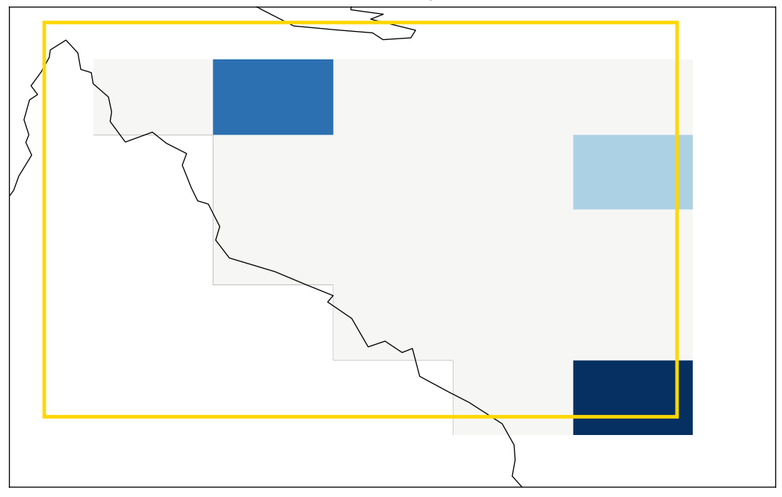
Actual DHW - Step 90

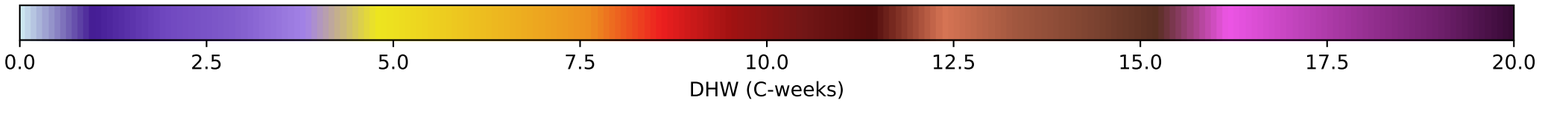


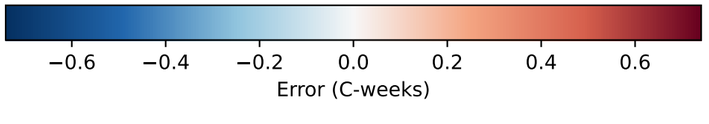
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

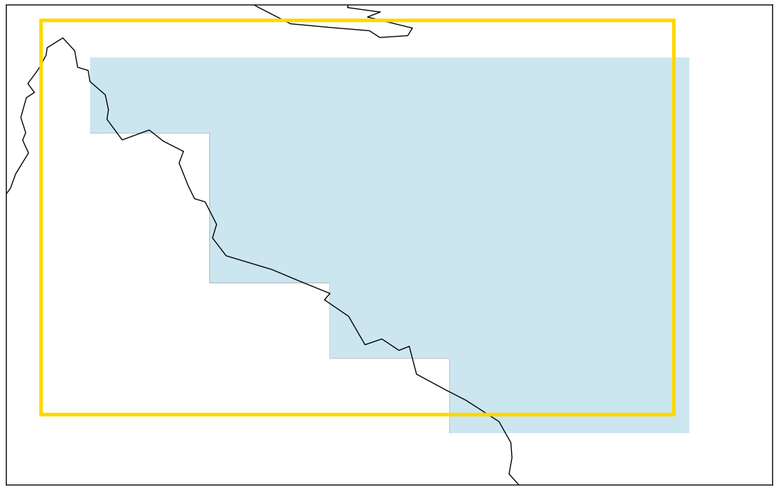




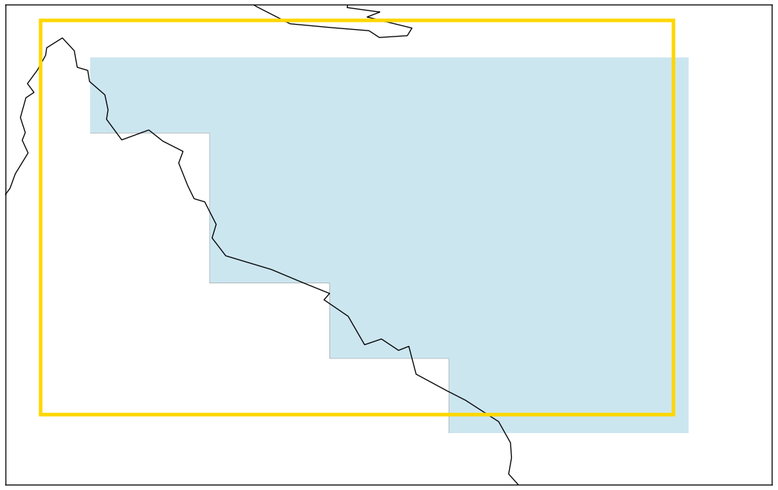


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

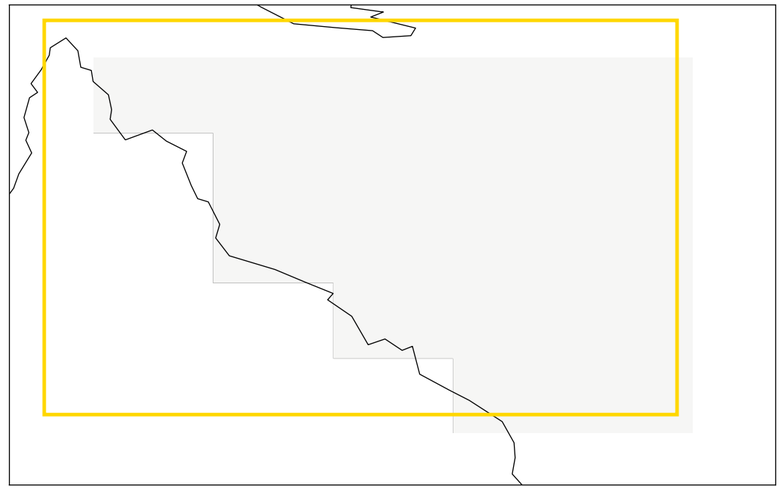
Actual DHW - Step 1



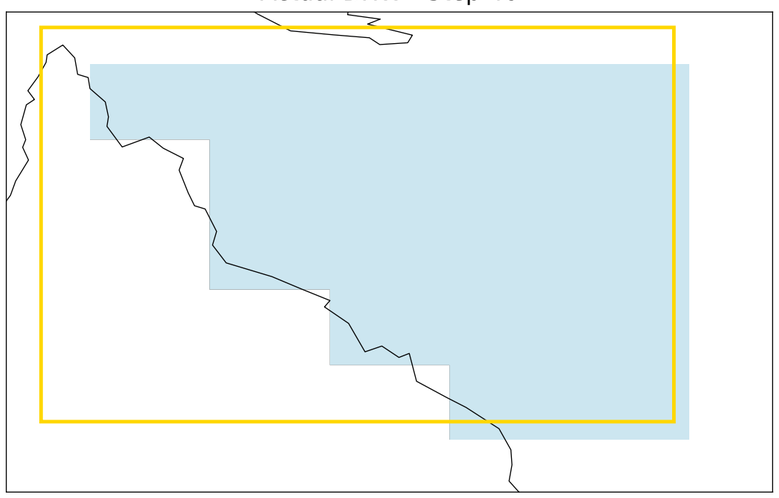
Predicted DHW - Step 1



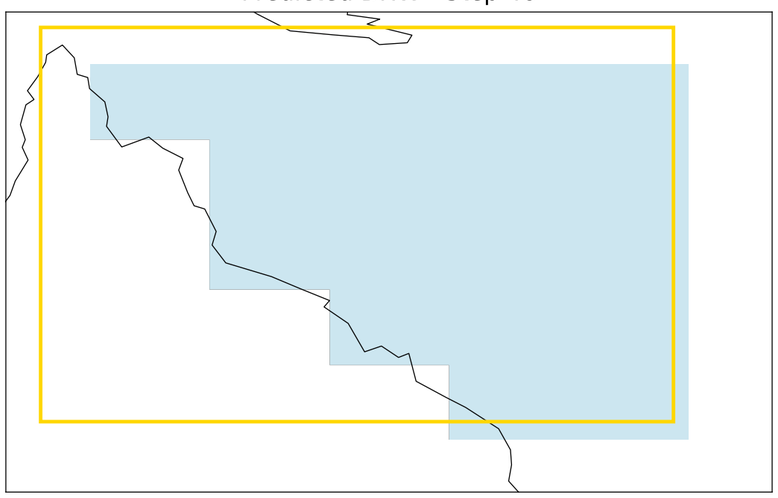
Error (Pred - Actual) - Step 1 [scale ±8]



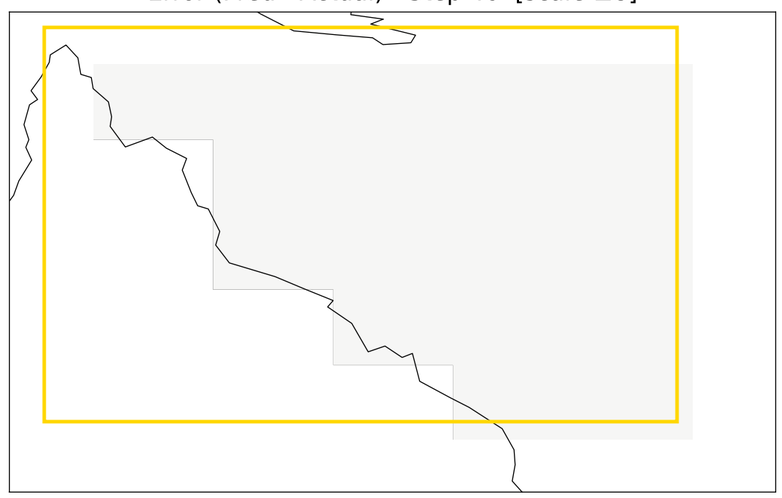
Actual DHW - Step 46



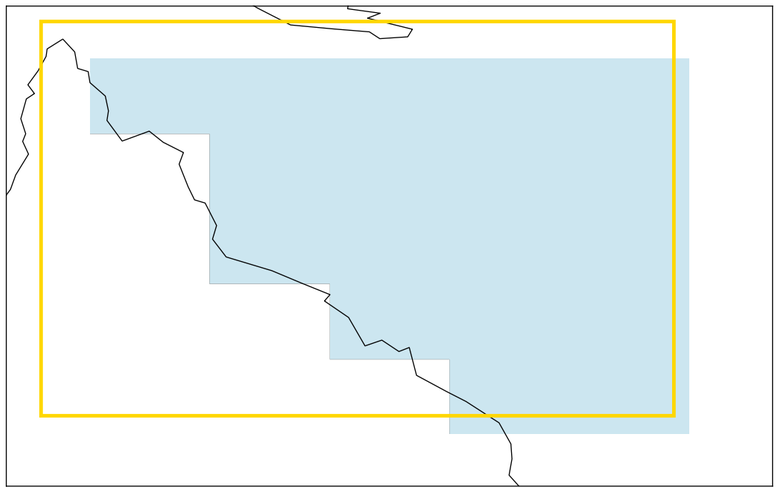
Predicted DHW - Step 46



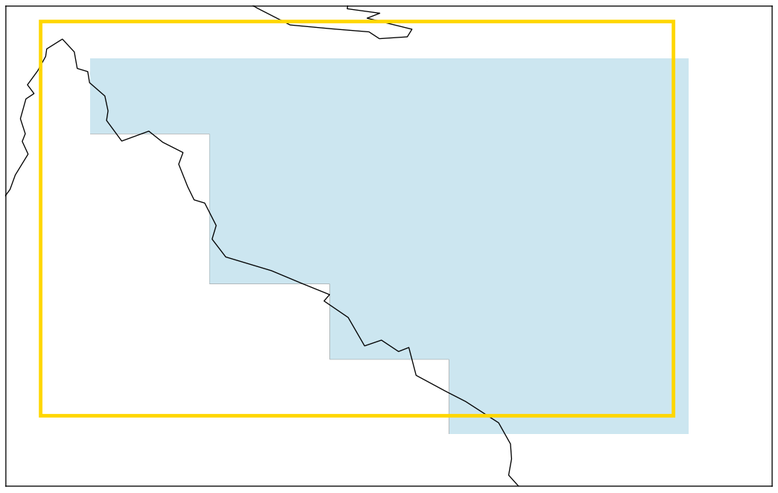
Error (Pred - Actual) - Step 46 [scale ±8]



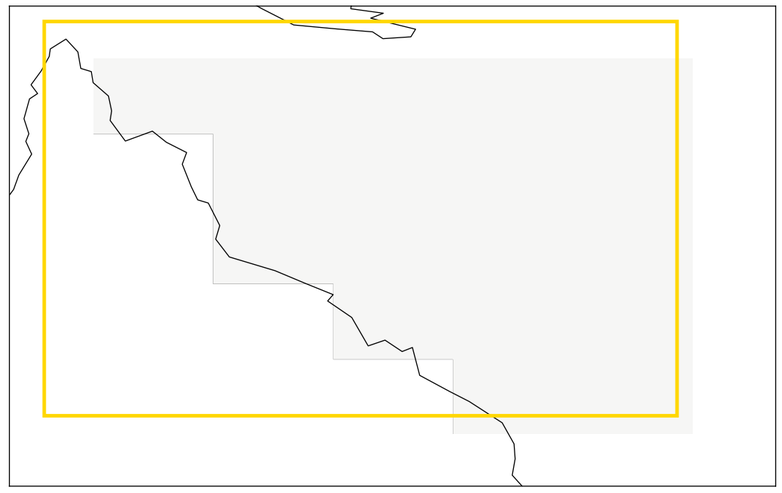
Actual DHW - Step 90

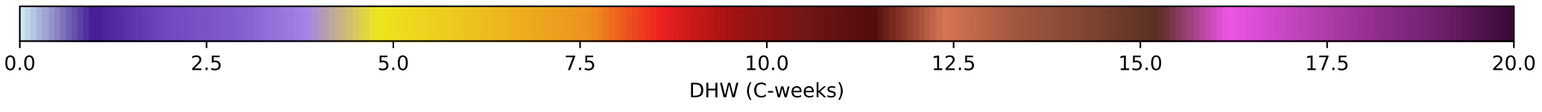


Predicted DHW - Step 90



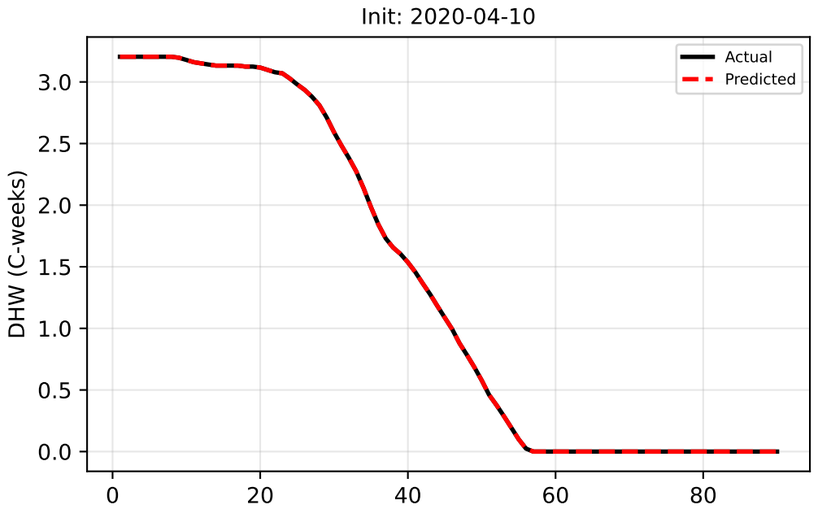
Error (Pred - Actual) - Step 90 [scale ±8]

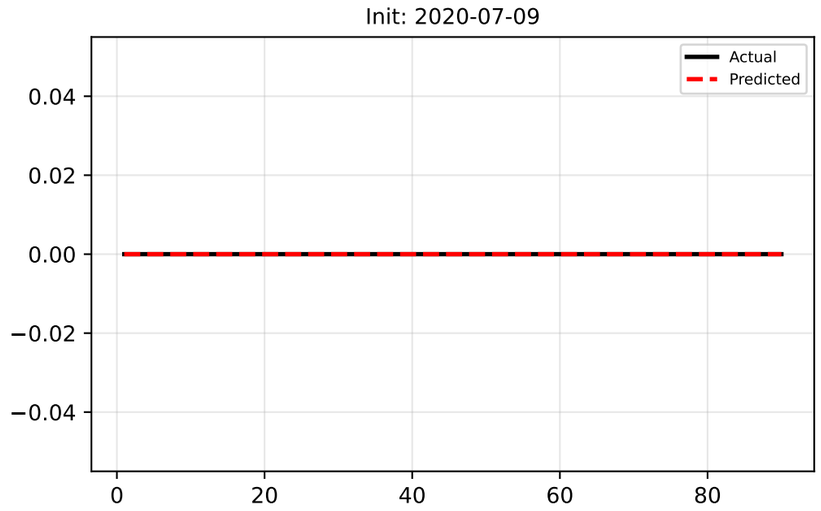


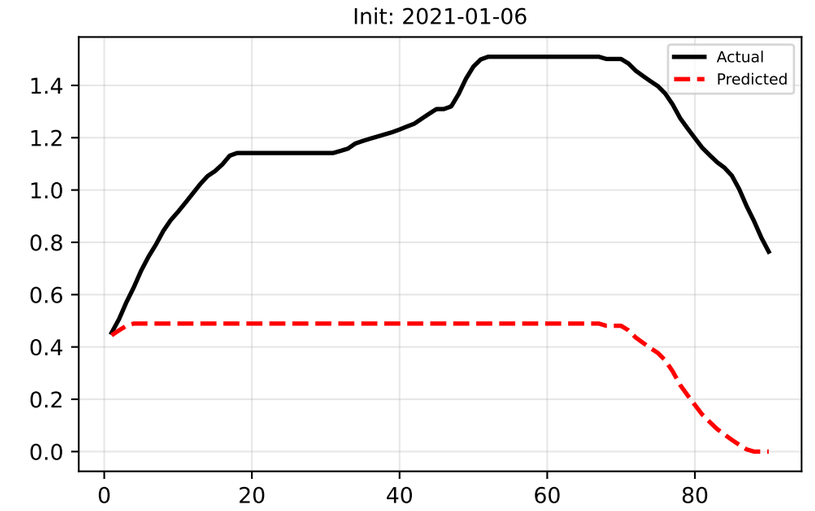


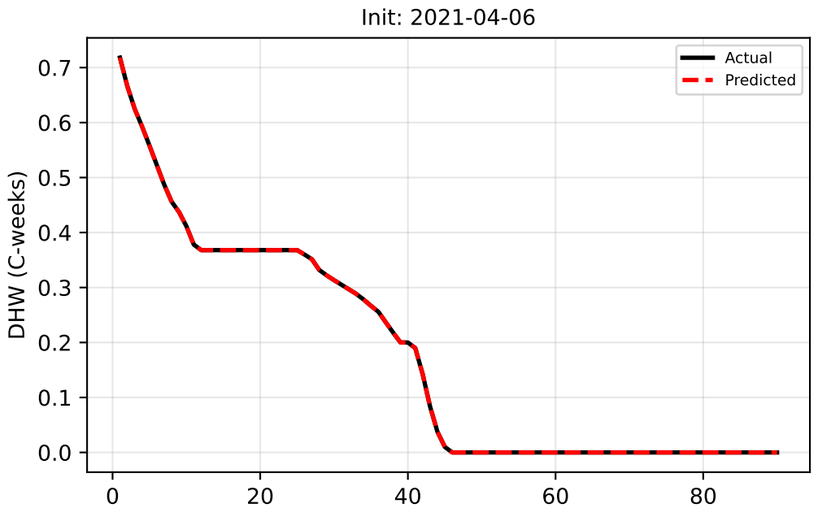


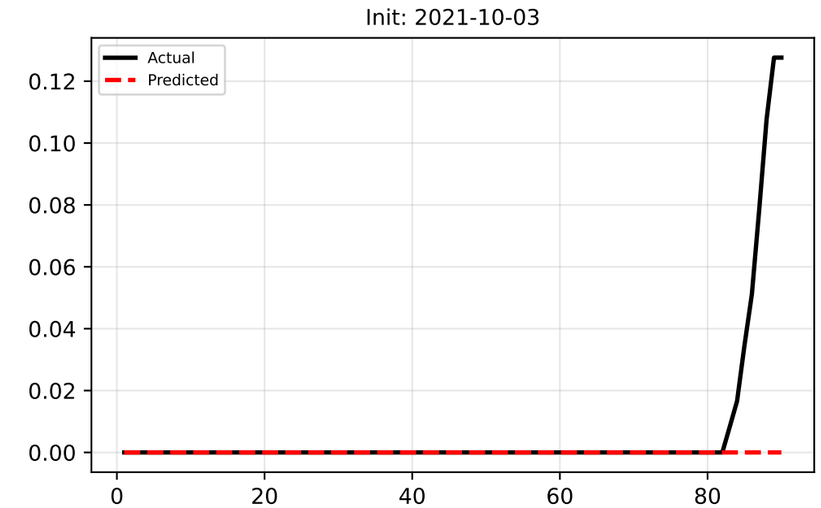
Global-Mean DHW - Test - GBR (C-weeks)

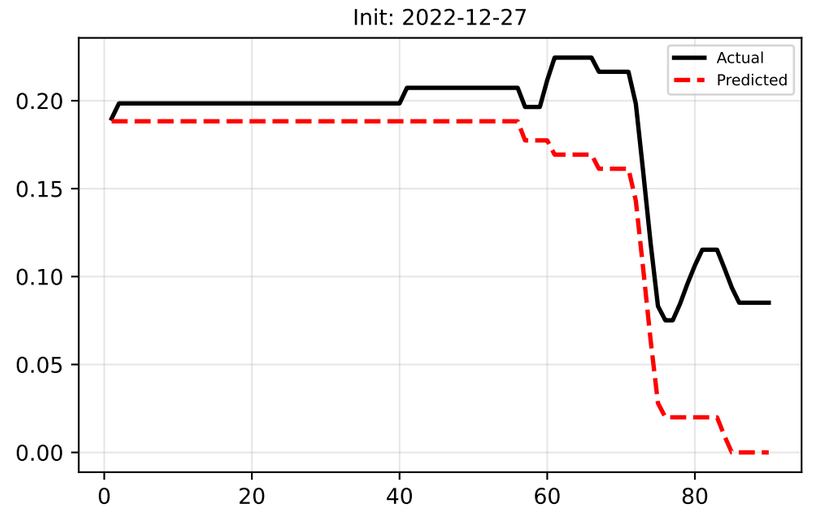


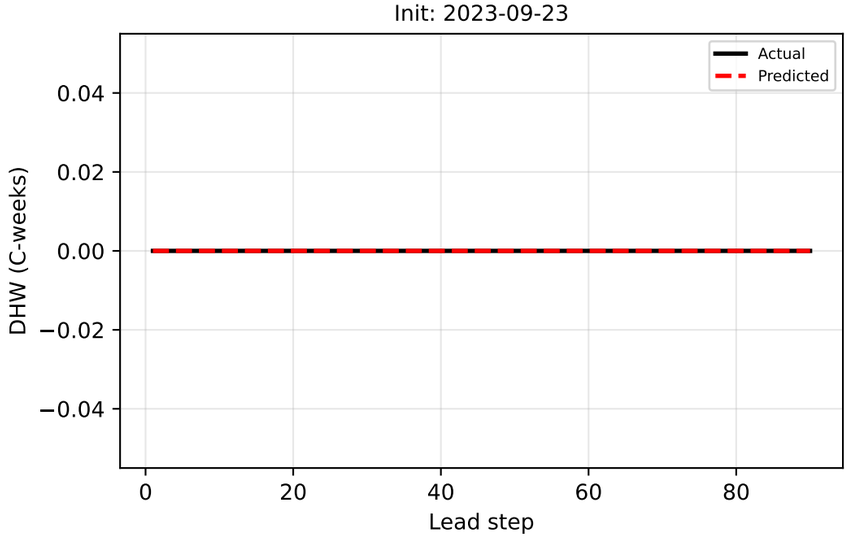


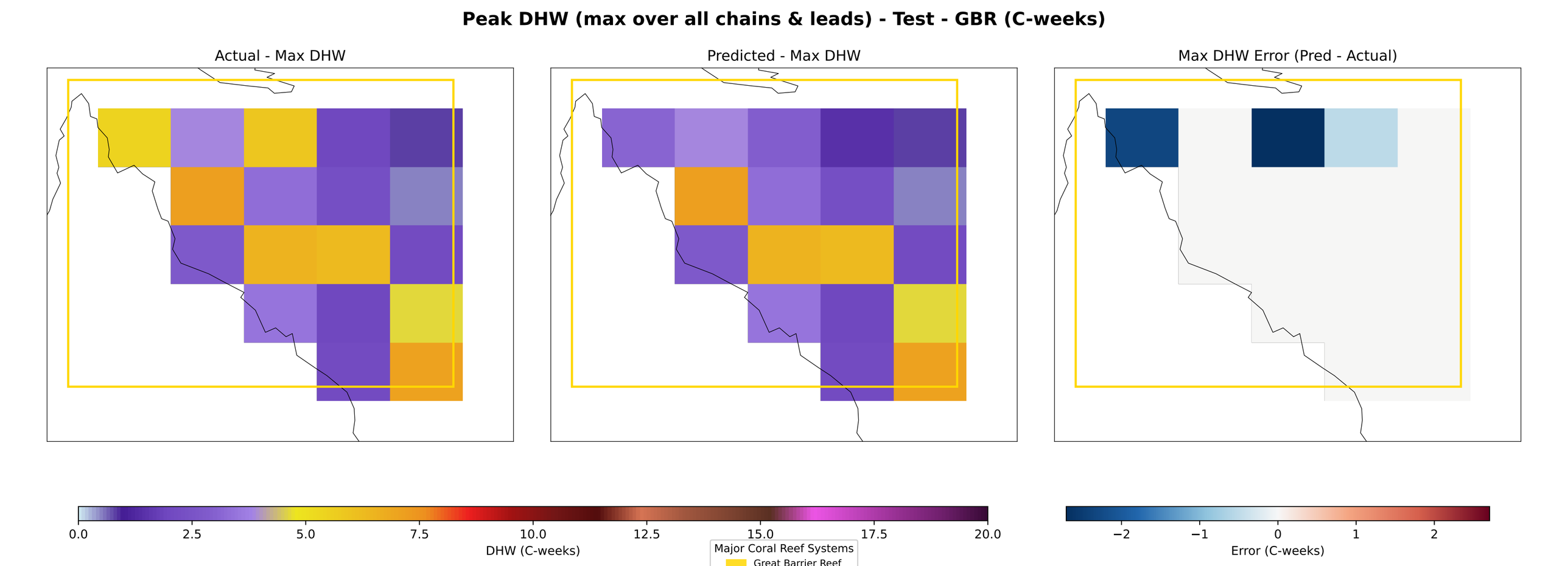


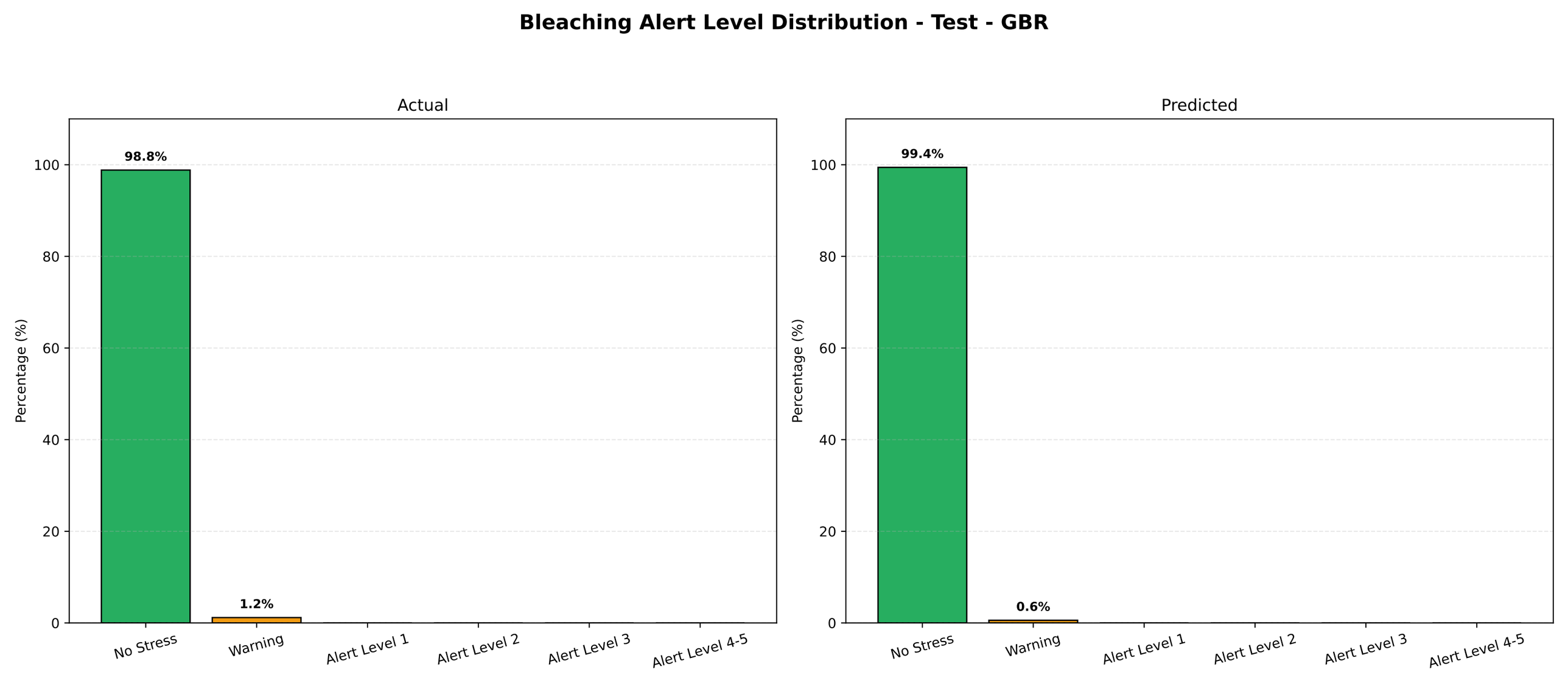




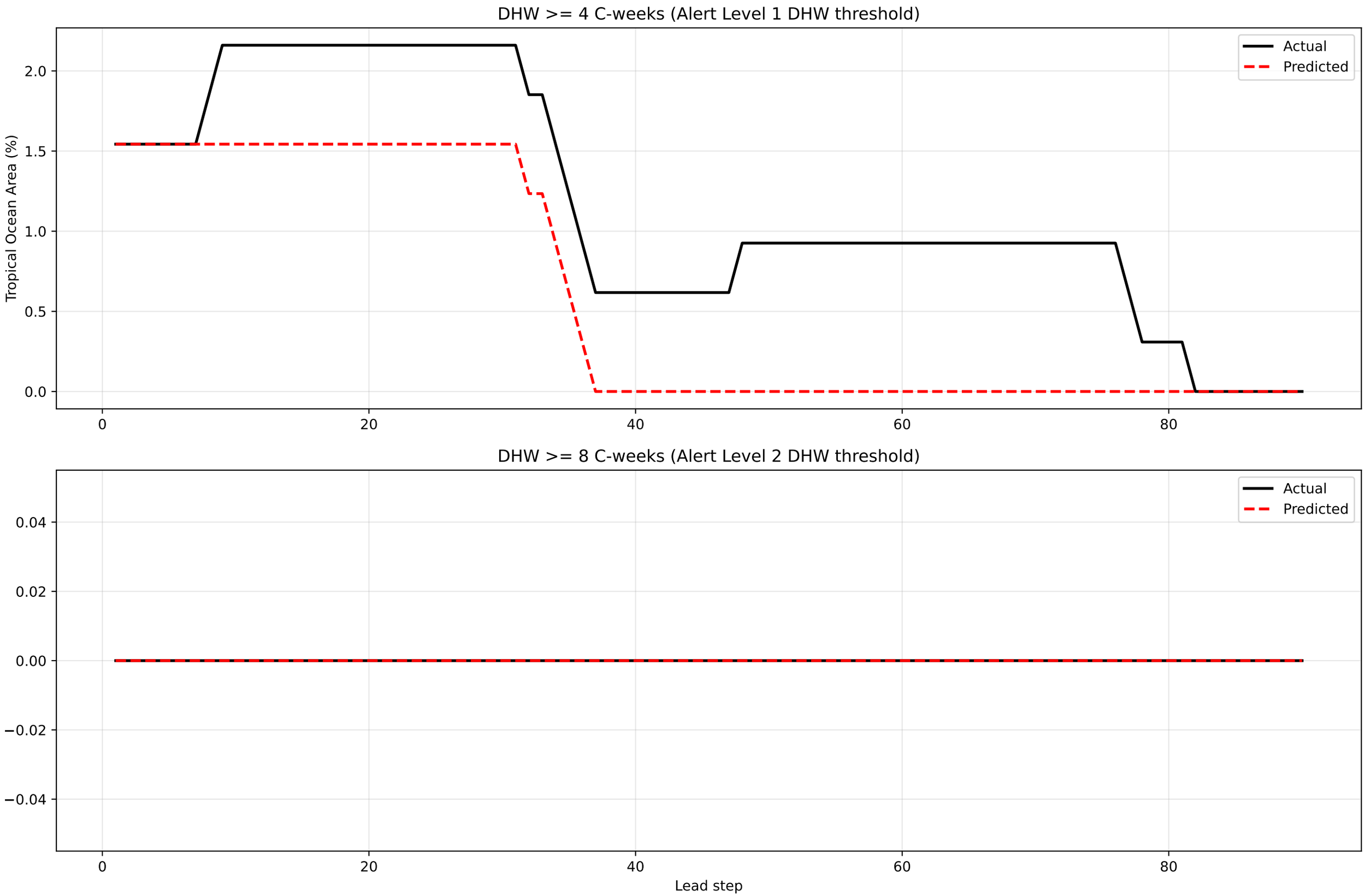





Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)


Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

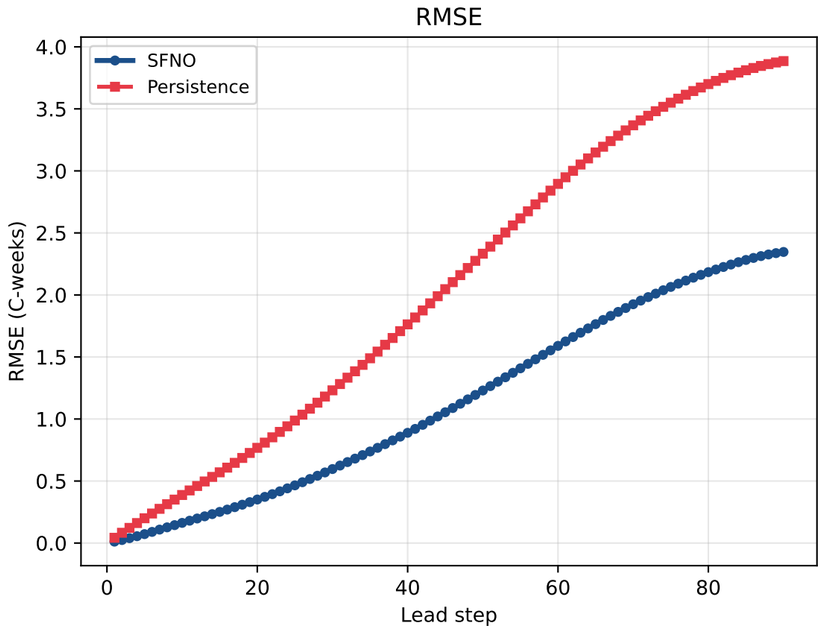
Tropical Ocean Area Under Bleaching Stress - Test - GBR

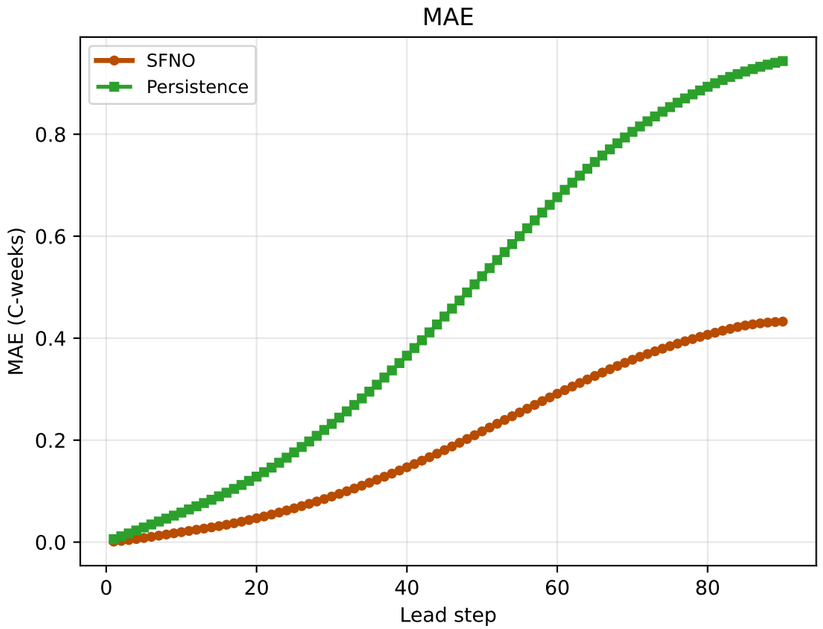


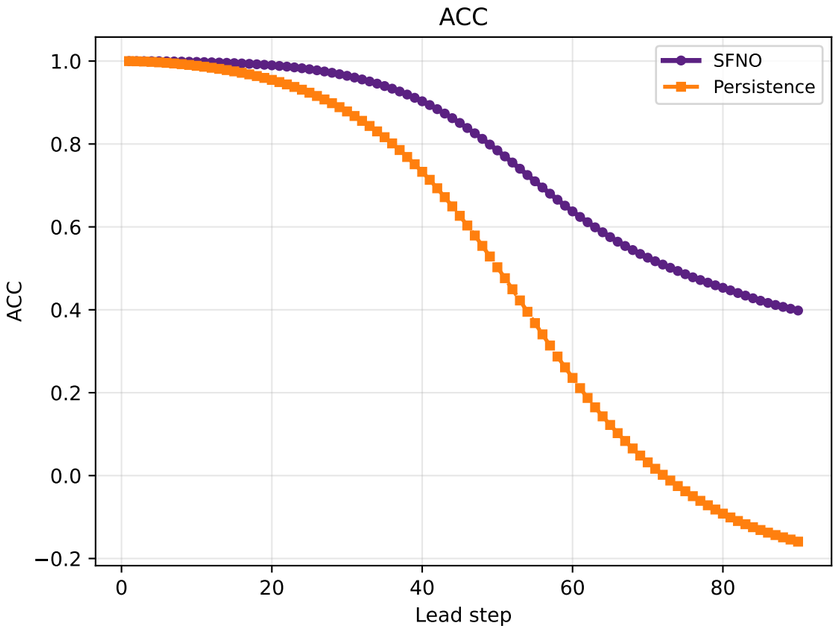
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

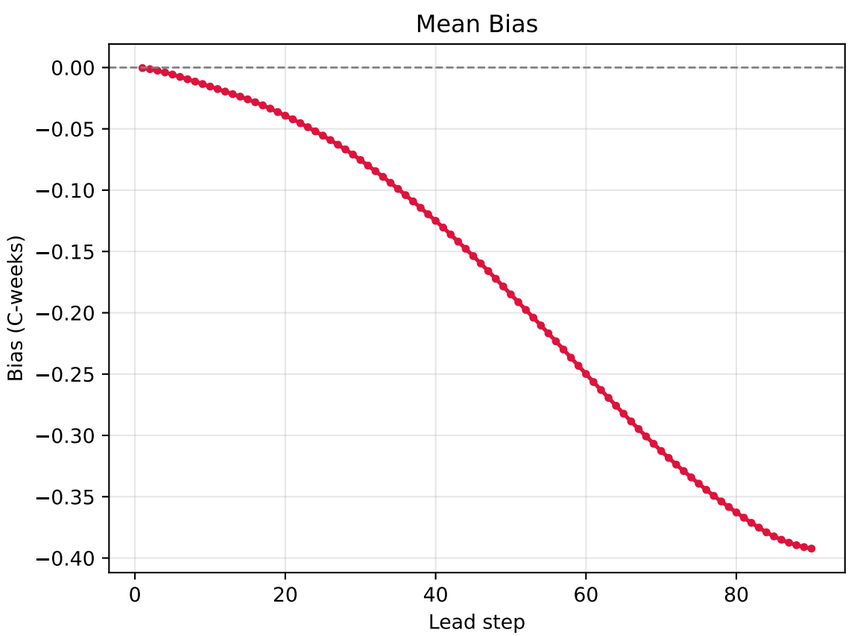
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0115	0.0425	72.9%	0.0010	0.0057	-0.0004	1.0000	0.9999	1.0000
2	0.0245	0.0829	70.5%	0.0024	0.0114	-0.0013	1.0000	0.9995	0.9999
3	0.0396	0.1222	67.6%	0.0042	0.0170	-0.0025	0.9999	0.9989	0.9998
4	0.0553	0.1613	65.7%	0.0060	0.0227	-0.0040	0.9998	0.9980	0.9996
5	0.0720	0.2000	64.0%	0.0081	0.0286	-0.0057	0.9996	0.9970	0.9994
6	0.0900	0.2381	62.2%	0.0104	0.0346	-0.0076	0.9994	0.9957	0.9990
7	0.1085	0.2758	60.6%	0.0127	0.0404	-0.0096	0.9991	0.9943	0.9985
8	0.1266	0.3133	59.6%	0.0149	0.0461	-0.0114	0.9988	0.9926	0.9980
9	0.1445	0.3504	58.7%	0.0172	0.0520	-0.0134	0.9984	0.9907	0.9974
10	0.1626	0.3874	58.0%	0.0196	0.0579	-0.0155	0.9980	0.9886	0.9967
11	0.1803	0.4241	57.5%	0.0219	0.0641	-0.0175	0.9975	0.9863	0.9959
12	0.1979	0.4601	57.0%	0.0242	0.0702	-0.0196	0.9970	0.9839	0.9951
13	0.2155	0.4965	56.6%	0.0266	0.0766	-0.0217	0.9964	0.9812	0.9941
14	0.2333	0.5332	56.2%	0.0289	0.0831	-0.0237	0.9958	0.9783	0.9930
15	0.2512	0.5701	55.9%	0.0314	0.0898	-0.0259	0.9951	0.9752	0.9919
16	0.2698	0.6080	55.6%	0.0341	0.0970	-0.0283	0.9943	0.9717	0.9906
17	0.2894	0.6467	55.2%	0.0370	0.1044	-0.0308	0.9933	0.9680	0.9890
18	0.3092	0.6861	54.9%	0.0400	0.1121	-0.0335	0.9923	0.9639	0.9874
19	0.3300	0.7264	54.6%	0.0433	0.1201	-0.0363	0.9911	0.9594	0.9855
20	0.3510	0.7676	54.3%	0.0468	0.1285	-0.0393	0.9897	0.9546	0.9834
21	0.3723	0.8096	54.0%	0.0503	0.1371	-0.0422	0.9883	0.9494	0.9811
22	0.3942	0.8524	53.7%	0.0541	0.1462	-0.0454	0.9867	0.9437	0.9785
23	0.4170	0.8961	53.5%	0.0581	0.1557	-0.0486	0.9848	0.9376	0.9757
24	0.4410	0.9412	53.1%	0.0621	0.1657	-0.0520	0.9827	0.9309	0.9724
25	0.4652	0.9875	52.9%	0.0663	0.1760	-0.0555	0.9804	0.9237	0.9688
26	0.4904	1.0350	52.6%	0.0706	0.1866	-0.0592	0.9778	0.9158	0.9649
27	0.5164	1.0830	52.3%	0.0751	0.1974	-0.0629	0.9749	0.9074	0.9606
28	0.5425	1.1316	52.1%	0.0797	0.2084	-0.0668	0.9718	0.8984	0.9558
29	0.5692	1.1811	51.8%	0.0844	0.2200	-0.0710	0.9683	0.8888	0.9507
30	0.5966	1.2311	51.5%	0.0894	0.2319	-0.0754	0.9645	0.8785	0.9451
31	0.6247	1.2818	51.3%	0.0946	0.2441	-0.0799	0.9602	0.8675	0.9389
32	0.6529	1.3332	51.0%	0.0998	0.2564	-0.0845	0.9557	0.8558	0.9323
33	0.6807	1.3846	50.8%	0.1051	0.2690	-0.0892	0.9508	0.8434	0.9254
34	0.7090	1.4364	50.6%	0.1106	0.2818	-0.0940	0.9456	0.8304	0.9179
35	0.7382	1.4894	50.4%	0.1164	0.2951	-0.0990	0.9397	0.8164	0.9097
36	0.7676	1.5434	50.3%	0.1223	0.3088	-0.1040	0.9335	0.8014	0.9010
37	0.7974	1.5981	50.1%	0.1282	0.3227	-0.1092	0.9266	0.7854	0.8916
38	0.8277	1.6531	49.9%	0.1343	0.3368	-0.1144	0.9192	0.7687	0.8816
39	0.8579	1.7081	49.8%	0.1405	0.3512	-0.1196	0.9114	0.7511	0.8711
40	0.8886	1.7629	49.6%	0.1467	0.3657	-0.1250	0.9030	0.7329	0.8599
41	0.9202	1.8185	49.4%	0.1531	0.3806	-0.1305	0.8939	0.7136	0.8480
42	0.9530	1.8750	49.2%	0.1598	0.3959	-0.1362	0.8840	0.6932	0.8352
43	0.9868	1.9323	48.9%	0.1666	0.4113	-0.1420	0.8734	0.6717	0.8217
44	1.0206	1.9898	48.7%	0.1734	0.4269	-0.1478	0.8623	0.6495	0.8077
45	1.0545	2.0466	48.5%	0.1804	0.4424	-0.1537	0.8507	0.6267	0.7933
46	1.0885	2.1032	48.2%	0.1875	0.4581	-0.1598	0.8385	0.6033	0.7783
47	1.1230	2.1599	48.0%	0.1948	0.4738	-0.1660	0.8258	0.5792	0.7627
48	1.1583	2.2173	47.8%	0.2022	0.4897	-0.1723	0.8124	0.5542	0.7466
49	1.1938	2.2749	47.5%	0.2096	0.5055	-0.1786	0.7986	0.5286	0.7302
50	1.2297	2.3325	47.3%	0.2172	0.5215	-0.1850	0.7845	0.5026	0.7134
51	1.2653	2.3896	47.0%	0.2247	0.5374	-0.1913	0.7700	0.4762	0.6963
52	1.3008	2.4464	46.8%	0.2321	0.5532	-0.1976	0.7553	0.4495	0.6792
53	1.3364	2.5032	46.6%	0.2396	0.5689	-0.2040	0.7403	0.4226	0.6620
54	1.3725	2.5604	46.4%	0.2469	0.5845	-0.2103	0.7251	0.3952	0.6447
55	1.4087	2.6175	46.2%	0.2543	0.6001	-0.2167	0.7100	0.3679	0.6276
56	1.4449	2.6741	46.0%	0.2617	0.6155	-0.2232	0.6950	0.3407	0.6108
57	1.4812	2.7302	45.7%	0.2691	0.6310	-0.2299	0.6801	0.3138	0.5944
58	1.5173	2.7859	45.5%	0.2765	0.6464	-0.2366	0.6656	0.2872	0.5784
59	1.5534	2.8409	45.3%	0.2838	0.6616	-0.2432	0.6512	0.2611	0.5628
60	1.5896	2.8953	45.1%	0.2910	0.6764	-0.2499	0.6374	0.2357	0.5476
61	1.6254	2.9487	44.9%	0.2981	0.6909	-0.2565	0.6240	0.2110	0.5331
62	1.6607	3.0007	44.7%	0.3051	0.7050	-0.2630	0.6113	0.1873	0.5193
63	1.6956	3.0513	44.4%	0.3121	0.7188	-0.2694	0.5989	0.1647	0.5060
64	1.7302	3.1006	44.2%	0.3190	0.7323	-0.2758	0.5870	0.1429	0.4931
65	1.7646	3.1484	44.0%	0.3259	0.7456	-0.2823	0.5753	0.1222	0.4806
66	1.7983	3.1948	43.7%	0.3326	0.7582	-0.2886	0.5643	0.1024	0.4687
67	1.8312	3.2402	43.5%	0.3391	0.7704	-0.2948	0.5539	0.0835	0.4574
68	1.8634	3.2842	43.3%	0.3454	0.7822	-0.3009	0.5441	0.0655	0.4467
69	1.8947	3.3266	43.0%	0.3517	0.7936	-0.3068	0.5347	0.0483	0.4364
70	1.9251	3.3677	42.8%	0.3577	0.8047	-0.3127	0.5257	0.0319	0.4266
71	1.9544	3.4069	42.6%	0.3635	0.8153	-0.3183	0.5171	0.0165	0.4173
72	1.9828	3.4445	42.4%	0.3689	0.8253	-0.3238	0.5091	0.0020	0.4086
73	2.0107	3.4811	42.2%	0.3742	0.8349	-0.3291	0.5013	-0.0119	0.4003
74	2.0380	3.5164	42.0%	0.3793	0.8443	-0.3343	0.4936	-0.0252	0.3921
75	2.0647	3.5499	41.8%	0.3843	0.8532	-0.3393	0.4859	-0.0377	0.3840
76	2.0910	3.5823	41.6%	0.3892	0.8619	-0.3444	0.4784	-0.0496	0.3762
77	2.1158	3.6139	41.5%	0.3939	0.8703	-0.3493	0.4718	-0.0609	0.3693
78	2.1392	3.6437	41.3%	0.3983	0.8782	-0.3539	0.4655	-0.0716	0.3628
79	2.1619	3.6724	41.1%	0.4026	0.8858	-0.3584	0.4592	-0.0819	0.3564
80	2.1839	3.6998	41.0%	0.4067	0.8931	-0.3628	0.4531	-0.0917	0.3502
81	2.2049	3.7255	40.8%	0.4107	0.9000	-0.3671	0.4468	-0.1009	0.3440
82	2.2255	3.7494	40.6%	0.4146	0.9064	-0.3713	0.4404	-0.1095	0.3377
83	2.2451	3.7716	40.5%	0.4183	0.9123	-0.3752	0.4343	-0.1173	0.3317
84	2.2643	3.7922	40.3%	0.4219	0.9179	-0.3790	0.4278	-0.1245	0.3255
85	2.2821	3.8114	40.1%	0.4249	0.9231	-0.3824	0.4219	-0.1313	0.3200
86	2.2983	3.8295	40.0%	0.4272	0.9278	-0.3851	0.4166	-0.1376	0.3150
87	2.3133	3.8461	39.9%	0.4291	0.9323	-0.3875	0.4116	-0.1434	0.3105
88	2.3263	3.8609	39.7%	0.4306	0.9365	-0.3895	0.4070	-0.1489	0.3062
89	2.3372	3.8738	39.7%	0.4317	0.9402	-0.3911	0.4026	-0.1542	0.3021
90	2.3466	3.8852	39.6%	0.4325	0.9435	-0.3923	0.3982	-0.1593	0.2982

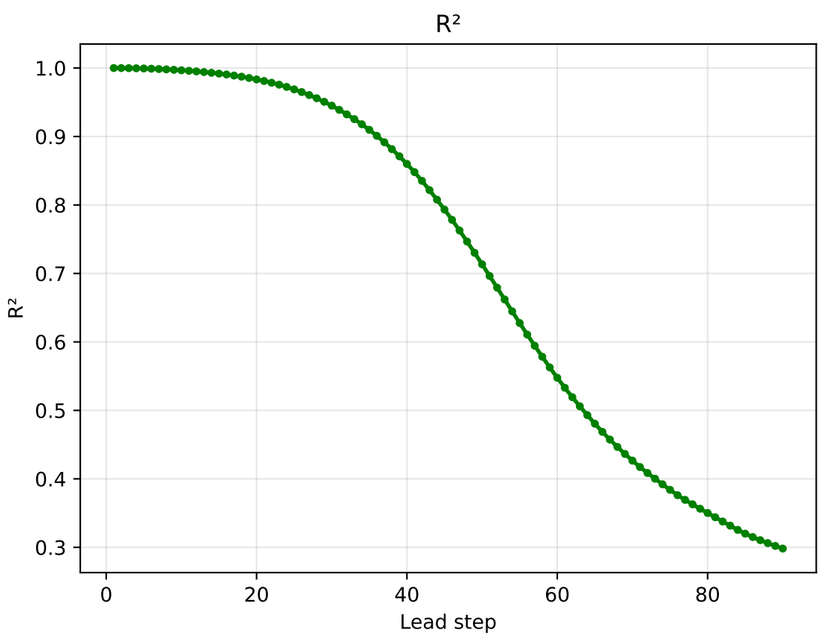
DHW Forecast Skill vs Lead Step - Val

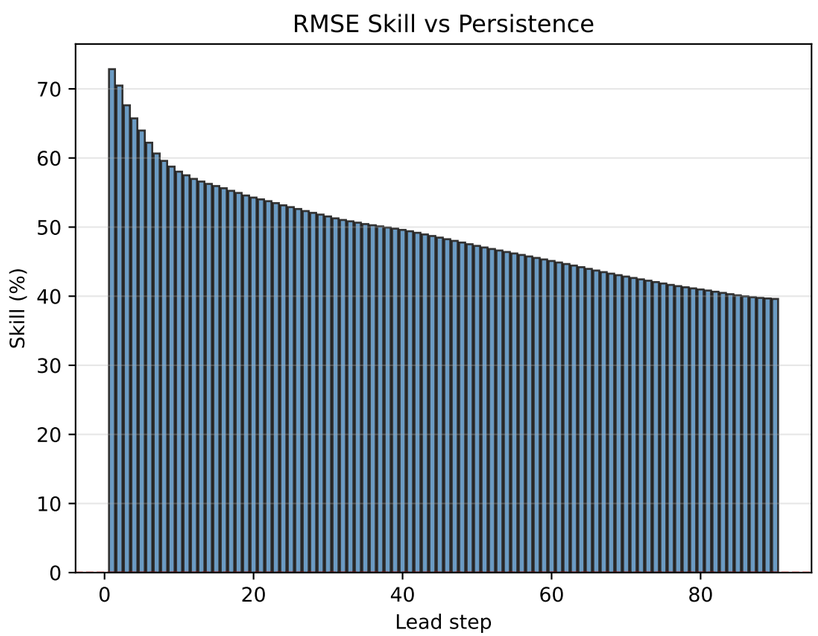


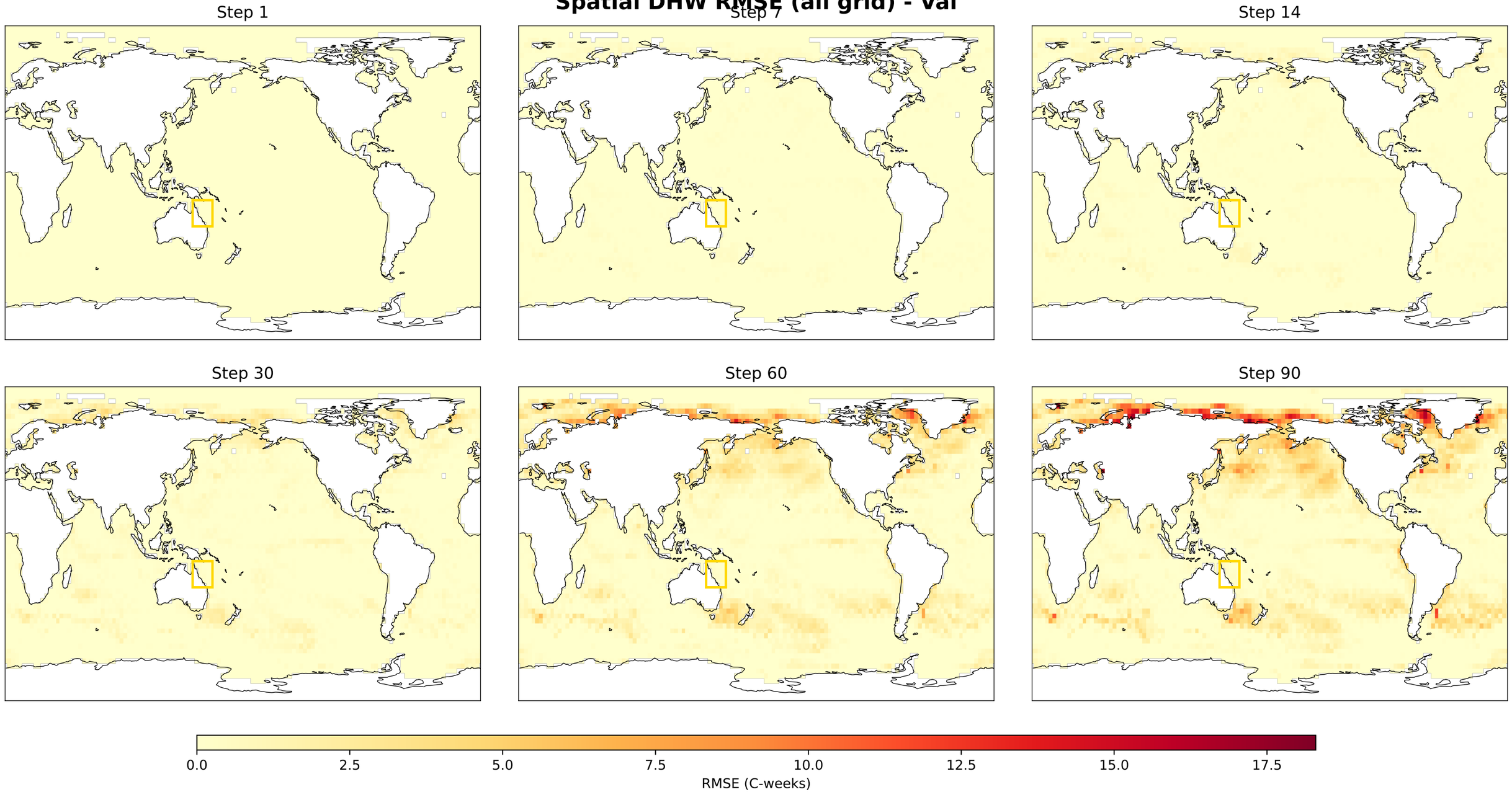


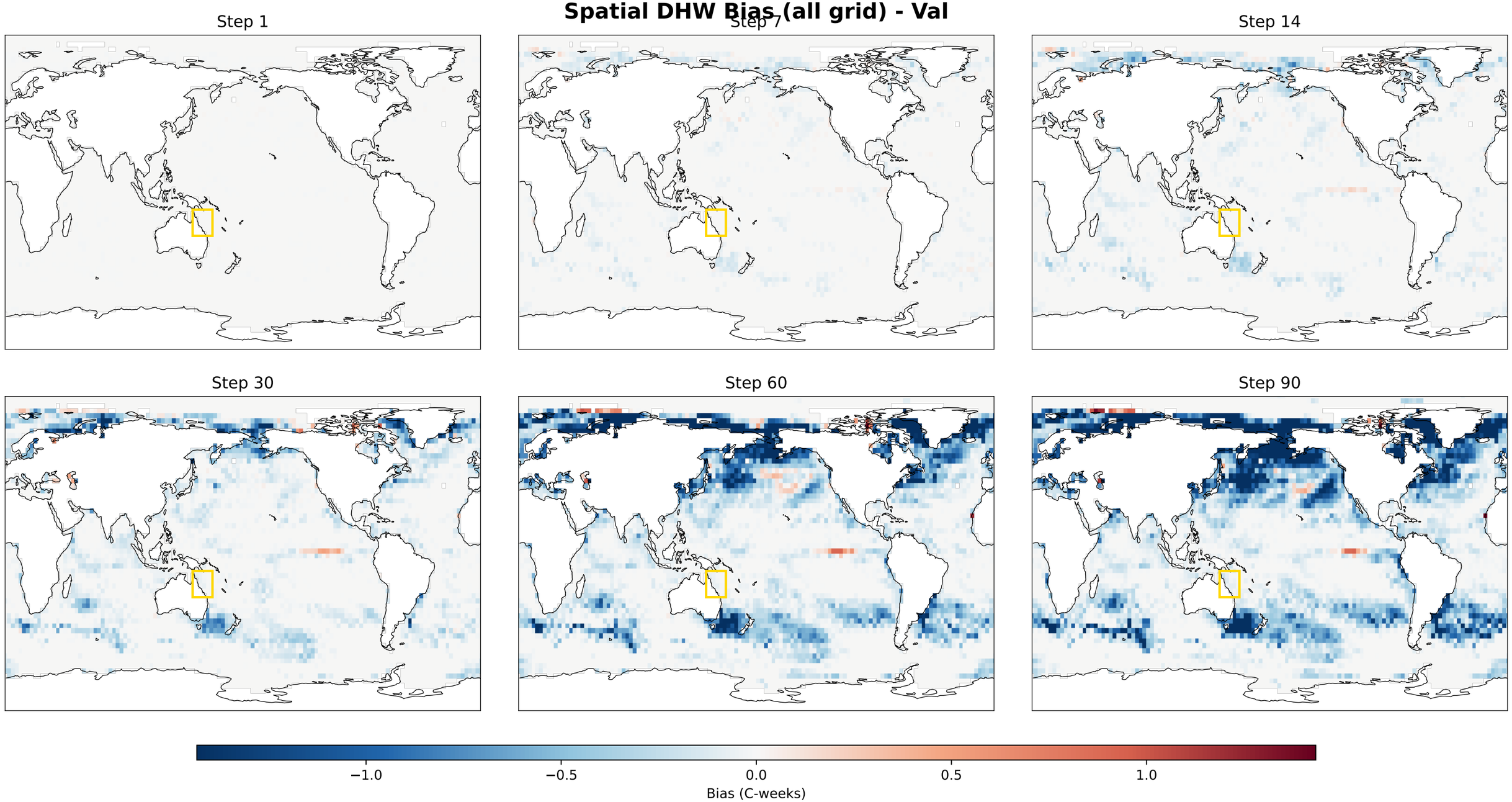


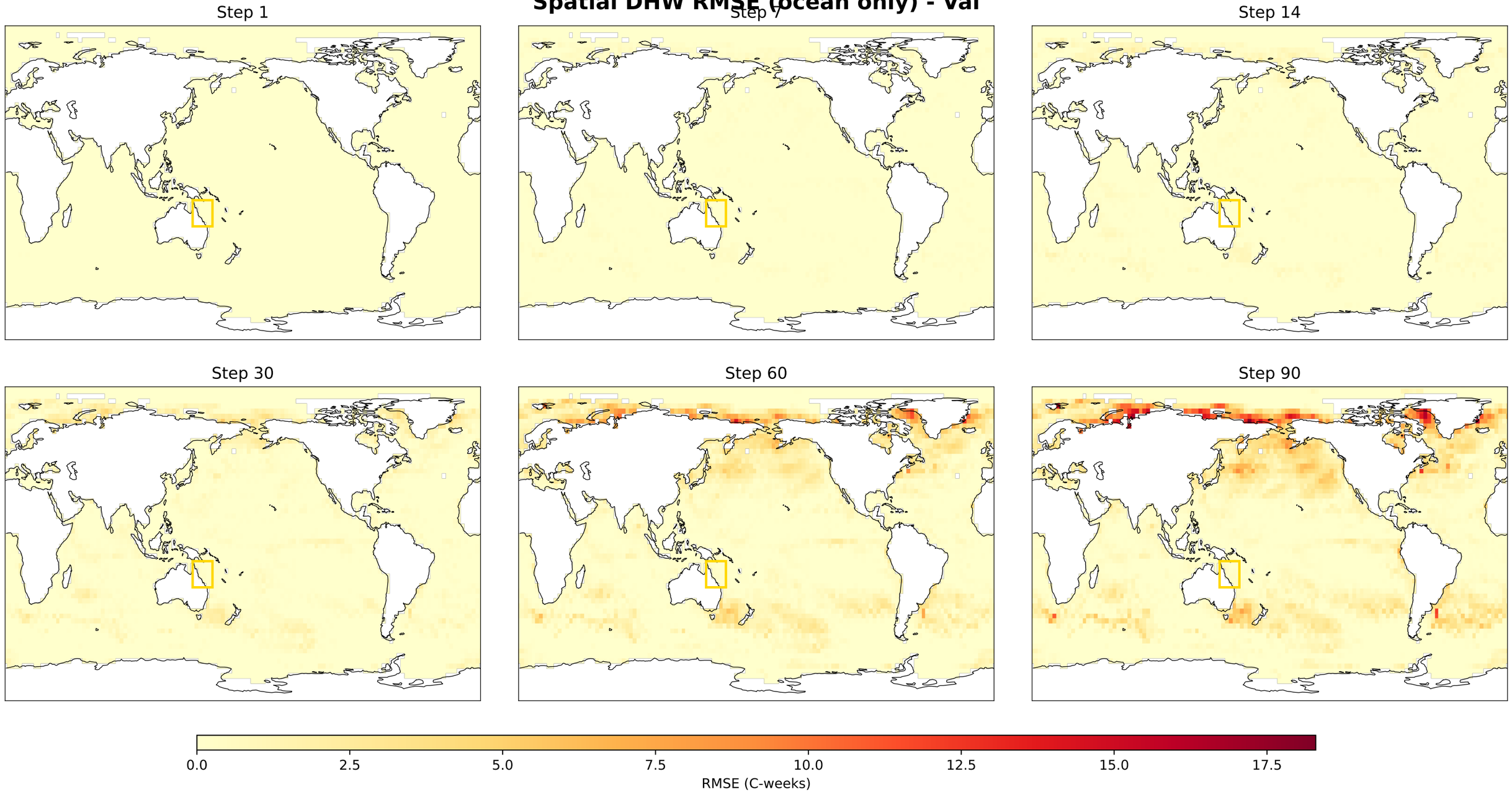


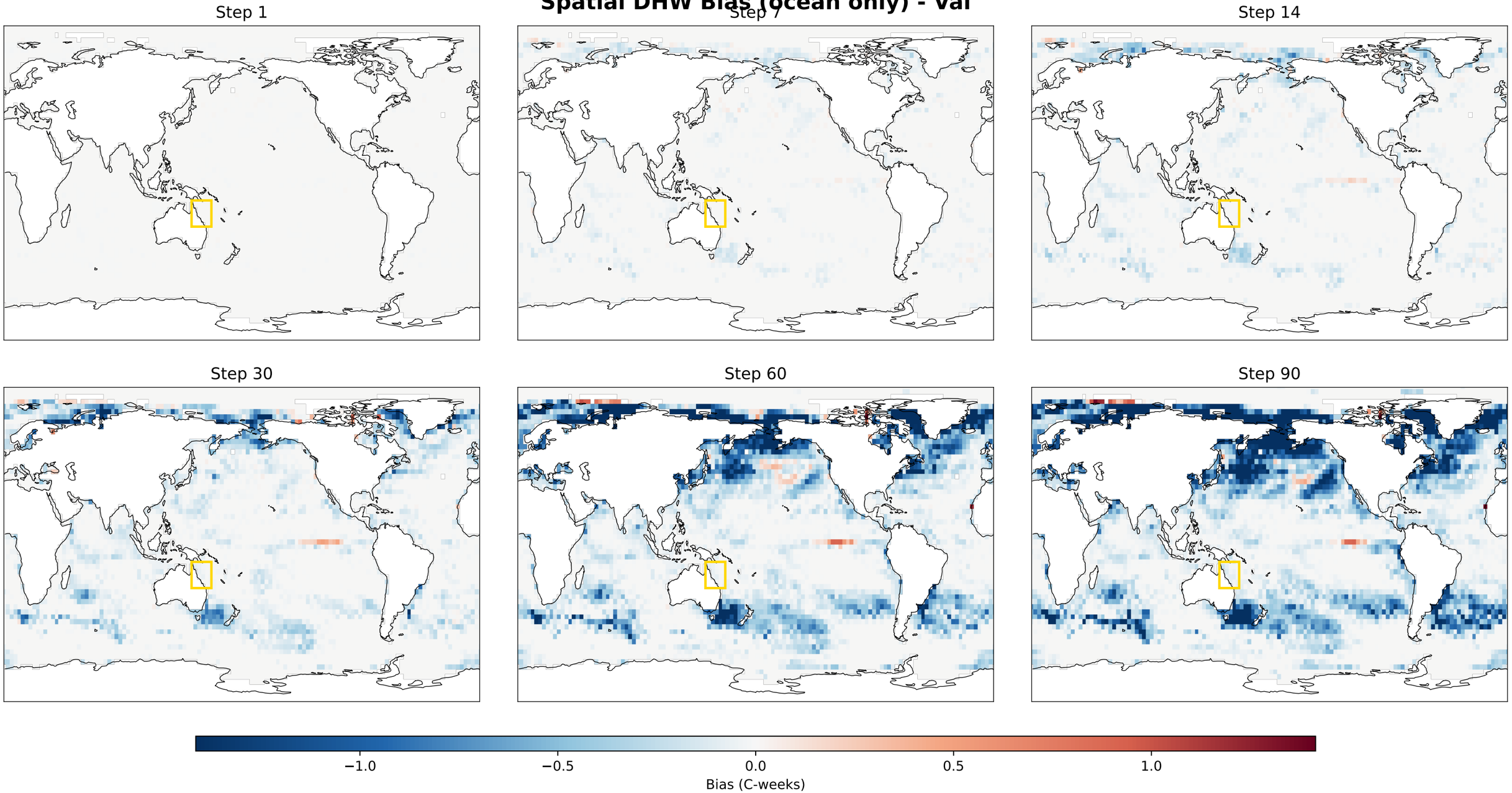





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
RMSE (C-weeks)

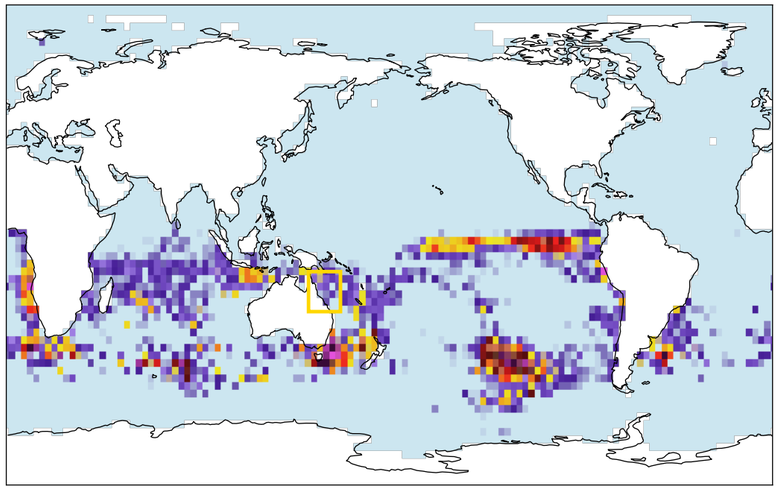

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
RMSE (C-weeks)

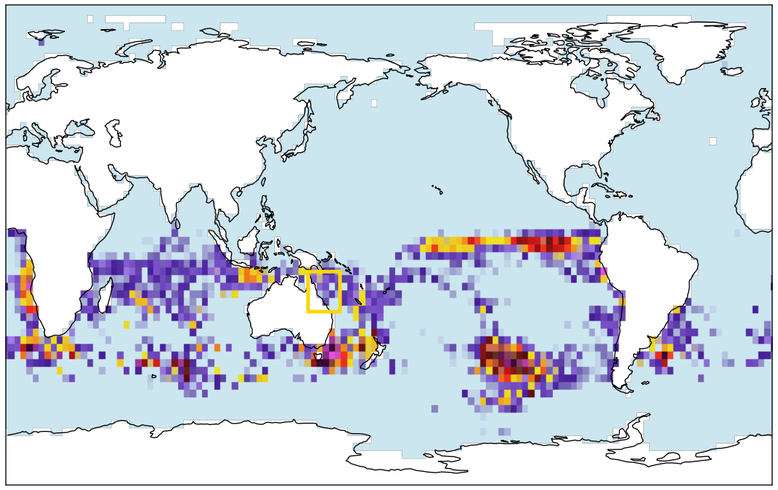

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

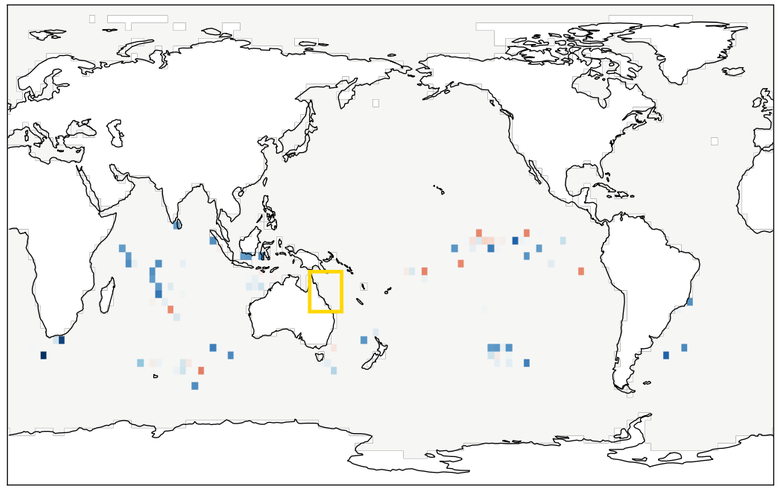
Actual DHW - Step 1



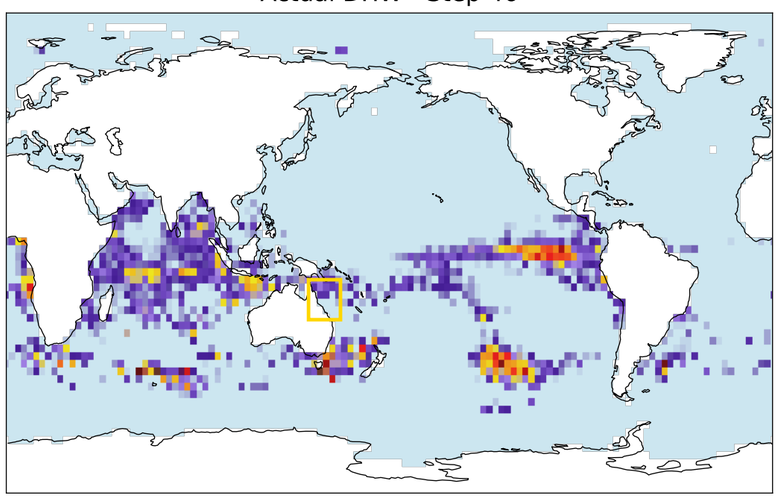
Predicted DHW - Step 1



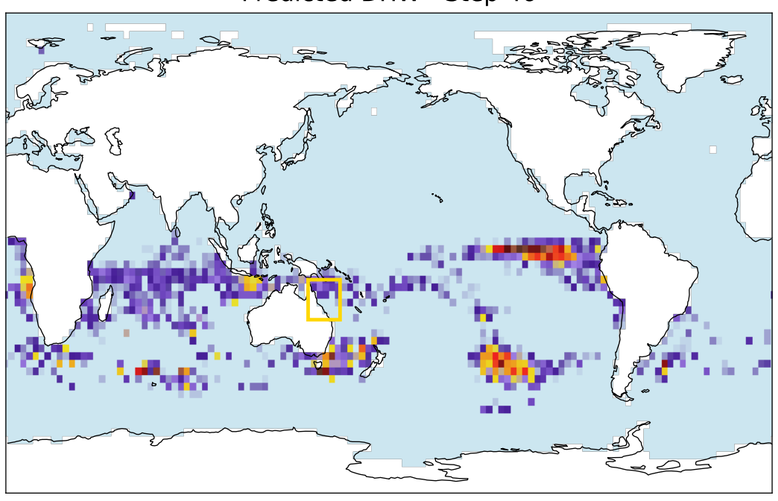
Error (Pred - Actual) - Step 1 [scale ±0.31]



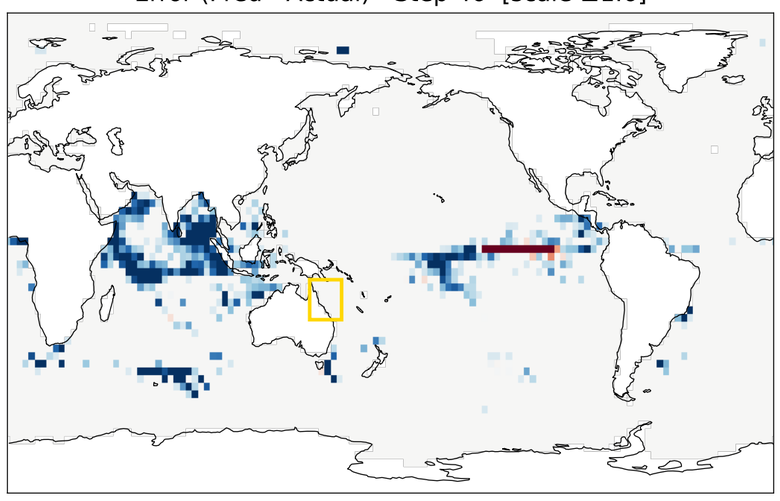
Actual DHW - Step 46



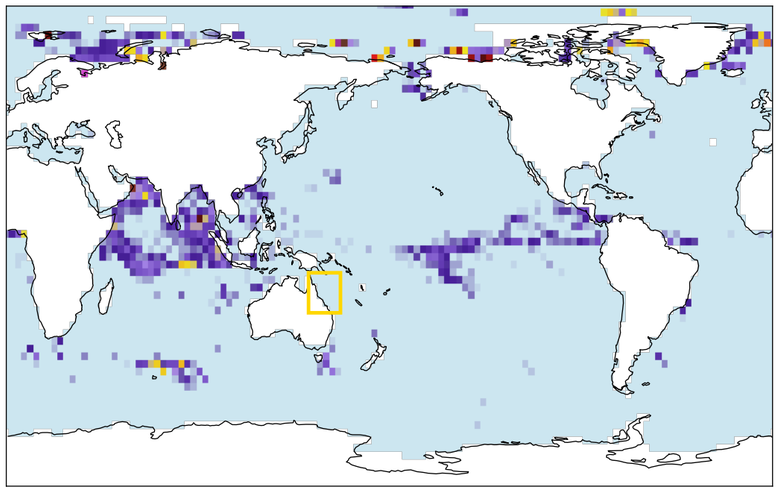
Predicted DHW - Step 46



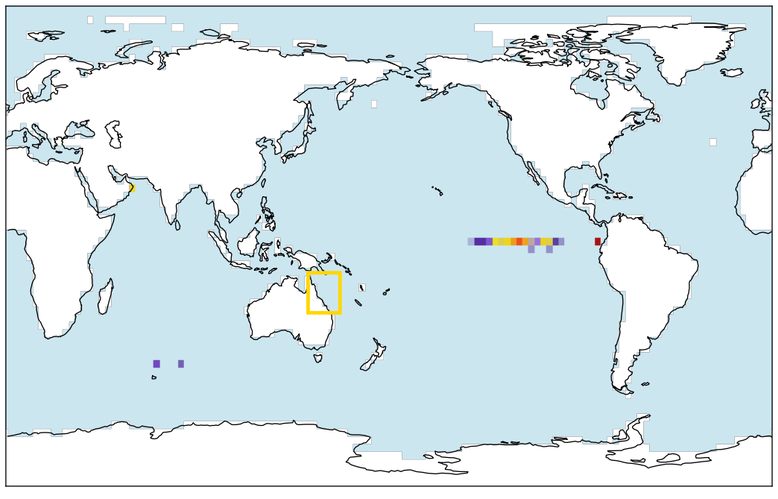
Error (Pred - Actual) - Step 46 [scale ±1.6]



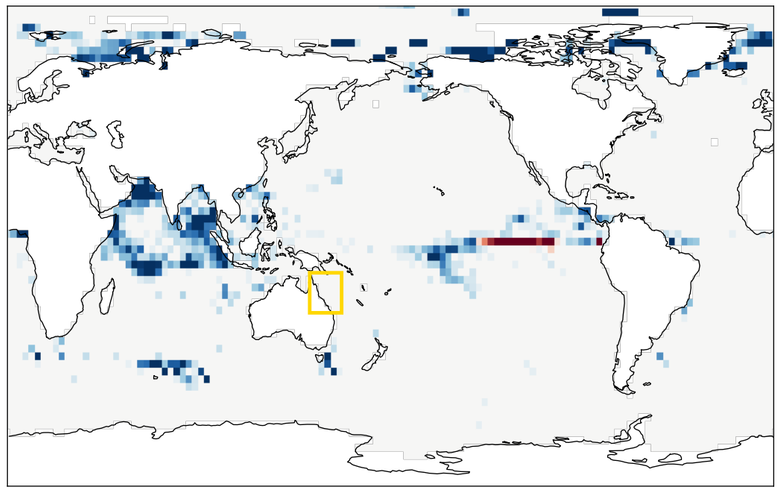
Actual DHW - Step 90

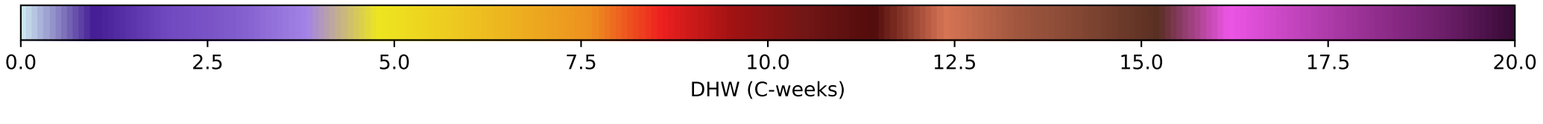


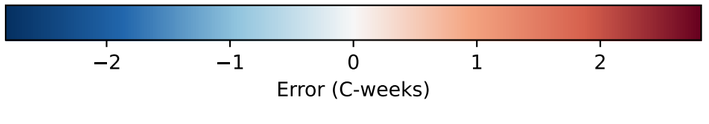
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.8]

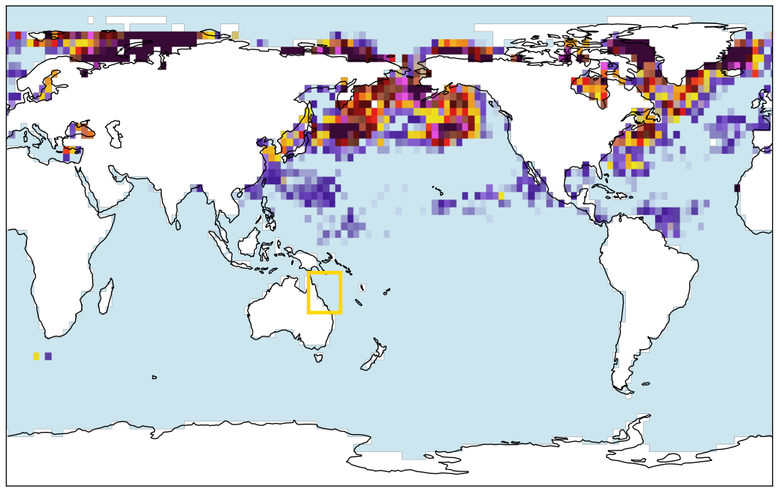




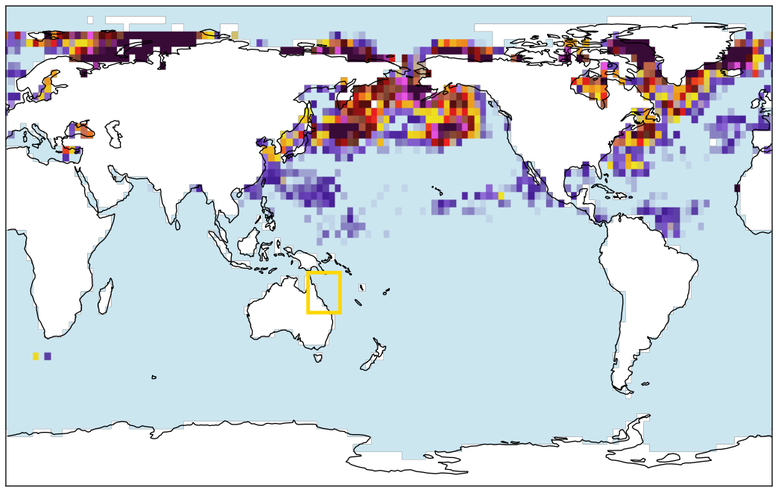


DHW Chain init: 2016-10-07 - Val (C-weeks)

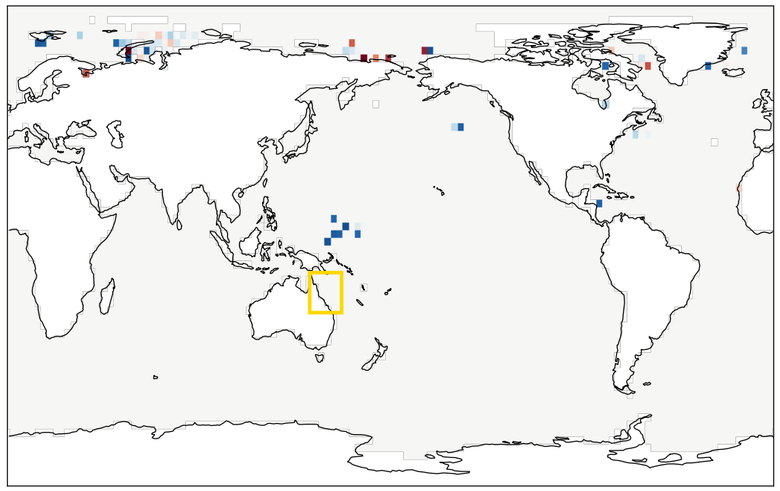
Actual DHW - Step 1



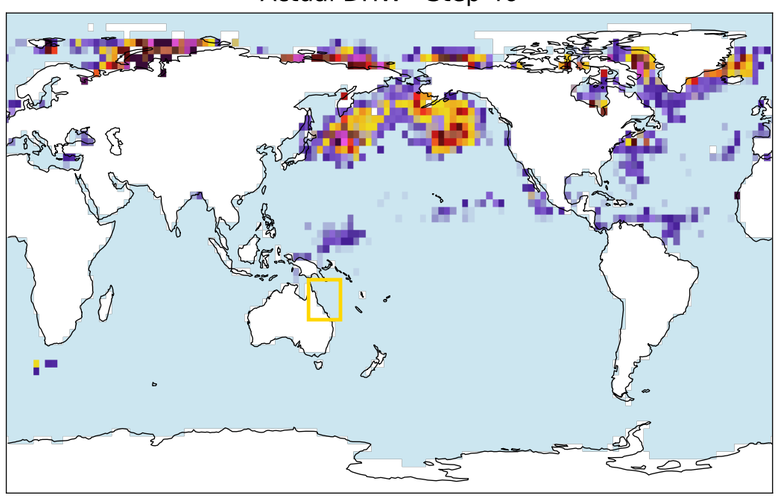
Predicted DHW - Step 1



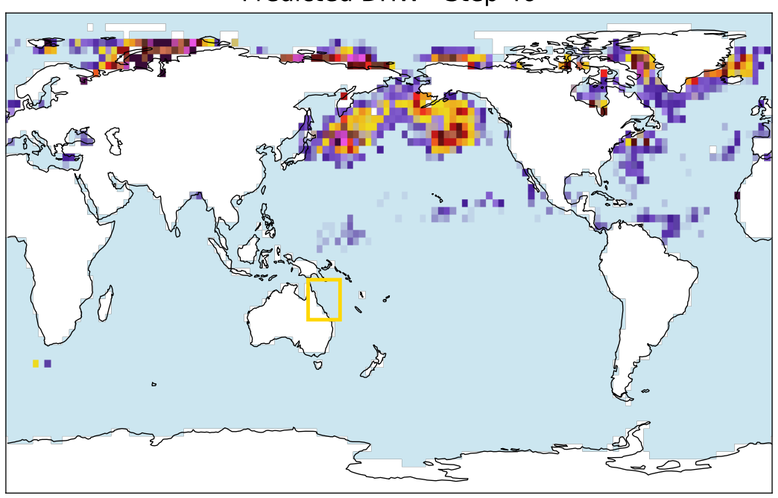
Error (Pred - Actual) - Step 1 [scale ±0.22]



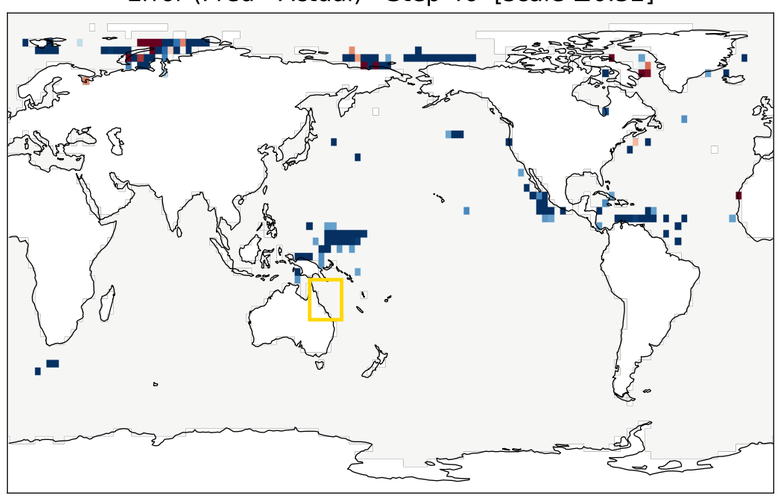
Actual DHW - Step 46



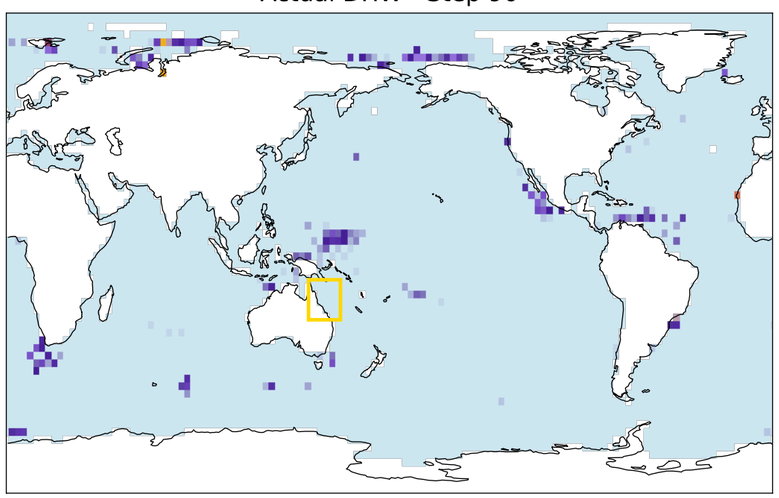
Predicted DHW - Step 46



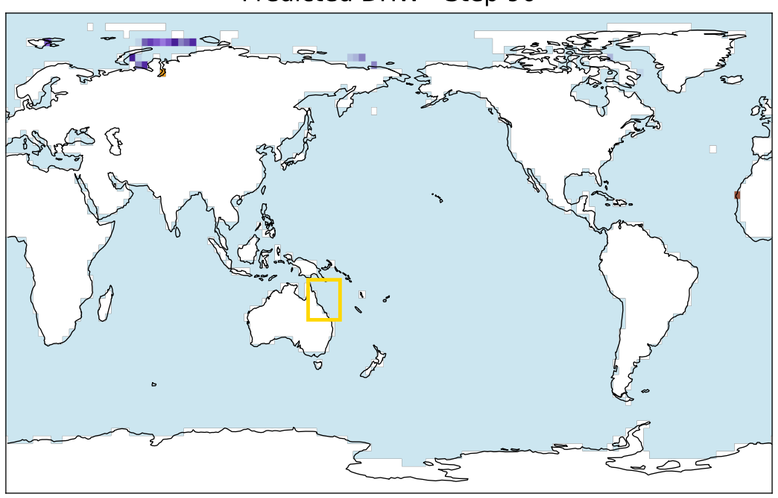
Error (Pred - Actual) - Step 46 [scale ±0.32]



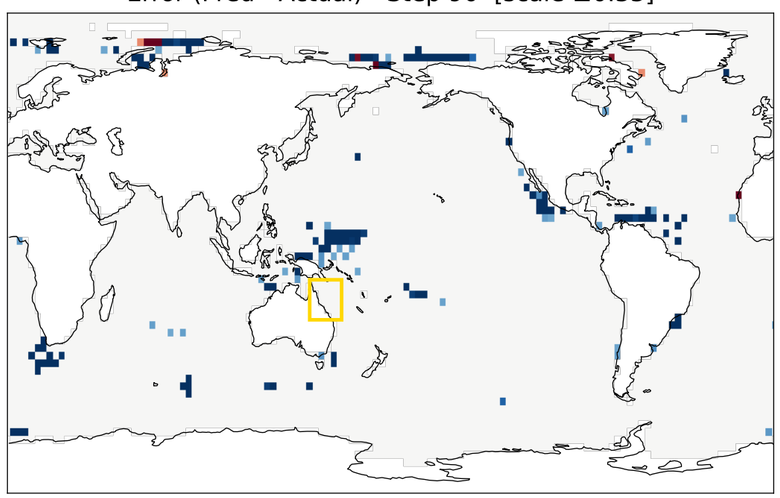
Actual DHW - Step 90

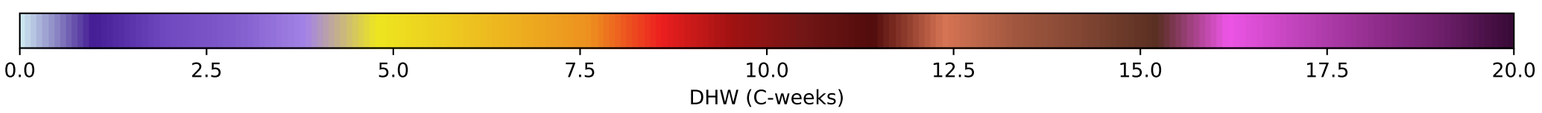


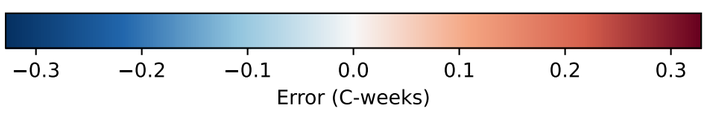
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

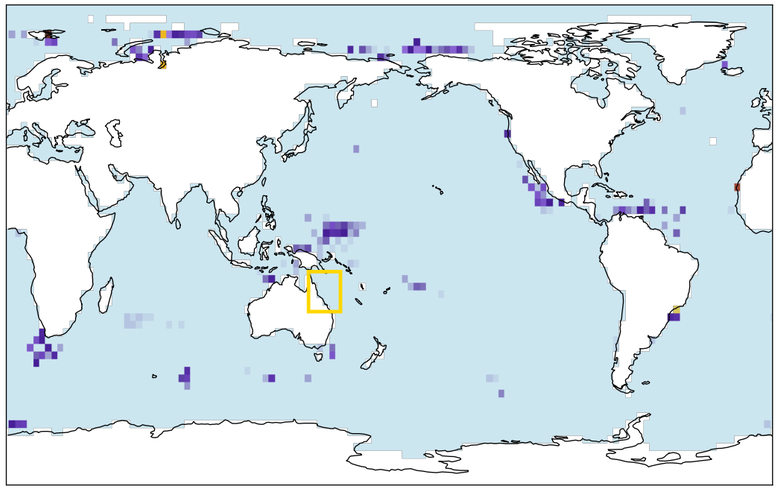




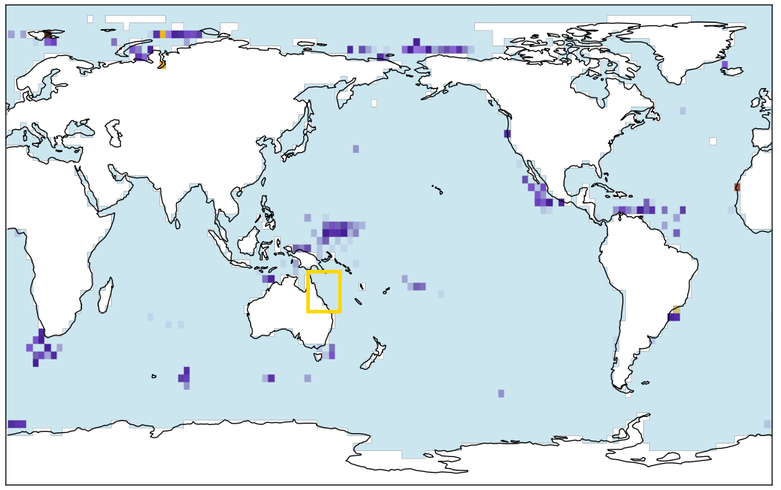


DHW Chain init: 2017-01-05 - Val (C-weeks)

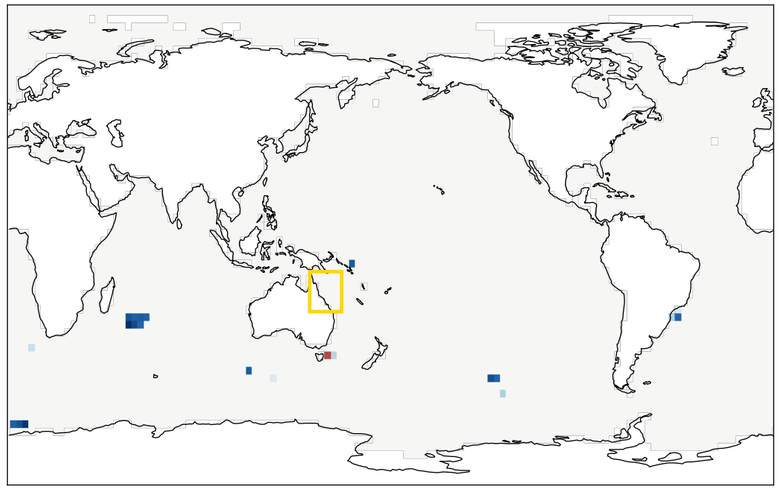
Actual DHW - Step 1



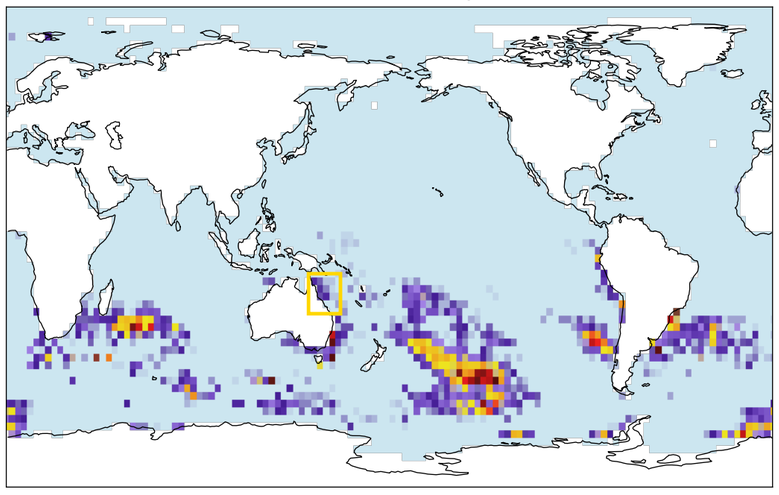
Predicted DHW - Step 1



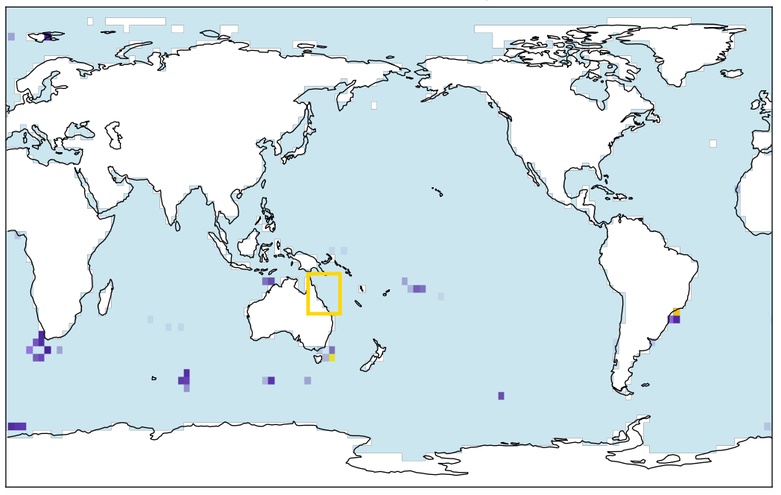
Error (Pred - Actual) - Step 1 [scale ±0.22]



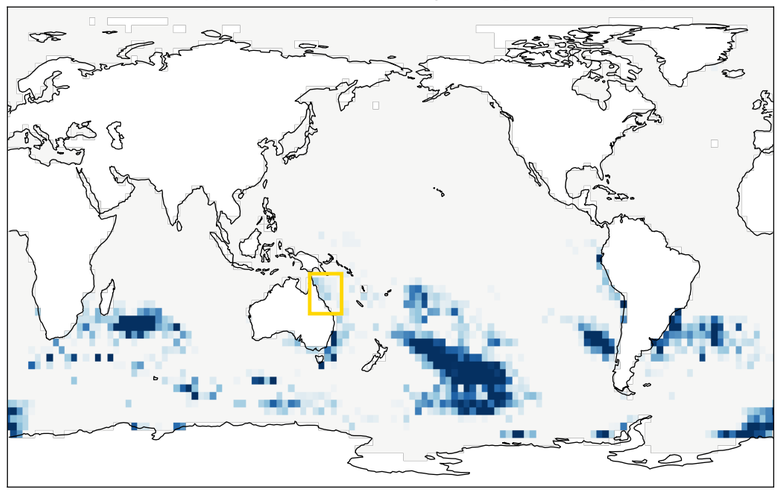
Actual DHW - Step 46



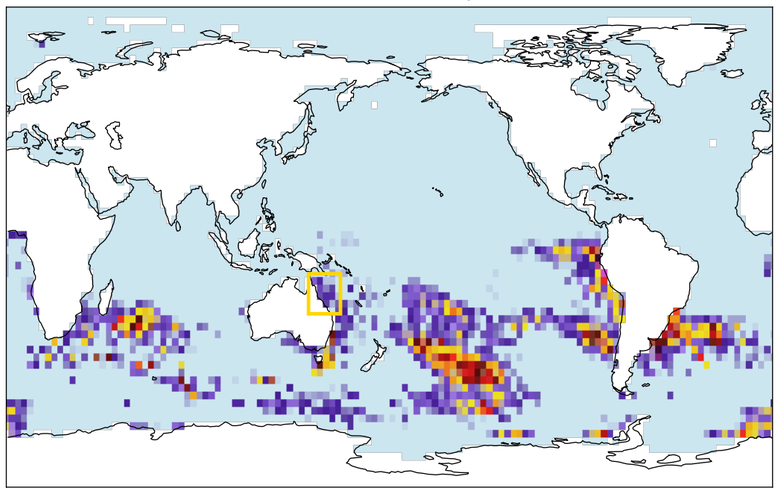
Predicted DHW - Step 46



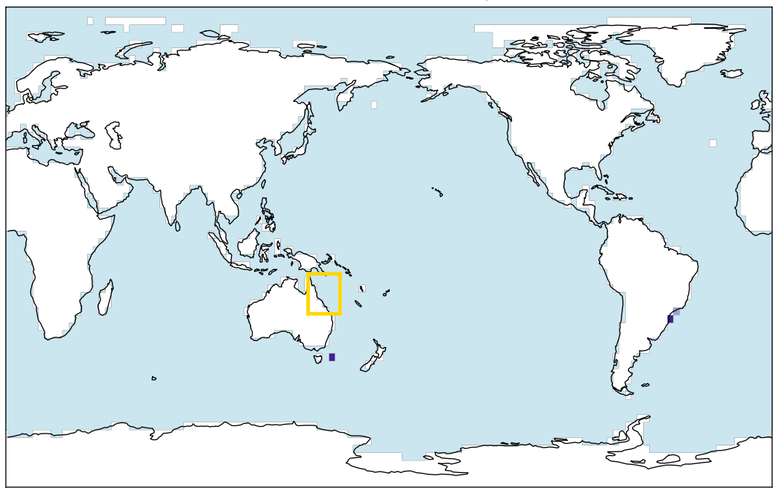
Error (Pred - Actual) - Step 46 [scale ±4.8]



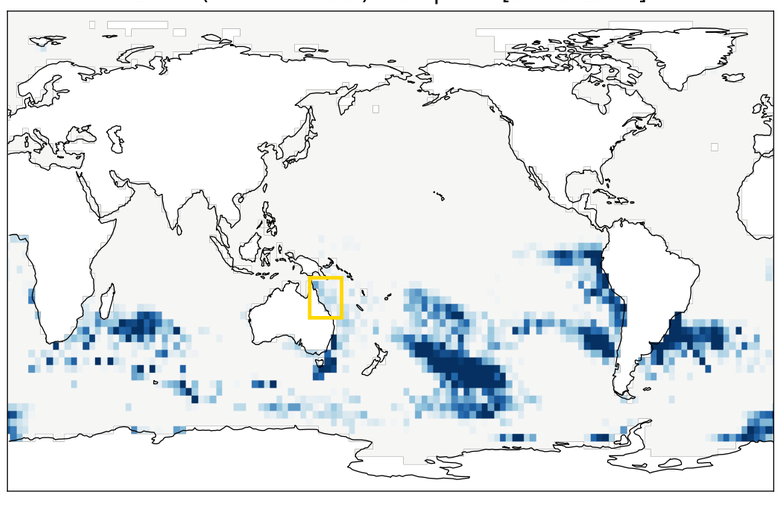
Actual DHW - Step 90

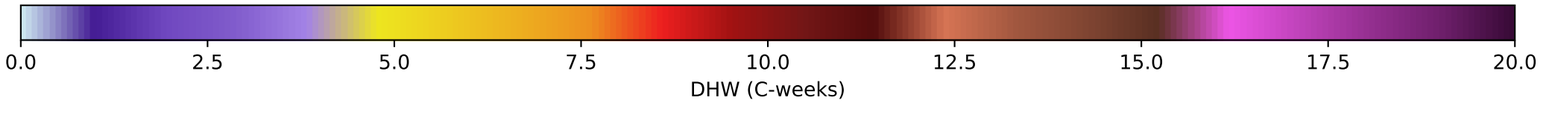


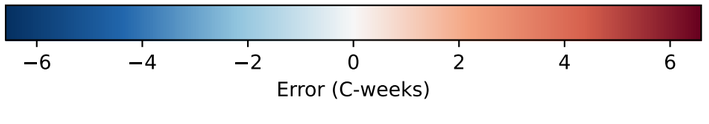
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

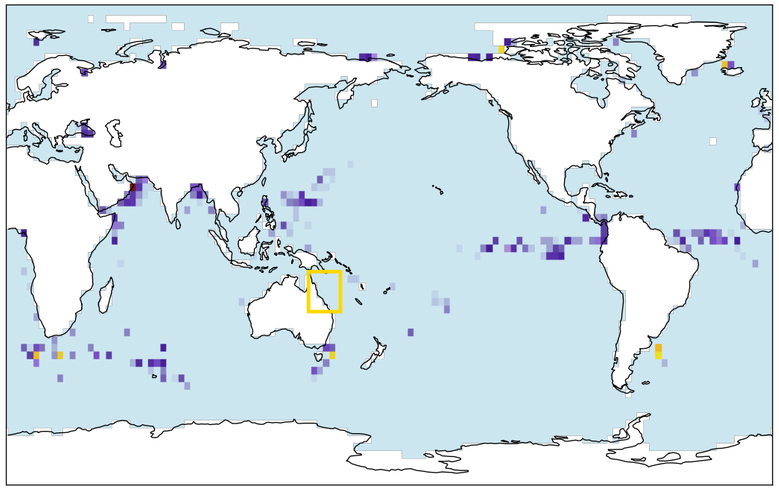




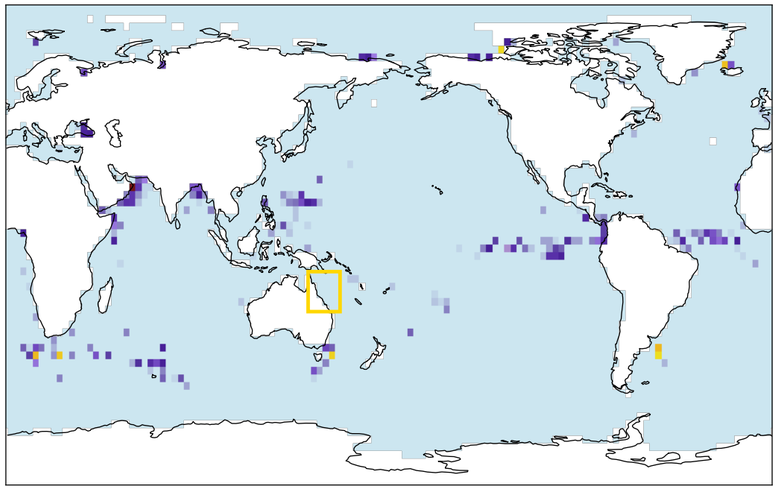


DHW Chain init: 2017-07-04 - Val (C-weeks)

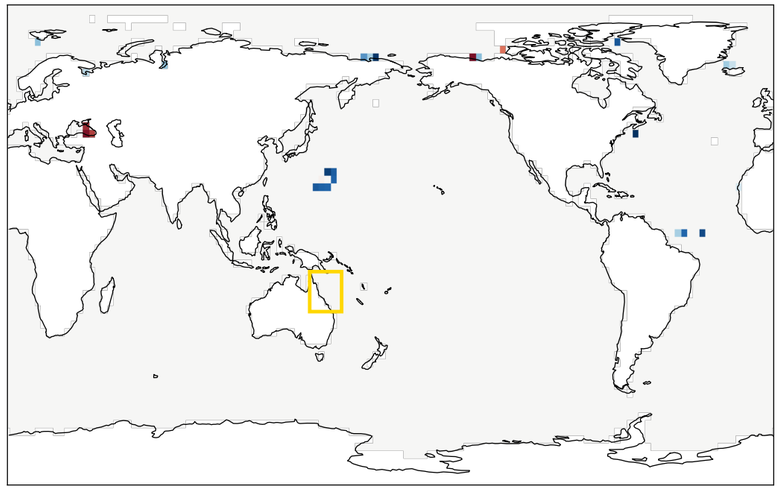
Actual DHW - Step 1



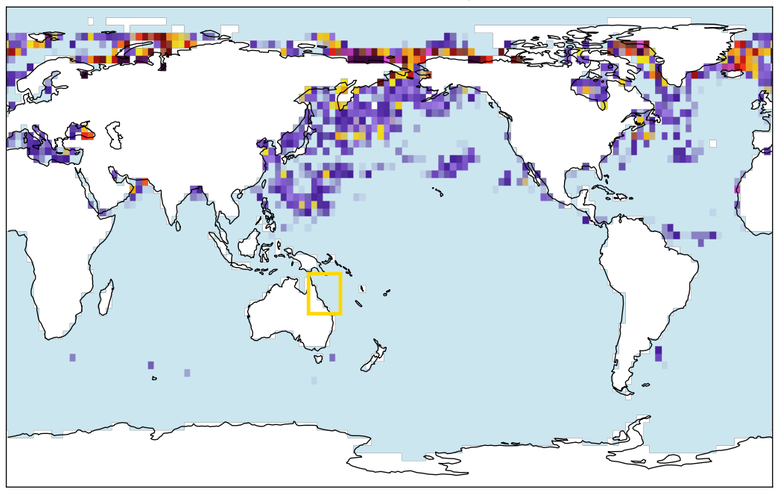
Predicted DHW - Step 1



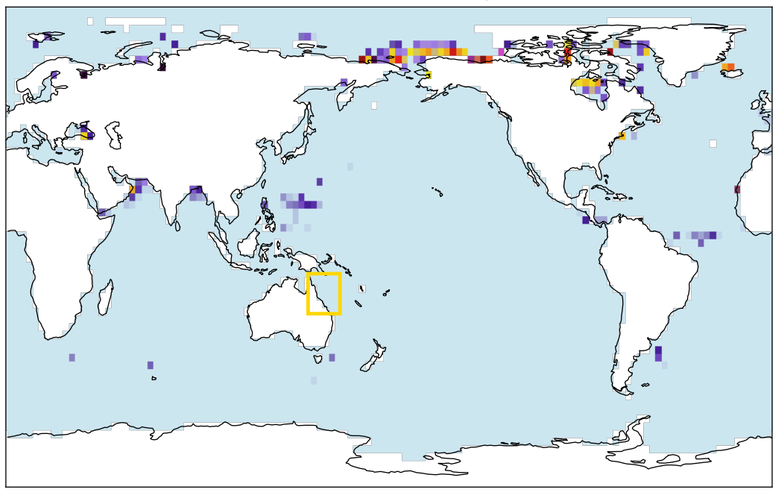
Error (Pred - Actual) - Step 1 [scale ±0.22]



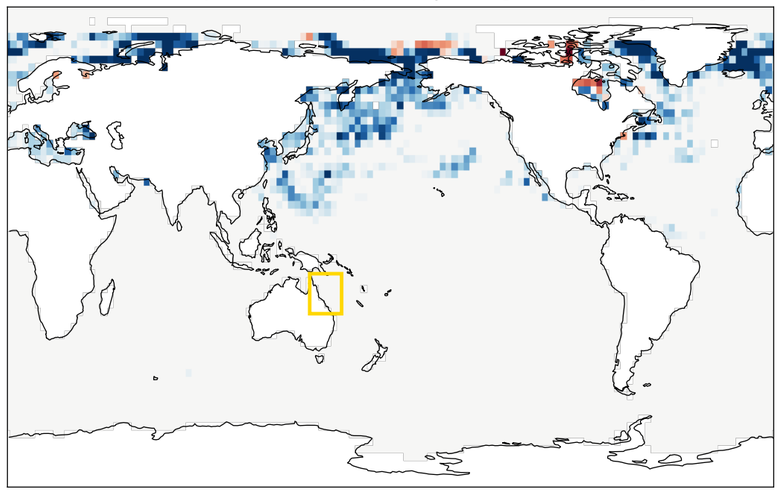
Actual DHW - Step 46



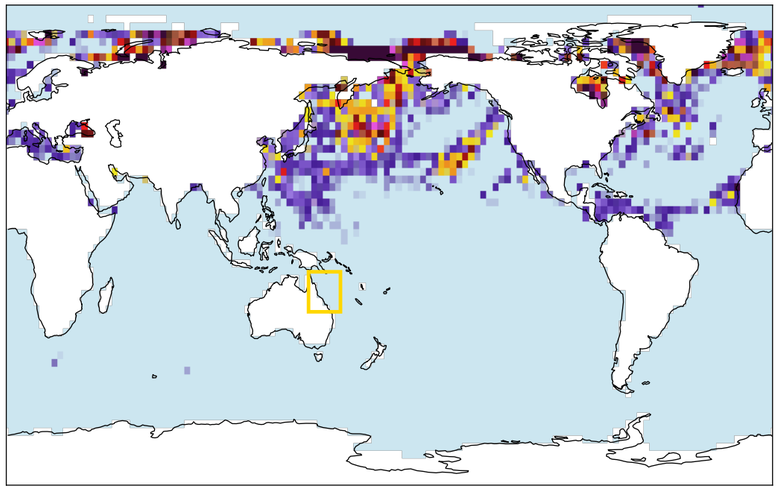
Predicted DHW - Step 46



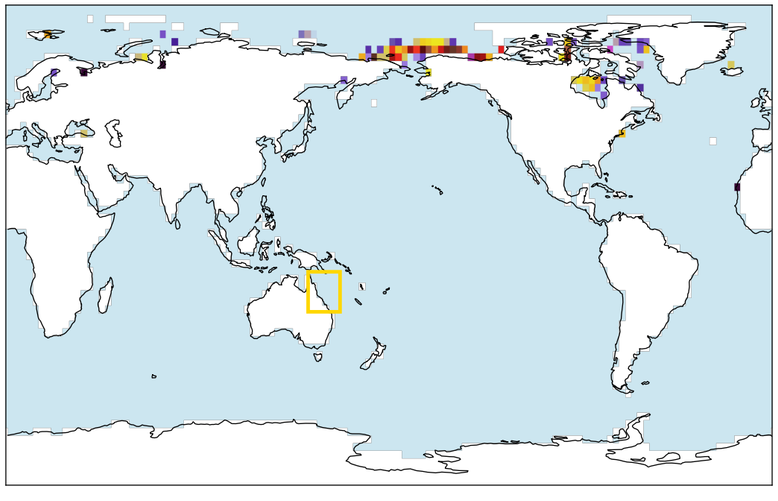
Error (Pred - Actual) - Step 46 [scale ±6.1]



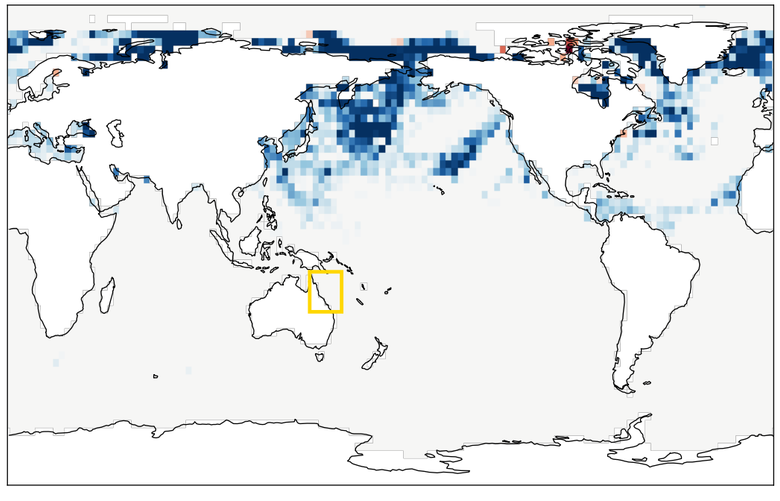
Actual DHW - Step 90

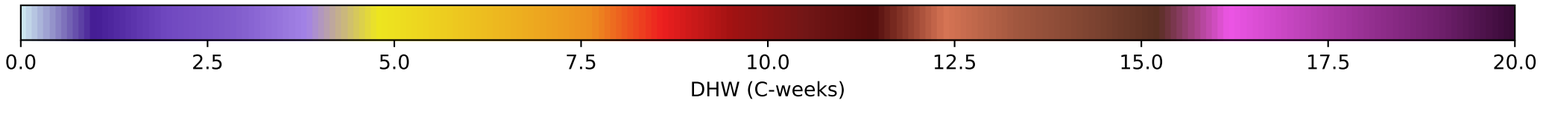


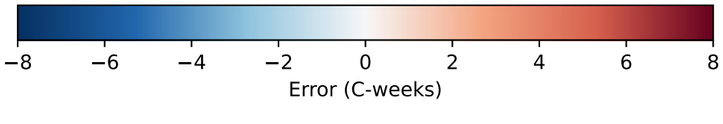
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

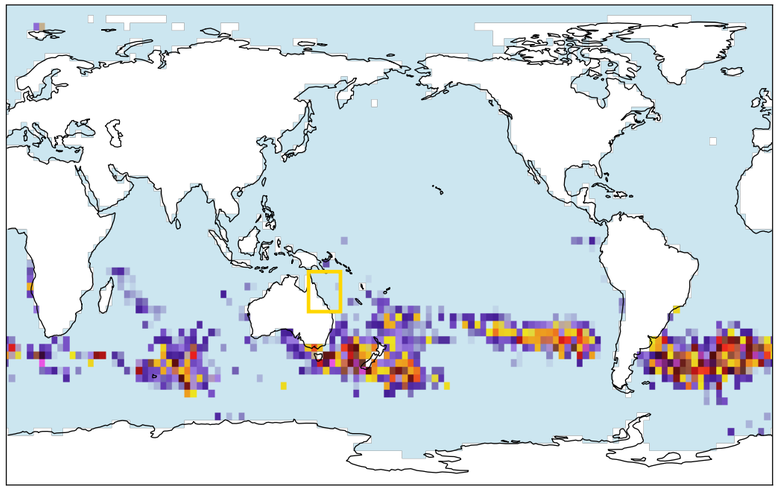




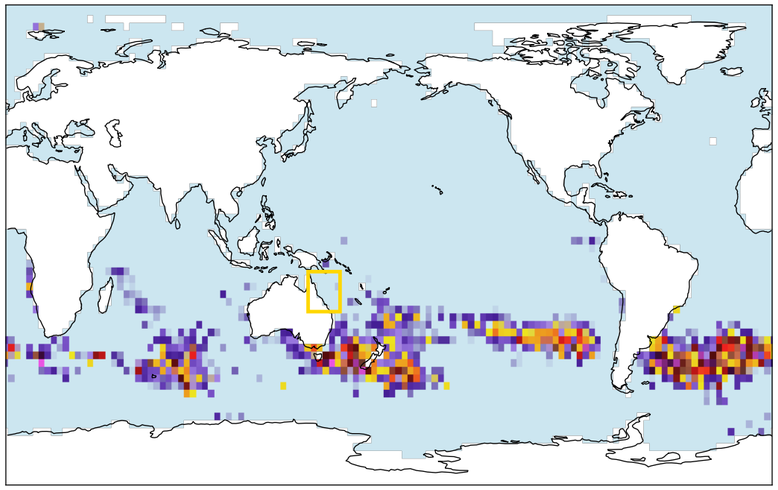


DHW Chain init: 2018-03-31 - Val (C-weeks)

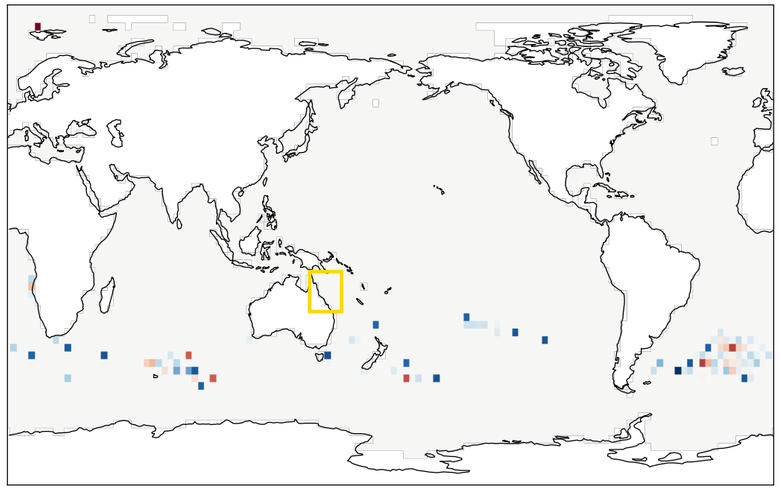
Actual DHW - Step 1



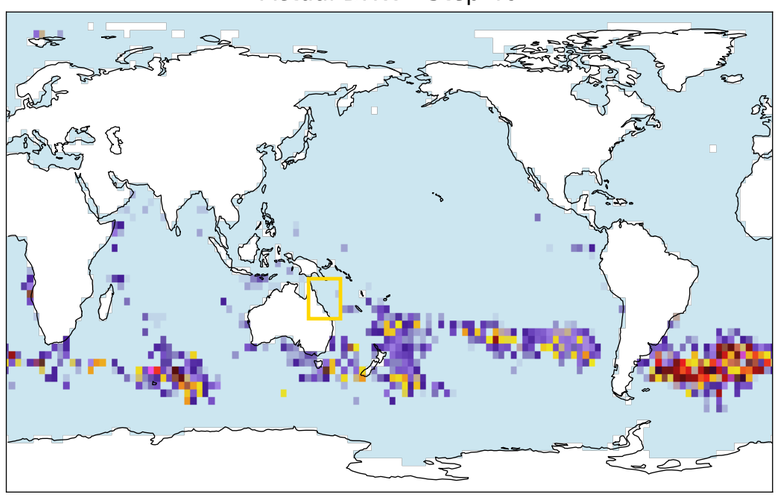
Predicted DHW - Step 1



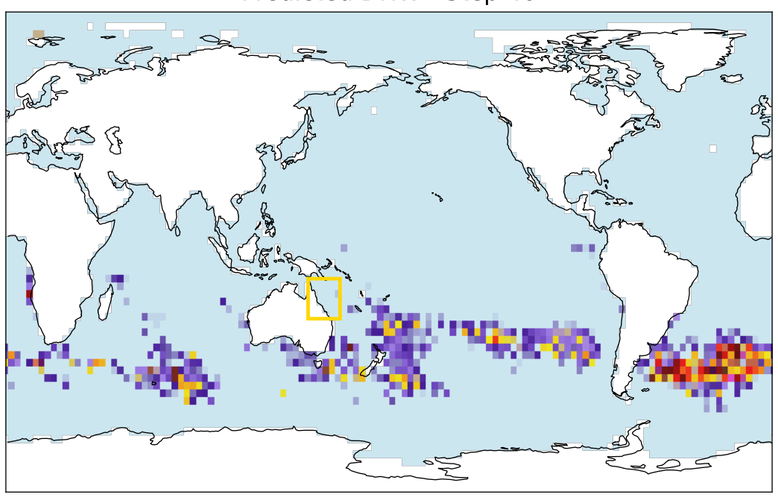
Error (Pred - Actual) - Step 1 [scale ±0.21]



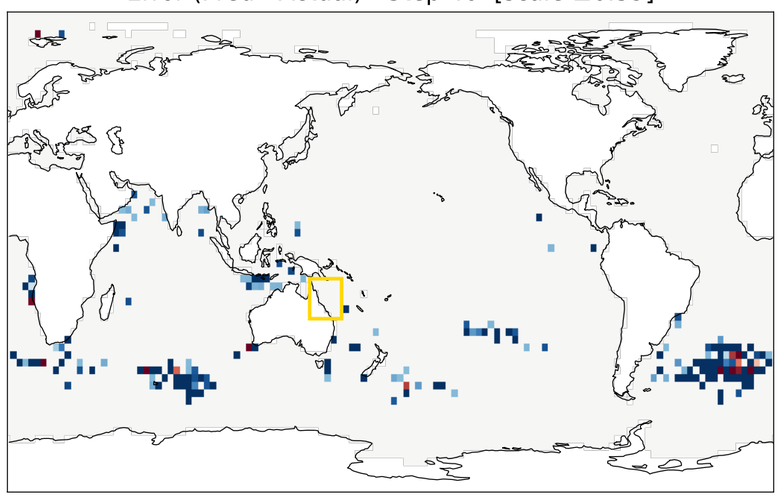
Actual DHW - Step 46



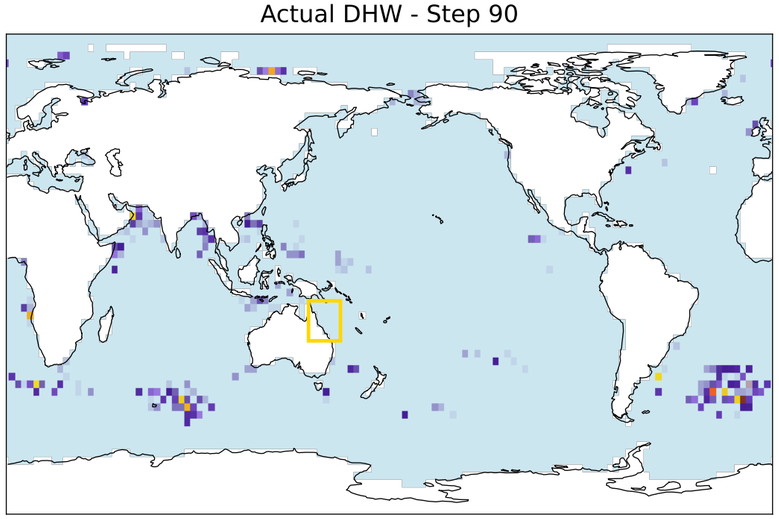
Predicted DHW - Step 46



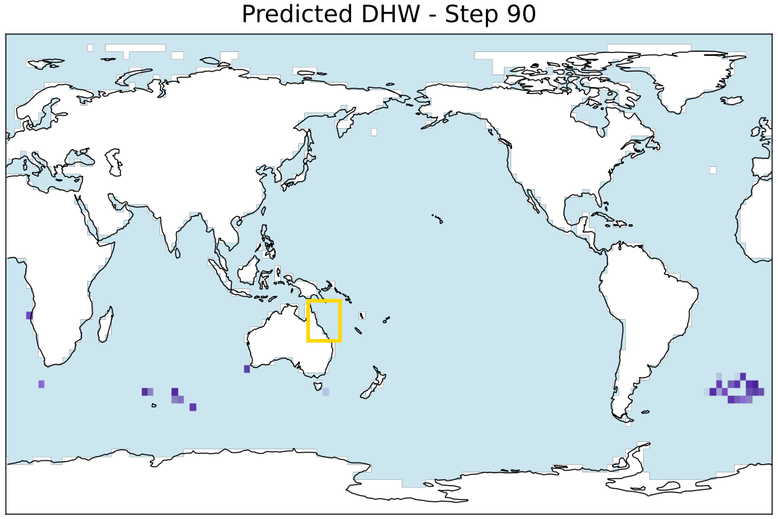
Error (Pred - Actual) - Step 46 [scale ±0.39]



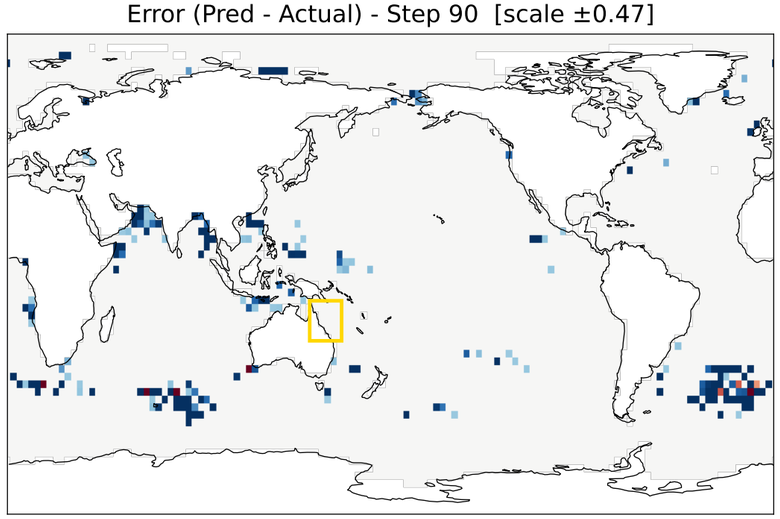
Actual DHW - Step 90

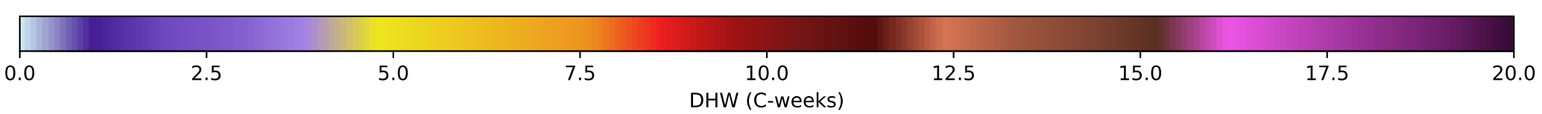


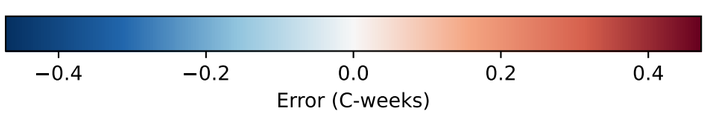
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.47]

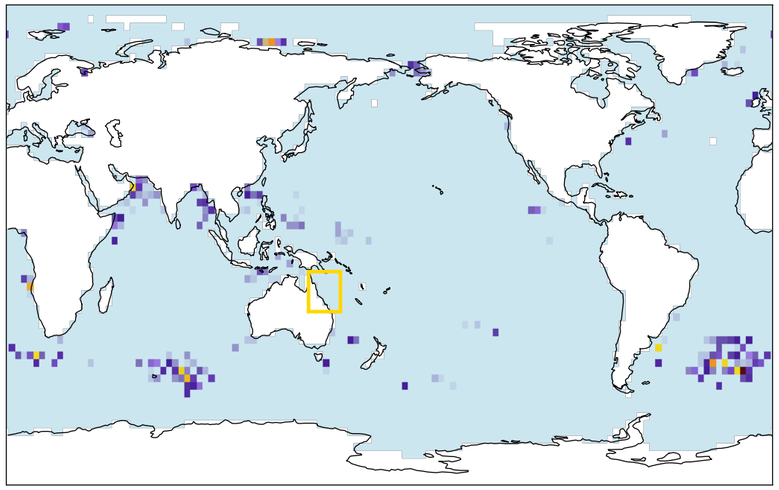




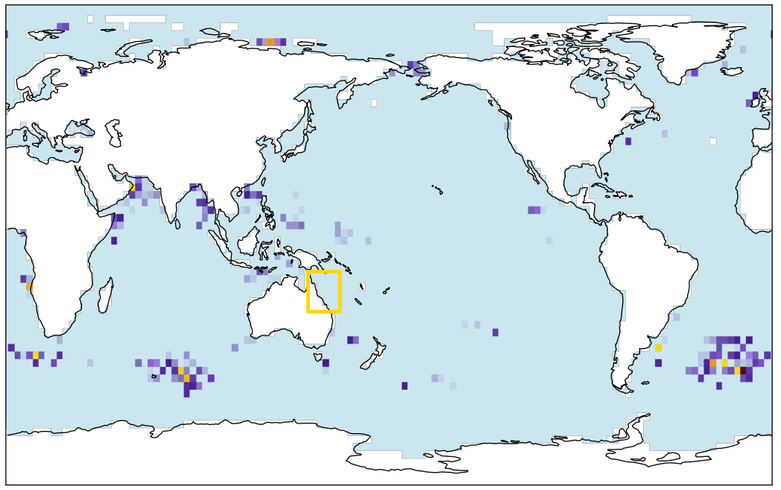


DHW Chain init: 2018-06-29 - Val (C-weeks)

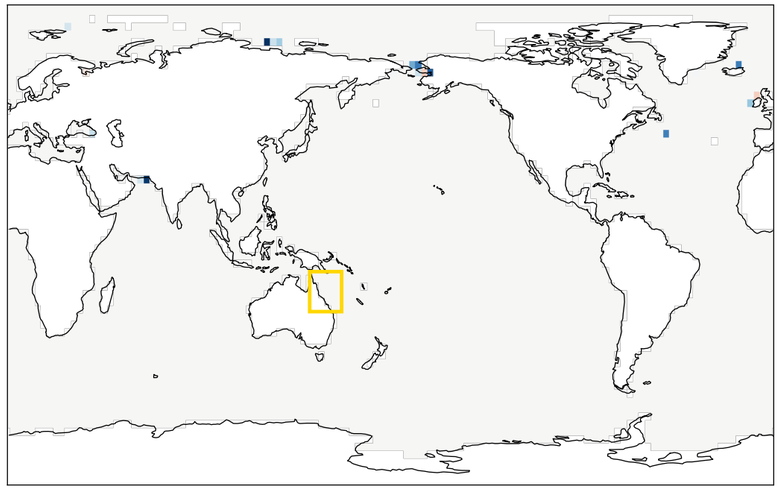
Actual DHW - Step 1



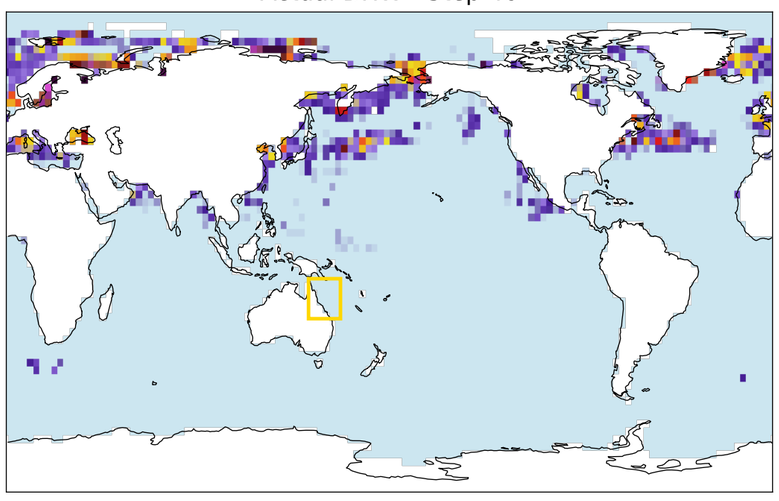
Predicted DHW - Step 1



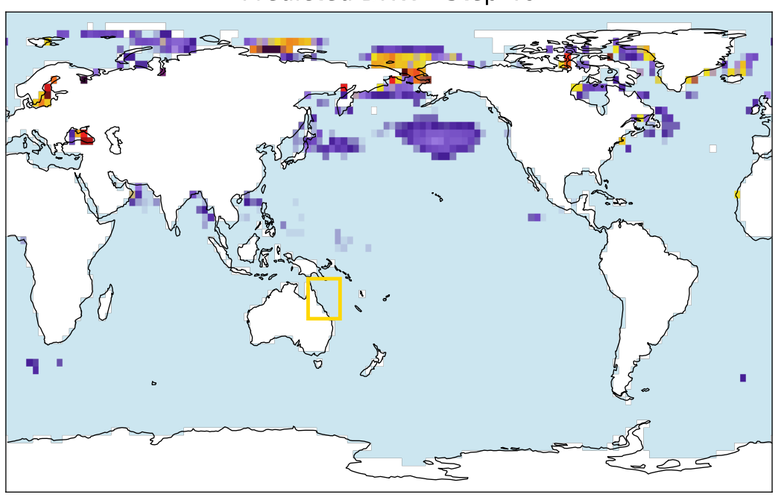
Error (Pred - Actual) - Step 1 [scale ±0.25]



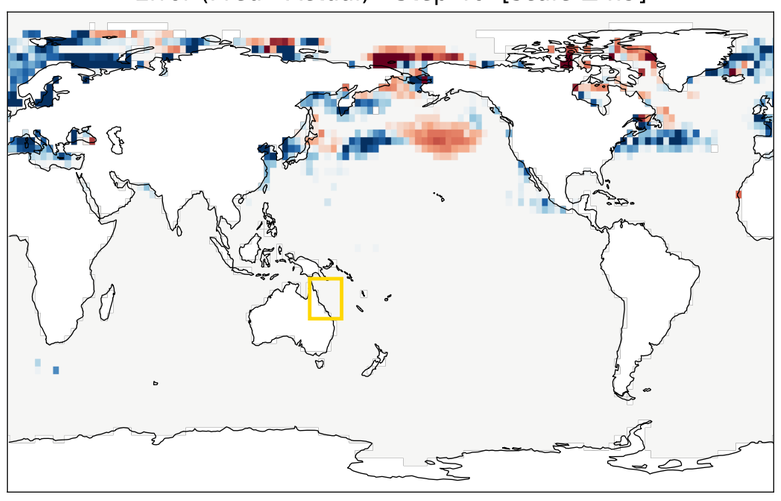
Actual DHW - Step 46



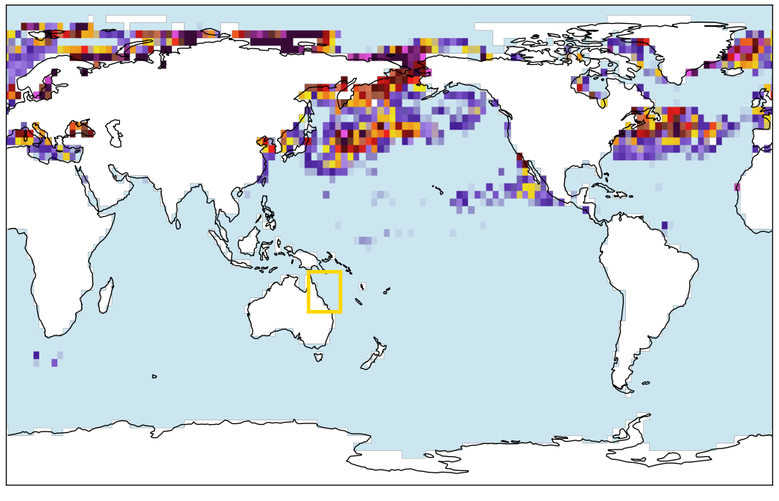
Predicted DHW - Step 46



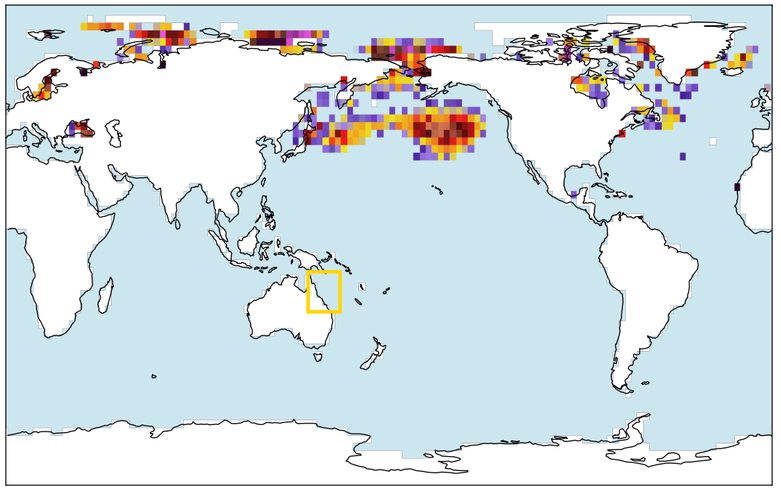
Error (Pred - Actual) - Step 46 [scale ±4.9]



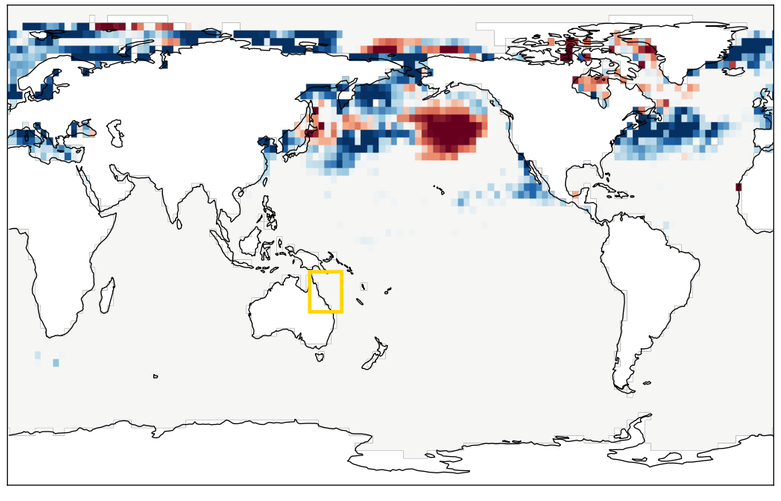
Actual DHW - Step 90

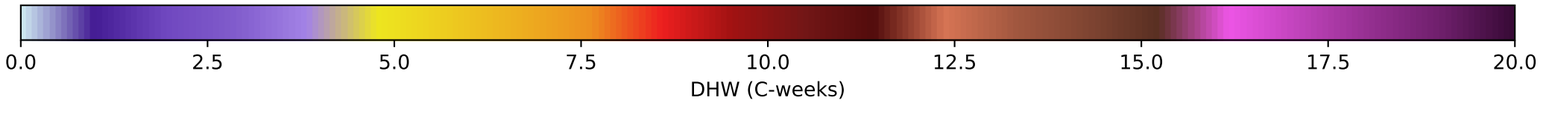


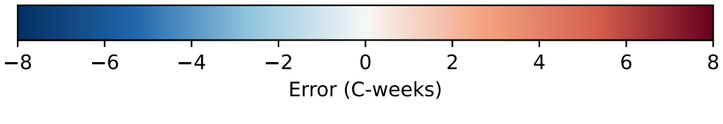
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

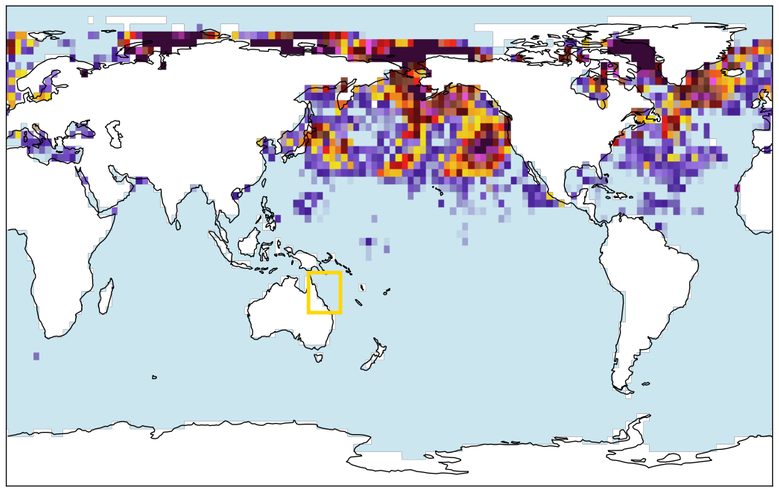




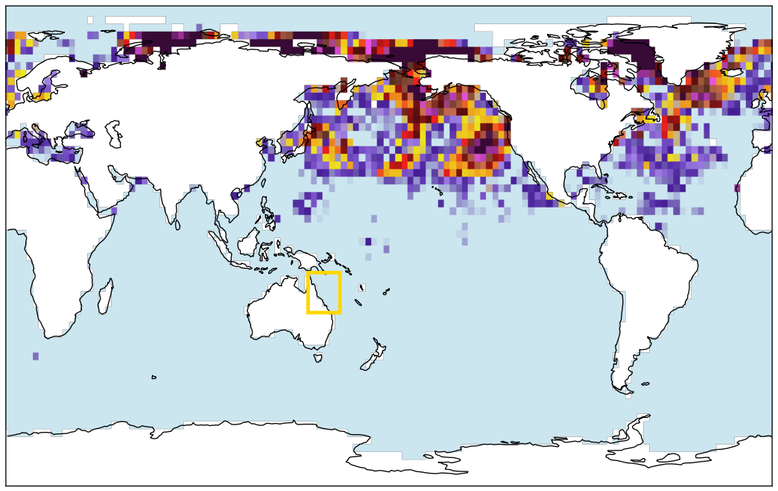


DHW Chain init: 2019-09-22 - Val (C-weeks)

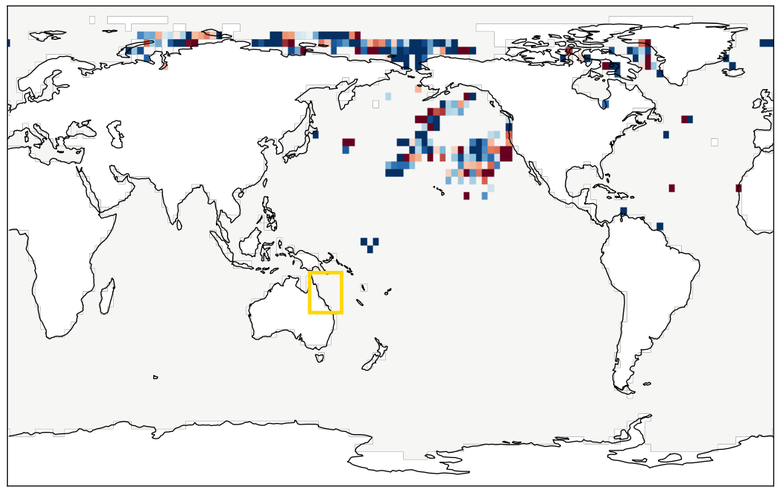
Actual DHW - Step 1



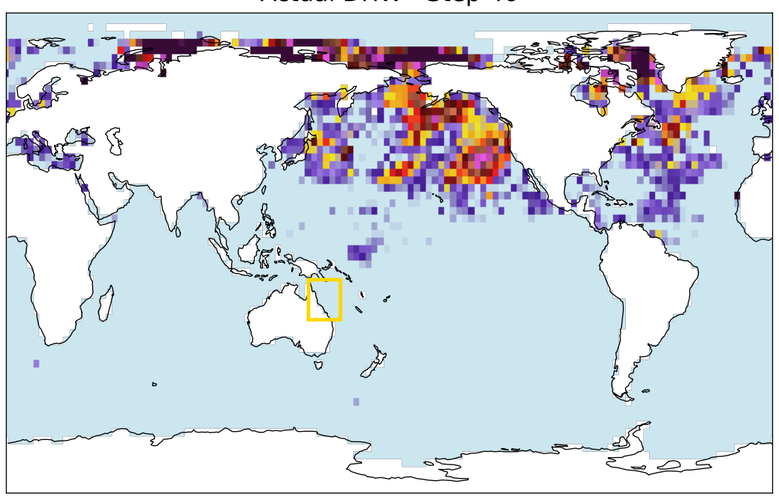
Predicted DHW - Step 1



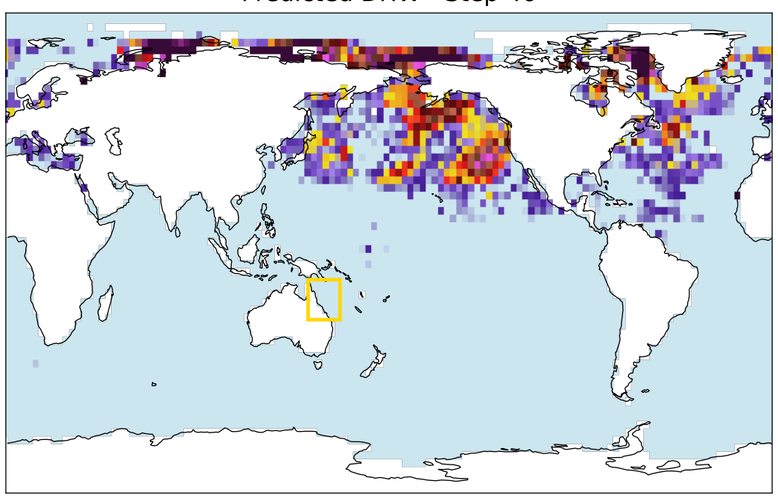
Error (Pred - Actual) - Step 1 [scale ±0.024]



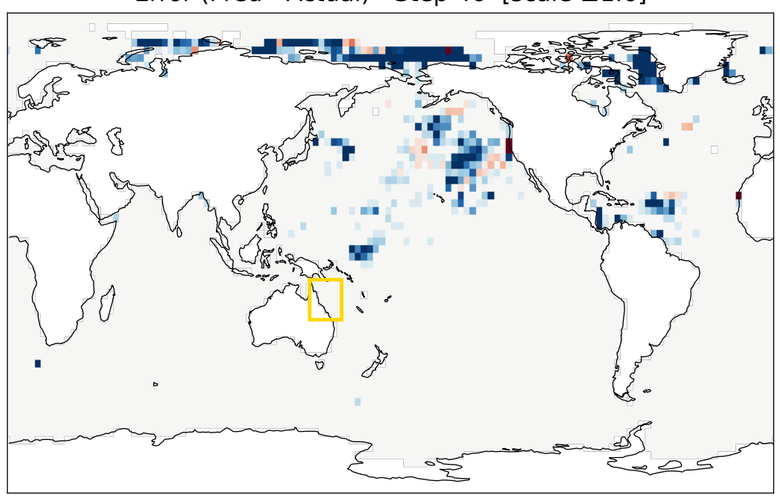
Actual DHW - Step 46



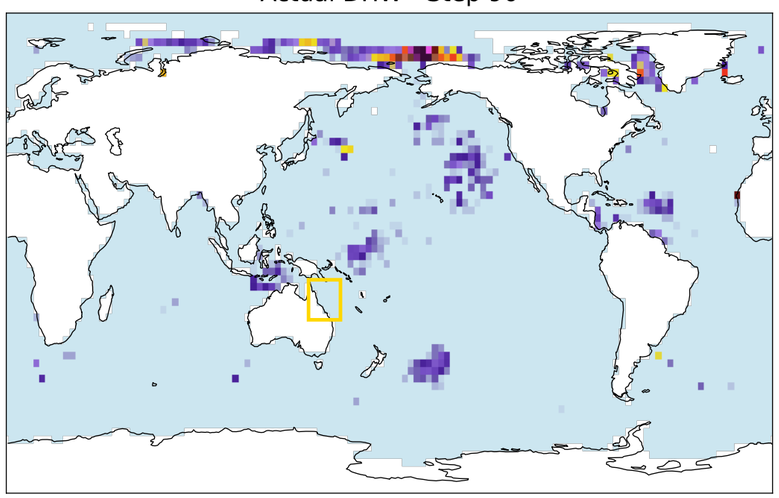
Predicted DHW - Step 46



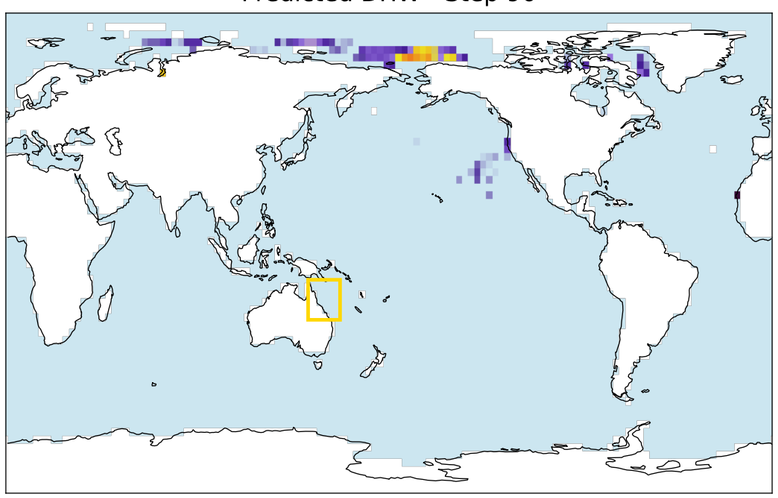
Error (Pred - Actual) - Step 46 [scale ±1.6]



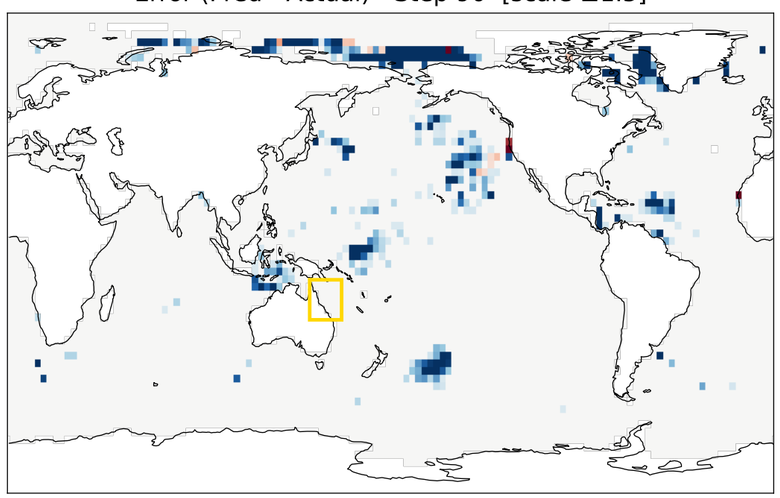
Actual DHW - Step 90

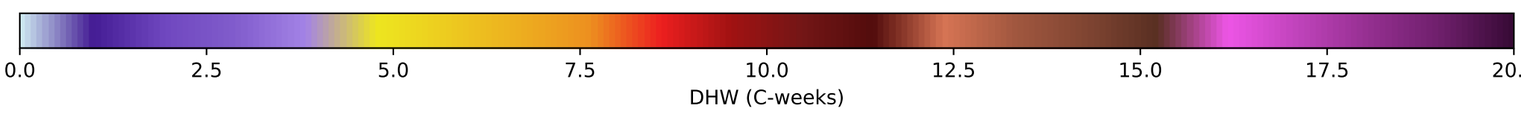


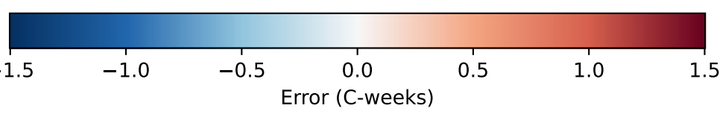
Predicted DHW - Step 90



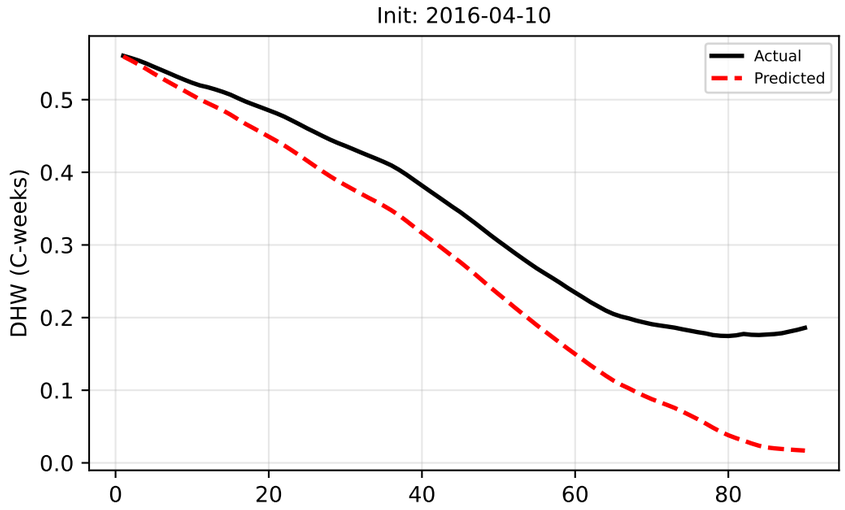
Error (Pred - Actual) - Step 90 [scale ±1.5]

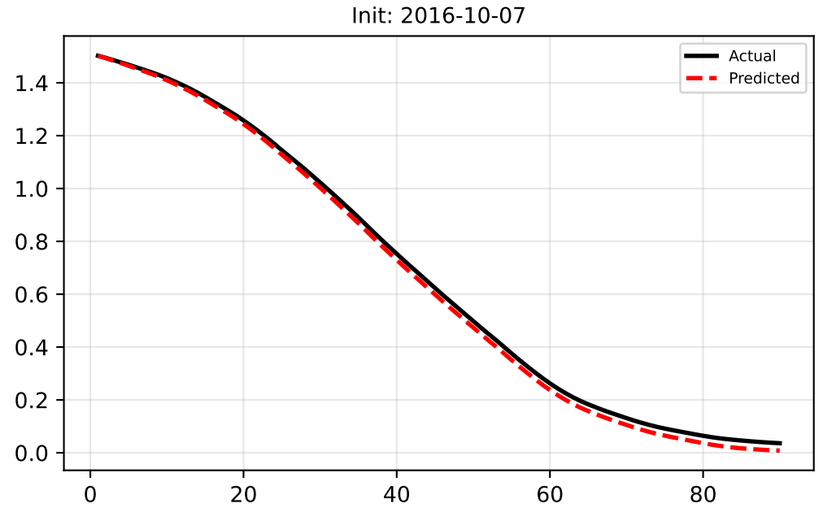


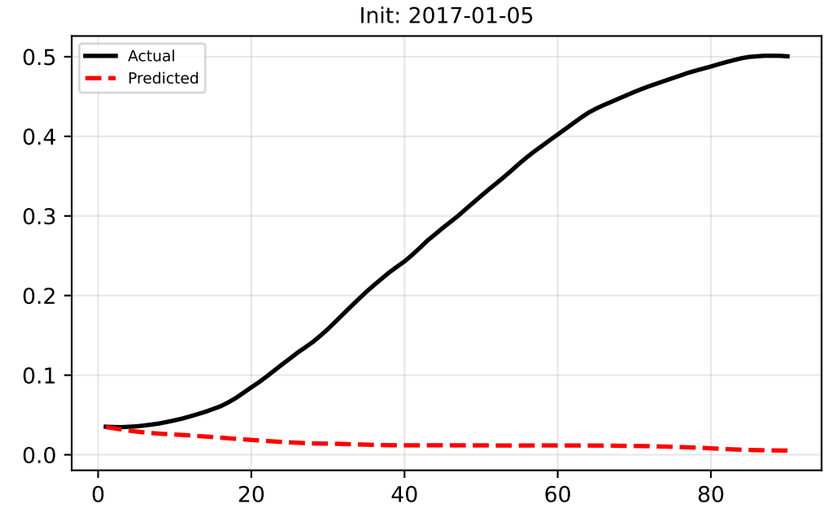


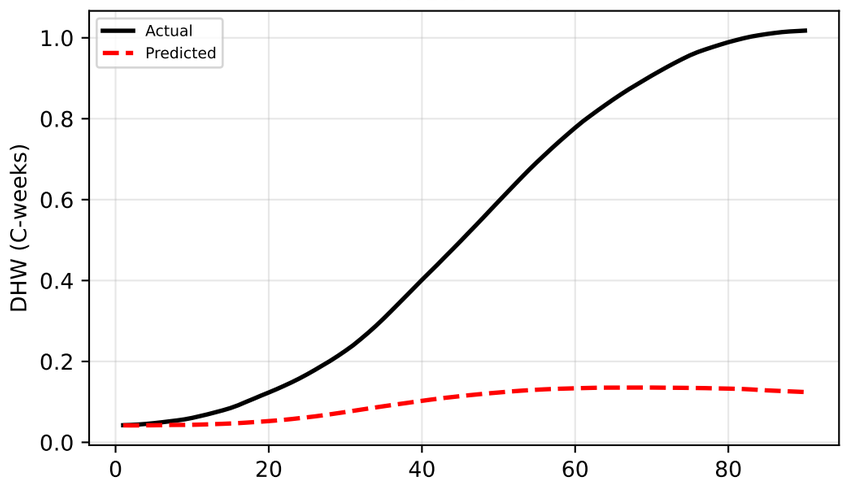


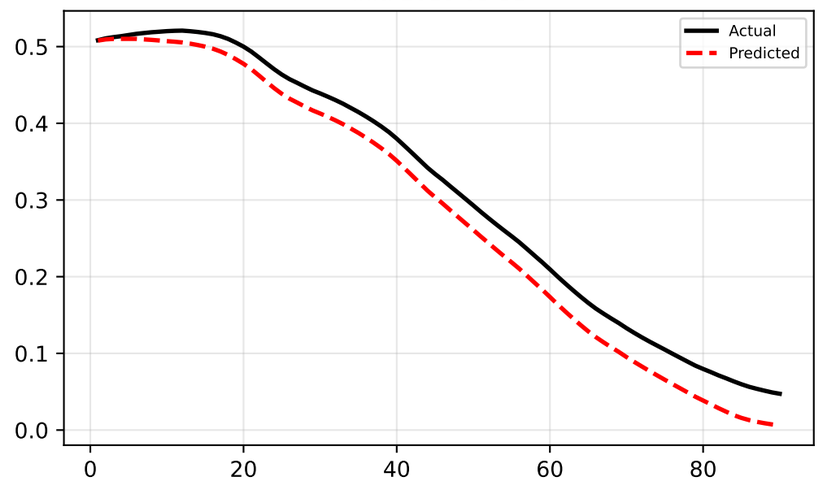
Global-Mean DHW - Val (C-weeks)

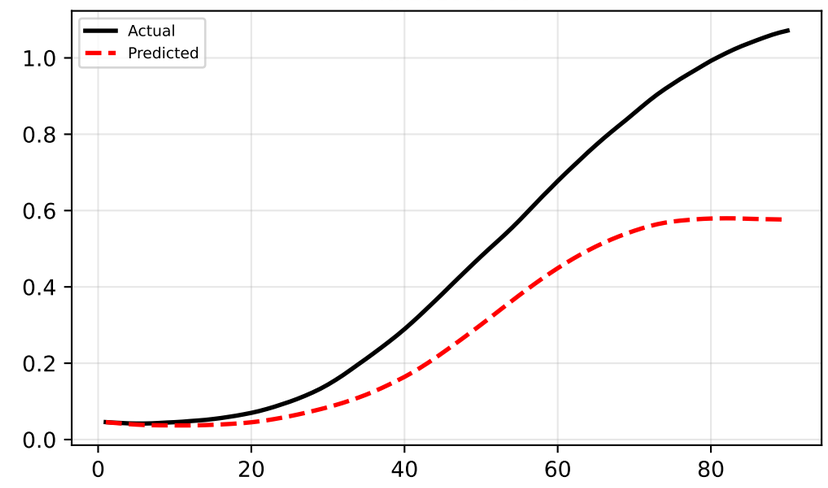


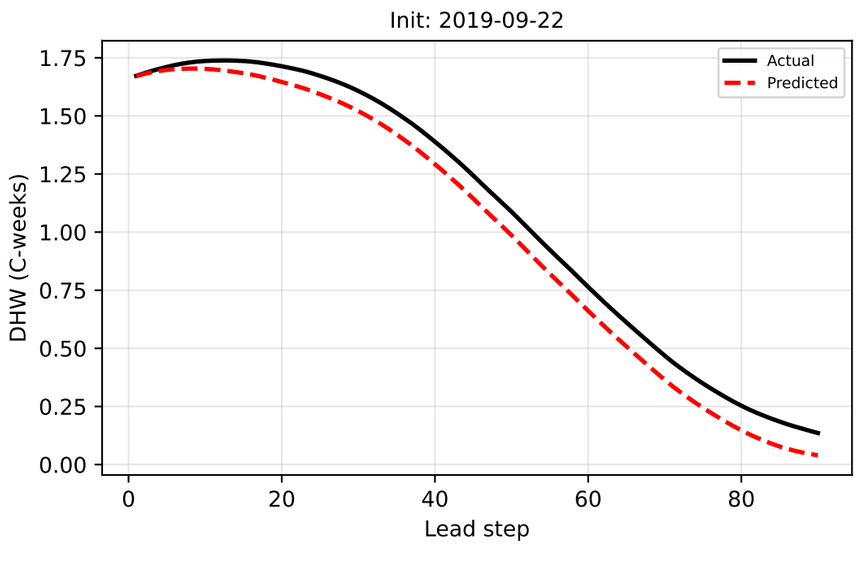




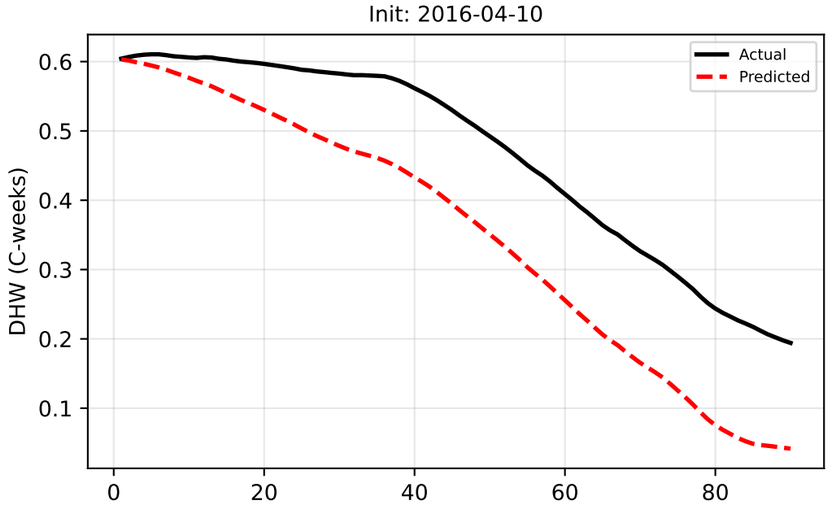


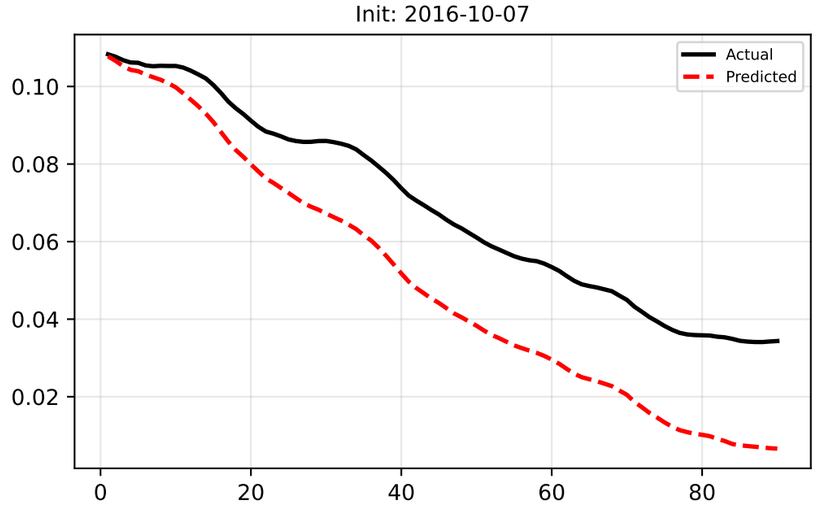


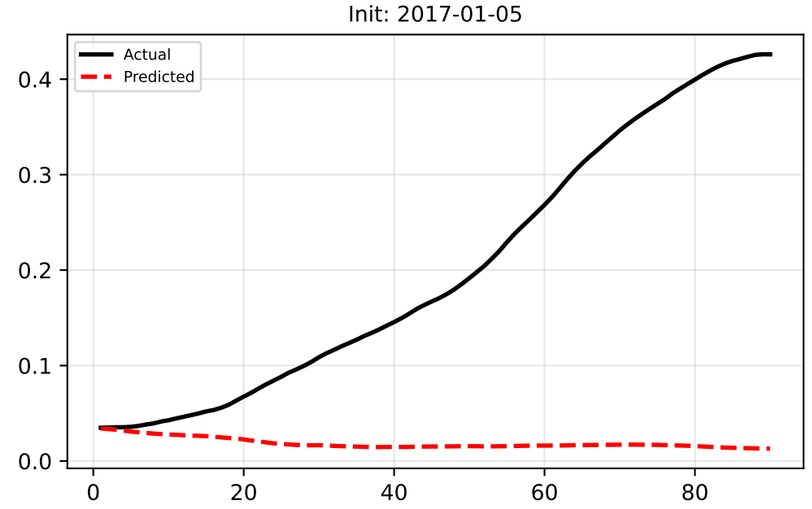


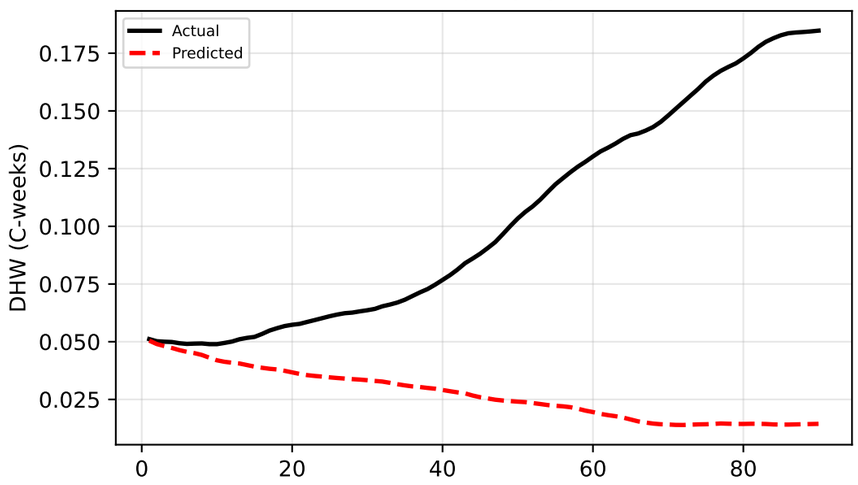


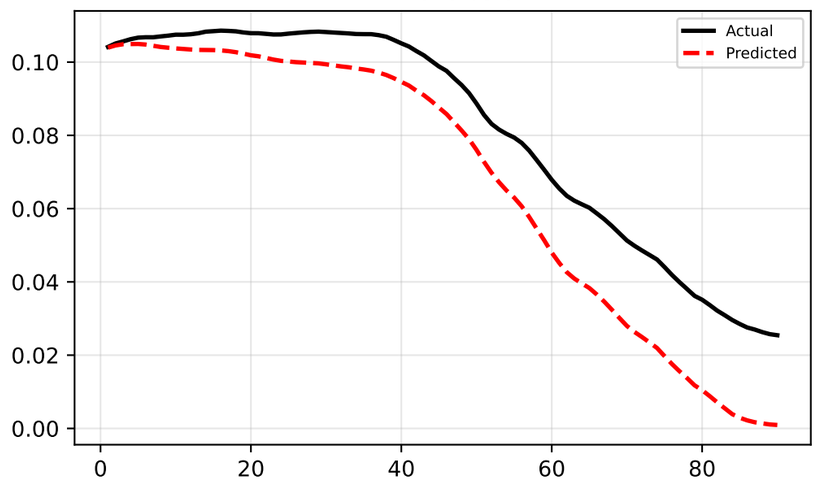
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

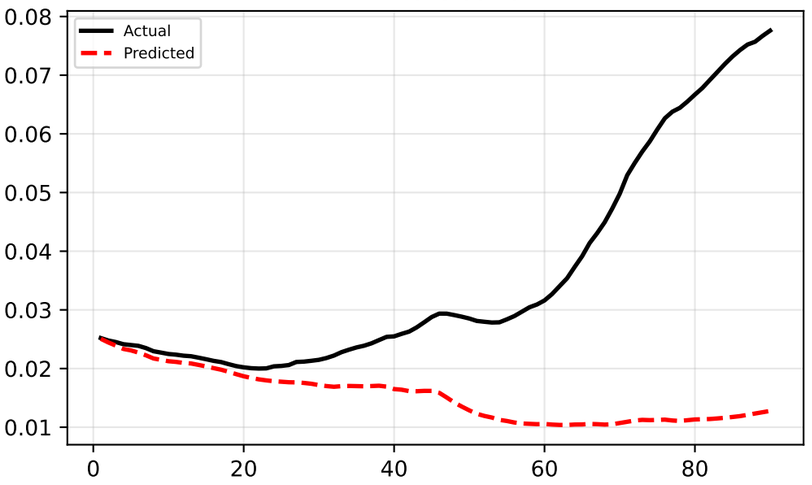


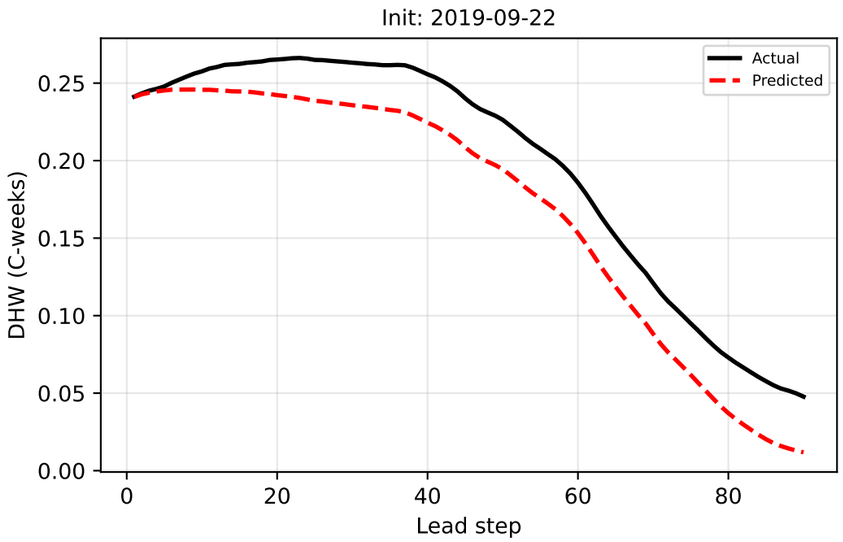




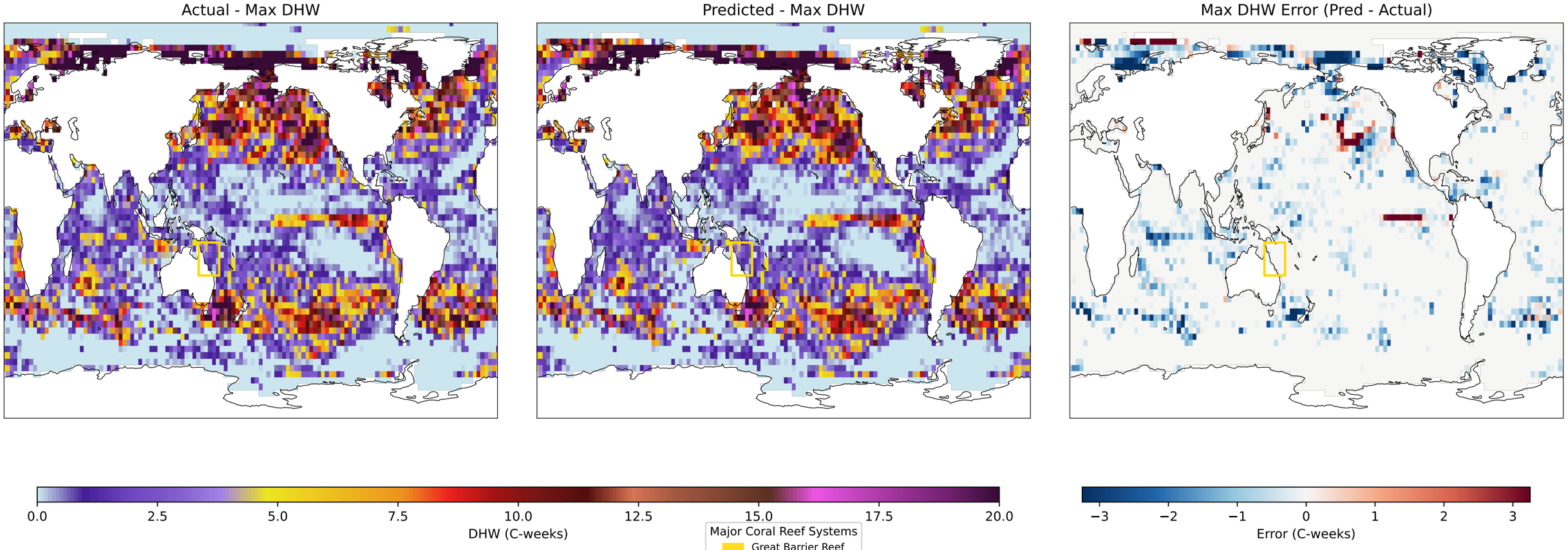


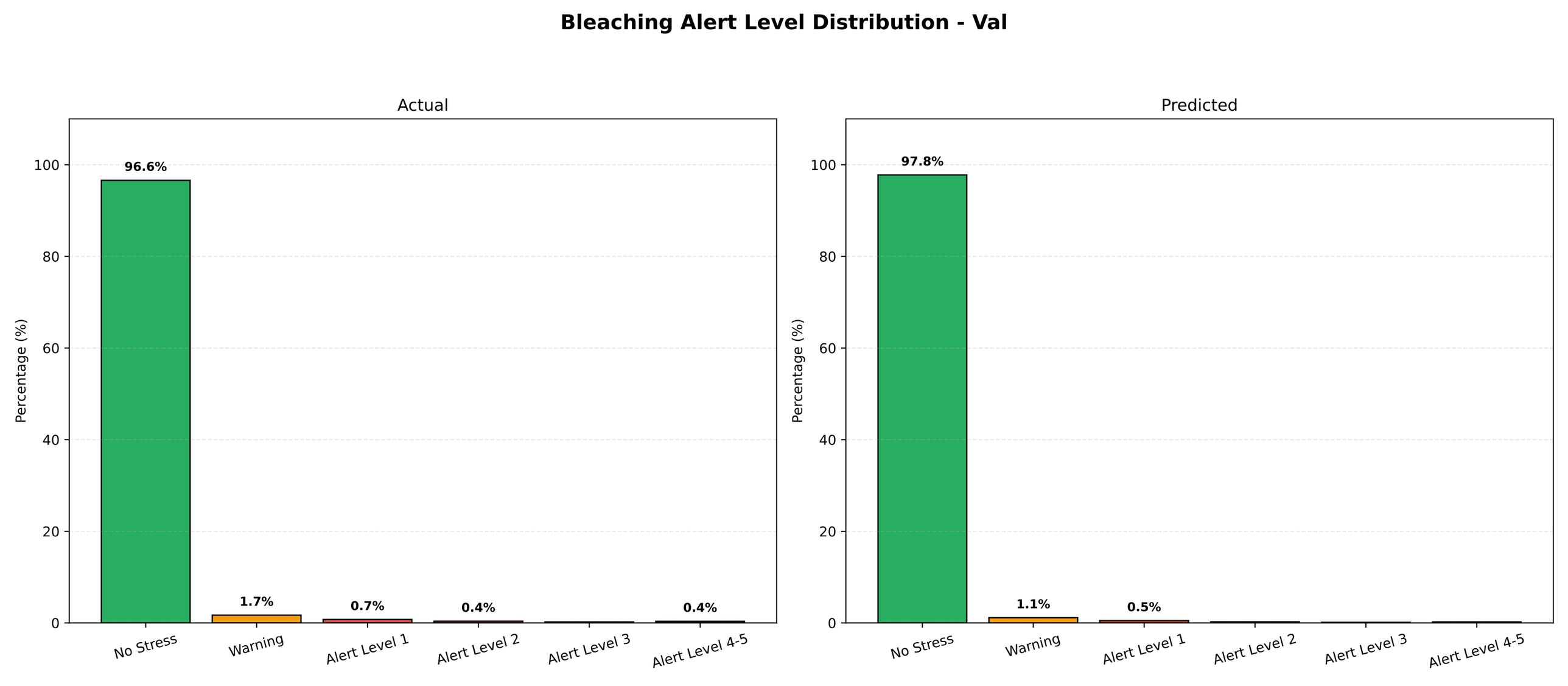






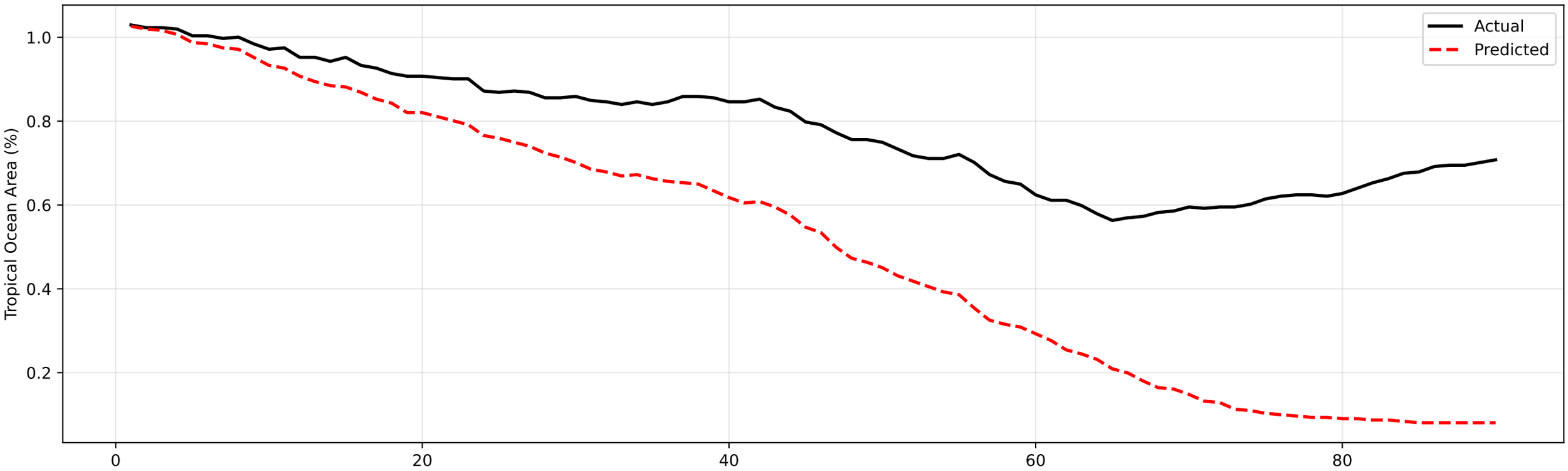
Peak DHW (max over all chains & leads) - Val (C-weeks)



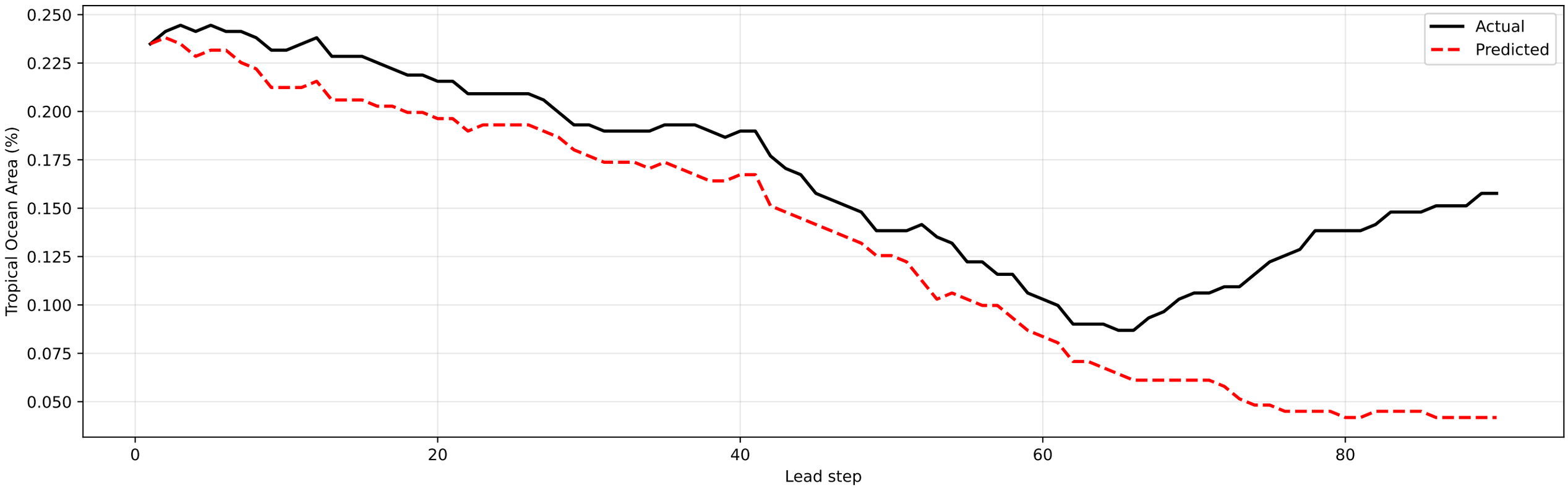

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.8%
1.1%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



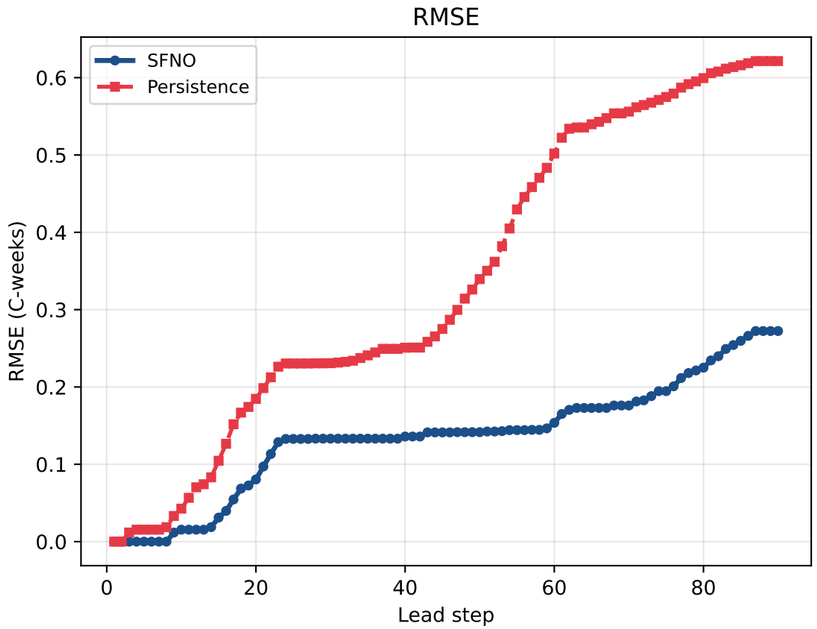
Great Barrier Reef Analysis - Val

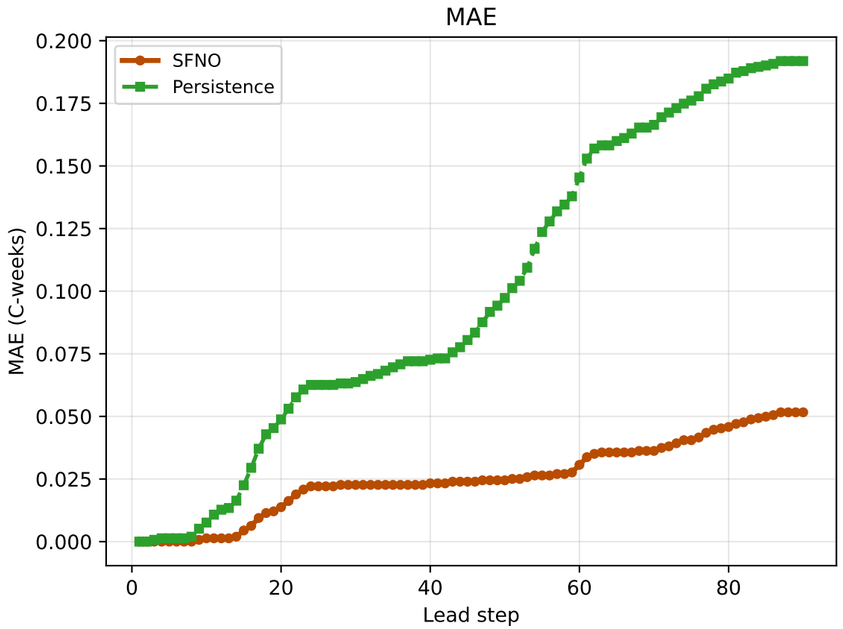
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

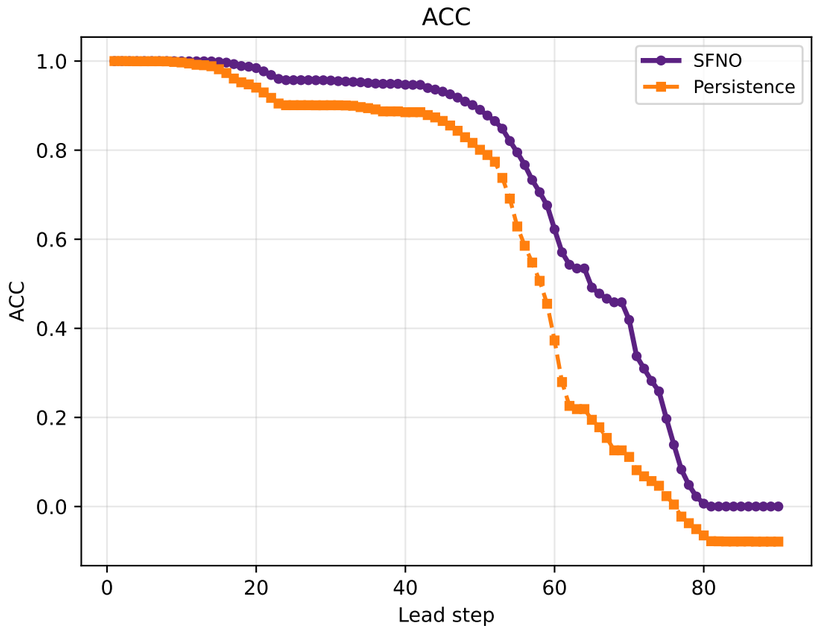
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

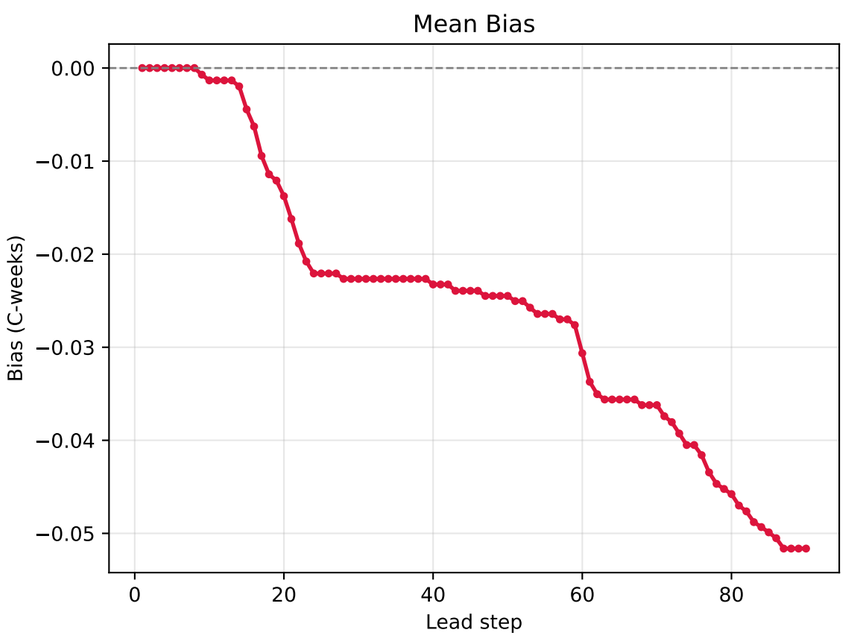
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

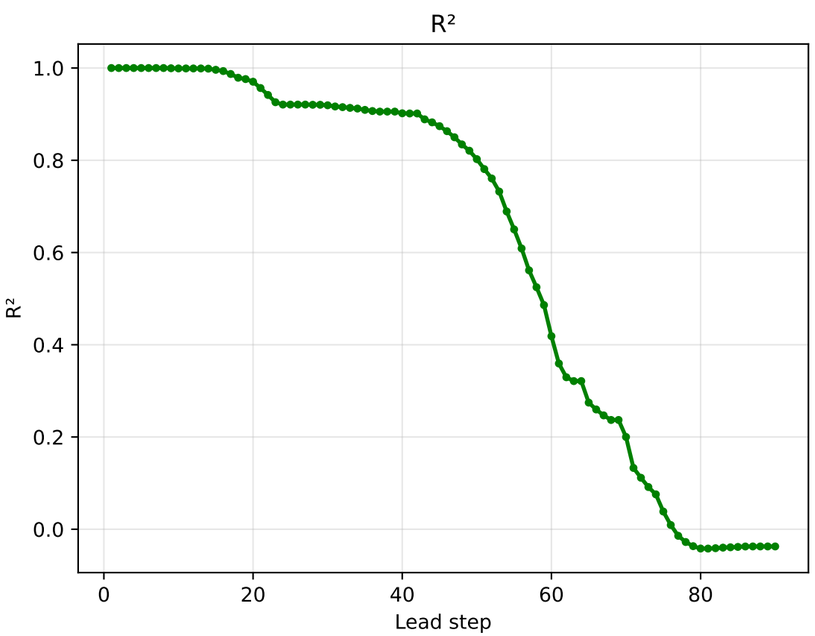
DHW Forecast Skill vs Lead Step - Val - GBR

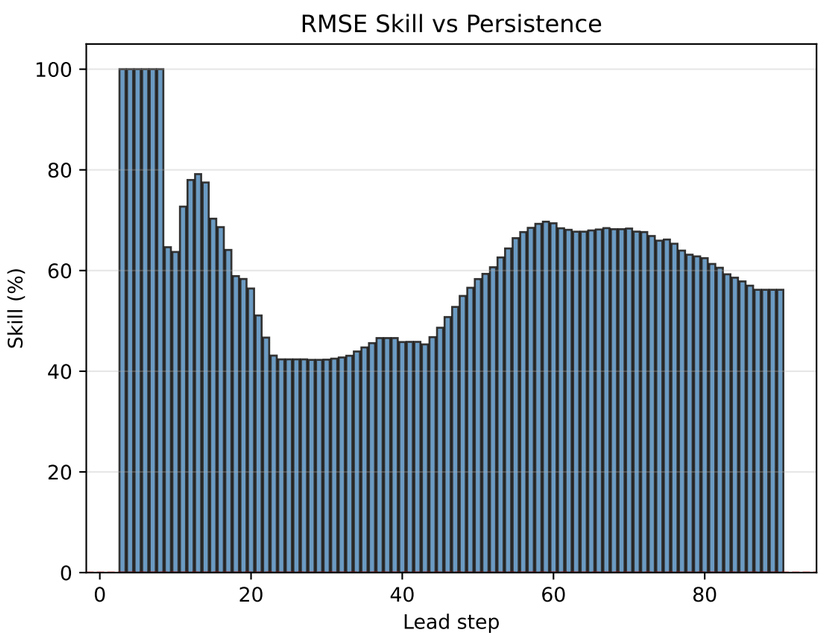


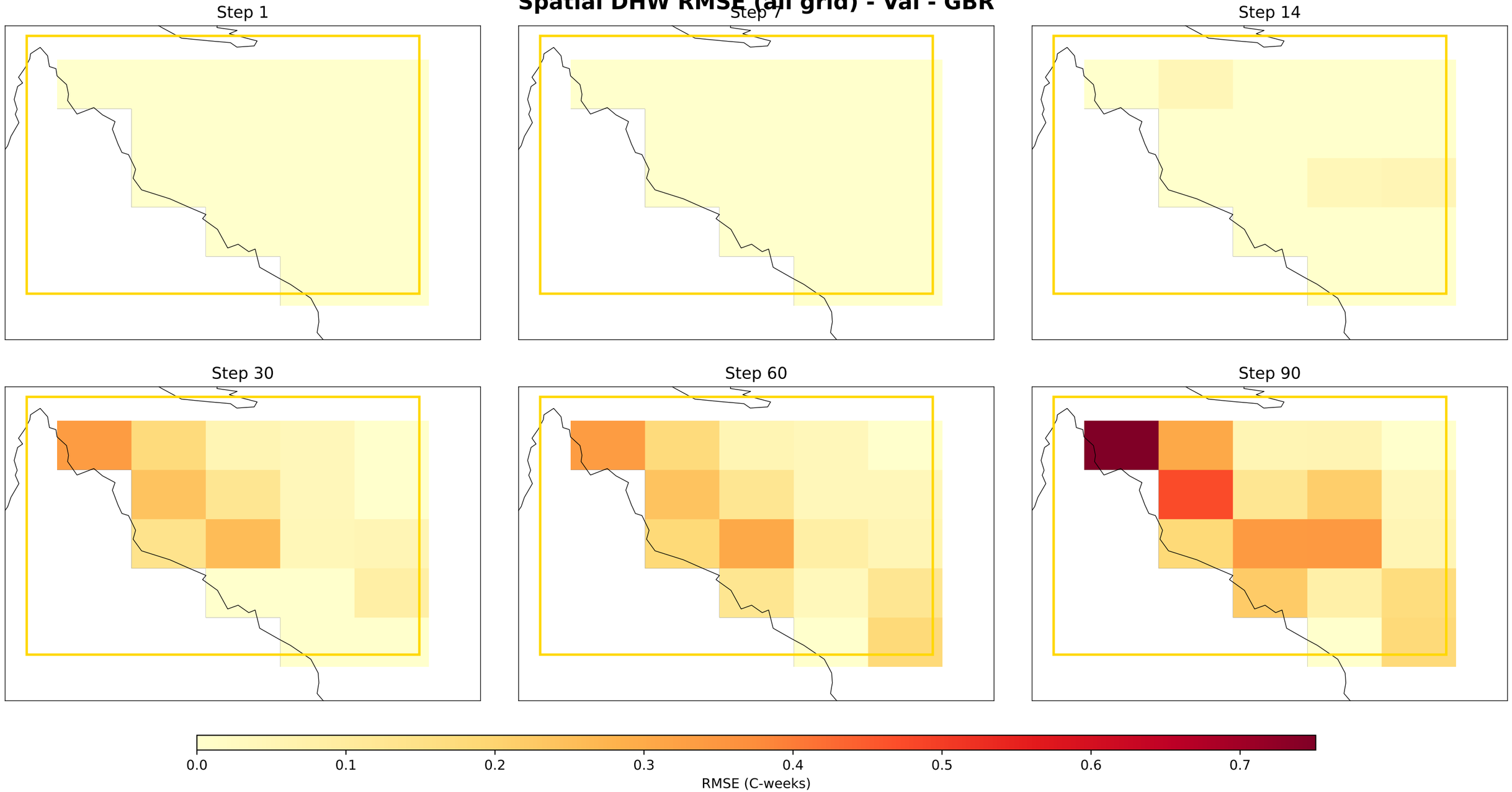


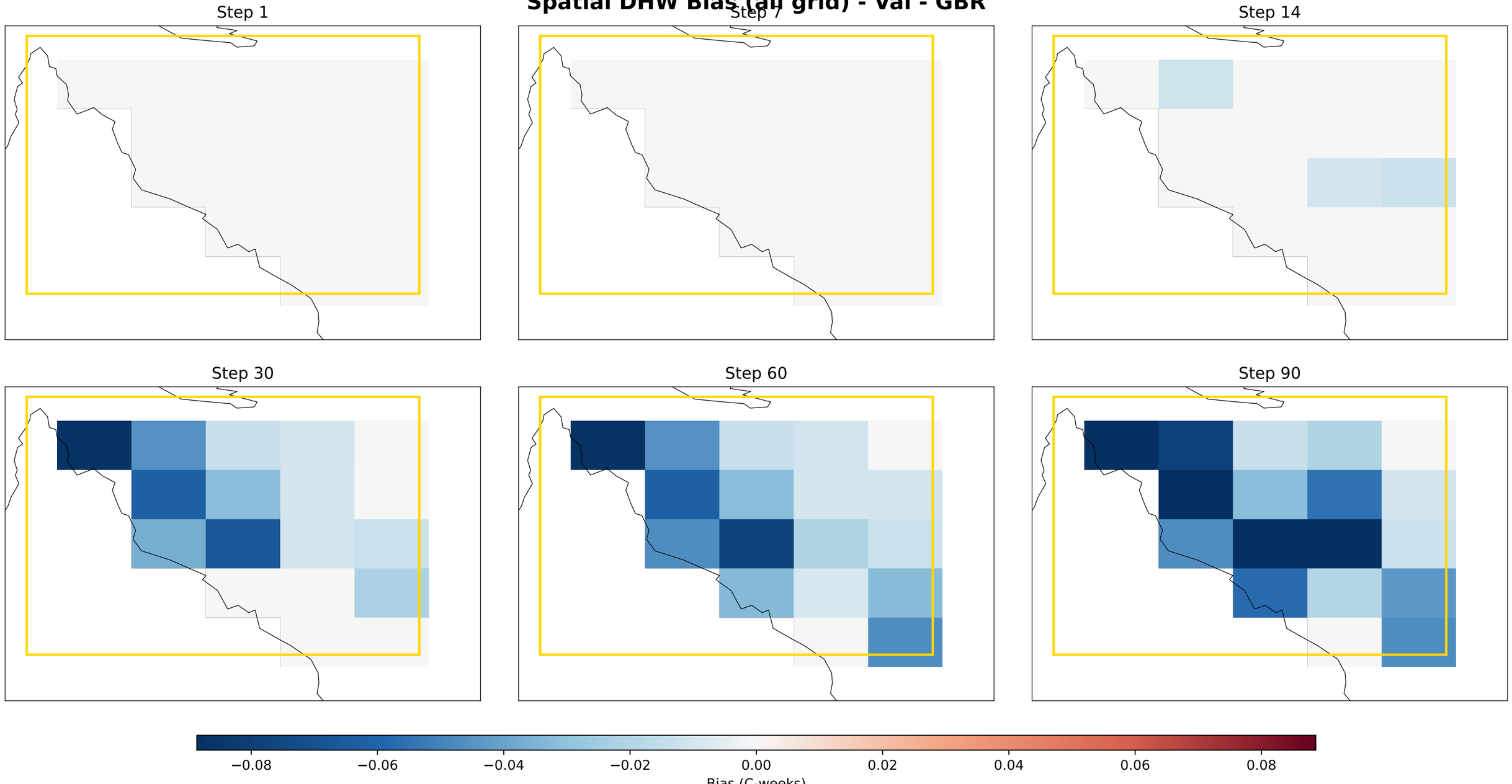


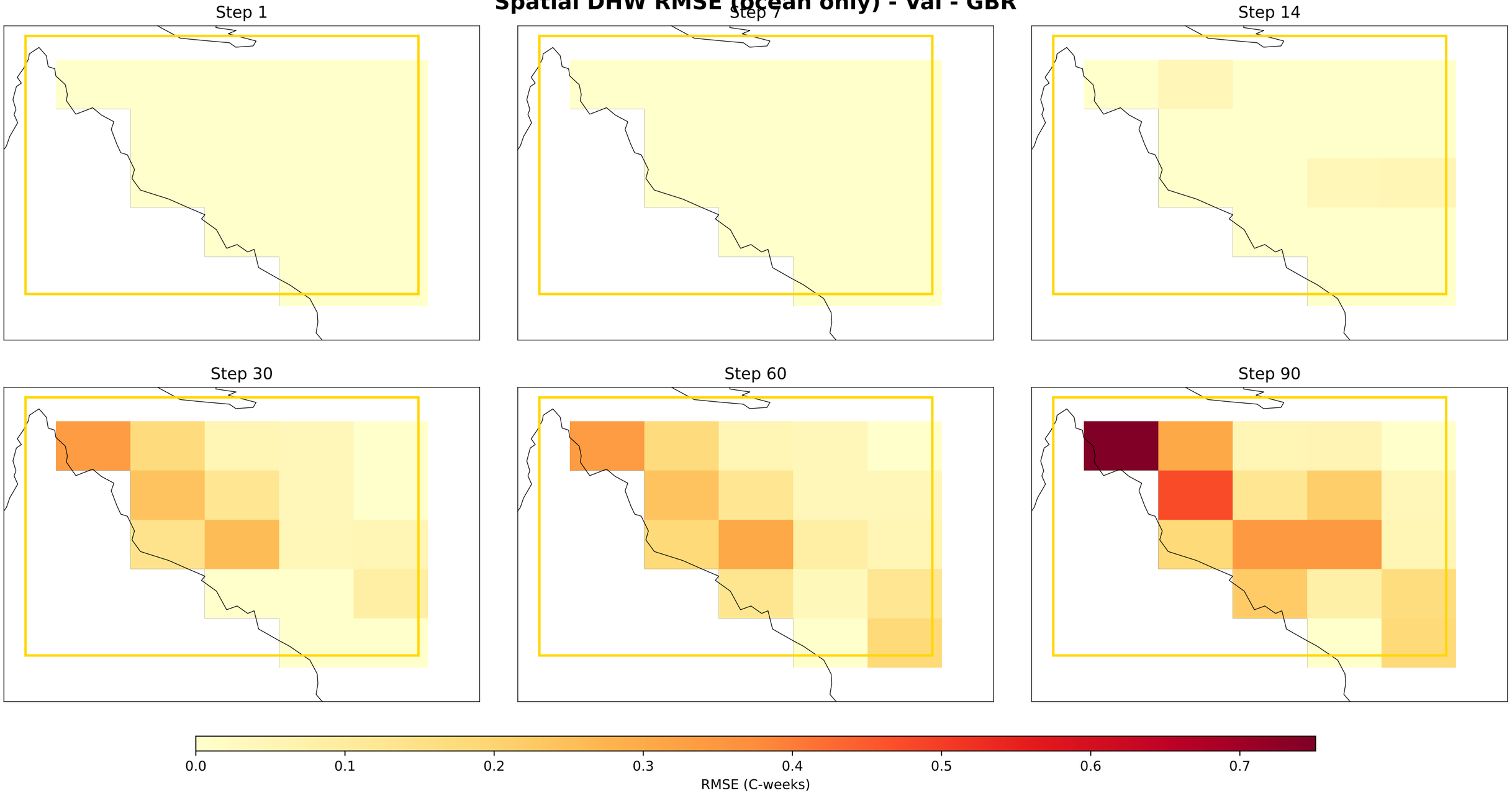


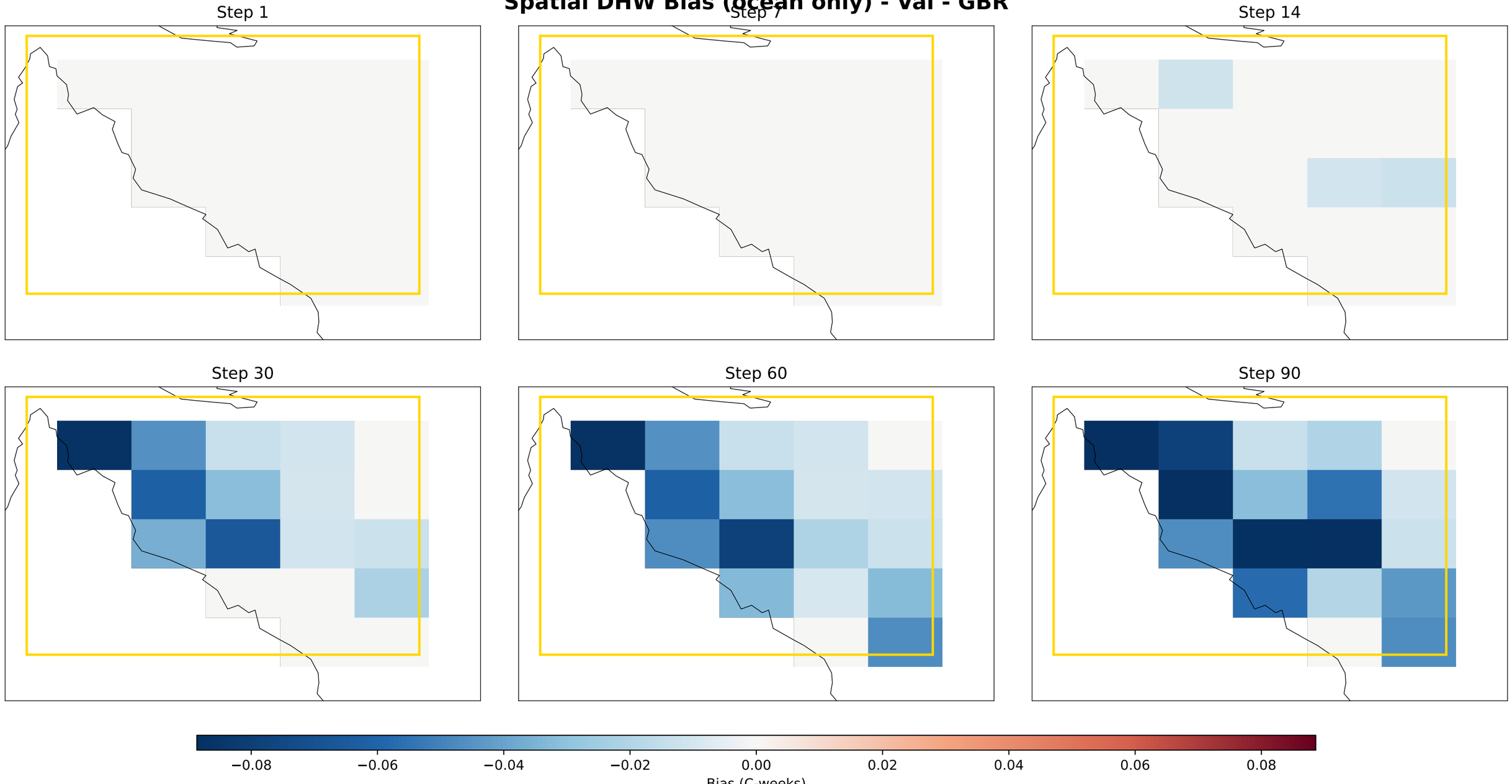





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

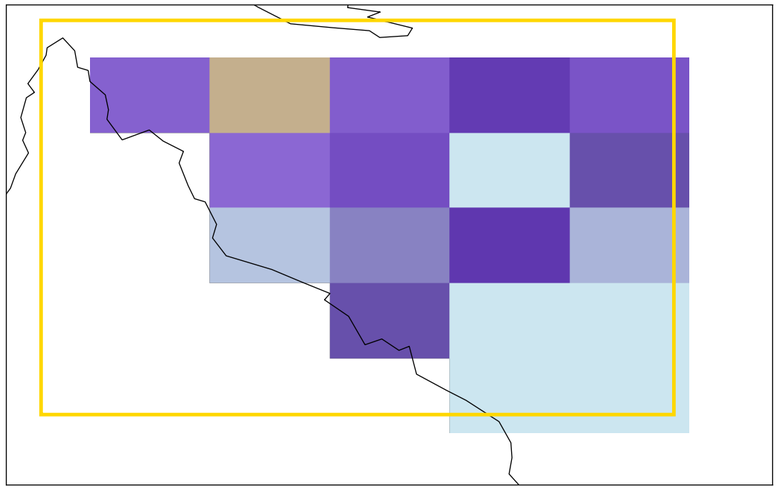

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

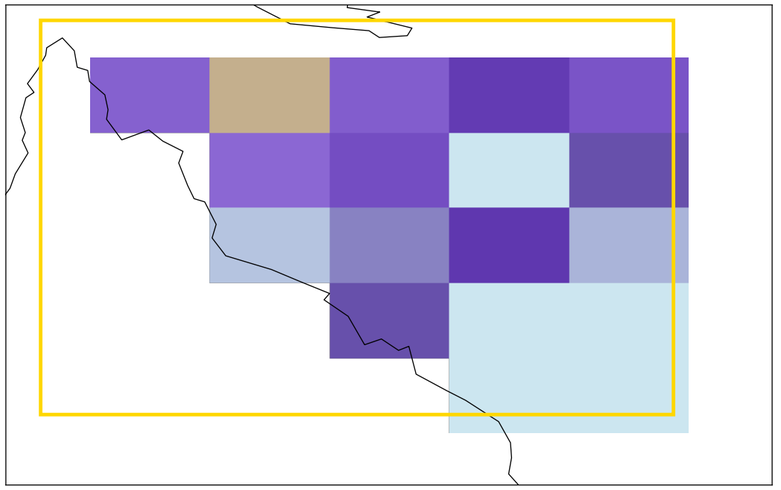

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

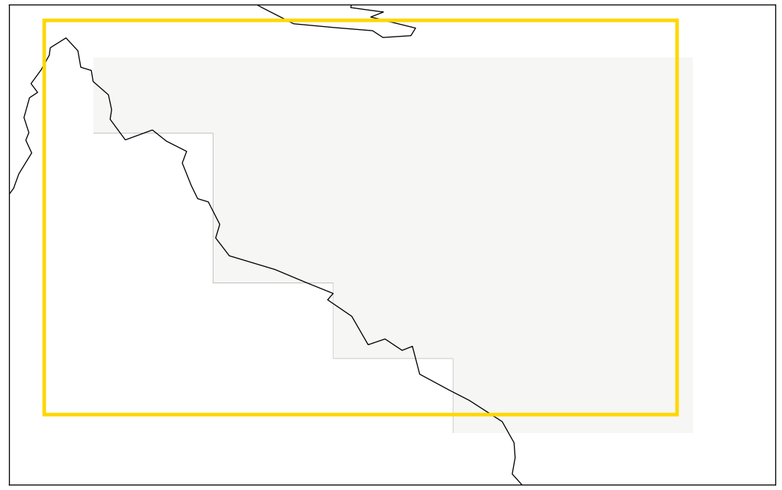
Actual DHW - Step 1



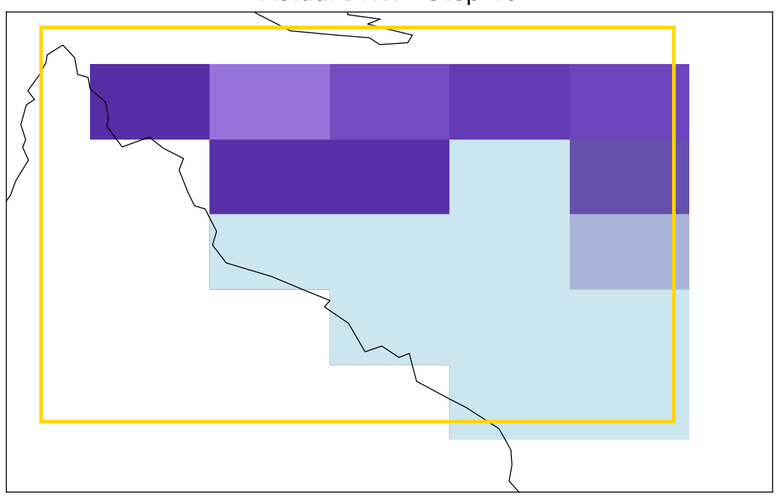
Predicted DHW - Step 1



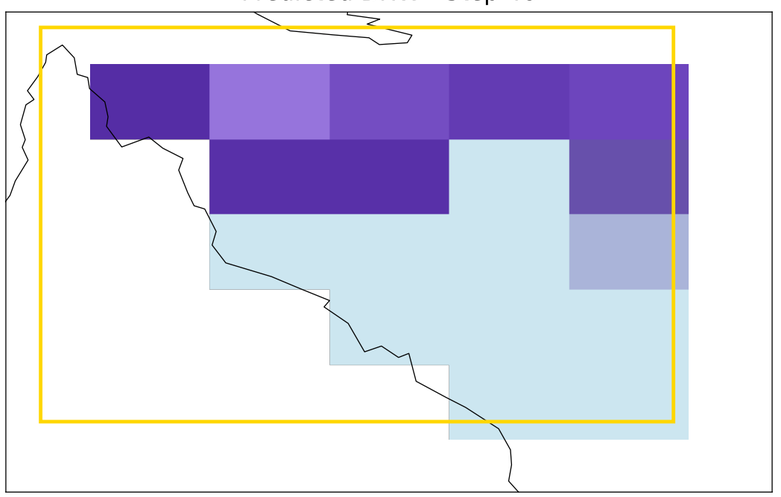
Error (Pred - Actual) - Step 1 [scale ±8]



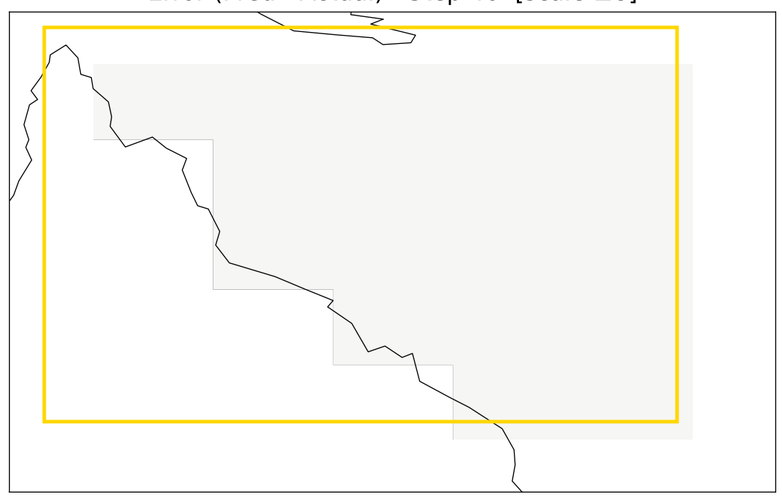
Actual DHW - Step 46



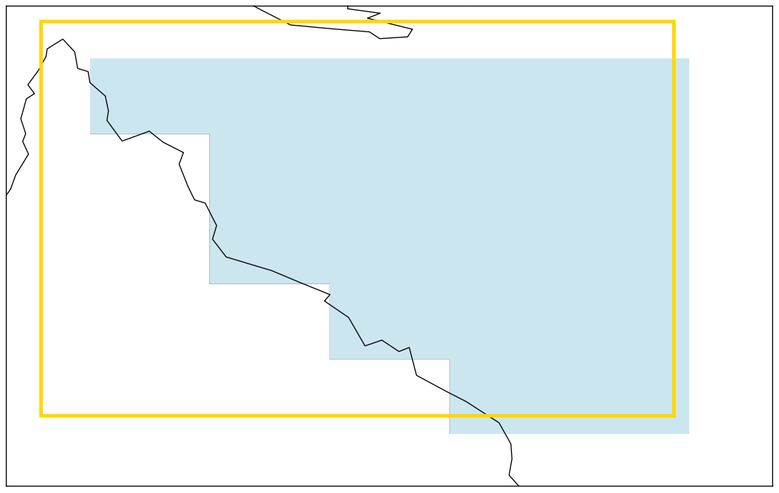
Predicted DHW - Step 46



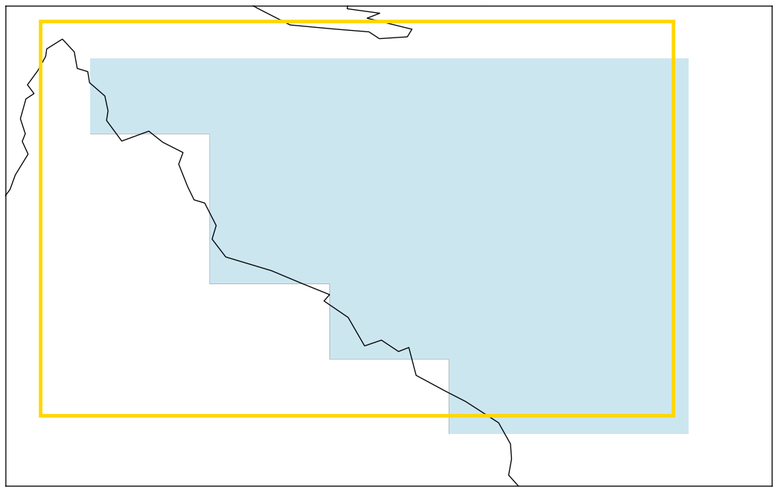
Error (Pred - Actual) - Step 46 [scale ±8]



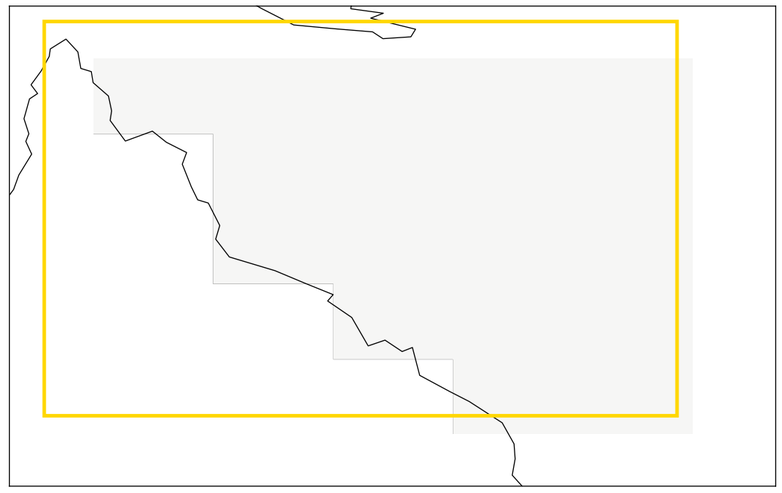
Actual DHW - Step 90

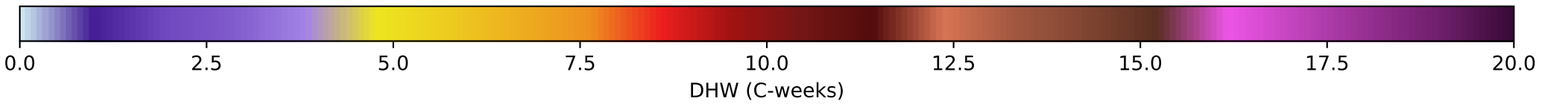


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

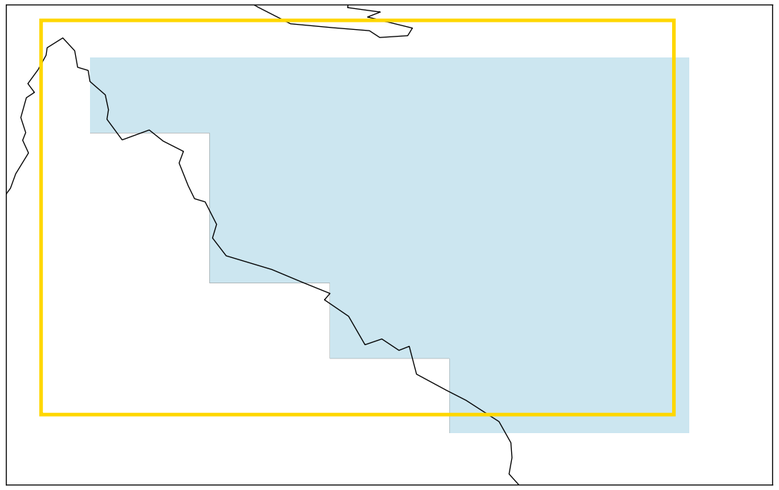




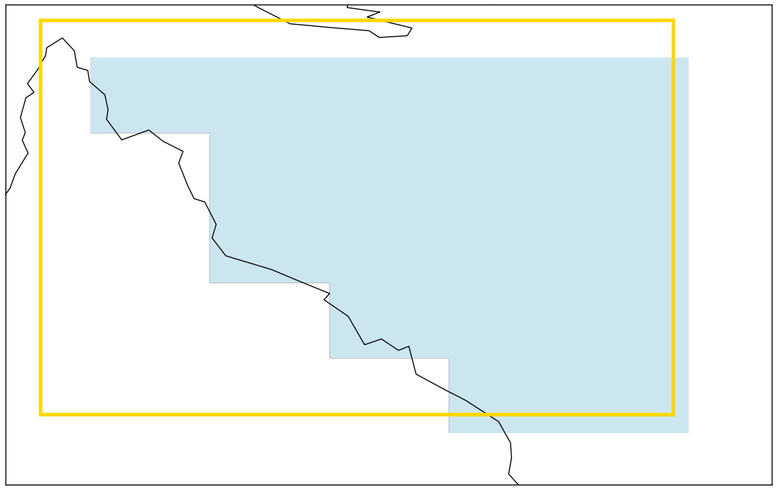


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

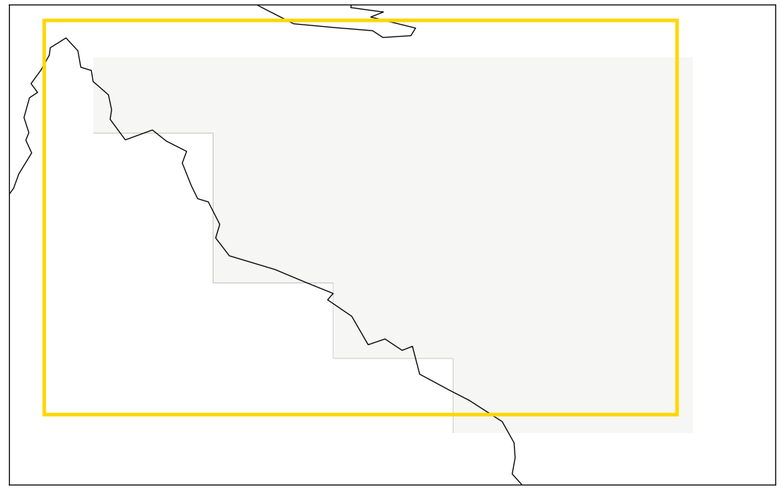
Actual DHW - Step 1



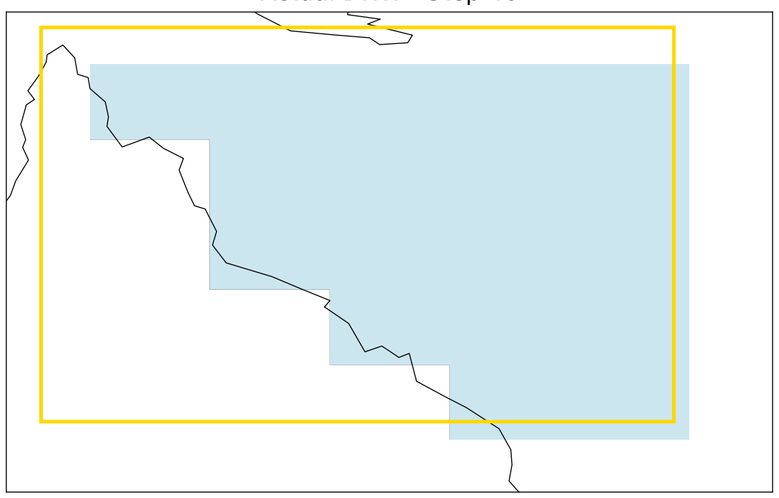
Predicted DHW - Step 1



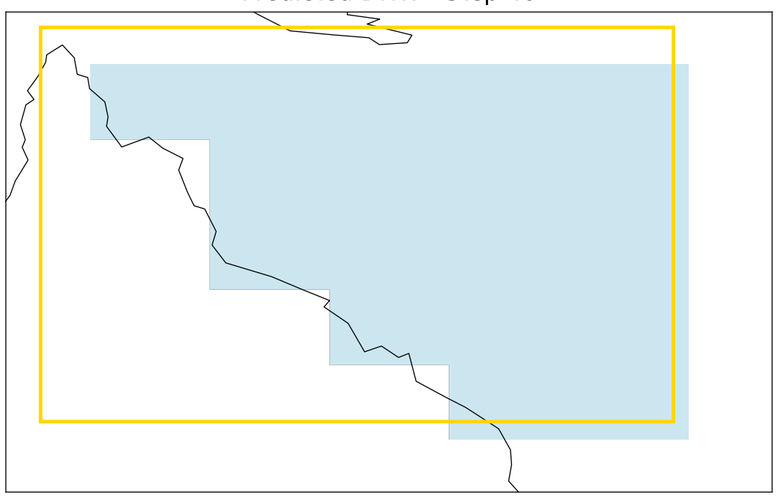
Error (Pred - Actual) - Step 1 [scale ±8]



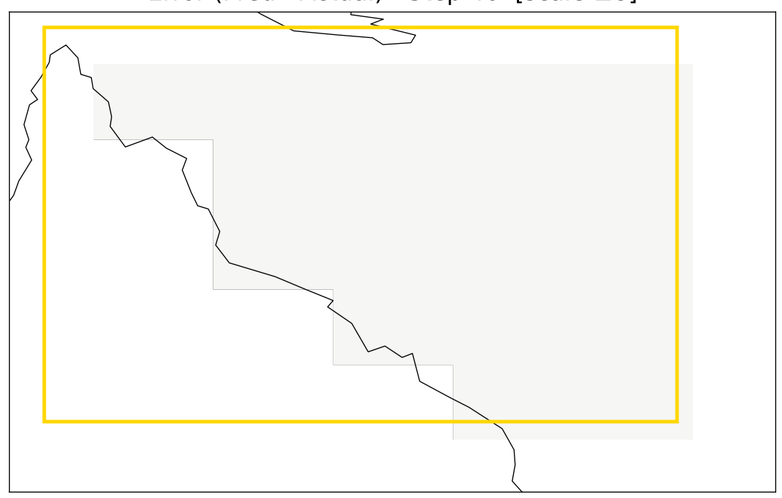
Actual DHW - Step 46



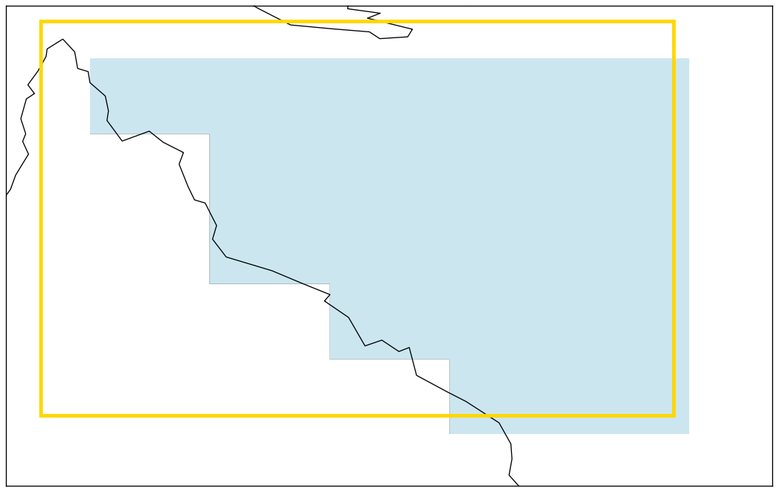
Predicted DHW - Step 46



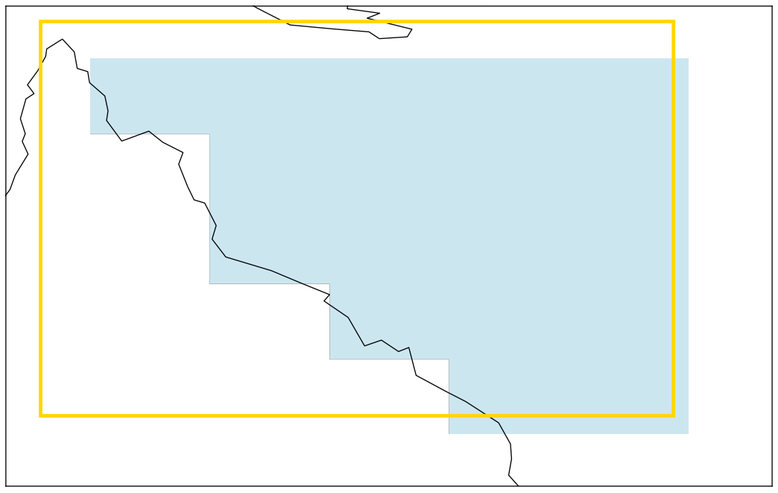
Error (Pred - Actual) - Step 46 [scale ±8]



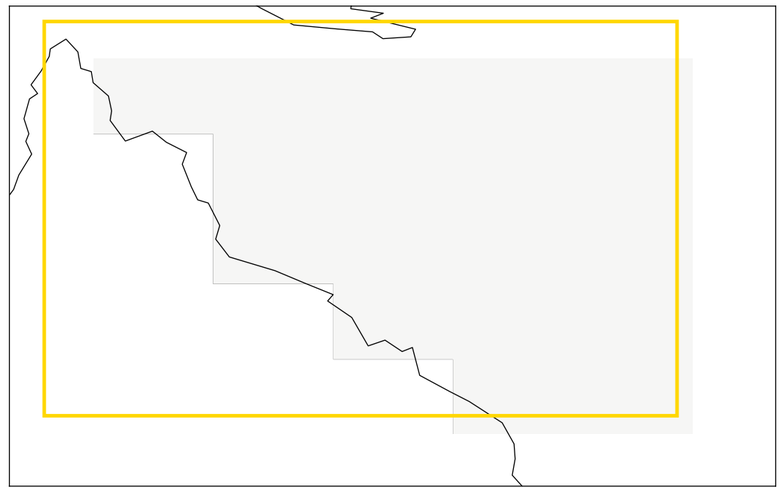
Actual DHW - Step 90

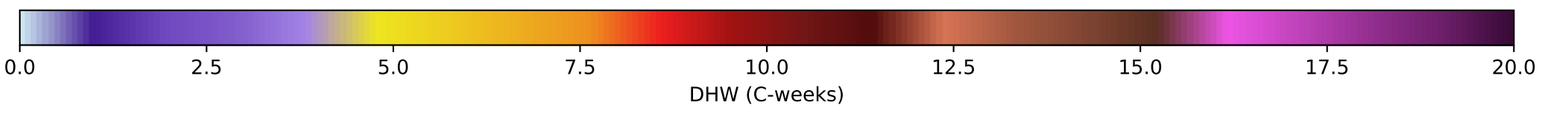


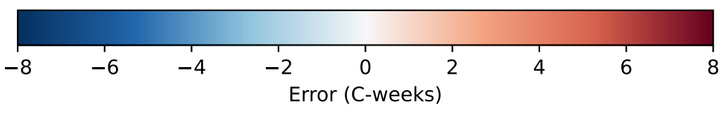
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

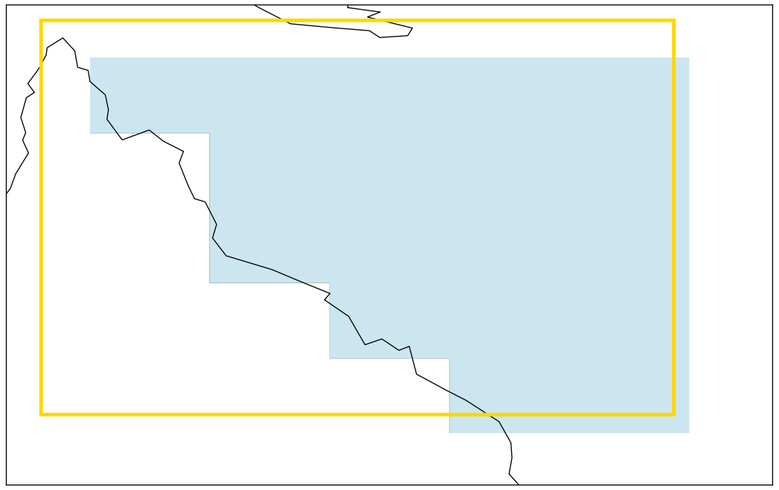




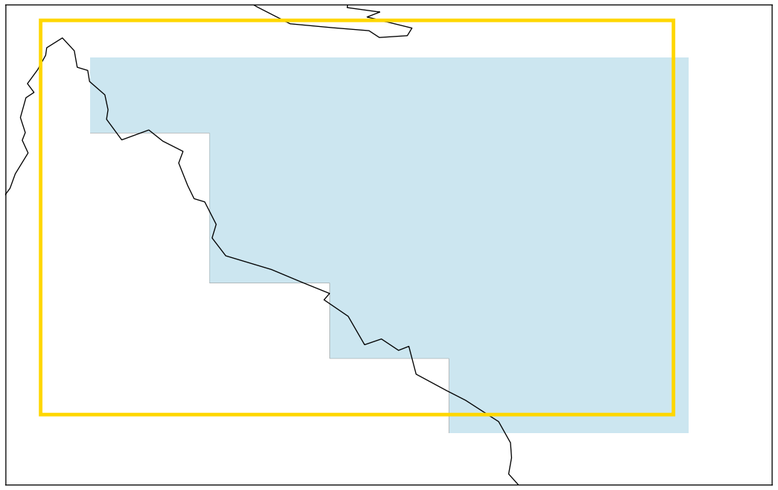


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

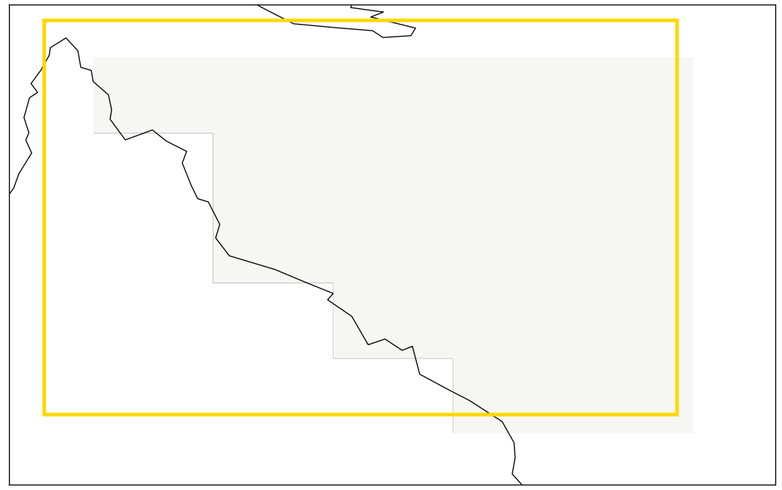
Actual DHW - Step 1



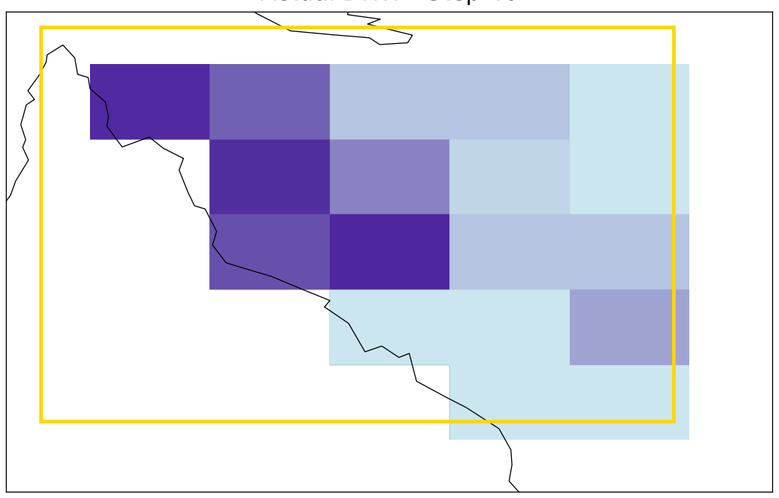
Predicted DHW - Step 1



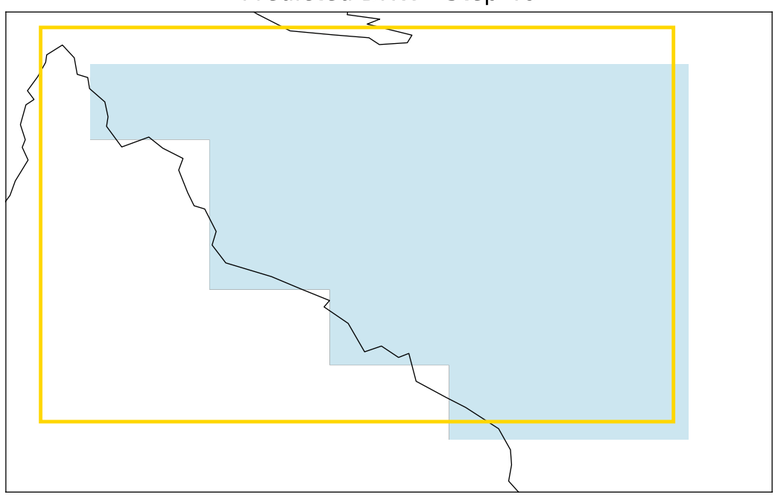
Error (Pred - Actual) - Step 1 [scale ±8]



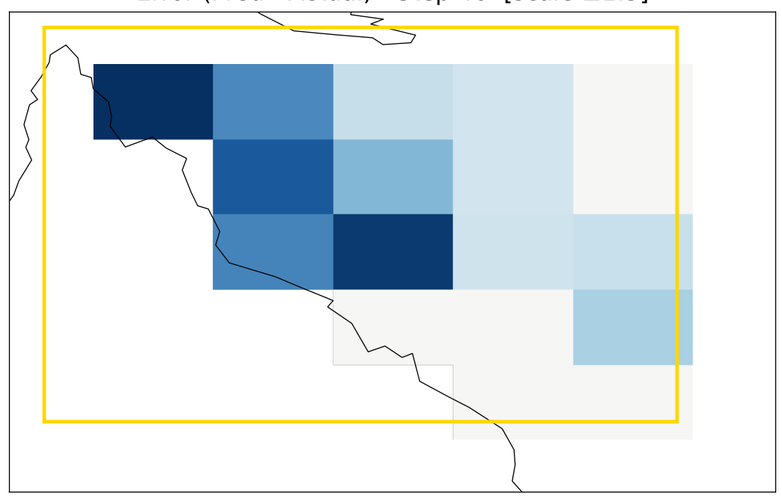
Actual DHW - Step 46



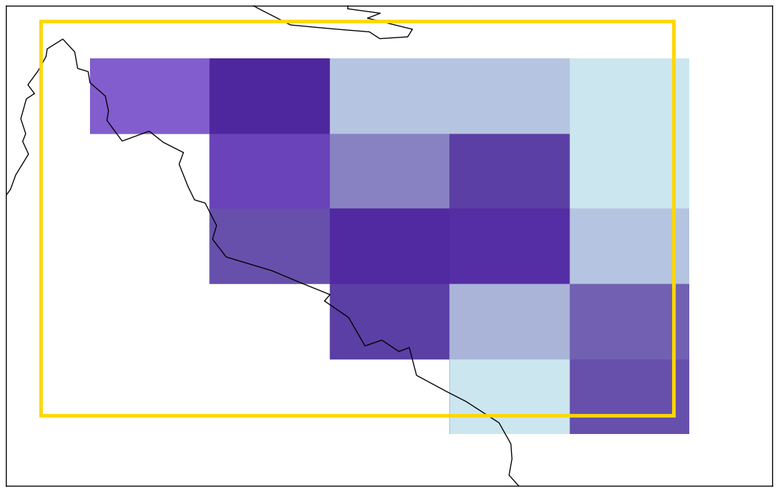
Predicted DHW - Step 46



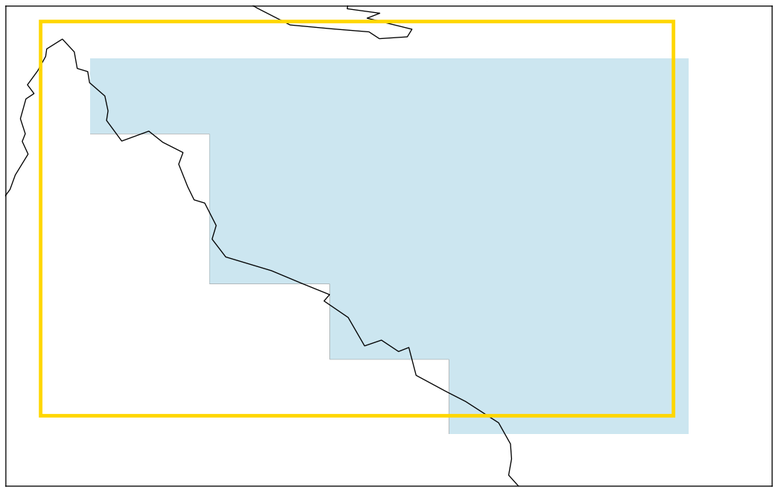
Error (Pred - Actual) - Step 46 [scale ±1.3]



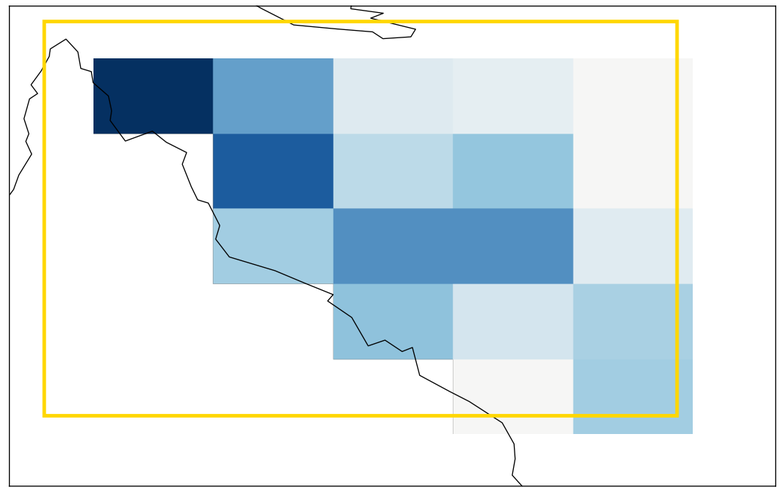
Actual DHW - Step 90

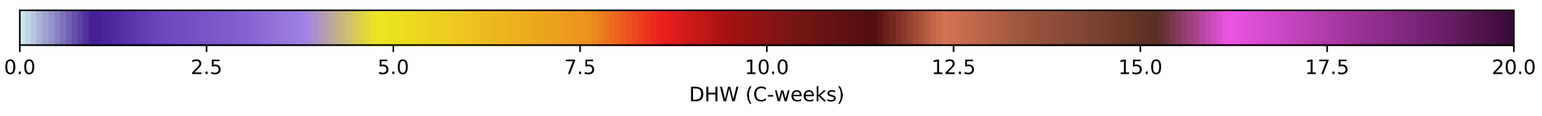


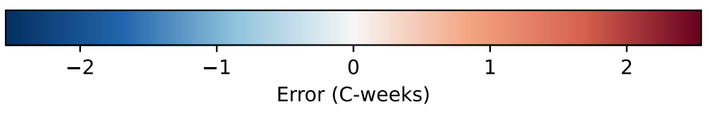
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

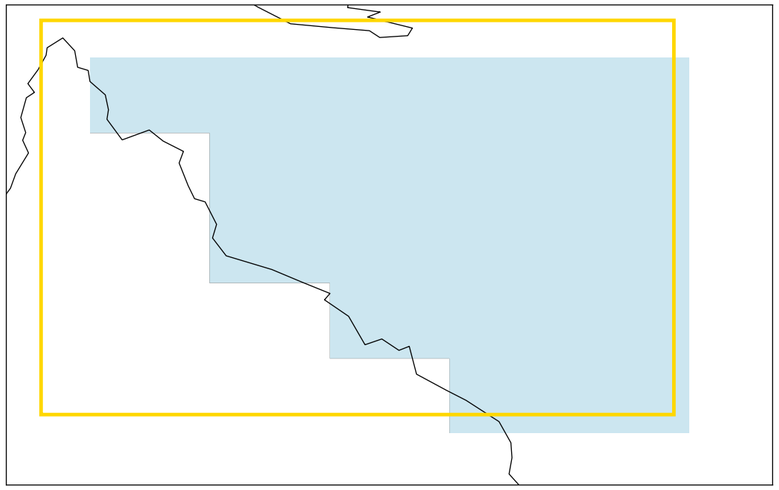




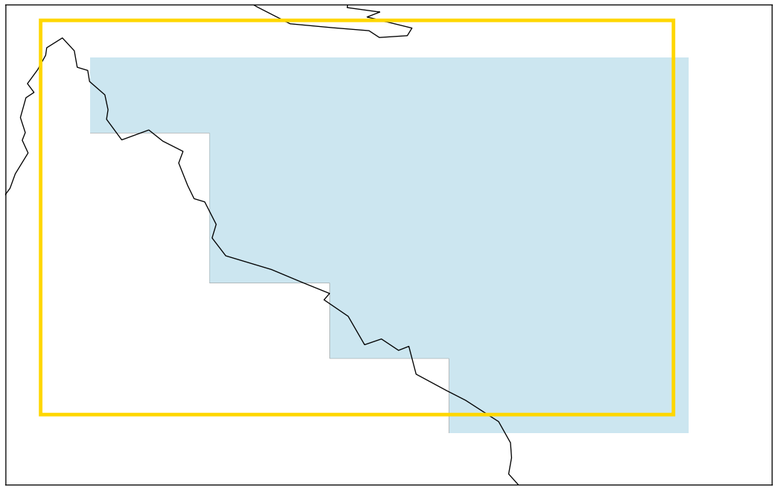


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

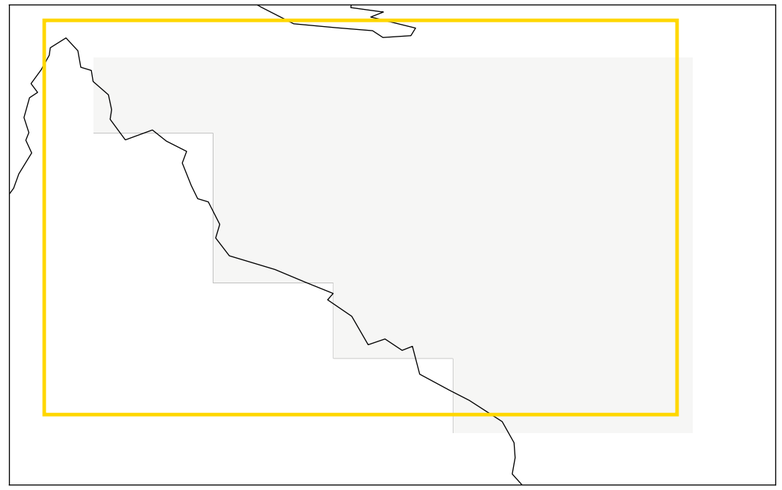
Actual DHW - Step 1



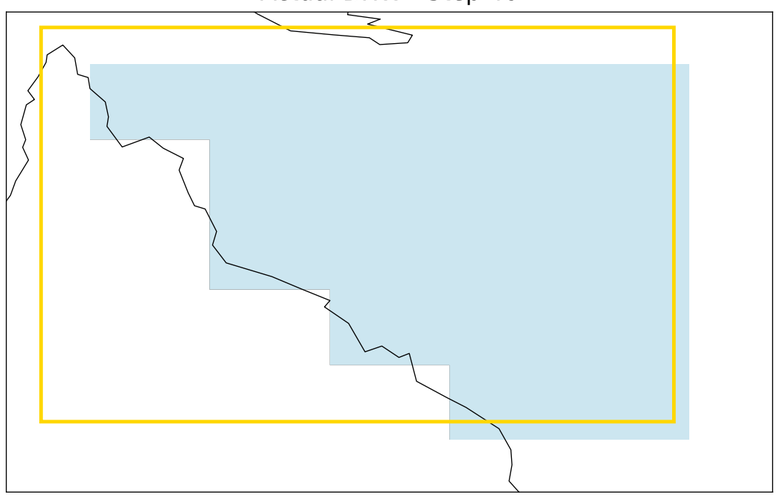
Predicted DHW - Step 1



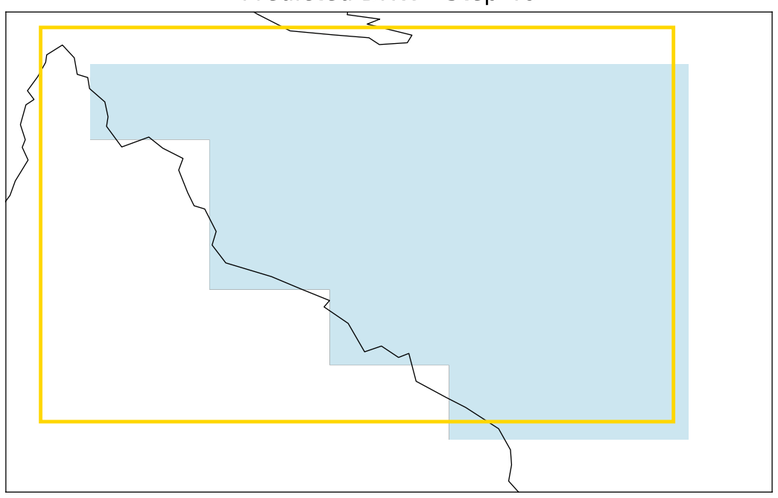
Error (Pred - Actual) - Step 1 [scale ±8]



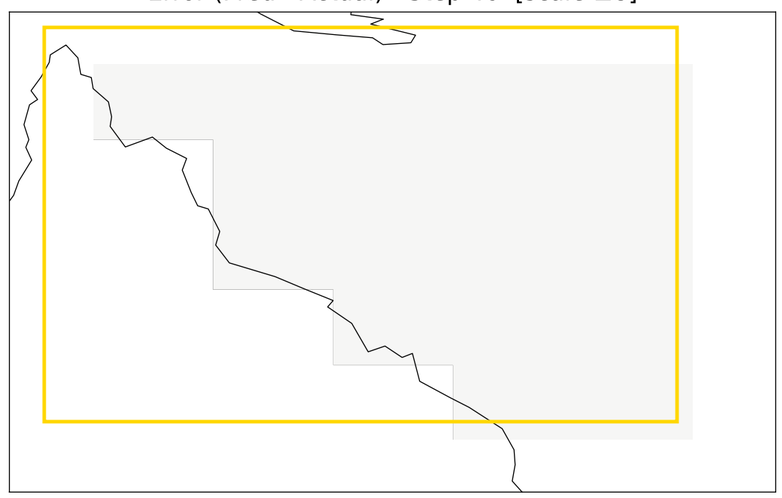
Actual DHW - Step 46



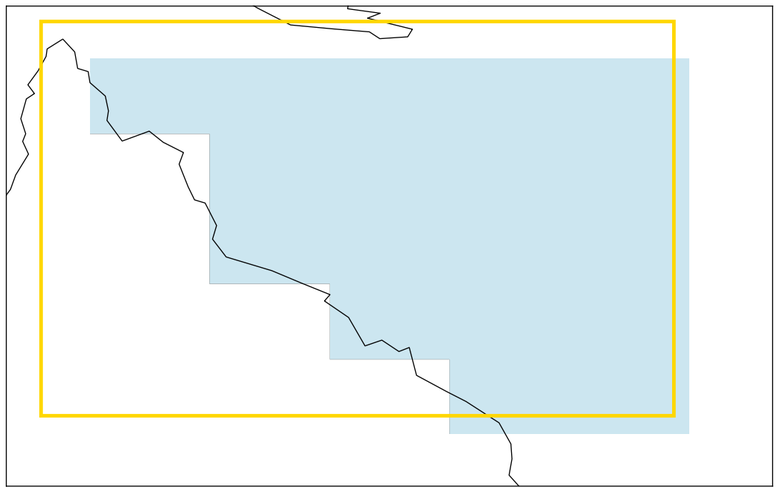
Predicted DHW - Step 46



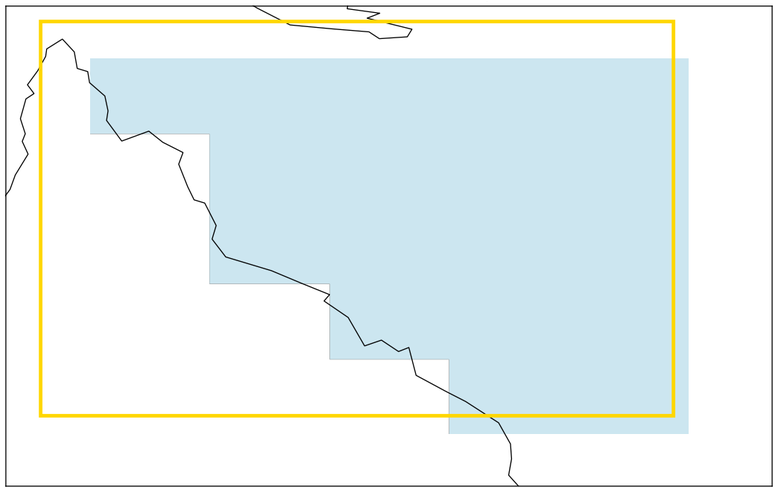
Error (Pred - Actual) - Step 46 [scale ±8]



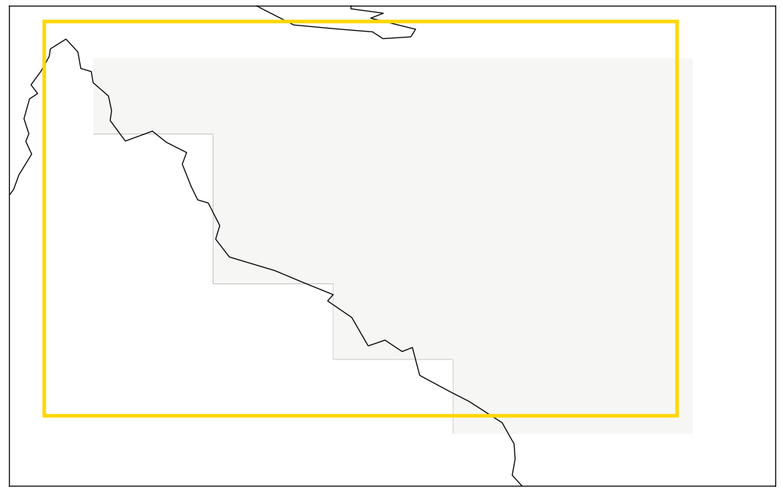
Actual DHW - Step 90

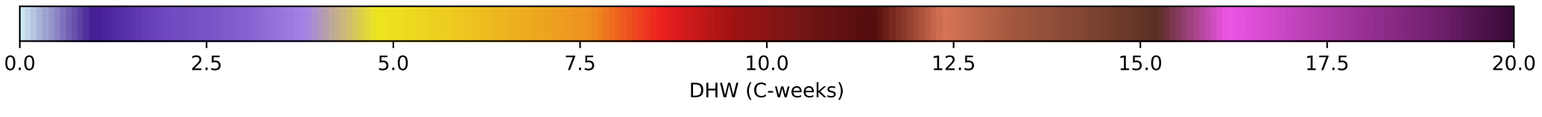


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

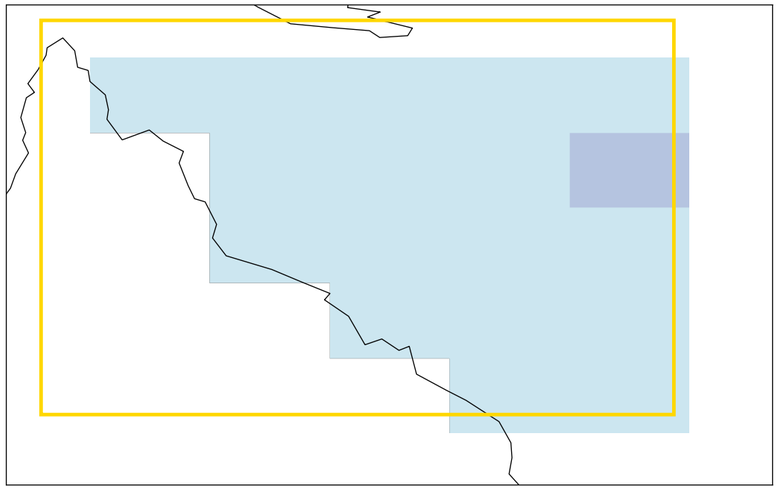




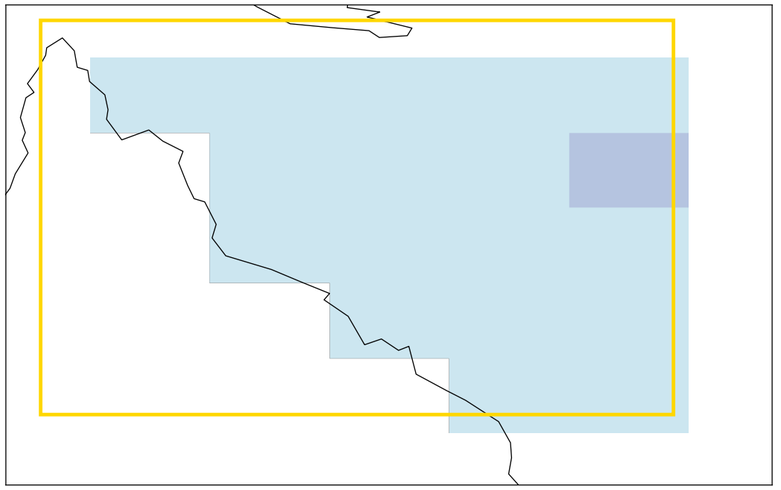


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

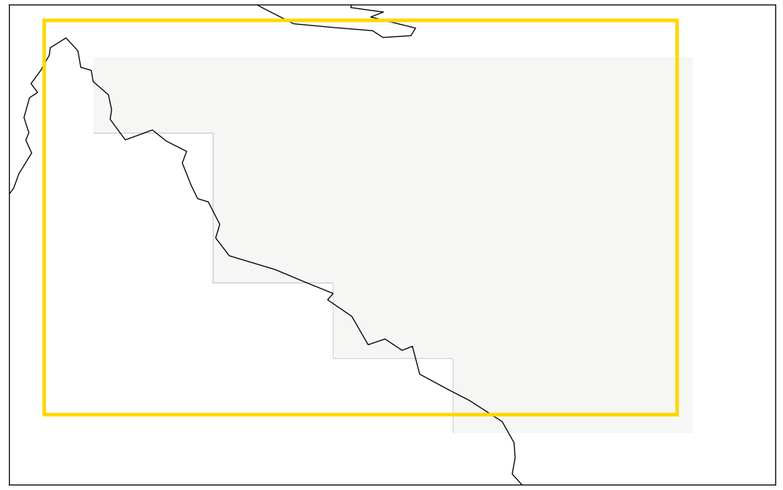
Actual DHW - Step 1



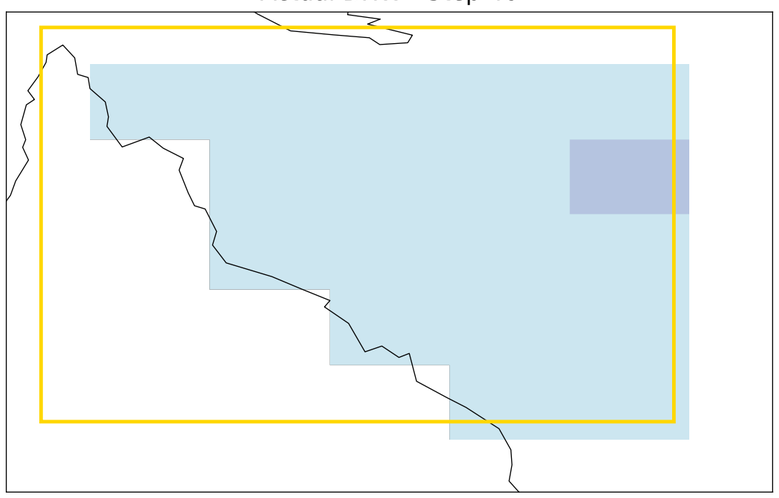
Predicted DHW - Step 1



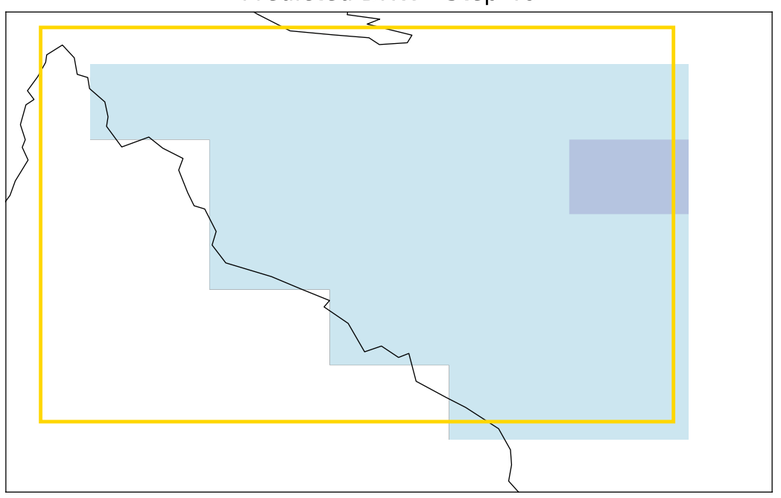
Error (Pred - Actual) - Step 1 [scale ±8]



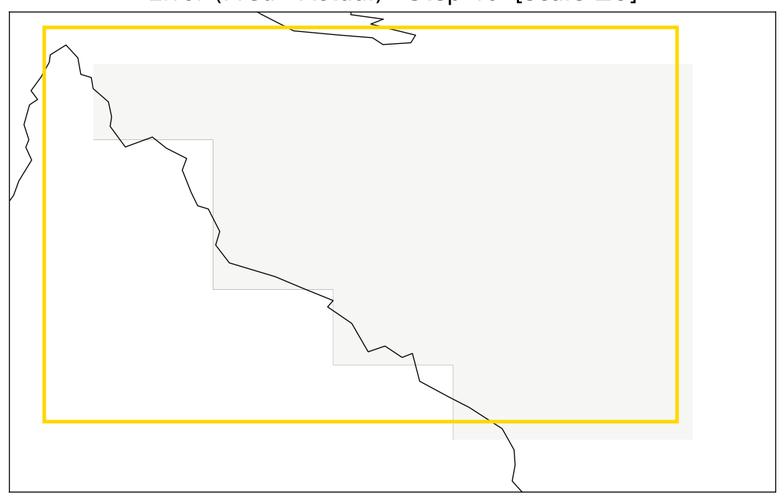
Actual DHW - Step 46



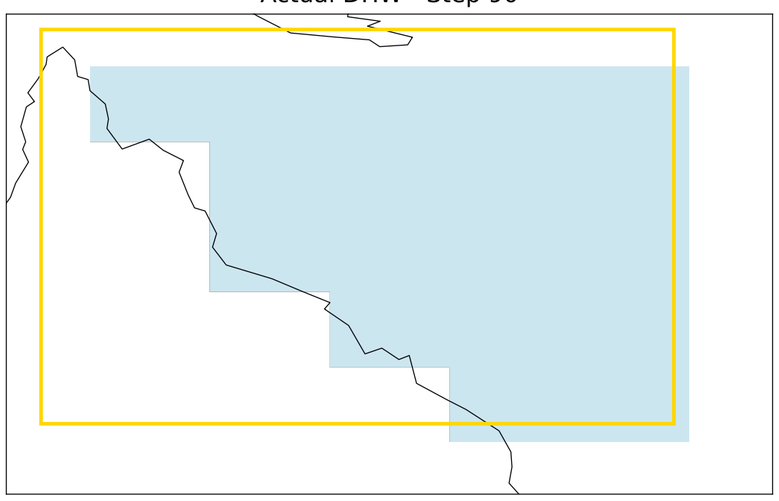
Predicted DHW - Step 46



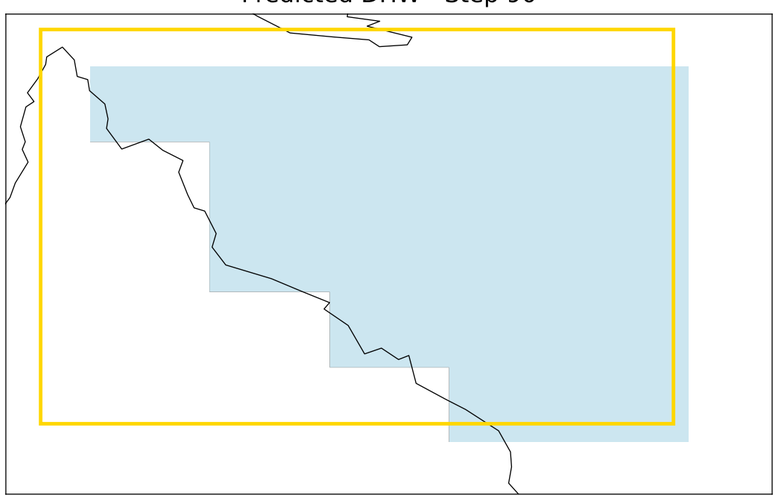
Error (Pred - Actual) - Step 46 [scale ±8]



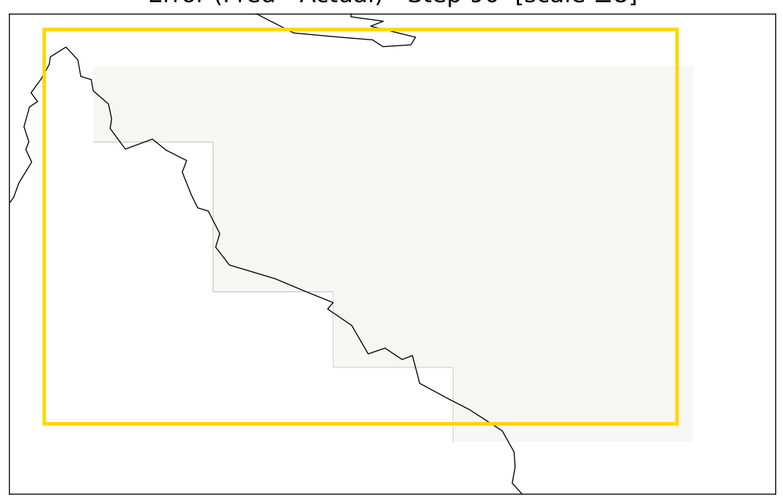
Actual DHW - Step 90

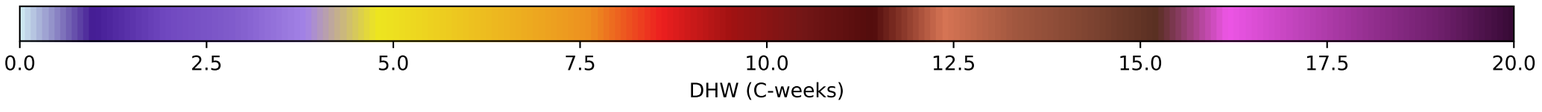


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

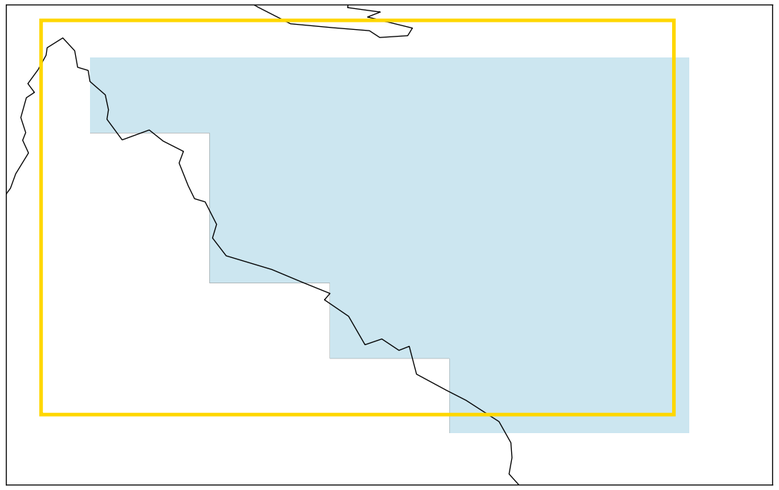




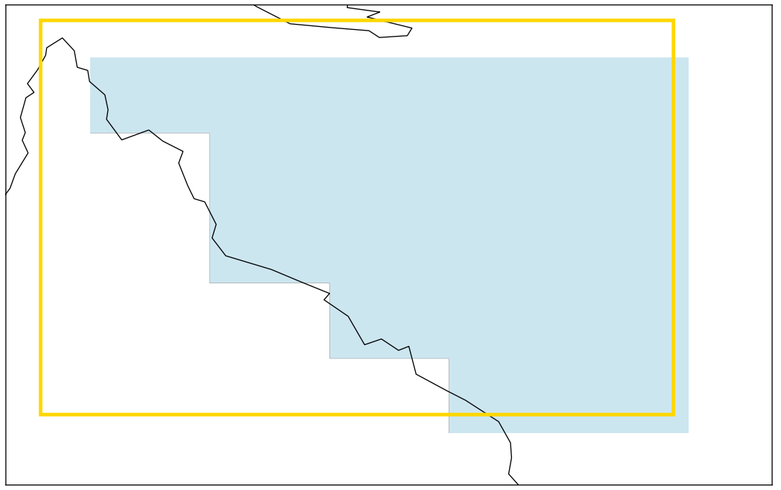


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

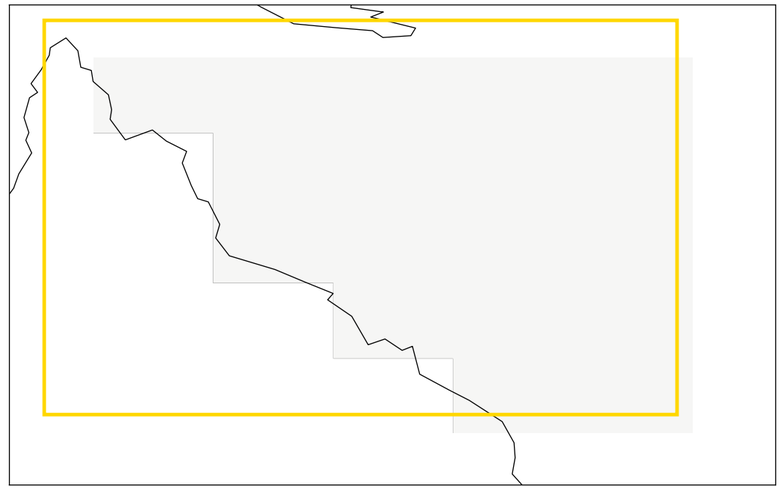
Actual DHW - Step 1



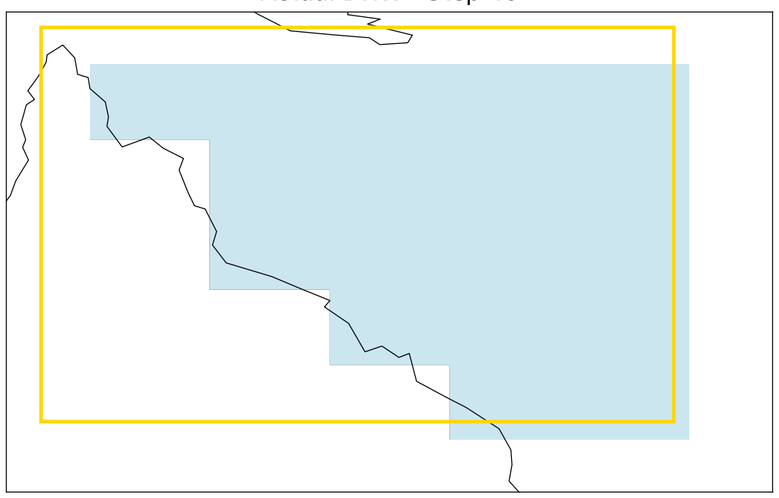
Predicted DHW - Step 1



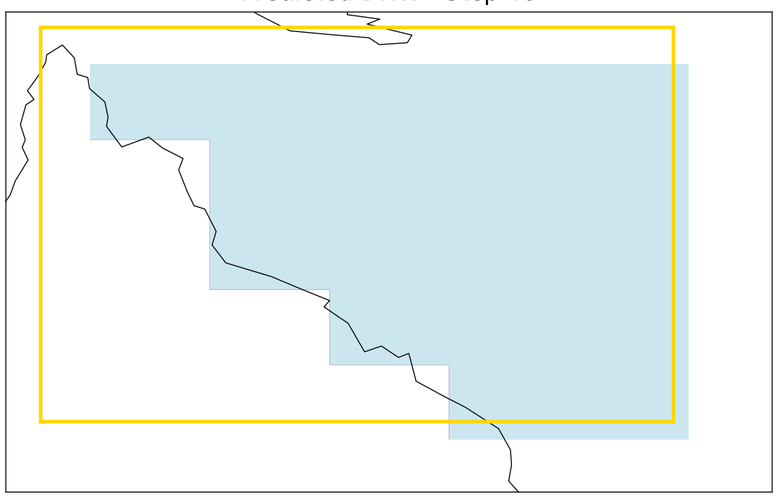
Error (Pred - Actual) - Step 1 [scale ±8]



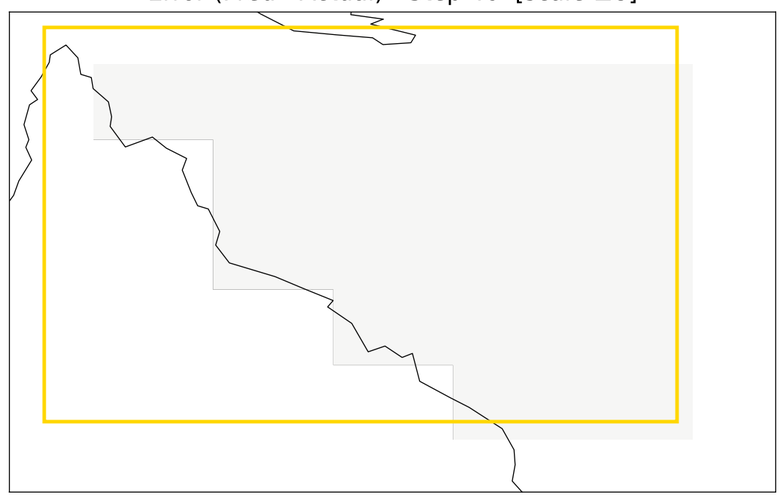
Actual DHW - Step 46



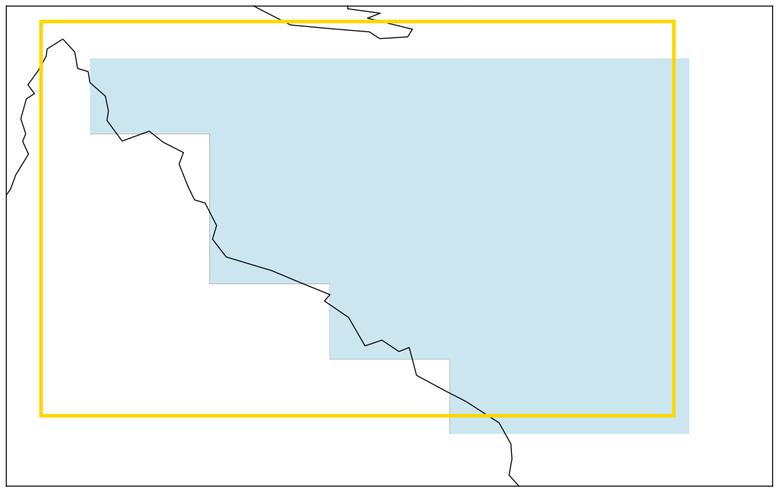
Predicted DHW - Step 46



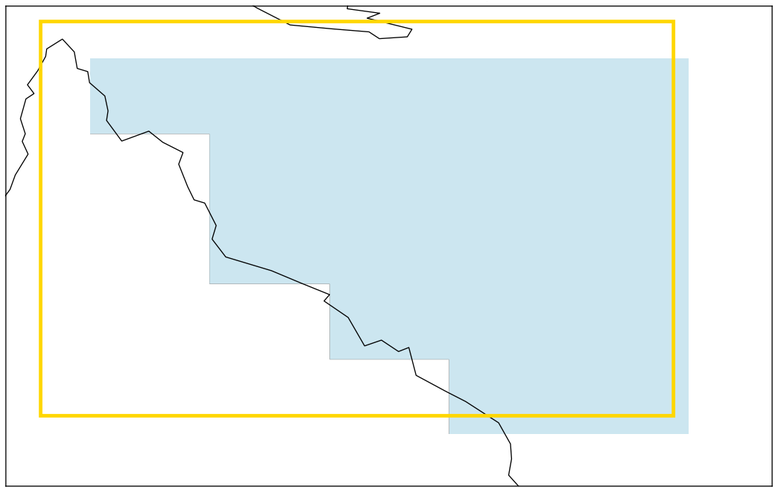
Error (Pred - Actual) - Step 46 [scale ±8]



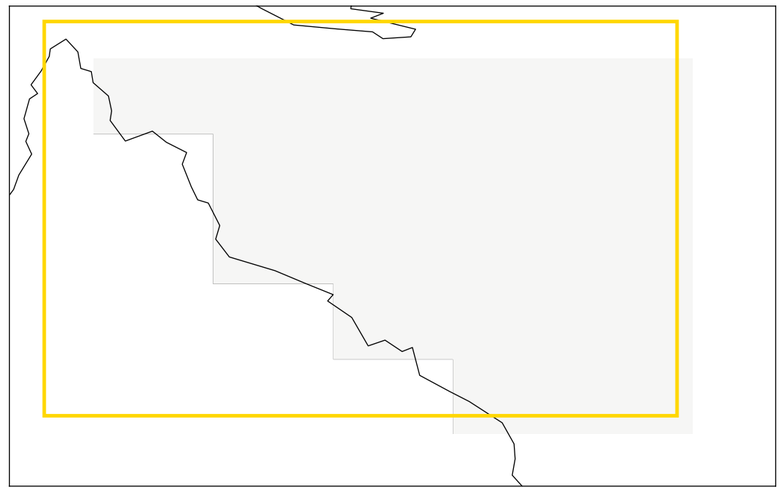
Actual DHW - Step 90

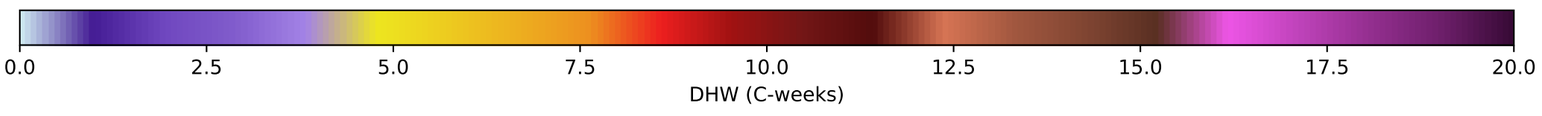


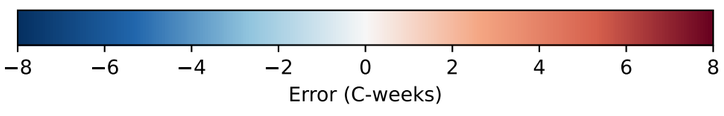
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

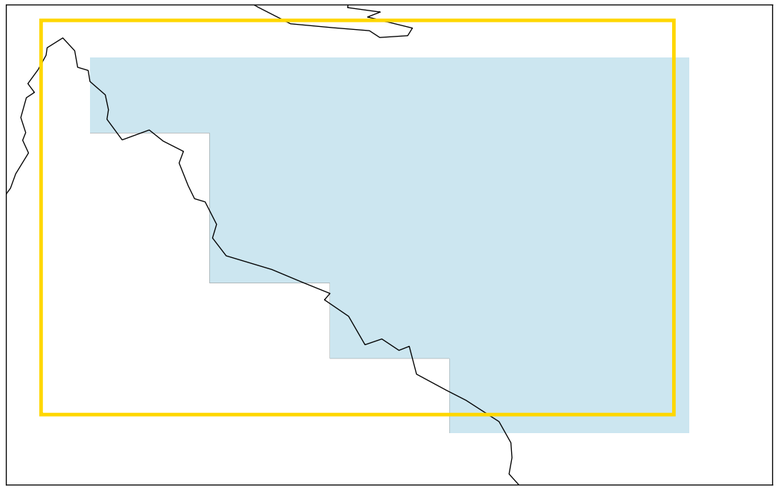




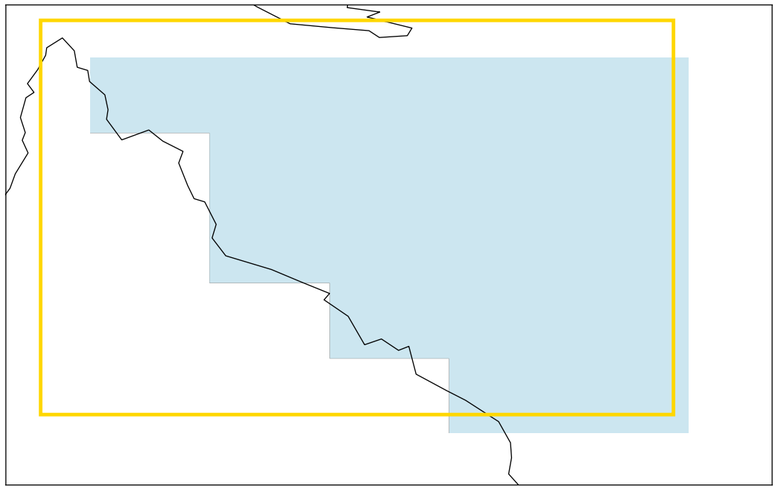


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

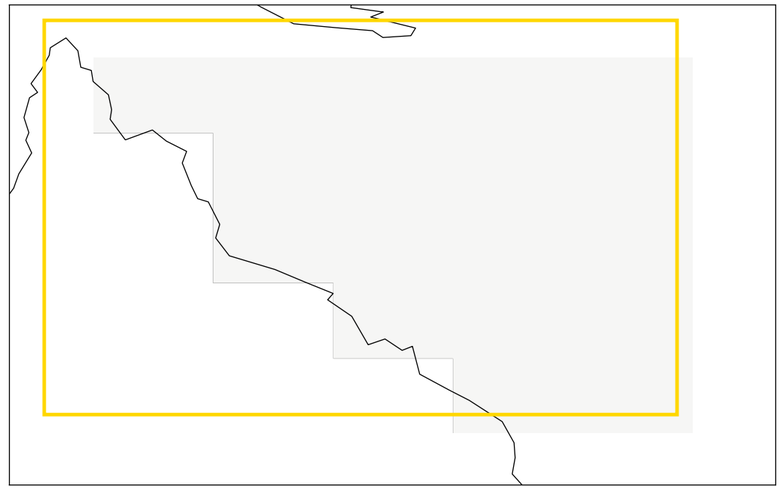
Actual DHW - Step 1



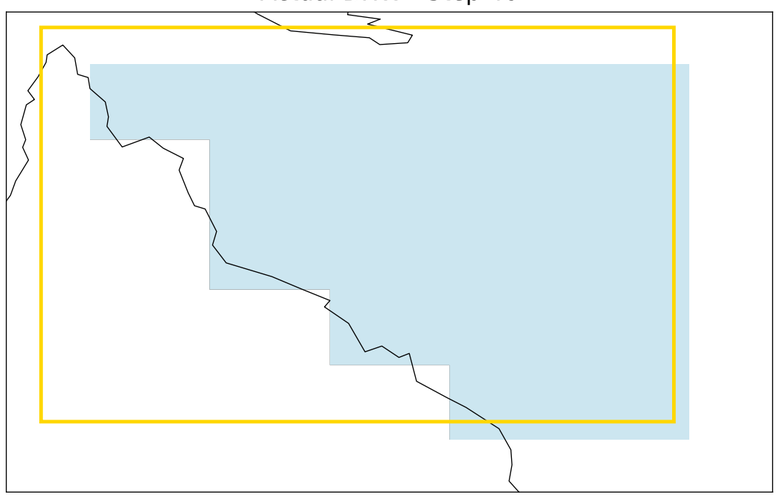
Predicted DHW - Step 1



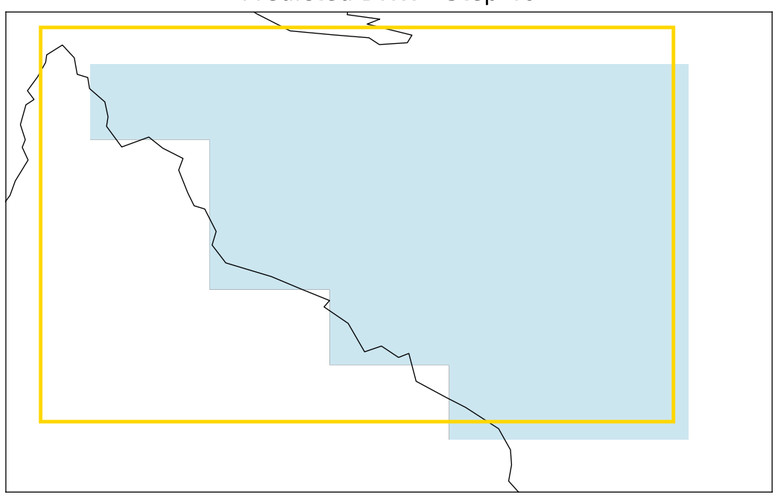
Error (Pred - Actual) - Step 1 [scale ±8]



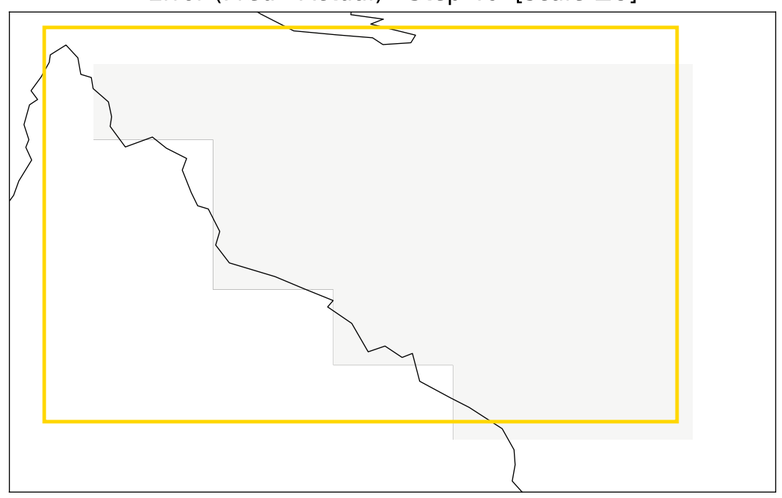
Actual DHW - Step 46



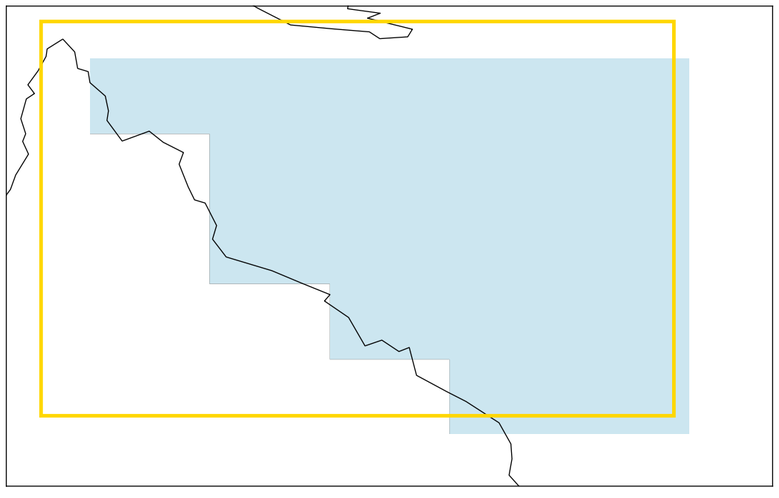
Predicted DHW - Step 46



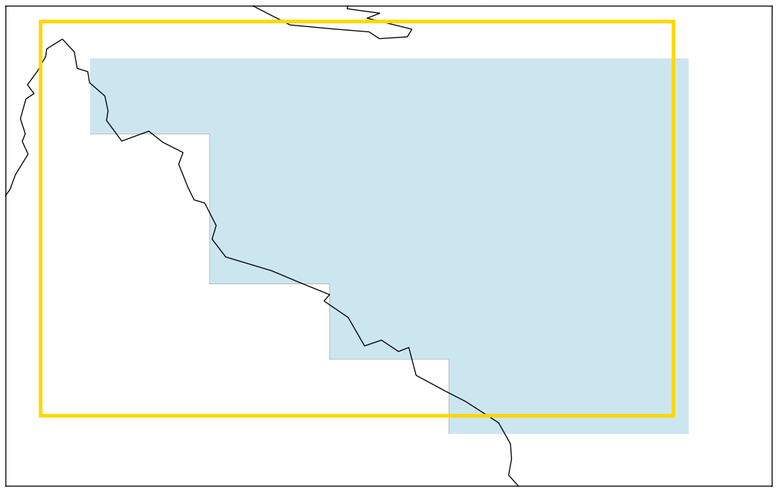
Error (Pred - Actual) - Step 46 [scale ±8]



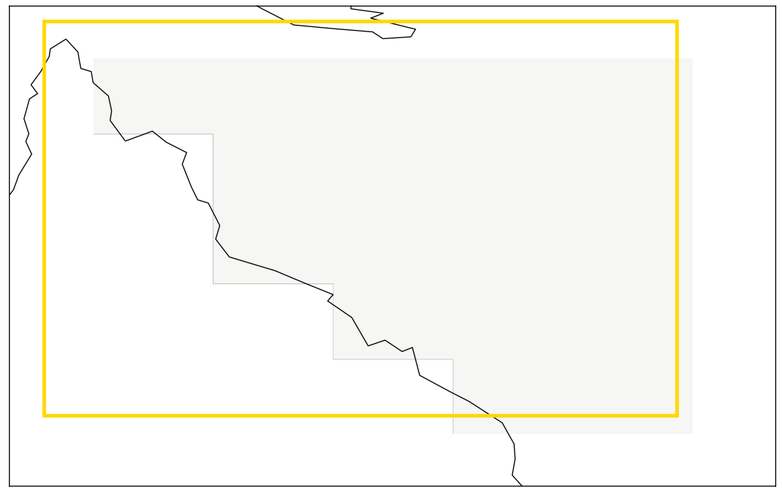
Actual DHW - Step 90

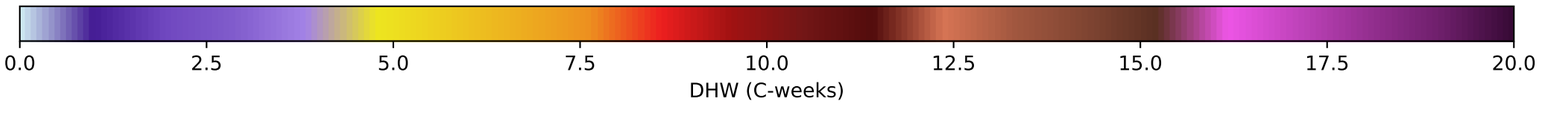


Predicted DHW - Step 90



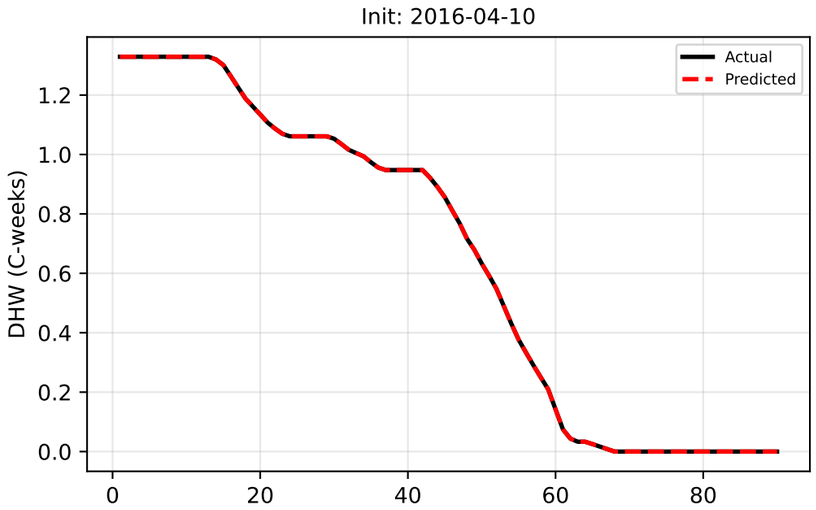
Error (Pred - Actual) - Step 90 [scale ±8]

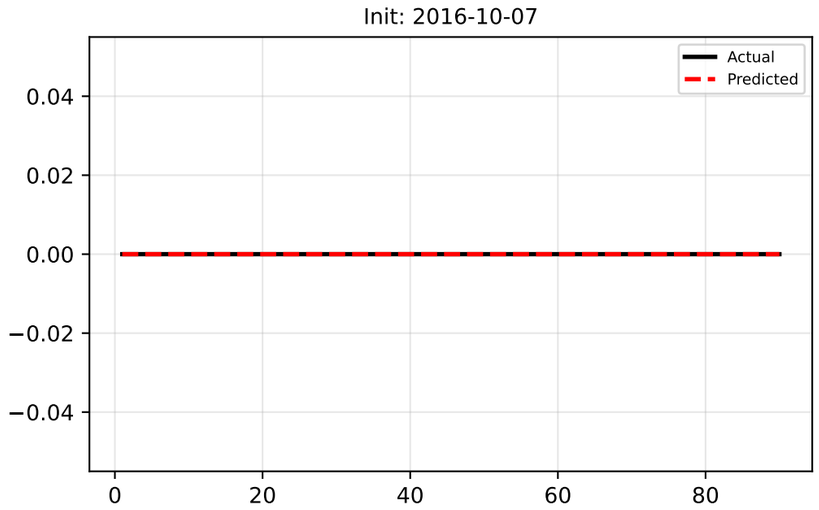


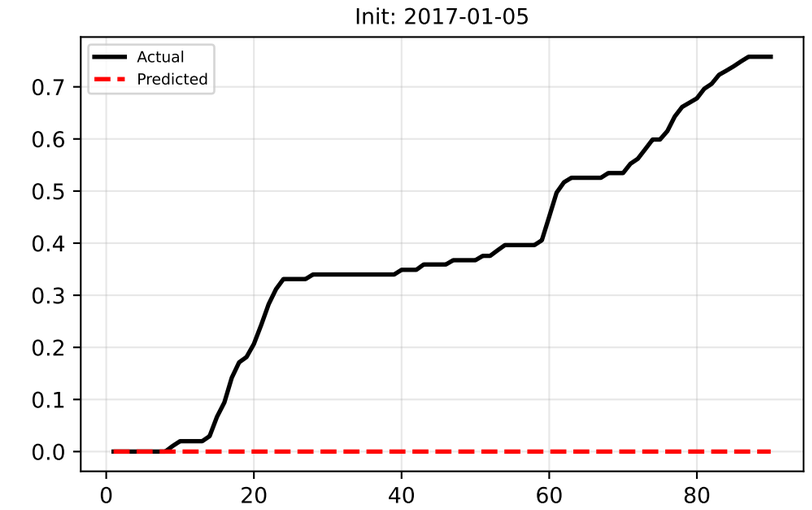


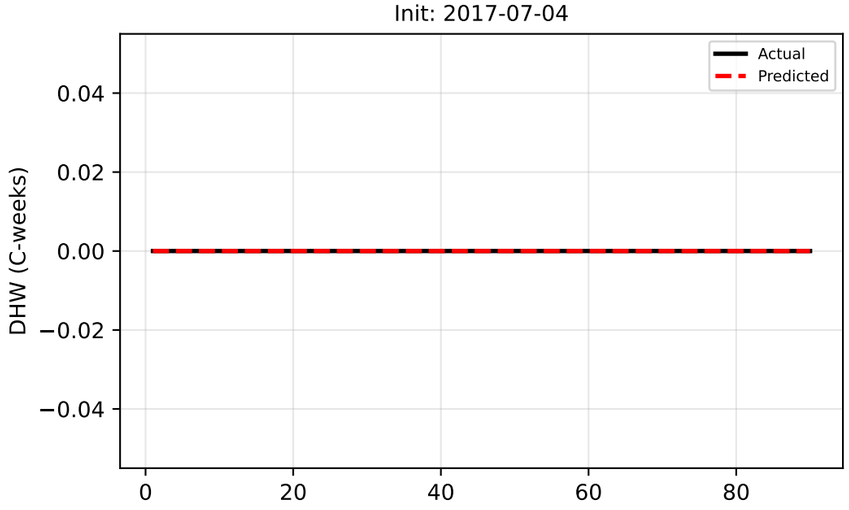


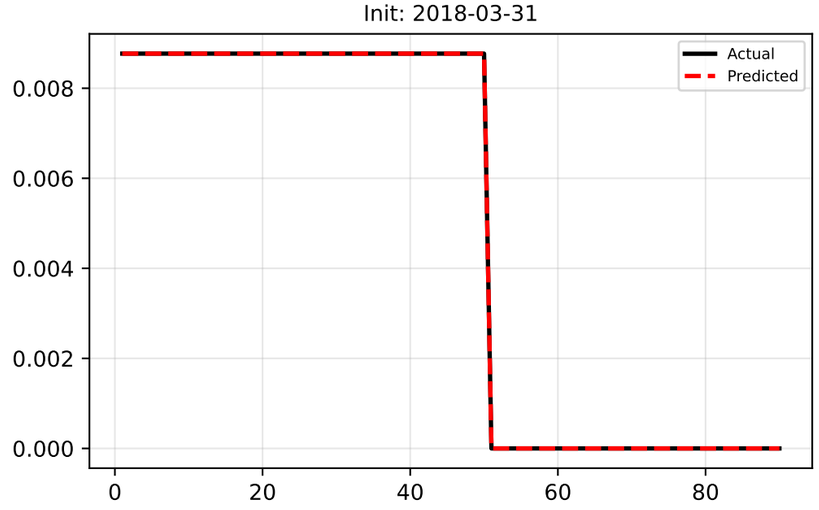
Global-Mean DHW - Val - GBR (C-weeks)

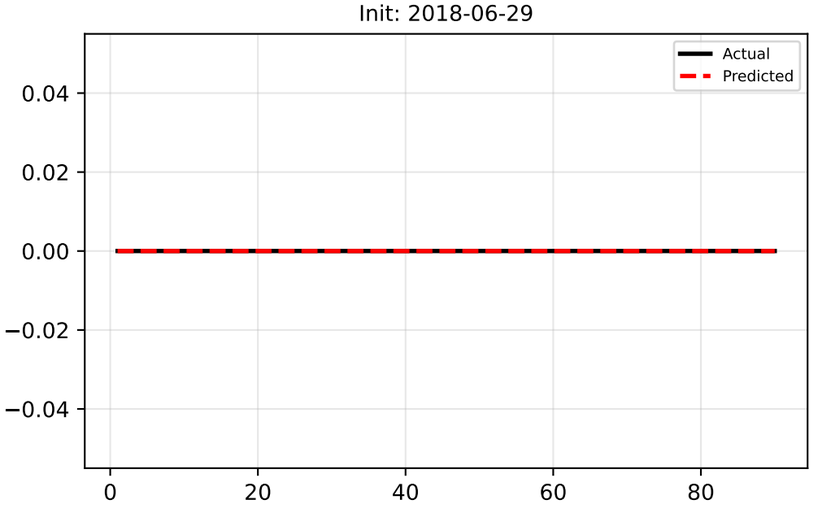


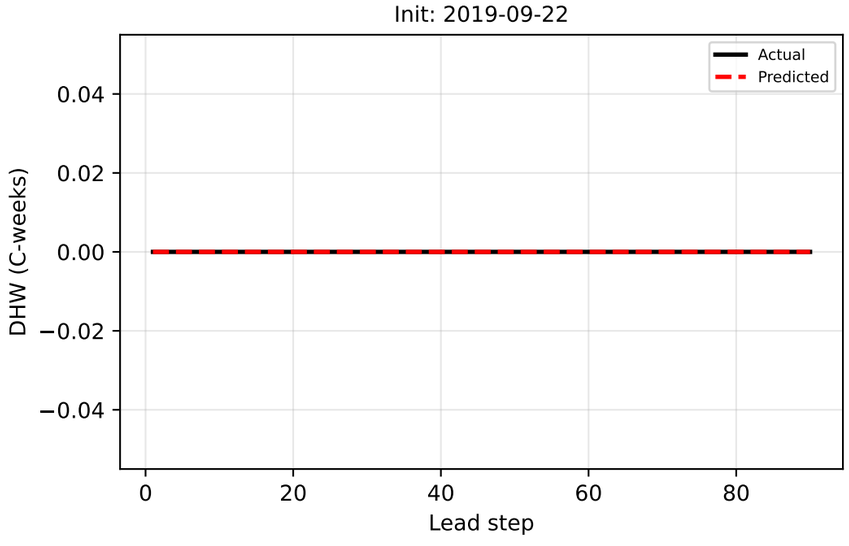


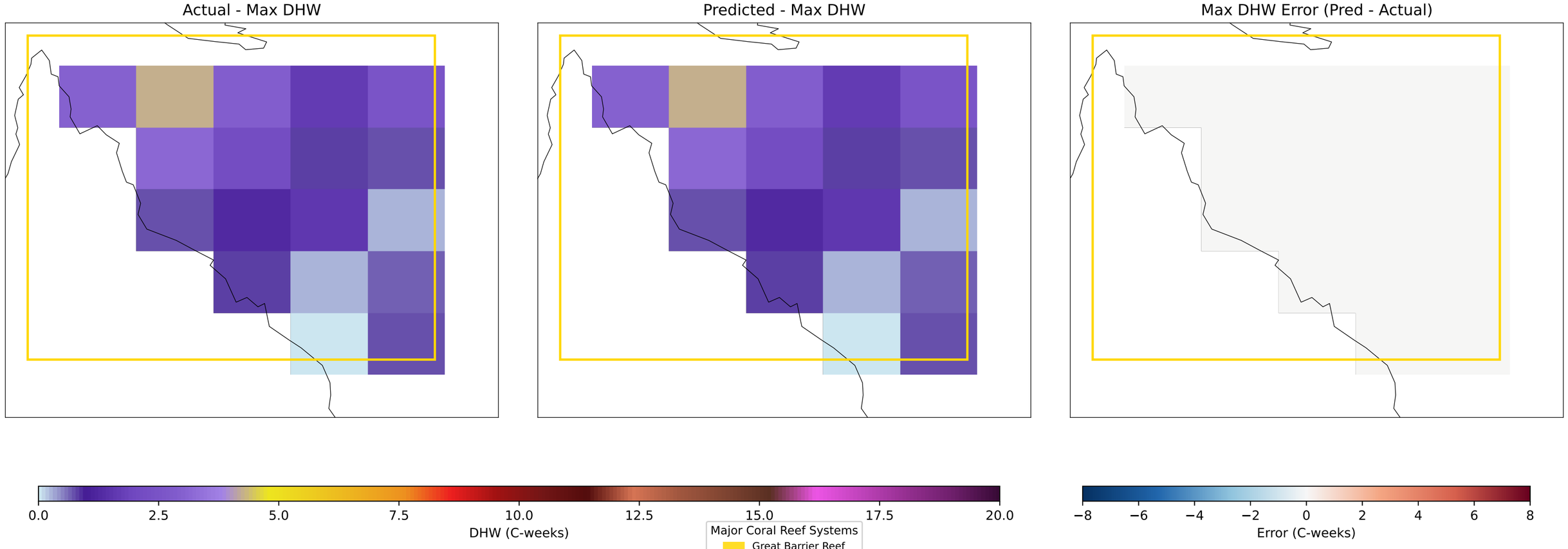


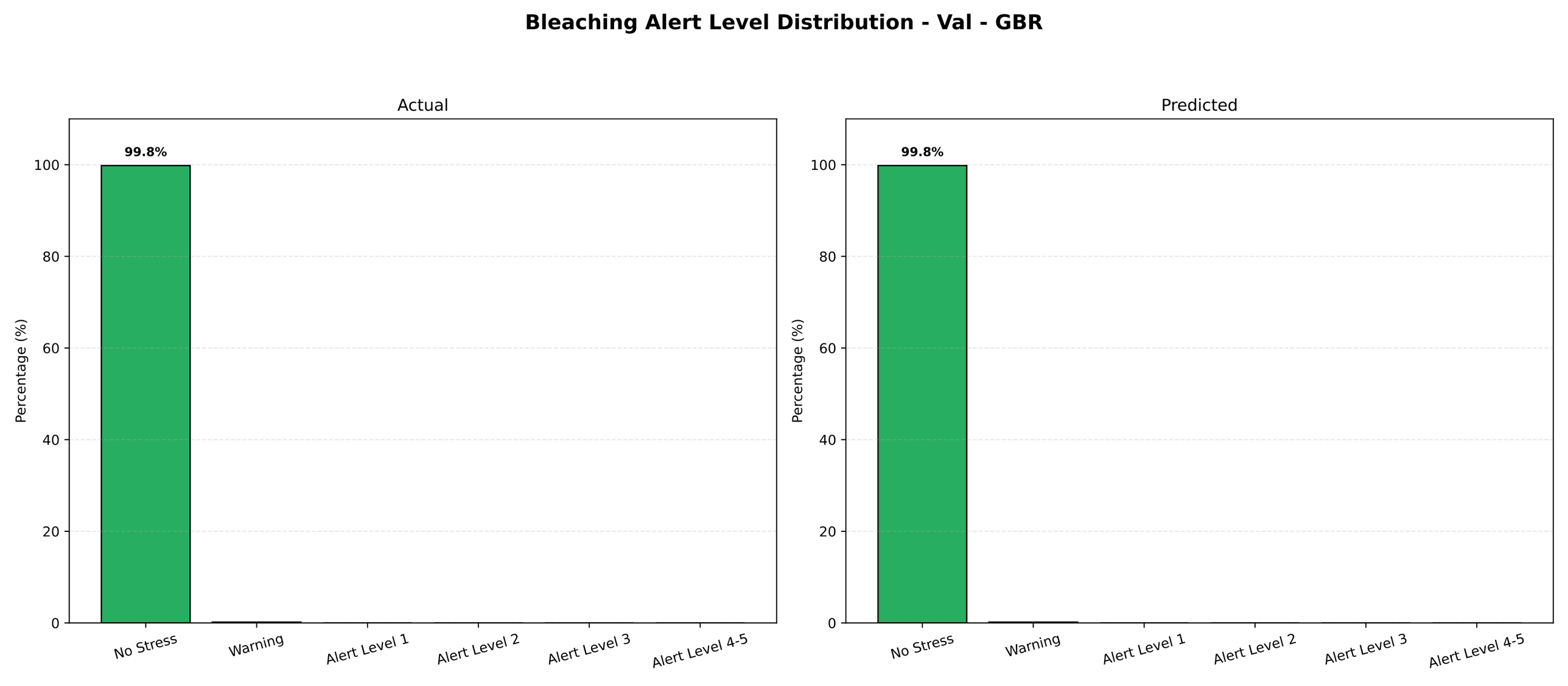






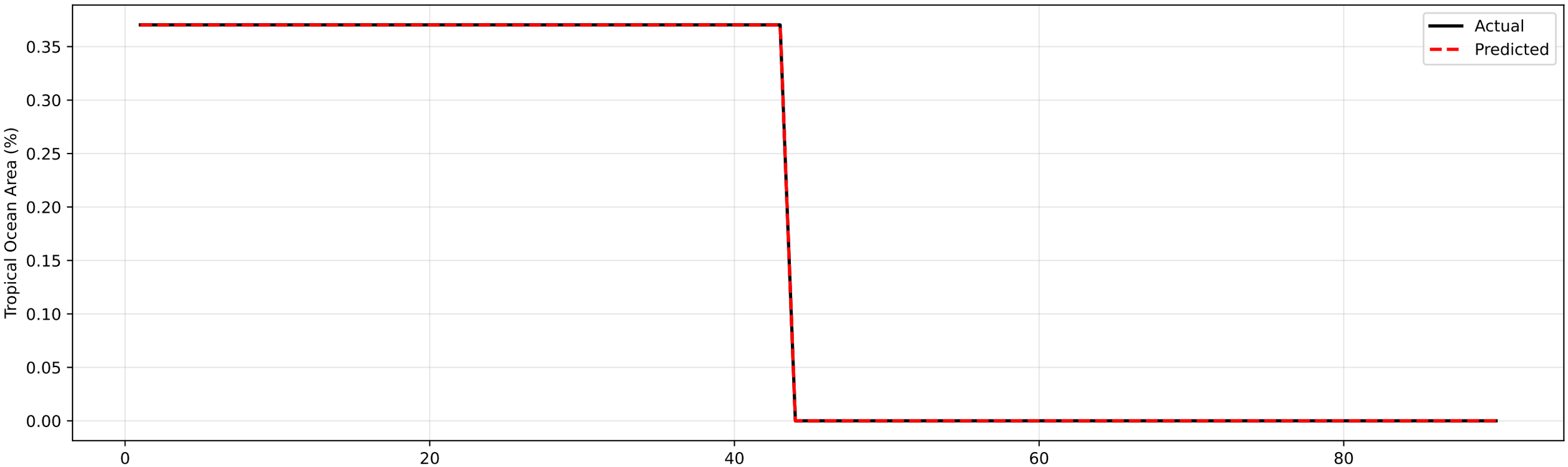



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

